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# MILITARY HANDBOOK

# RELIABILITY PREDICTION OF ELECTRONIC EQUIPMENT




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DEPARTMENT OF DEFENSE
WASHINGTON DC 20301

# RELIABILITY PREDICTION OF ELECTRONIC EQUIPMENT

1. This standardization handbook was developed by the Department of Defense with the assistance of the military departments, federal agencies, and industry.

2. Every effort has been made to reflect the latest information on reliability prediction procedures. It is the intent to review this handbook periodically to ensure its completeness and currency.

3. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Rome Laboratory, AFSC, ATTN: ERSS, Griffiss Air Force Base, New York 13441-5700, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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## FOREWORD

This revision to MIL-HDBK-217 provides the following changes based upon recently completed studies (see Ref. 30 and 32 listed in Appendix C):

1. New failure rate prediction models are provided for the following nine major classes of microcircuits:
    - Monolithic Bipolar Digital and Linear Gate/Logic Array Devices
    - Monolithic MOS Digital and Linear Gate/Logic Array Devices
    - Monolithic Bipolar and MOS Digital Microprocessor Devices (Including Controllers)
    - Monolithic Bipolar and MOS Memory Devices
    - Monolithic GaAs Digital Devices
    - Monolithic GaAs MMIC Devices
    - Hybrid Microcircuits
    - Magnetic Bubble Memories
    - Surface Acoustic Wave Devices

This revision provides new prediction models for bipolar and MOS microcircuits with gate counts up to 60,000, linear microcircuits with up to 3000 transistors, bipolar and MOS digital microprocessor and co-processors up to 32 bits, memory devices with up to 1 million bits, GaAs monolithic microwave integrated circuits (MMICs) with up to 1,000 active elements, and GaAs digital ICs with up to 10,000 transistors. The $C_1$ factors have been extensively revised to reflect new technology devices with improved reliability, and the activation energies representing the temperature sensitivity of the dice ($\pi_T$) have been changed for MOS devices and for memories. The $C_2$ factor remains unchanged from the previous Handbook version, but includes pin grid arrays and surface mount packages using the same model as hermetic, solder-sealed dual in-line packages. New values have been included for the quality factor ($\pi_Q$), the learning factor ($\pi_L$), and the environmental factor ($\pi_E$). The model for hybrid microcircuits has been revised to be simpler to use, to delete the temperature dependence of the seal and interconnect failure rate contributions, and to provide a method of calculating chip junction temperatures.

2. A new model for Very High Speed Integrated Circuits (VHSIC/VHSIC Like) and Very Large Scale Integration (VLSI) devices (gate counts above 60,000).
3. The reformatting of the entire handbook to make it easier to use.
4. A reduction in the number of environmental factors ($\pi_E$) from 27 to 14.
5. A revised failure rate model for Network Resistors.
6. Revised models for TWTs and Klystrons based on data supplied by the Electronic Industries Association Microwave Tube Division.



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## 1.0 SCOPE

**1.1 Purpose** - The purpose of this handbook is to establish and maintain consistent and uniform methods for estimating the inherent reliability (i.e., the reliability of a mature design) of military electronic equipment and systems. It provides a common basis for reliability predictions during acquisition programs for military electronic systems and equipment. It also establishes a common basis for comparing and evaluating reliability predictions of related or competitive designs. The handbook is intended to be used as a tool to increase the reliability of the equipment being designed.

**1.2 Application** - This handbook contains two methods of reliability prediction - "Part Stress Analysis" in Sections 5 through 23 and "Parts Count" in Appendix A. These methods vary in degree of information needed to apply them. The Part Stress Analysis Method requires a greater amount of detailed information and is applicable during the later design phase when actual hardware and circuits are being designed. The Parts Count Method requires less information, generally part quantities, quality level, and the application environment. This method is applicable during the early design phase and during proposal formulation. In general, the Parts Count Method will usually result in a more conservative estimate (i.e., higher failure rate) of system reliability than the Parts Stress Method.

**1.3 Computerized Reliability Prediction** - Rome Laboratory - ORACLE is a computer program developed to aid in applying the part stress analysis procedure of MIL-HDBK-217. Based on environmental use characteristics, piece part count, thermal and electrical stresses, subsystem repair rates and system configuration, the program calculates piece part, assembly and subassembly failure rates. It also flags overstressed parts, allows the user to perform tradeoff analyses and provides system mean-time-to-failure and availability. The ORACLE computer program software (available in both VAX and IBM compatible PC versions) is available at replacement tape/disc cost to all DoD organizations, and to contractors for application on specific DoD contracts as government furnished property (GFP). A statement of terms and conditions may be obtained upon written request to: Rome Laboratory/ERSR, Griffiss AFB, NY 13441-5700.



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## 2.0 REFERENCE DOCUMENTS

This handbook cites some specifications which have been cancelled or which describe devices that are not to be used for new design. This information is necessary because some of these devices are used in so-called "off-the-shelf" equipment which the Department of Defense purchases. The documents cited in this section are for guidance and information.

| SPECIFICATION | SECTION # | TITLE |
|---|---|---|
| MIL-C-5 | 10.7 | Capacitors, Fixed, Mica-Dielectric, General Specification for |
| MIL-R-11 | 9.1 | Resistor, Fixed, Composition (Insulated) General Specification for |
| MIL-R-19 | 9.11 | Resistor, Variable, Wirewound (Low Operating Temperature) General Specification for |
| MIL-C-20 | 10.11 | Capacitor, Fixed, Ceramic Dielectric (Temperature Compensating) Established and Nonestablished Reliability, General Specification for |
| MIL-R-22 | 9.12 | Resistor, Wirewound, Power Type, General Specification for |
| MIL-C-25 | 10.1 | Capacitor, Fixed, Paper-Dielectric, Direct Current (Hermetically Sealed in Metal Cases), General Specification for |
| MIL-R-26 | 9.6 | Resistor, Fixed, Wirewound (Power Type), General Specification for |
| MIL-T-27 | 11.1 | Transformer and Inductor (Audio, Power, High Power, High Power Pulse), General Specification for |
| MIL-C-62 | 10.15 | Capacitor, Fixed Electrolytic (DC, Aluminum, Dry Electrolyte, Polarized), General Specification for |
| MIL-C-81 | 10.16 | Capacitor, Variable, Ceramic Dielectric (Trimmer), General Specification for |
| MIL-C-92 | 10.18 | Capacitor, Variable, Air Dielectric (Trimmer), General Specification for |
| MIL-R-93 | 9.5 | Resistor, Fixed, Wirewound (Accurate), General Specification for |
| MIL-R-94 | 9.14 | Resistor, Variable, Composition, General Specification for |
| MIL-V-95 | 23.1 | Vibrator, Interrupter and Self-Rectifying, General Specification for |
| W-L-111 | 20.1 | Lamp, Incandescent Miniature, Tungsten Filament |
| W-C-375 | 14.5 | Circuit Breaker, Molded Case, Branch Circuit and Service |
| W-F-1726 | 22.1 | Fuse, Cartridge, Class H (This covers renewable and nonrenewable) |
| W-F-1814 | 22.1 | Fuse, Cartridge, High Interrupting Capacity |
| MIL-C-3098 | 19.1 | Crystal Unit, Quartz, General Specification for |
| MIL-C-3607 | 15.1 | Connector, Coaxial, Radio Frequency, Series Pulse, General Specifications for |
| MIL-C-3643 | 15.1 | Connector, Coaxial, Radio Frequency, Series NH, Associated Fittings, General Specification for |
| MIL-C-3650 | 15.1 | Connector, Coaxial, Radio Frequency, Series LC |



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| SPECIFICATION | SECTION # | TITLE |
|---|---|---|
| MIL-C-3655 | 15.1 | Connector, Plug and Receptacle, Electrical (Coaxial Series Twin) and Associated Fittings, General Specification for |
| MIL-C-3767 | 15.1 | Connector, Plug and Receptacle (Power, Bladed Type) General Specification for |
| MIL-S-3786 | 14.3 | Switch, Rotary (Circuit Selector, Low-Current (Capacity)), General Specification for |
| MIL-C-3950 | 14.1 | Switch, Toggle, Environmentally Sealed, General Specification for |
| MIL-C-3965 | 10.13 | Capacitor, Fixed, Electrolytic (Nonsolid Electrolyte), Tantalum, General Specification for |
| MIL-C-5015 | 15.1 | Connector, Electrical, Circular Threaded, AN Type, General Specification for |
| MIL-F-5372 | 22.1 | Fuse, Current Limiter Type, Aircraft |
| MIL-R-5757 | 13.1 | Relay, Electrical (For Electronic and Communication Type Equipment), General Specification for |
| MIL-R-6106 | 13.1 | Relay, Electromagnetic (Including Established Reliability (ER) Types), General Specification for |
| MIL-L-6363 | 20.1 | Lamp, Incandescent, Aviation Service, General Requirement for |
| MIL-S-8805 | 14.1, 14.2 | Switches and Switch Assemblies, Sensitive and Push, (Snap Action) General Specification for |
| MIL-S-8834 | 14.1 | Switches, Toggle, Positive Break, General Specification for |
| MIL-M-10304 | 18.1 | Meter, Electrical Indicating, Panel Type, Ruggedized, General Specification for |
| MIL-R-10509 | 9.2 | Resistor, Fixed Film (High Stability), General Specification for |
| MIL-C-10950 | 10.8 | Capacitor, Fixed, Mica Dielectric, Button Style, General Specification for |
| MIL-C-11015 | 10.10 | Capacitor, Fixed, Ceramic Dielectric (General Purpose), General Specification for |
| MIL-C-11272 | 10.9 | Capacitor, Fixed, Glass Dielectric, General Specification for |
| MIL-C-11693 | 10.2 | Capacitor, Feed Through, Radio Interference Reduction AC and DC, (Hermetically Sealed in Metal Cases) Established and Nonestablished Reliability, General Specification for |
| MIL-R-11804 | 9.3 | Resistor, Fixed, Film (Power Type), General Specification for |
| MIL-C-12889 | 10.1 | Capacitor, By-Pass, Radio - Interference Reduction, Paper Dielectric, AC and DC, (Hermetically Sealed in Metallic Cases), General Specification for |
| MIL-R-12934 | 9.10 | Resistor, Variable, Wirewound, Precision, General Specification for |



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| SPECIFICATION | SECTION # | TITLE |
|---|---|---|
| MIL-C-14157 | 10.3 | Capacitor, Fixed, Paper (Paper Plastic) or Plastic Dielectric, Direct Current (Hermetically Sealed in Metal Cases) Established Reliability, General Specification for |
| MIL-C-14409 | 10.17 | Capacitor, Variable (Piston Type, Tubular Trimmer), General Specification for |
| MIL-F-15160 | 22.1 | Fuse, Instrument, Power and Telephone |
| MIL-C-15305 | 11.2 | Coil, Fixed and Variable, Radio Frequency, General Specification for |
| MIL-F-15733 | 21.1 | Filter, Radio Interference, General Specification for |
| MIL-C-18312 | 10.4 | Capacitor, Fixed, Metallized (Paper, Paper Plastic or Plastic Film) Dielectric, Direct Current (Hermetically Sealed in Metal Cases), General Specification for |
| MIL-F-18327 | 21.1 | Filter, High Pass, Low Pass, Band Pass, Band Suppression and Dual Functioning, General Specification for |
| MIL-R-18546 | 9.7 | Resistor, Fixed, Wirewound (Power Type, Chassis Mounted), General Specification for |
| MIL-S-19500 | 6.0 | Semiconductor Device, General Specification for |
| MIL-R-19523 | 13.1 | Relay, Control, Naval Shipboard |
| MIL-R-19648 | 13.1 | Relay, Time, Delay, Thermal, General Specification for |
| MIL-C-19978 | 10.3 | Capacitor, Fixed Plastic (or Paper-Plastic) Dielectric (Hermetically Sealed in Metal, Ceramic or Glass Cases), Established and Nonestablished Reliability, General Specification for |
| MIL-T-21038 | 11.1 | Transformer, Pulse, Low Power, General Specification for |
| MIL-C-21097 | 15.2 | Connector, Electrical, Printed Wiring Board, General Purpose, General Specification for |
| MIL-R-22097 | 9.13 | Resistor, Variable, Nonwirewound (Adjustment Types), General Specification for |
| MIL-R-22684 | 9.2 | Resistor, Fixed, Film, Insulated, General Specification for |
| MIL-S-22710 | 14.4 | Switch, Rotary (Printed Circuit), (Thumbwheel, In-line and Pushbutton), General Specification for |
| MIL-S-22885 | 14.1 | Switches, Pushbutton, Illuminated, General Specification for |
| MIL-C-22992 | 15.1 | Connector, Cylindrical, Heavy Duty, General Specification for |
| MIL-C-23183 | 10.19 | Capacitor, Fixed or Variable, Vacuum Dielectric, General Specification for |
| MIL-C-23269 | 10.9 | Capacitor, Fixed, Glass Dielectric, Established Reliability, General Specification for |
| MIL-R-23285 | 9.15 | Resistor, Variable, Nonwirewound, General Specification for |



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| SPECIFICATION | SECTION # | TITLE |
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| MIL-F-23419 | 22.1 | Fuse, Instrument Type, General Specification for |
| MIL-T-23648 | 9.8 | Thermistor, (Thermally Sensitive Resistor), Insulated, General Specification for |
| MIL-C-24308 | 15.1 | Connector, Electric, Rectangular, Miniature Polarized Shell, Rack and Panel, General Specification for |
| MIL-C-25516 | 15.1 | Connector, Electrical, Miniature, Coaxial, Environment Resistant Type, General Specification for |
| MIL-C-26482 | 15.1 | Connector, Electrical (Circular, Miniature, Quick Disconnect, Environment Resisting) Receptacles and Plugs, General Specification for |
| MIL-R-27208 | 9.9 | Resistor, Variable, Wirewound, (Lead Screw Activated) General Specification for |
| MIL-C-28748 | 15.1 | Connector, Electrical, Rectangular, Rack and Panel, Solder Type and Crimp Type Contacts, General Specification for |
| MIL-R-28750 | 13.2 | Relay, Solid State, General Specification for |
| MIL-C-28804 | 15.1 | Connector, Electric Rectangular, High Density, Polarized Central Jackscrew, General Specification for, Inactive for New Designs |
| MIL-C-28840 | 15.1 | Connector, Electrical, Circular Threaded, High Density, High Shock Shipboard, Class D, General Specification for |
| MIL-M-38510 | 5.0 | Microcircuits, General Specification for |
| MIL-H-38534 | 5.0 | Hybrid Microcircuits, General Specification for |
| MIL-I-38535 | 5.0 | Integrated Circuits (Microcircuits) Manufacturing, General Specification for |
| MIL-C-38999 | 15.1 | Connector, Electrical, Circular, Miniature, High Density, Quick Disconnect, (Bayonet, Threaded, and Breech Coupling) Environment Resistant, Removable Crimp and Hermetic Solder Contacts, General Specification for |
| MIL-C-39001 | 10.7 | Capacitor, Fixed, Mica Dielectric, Established Reliability, General Specification for |
| MIL-R-39002 | 9.11 | Resistor, Variable, Wirewound, Semi-Precision, General Specification for |
| MIL-C-39003 | 10.12 | Capacitor, Fixed, Electrolytic, (Solid Electrolyte), Tantalum, Established Reliability, General Specification for |
| MIL-R-39005 | 9.5 | Resistor, Fixed, Wirewound, (Accurate) Established Reliability, General Specification for |
| MIL-C-39006 | 10.13 | Capacitor, Fixed, Electrolytic (Nonsolid Electrolyte) Tantalum Established Reliability, General Specification for |
| MIL-R-39007 | 9.6 | Resistor, Fixed, Wirewound (Power Type) Established Reliability, General Specification for |



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## 2.0 REFERENCE DOCUMENTS

| SPECIFICATION | SECTION # | TITLE |
|---|---|---|
| MIL-R-39008 | 9.1 | Resistor, Fixed, Composition, (Insulated) Established Reliability, General Specification for |
| MIL-R-39009 | 9.7 | Resistor, Fixed, Wirewound (Power Type, Chassis Mounted) Established Reliability, General Specification for |
| MIL-C-39010 | 11.2 | Coil, Fixed, Radio Frequency, Molded, Established Reliability, General Specification for |
| MIL-C-39012 | 15.1 | Connector, Coaxial, Radio Frequency, General Specification for |
| MIL-C-39014 | 10.10 | Capacitor, Fixed, Ceramic Dielectric (General Purpose) Established Reliability, General Specification for |
| MIL-C-39015 | 9.9 | Resistor, Variable, Wirewound (Lead Screw Actuated) Established Reliability, General Specification for |
| MIL-R-39016 | 13.1 | Relay, Electromagnetic, Established Reliability, General Specification for |
| MIL-R-39017 | 9.2 | Resistor, Fixed, Film (Insulated), Established Reliability, General Specification for |
| MIL-C-39018 | 10.14 | Capacitor, Fixed, Electrolytic (Aluminum Oxide) Established Reliability and Nonestablished Reliability, General Specification for |
| MIL-C-39019 | 14.5 | Circuit Breakers, Magnetic, Low Power, Sealed, Trip-Free, General Specification for |
| MIL-C-39022 | 10.4 | Capacitor, Fixed, Metallized Paper, Paper-Plastic Film, or Plastic Film Dielectric, Direct and Alternating Current (Hermetically Sealed in Metal Cases) Established Reliability, General Specification for |
| MIL-R-39023 | 9.15 | Resistor, Variable, Nonwirewound, Precision, General Specification for |
| MIL-R-39035 | 9.13 | Resistor, Variable, Nonwirewound, (Adjustment Type) Established Reliability, General Specification for |
| MIL-C-49142 | 15.1 | Connector, Triaxial, RF, General Specification for |
| MIL-P-55110 | 15.2 | Printed Wiring Boards |
| MIL-R-55182 | 9.2 | Resistor, Fixed, Film, Established Reliability, General Specification for |
| MIL-C-55235 | 15.1 | Connector, Coaxial, RF, General Specification for |
| MIL-C-55302 | 15.2 | Connector, Printed Circuit, Subassembly and Accessories |
| MIL-C-55339 | 15.1 | Adapter, Coaxial, RF, General Specification for |
| MIL-C-55514 | 10.5 | Capacitor, Fixed, Plastic (or Metallized Plastic) Dielectric, Direct Current, In Non-Metal Cases, General Specification for |
| MIL-C-55629 | 14.5 | Circuit Breaker, Magnetic, Unsealed, Trip-Free, General Specification for |
| MIL-T-55631 | 11.1 | Transformer, Intermediate Frequency, Radio Frequency, and Discriminator, General Specification for |



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## 2.0 REFERENCE DOCUMENTS

| SPECIFICATION | SECTION # | TITLE |
|---|---|---|
| MIL-C-55681 | 10.11 | Capacitor, Chip, Multiple Layer, Fixed, Ceramic Dielectric, Established Reliability, General Specification for |
| MIL-C-81511 | 15.1 | Connector, Electrical, Circular, High Density, Quick Disconnect, Environment Resisting, and Accessories, General Specification for |
| MIL-C-83383 | 14.5 | Circuit Breaker, Remote Control, Thermal, Trip-Free, General Specification for |
| MIL-R-83401 | 9.4 | Resistor Networks, Fixed, Film, General Specification for |
| MIL-C-83421 | 10.6 | Capacitor, Fixed Supermetallized Plastic Film Dielectric (DC, AC or DC and AC) Hermetically Sealed in Metal Cases, Established Reliability, General Specification for |
| MIL-C-83513 | 15.1 | Connector, Electrical, Rectangular, Microminiature, Polarized Shell, General Specification for |
| MIL-C-83723 | 15.1 | Connector, Electrical (Circular Environment Resisting), Receptacles and Plugs, General Specification for |
| MIL-R-83725 | 13.1 | Relay, Vacuum, General Specification for |
| MIL-R-83726 | 13.1, 13.2, 13.3 | Relay, Time Delay, Electric and Electronic, General Specification for |
| MIL-S-83731 | 14.1 | Switch, Toggle, Unsealed and Sealed Toggle, General Specification for |
| MIL-C-83733 | 15.1 | Connector, Electrical, Miniature, Rectangular Type, Rack to Panel, Environment Resisting, 200 Degrees C Total Continuous Operating Temperature, General Specification for |
| MIL-S-83734 | 15.3 | Socket, Plug-in Electronic Components, General Specification for |

| STANDARD | TITLE |
|---|---|
| MIL-STD-756 | Reliability Modeling and Prediction |
| MIL-STD-883 | Test Methods and Procedures for Microelectronics |
| MIL-STD-975 | NASA Standard Electrical, Electronic and Electromechanical Parts List |
| MIL-STD-1547 | Parts, Materials and Processes for Space Launch Vehicles, Technical Requirements for |
| MIL-STD-1772 | Certification Requirements for Hybrid Microcircuit Facilities and Lines |

Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer. Single copies are also available (without charge) upon written request to:

Standardization Document Order Desk
700 Robins Ave.
Building 4, Section D
Philadelphia, PA 19111-5094
(215) 697-2667



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## 3.0 INTRODUCTION

**3.1 Reliability Engineering** - Reliability is currently recognized as an essential need in military electronic systems. It is looked upon as a means for reducing costs from the factory, where rework of defective components adds a non-productive overhead expense, to the field, where repair costs include not only parts and labor but also transportation and storage. More importantly, reliability directly impacts force effectiveness, measured in terms of availability or sortie rates, and determines the size of the "logistics tail" inhibiting force utilization.

The achievement of reliability is the function of reliability engineering. Every aspect of an electronic system, from the purity of materials used in its component devices to the operator's interface, has an impact on reliability. Reliability engineering must, therefore, be applied throughout the system's development in a diligent and timely fashion, and integrated with other engineering disciplines.

A variety of reliability engineering tools have been developed. This handbook provides the models supporting a basic tool, reliability prediction.

**3.2 The Role of Reliability Prediction** - Reliability prediction provides the quantitative baseline needed to assess progress in reliability engineering. A prediction made of a proposed design may be used in several ways.

A characteristic of Computer Aided Design is the ability to rapidly generate alternative solutions to a particular problem. Reliability predictions for each design alternative provide one measure of relative worth which, combined with other considerations, will aid in selecting the best of the available options.

Once a design is selected, the reliability prediction may be used as a guide to improvement by showing the highest contributors to failure. If the part stress analysis method is used, it may also reveal other fruitful areas for change (e.g., over stressed parts).

The impact of proposed design changes on reliability can be determined only by comparing the reliability predictions of the existing and proposed designs.

The ability of the design to maintain an acceptable reliability level under environmental extremes may be assessed through reliability predictions. The predictions may be used to evaluate the need for environmental control systems.

The effects of complexity on the probability of mission success can be evaluated through reliability predictions. The need for redundant or back-up systems may be determined with the aid of reliability predictions. A tradeoff of redundancy against other reliability enhancing techniques (e.g.: more cooling, higher part quality, etc.) must be based on reliability predictions coupled with other pertinent considerations such as cost, space limitations, etc.

The prediction will also help evaluate the significance of reported failures. For example, if several failures of one type or component occur in a system, the predicted failure rate can be used to determine whether the number of failures is commensurate with the number of components used in the system, or, that it indicates a problem area.

Finally, reliability predictions are useful to various other engineering analyses. As examples, the location of built-in-test circuitry should be influenced by the predicted failure rates of the circuitry monitored, and maintenance strategy planners can make use of the relative probability of a failure's location, based on predictions, to minimize downtime. Reliability predictions are also used to evaluate the probabilities of failure events described in a failure modes, effects and criticality analysis (FMECAs).



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## 3.0 INTRODUCTION

**3.3 Limitations of Reliability Predictions** - This handbook provides a common basis for reliability predictions, based on analysis of the best available data at the time of issue. It is intended to make reliability prediction as good a tool as possible. However, like any tool, reliability prediction must be used intelligently, with due consideration of its limitations.

The first limitation is that the failure rate models are point estimates which are based on available data. Hence, they are valid for the conditions under which the data was obtained, and for the devices covered. Some extrapolation during model development is possible, but the inherently empirical nature of the models can be severely restrictive. For example, none of the models in this handbook predict nuclear survivability or the effects of ionizing radiation.

Even when used in similar environments, the differences between system applications can be significant. Predicted and achieved reliability have always been closer for ground electronic systems than for avionic systems, because the environmental stresses vary less from system to system on the ground and hence the field conditions are in general closer to the environment under which the data was collected for the prediction model. However, failure rates are also impacted by operational scenarios, operator characteristics, maintenance practices, measurement techniques and differences in definition of failure. Hence, a reliability prediction should never be assumed to represent the expected field reliability as measured by the user (i.e., Mean-Time-Between-Maintenance, Mean-Time-Between-Removals, etc.). This does not negate its value as a reliability engineering tool; note that none of the applications discussed above requires the predicted reliability to match the field measurement.

Electronic technology is noted for its dynamic nature. New types of devices and new processes are continually introduced, compounding the difficulties of predicting reliability. Evolutionary changes may be handled by extrapolation from the existing models; revolutionary changes may defy analysis.

Another limitation of reliability predictions is the mechanics of the process. The part stress analysis method requires a significant amount of design detail. This naturally imposes a time and cost penalty. More significantly, many of the details are not available in the early design stages. For this reason this handbook contains both the part stress analysis method (Sections 5 through 23) and a simpler parts count method (Appendix A) which can be used in early design and bid formulation stages.

Finally, a basic limitation of reliability prediction is its dependence on correct application by the user. Those who correctly apply the models and use the information in a conscientious reliability program will find the prediction a useful tool. Those who view the prediction only as a number which must exceed a specified value can usually find a way to achieve their goal without any impact on the system.

### 3.4 Part Stress Analysis Prediction

**3.4.1 Applicability** - This method is applicable when most of the design is completed and a detailed parts list including part stresses is available. It can also be used during later design phases for reliability trade-offs vs. part selection and stresses. Sections 5 through 23 contain failure rate models for a broad variety of parts used in electronic equipment. The parts are grouped by major categories and, where appropriate, are subgrouped within categories. For mechanical and electromechanical parts not covered by this Handbook, refer to Bibliography items 20 and 36 (Appendix C).

The failure rates presented apply to equipment under normal operating conditions, i.e., with power on and performing its intended functions in its intended environment. Extrapolation of any of the base failure rate models beyond the tabulated values such as high or sub-zero temperature, electrical stress values above 1.0, or extrapolation of any associated model modifiers is completely invalid. Base failure rates can be interpolated between electrical stress values from 0 to 1 using the underlying equations.

The general procedure for determining a board level (or system level) failure rate is to sum individually calculated failure rates for each component. This summation is then added to a failure rate for the circuit board (which includes the effects of soldering parts to it) using Section 16, Interconnection Assemblies.



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For parts or wires soldered together (e.g., a jumper wire between two parts), the connections model appearing in Section 17 is used. Finally, the effects of connecting circuit boards together is accounted for by adding in a failure rate for each connector (Section 15, Connectors). The wire between connectors is assumed to have a zero failure rate. For various service use profiles, duty cycles and redundancies the procedures described in MIL-STD-756, Reliability Modeling and Prediction, should be used to determine an effective system level failure rate.

**3.4.2 Part Quality** - The quality of a part has a direct effect on the part failure rate and appears in the part models as a factor, $\pi_Q$. Many parts are covered by specifications that have several quality levels, hence, the part models have values of $\pi_Q$ that are keyed to these quality levels. Such parts with their quality designators are shown in Table 3-1. The detailed requirements for these levels are clearly defined in the applicable specification, except for microcircuits. Microcircuits have quality levels which are dependent on the number of MIL-STD-883 screens (or equivalent) to which they are subjected.

**Table 3-1: Parts With Multi-Level Quality Specifications**

| Part | Quality Designators |
|---|---|
| Microcircuits | S, B, B-1, Other: Quality Judged by Screening Level |
| Discrete Semiconductors | JANTXV, JANTX, JAN |
| Capacitors, Established Reliability (ER) | D, C, S, R, B, P, M, L |
| Resistors, Established Reliability (ER) | S, R, P, M |
| Coils, Molded, R.F., Reliability (ER) | S, R, P, M |
| Relays, Established Reliability (ER) | R, P, M, L |

Some parts are covered by older specifications, usually referred to as Nonestablished Reliability (Non-ER), that do not have multi-levels of quality. These part models generally have two quality levels designated as "MIL-SPEC.", and "Lower". If the part is procured in complete accordance with the applicable specification, the $\pi_Q$ value for MIL-SPEC should be used. If any requirements are waived, or if a commercial part is procured, the $\pi_Q$ value for Lower should be used.

The foregoing discussion involves the "as procured" part quality. Poor equipment design, production, and testing facilities can degrade part quality. The use of the higher quality parts requires a total equipment design and quality control process commensurate with the high part quality. It would make little sense to procure high quality parts only to have the equipment production procedures damage the parts or introduce latent defects. Total equipment program descriptions as they might vary with different part quality mixes is beyond the scope of this Handbook. Reliability management and quality control procedures are described in other DoD standards and publications. Nevertheless, when a proposed equipment development is pushing the state-of-the-art and has a high reliability requirement necessitating high quality parts, the total equipment program should be given careful scrutiny and not just



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the parts quality. Otherwise, the low failure rates as predicted by the models for high quality parts will not be realized.

**3.4.3 Use Environment** - All part reliability models include the effects of environmental stresses through the environmental factor, $\pi_E$, except for the effects of ionizing radiation. The descriptions of these environments are shown in Table 3-2. The $\pi_E$ factor is quantified within each part failure rate model. These environments encompass the major areas of equipment use. Some equipment will experience more than one environment during its normal use, e.g., equipment in spacecraft. In such a case, the reliability analysis should be segmented, namely, missile launch ($M_L$) conditions during boost into and return from orbit, and space flight ($S_F$) while in orbit.

**Table 3-2: Environmental Symbol and Description**

| Environment | $\pi_E$ Symbol | Equivalent MIL-HDBK-217E, Notice 1 $\pi_E$ Symbol | Description |
|---|---|---|---|
| Ground, Benign | $G_B$ | $G_B$<br>$G_{MS}$ | Nonmobile, temperature and humidity controlled environments readily accessible to maintenance; includes laboratory instruments and test equipment, medical electronic equipment, business and scientific computer complexes, and missiles and support equipment in ground silos. |
| Ground, Fixed | $G_F$ | $G_F$ | Moderately controlled environments such as installation in permanent racks with adequate cooling air and possible installation in unheated buildings; includes permanent installation of air traffic control radar and communications facilities. |
| Ground, Mobile | $G_M$ | $G_M$<br>$M_P$ | Equipment installed on wheeled or tracked vehicles and equipment manually transported; includes tactical missile ground support equipment, mobile communication equipment, tactical fire direction systems, handheld communications equipment, laser designations and range finders. |
| Naval, Sheltered | $N_S$ | $N_S$<br>$N_{SB}$ | Includes sheltered or below deck conditions on surface ships and equipment installed in submarines. |
| Naval, Unsheltered | $N_U$ | $N_U$<br>$N_{UU}$<br>$N_H$ | Unprotected surface shipborne equipment exposed to weather conditions and equipment immersed in salt water. Includes sonar equipment and equipment installed on hydrofoil vessels. |



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**Table 3-2: Environmental Symbol and Description (cont'd)**

| Environment | $\pi_E$ Symbol | Equivalent MIL-HDBK-217E, Notice 1 $\pi_E$ Symbol | Description |
|---|---|---|---|
| Airborne, Inhabited, Cargo | $A_{IC}$ | $A_{IC}$<br>$A_{IT}$<br>$A_{IB}$ | Typical conditions in cargo compartments which can be occupied by an aircrew. Environment extremes of pressure, temperature, shock and vibration are minimal. Examples include long mission aircraft such as the C130, C5, B52, and C141. This category also applies to inhabited areas in lower performance smaller aircraft such as the T38. |
| Airborne, Inhabited, Fighter | $A_{IF}$ | $A_{IF}$<br>$A_{IA}$ | Same as $A_{IC}$ but installed on high performance aircraft such as fighters and interceptors. Examples include the F15, F16, F111, F/A 18 and A10 aircraft. |
| Airborne, Uninhabited, Cargo | $A_{UC}$ | $A_{UC}$<br>$A_{UT}$<br>$A_{UB}$ | Environmentally uncontrolled areas which cannot be inhabited by an aircrew during flight. Environmental extremes of pressure, temperature and shock may be severe. Examples include uninhabited areas of long mission aircraft such as the C130, C5, B52 and C141. This category also applies to uninhabited area of lower performance smaller aircraft such as the T38. |
| Airborne, Uninhabited, Fighter | $A_{UF}$ | $A_{UF}$<br>$A_{UA}$ | Same as $A_{UC}$ but installed on high performance aircraft such as fighters and interceptors. Examples include the F15, F16, F111 and A10 aircraft. |
| Airborne, Rotary Winged | $A_{RW}$ | $A_{RW}$ | Equipment installed on helicopters. Applies to both internally and externally mounted equipment such as laser designators, fire control systems, and communications equipment. |
| Space, Flight | $S_F$ | $S_F$ | Earth orbital. Approaches benign ground conditions. Vehicle neither under powered flight nor in atmospheric reentry; includes satellites and shuttles. |



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**Table 3-2: Environmental Symbol and Description (cont'd)**

| Environment | $\pi_E$ Symbol | Equivalent MIL-HDBK-217E, Notice 1 $\pi_E$ Symbol | Description |
|---|---|---|---|
| Missile, Flight | $M_F$ | $M_{FF}$<br>$M_{FA}$ | Conditions related to powered flight of air breathing missiles, cruise missiles, and missiles in unpowered free flight. |
| Missile, Launch | $M_L$ | $M_L$<br>$U_{SL}$ | Severe conditions related to missile launch (air, ground and sea), space vehicle boost into orbit, and vehicle re-entry and landing by parachute. Also applies to solid rocket motor propulsion powered flight, and torpedo and missile launch from submarines. |
| Cannon, Launch | $C_L$ | $C_L$ | Extremely severe conditions related to cannon launching of 155 mm. and 5 inch guided projectiles. Conditions apply to the projectile from launch to target impact. |

**3.4.4 Part Failure Rate Models** - Part failure rate models for microelectronic parts are significantly different from those for other parts and are presented entirely in Section 5.0. A typical example of the type of model used for most other part types is the following one for discrete semiconductors:

$$\lambda_p = \lambda_b \pi_T \pi_A \pi_R \pi_S \pi_C \pi_Q \pi_E$$

where:

$\lambda_p$ is the part failure rate,

$\lambda_b$ is the base failure rate usually expressed by a model relating the influence of electrical and temperature stresses on the part,

$\pi_E$ and the other $\pi$ factors modify the base failure rate for the category of environmental application and other parameters that affect the part reliability.

The $\pi_E$ and $\pi_Q$ factors are used in most all models and other $\pi$ factors apply only to specific models. The applicability of $\pi$ factors is identified in each section.

The base failure rate ($\lambda_b$) models are presented in each part section along with identification of the applicable model factors. Tables of calculated $\lambda_b$ values are also provided for use in manual calculations. The model equations can, of course, be incorporated into computer programs for machine processing. The tabulated values of $\lambda_b$ are cut off at the part ratings with regard to temperature and stress, hence, use of parts beyond these cut off points will overstress the part. The use of the $\lambda_b$ models in a computer



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program should take the part rating limits into account. The $\lambda_b$ equations are mathematically continuous beyond the part ratings but such failure rate values are invalid in the overstressed regions.

All the part models include failure data from both catastrophic and permanent drift failures (e.g., a resistor permanently falling out of rated tolerance bounds) and are based upon a constant failure rate, except for motors which show an increasing failure rate over time. Failures associated with connection of parts into circuit assemblies are not included within the part failure rate models. Information on connection reliability is provided in Sections 16 and 17.

**3.4.5 Thermal Aspects** - The use of this prediction method requires the determination of the temperatures to which the parts are subjected. Since parts reliability is sensitive to temperature, the thermal analysis of any design should fairly accurately provide the ambient temperatures needed in using the part models. Of course, lower temperatures produce better reliability but also can produce increased penalties in terms of added loads on the environmental control system, unless achieved through improved thermal design of the equipment. The thermal analysis should be part of the design process and included in all the trade-off studies covering equipment performance, reliability, weight, volume, environmental control systems, etc. References 17 and 34 listed in Appendix C may be used as guides in determining component temperatures.



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## 4.0 RELIABILITY ANALYSIS EVALUATION

Table 4-1 provides a general checklist to be used as a guide for evaluating a reliability prediction report. For completeness, the checklist includes categories for reliability modeling and allocation, which are sometimes delivered as part of a prediction report. It should be noted that the scope of any reliability analysis depends on the specific requirements called out in a statement-of-work (SOW) or system specification. The inclusion of this checklist is not intended to change the scope of these requirements.

**Table 4-1: Reliability Analysis Checklist**

| Major Concerns | Comments |
|---|---|
| **MODELS** | |
| Are all functional elements included in the reliability block diagram /model? | System design drawings/diagrams must be reviewed to be sure that the reliability model/diagram agrees with the hardware. |
| Are all modes of operation considered in the math model? | Duty cycles, alternate paths, degraded conditions and redundant units must be defined and modeled. |
| Do the math model results show that the design achieves the reliability requirement? | Unit failure rates and redundancy equations are used from the detailed part predictions in the system math model (See MIL-STD-756, Reliability Prediction and Modeling). |
| **ALLOCATION** | |
| Are system reliability requirements allocated (subdivided) to useful levels? | Useful levels are defined as: equipment for subcontractors, assemblies for sub-subcontractors, circuit boards for designers. |
| Does the allocation process consider complexity, design flexibility, and safety margins? | Conservative values are needed to prevent reallocation at every design change. |
| **PREDICTION** | |
| Does the sum of the parts equal the value of the module or unit? | Many predictions neglect to include all the parts producing optimistic results (check for solder connections, connectors, circuit boards). |
| Are environmental conditions and part quality representative of the requirements? | Optimistic quality levels and favorable environmental conditions are often assumed causing optimistic results. |
| Are the circuit and part temperatures defined and do they represent the design? | Temperature is the biggest driver of part failure rates; low temperature assumptions will cause optimistic results. |
| Are equipment, assembly, subassembly and part reliability drivers identified? | Identification is needed so that corrective actions for reliability improvement can be considered. |
| Are alternate (Non MIL-HDBK-217) failure rates highlighted along with the rationale for their use? | Use of alternate failure rates, if deemed necessary, require submission of backup data to provide credence in the values. |
| Is the level of detail for the part failure rate models sufficient to reconstruct the result? | Each component type should be sampled and failure rates completely reconstructed for accuracy. |
| Are critical components such as VHSIC, Monolithic Microwave Integrated Circuits (MMIC), Application Specific Integrated Circuits (ASIC) or Hybrids highlighted? | Prediction methods for advanced technology parts should be carefully evaluated for impact on the module and system. |



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## 5.0 MICROCIRCUITS, INTRODUCTION

This section presents failure rate prediction models for the following ten major classes of microelectronic devices:

| Section | |
|---|---|
| 5.1 | Monolithic Bipolar Digital and Linear Gate/Logic Array Devices |
| 5.1 | Monolithic MOS Digital and Linear Gate/Logic Array Devices |
| 5.1 | Monolithic Bipolar and MOS Digital Microprocessor Devices |
| 5.2 | Monolithic Bipolar and MOS Memory Devices |
| 5.3 | Very High Speed Integrated Circuit (VHSIC/VHSIC-Like and VLSI) CMOS Devices (> 60K Gates) |
| 5.4 | Monolithic GaAs Digital Devices |
| 5.4 | Monolithic GaAs MMIC |
| 5.5 | Hybrid Microcircuits |
| 5.6 | Surface Acoustic Wave Devices |
| 5.7 | Magnetic Bubble Memories |

In the title description of each monolithic device type, Bipolar represents all TTL, ASTTL, DTL, ECL, CML, ALSTTL, HTTL, FTTL, F, LTTL, STTL, BiCMOS, LSTTL, IIL, $I^3L$ and ISL devices. MOS represents all metal-oxide microcircuits, which includes NMOS, PMOS, CMOS and MNOS fabricated on various substrates such as sapphire, polycrystalline or single crystal silicon. The hybrid model is structured to accommodate all of the monolithic chip device types and various complexity levels.

Monolithic memory complexity factors are expressed in the number of bits in accordance with JEDEC STD 21A. This standard, which is used by all government and industry agencies that deal with microcircuit memories, states that memories of 1024 bits and greater shall be expressed as K bits, where 1K = 1024 bits. For example, a 16K memory has 16,384 bits, a 64K memory has 65,536 bits and a 1M memory has 1,048,576 bits. Exact numbers of bits are not used for memories of 1024 bits and greater.

For devices having both linear and digital functions not covered by MIL-M-38510 or MIL-I-38535, use the linear model. Line drivers and line receivers are considered linear devices. For linear devices not covered by MIL-M-38510 or MIL-I-38535, use the transistor count from the schematic diagram of the device to determine circuit complexity.

For digital devices not covered by MIL-M-38510 or MIL-I-38535, use the gate count as determined from the logic diagram. A J-K or R-S flip flop is equivalent to 6 gates when used as part of an LSI circuit. For the purpose of this Handbook, a gate is considered to be any one of the following functions; AND, OR, exclusive OR, NAND, NOR and inverter. When a logic diagram is unavailable, use device transistor count to determine gate count using the following expressions:

| Technology | Gate Approximation |
|---|---|
| Bipolar | No. Gates = No. Transistors/3.0 |
| CMOS | No. Gates = No. Transistors/4.0 |
| All other MOS except CMOS | No. Gates = No. Transistors/3.0 |



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## 5.0 MICROCIRCUITS, INTRODUCTION

A detailed form of the Section 5.3 VHSIC/VHSIC-Like model is included as Appendix B to allow more detailed trade-offs to be performed. Reference 30 should be consulted for more information about this model.

Reference 32 should be consulted for more information about the models appearing in Sections 5.1, 5.2, 5.4, 5.5, and 5.6. Reference 13 should be consulted for additional information on Section 5.7.



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## 5.1 MICROCIRCUITS, GATE/LOGIC ARRAYS AND MICROPROCESSORS

**DESCRIPTION**

1. Bipolar Devices, Digital and Linear Gate/Logic Arrays
2. MOS Devices, Digital and Linear Gate/Logic Arrays
3. Field Programmable Logic Array (PLA) and Programmable Array Logic (PAL)
4. Microprocessors

$$\lambda_p = (C_1\pi_T + C_2\pi_E)\,\pi_Q\pi_L \text{ Failures/}10^6 \text{ Hours}$$

Bipolar Digital and Linear Gate/Logic Array Die Complexity Failure Rate - $C_1$

| Digital | | Linear | | PLA/PAL | |
|---|---|---|---|---|---|
| No. Gates | $C_1$ | No. Transistors | $C_1$ | No. Gates | $C_1$ |
| 1 to 100 | .0025 | 1 to 100 | .010 | Up to 200 | .010 |
| 101 to 1,000 | .0050 | 101 to 300 | .020 | 201 to 1,000 | .021 |
| 1,001 to 3,000 | .010 | 301 to 1,000 | .040 | 1,001 to 5,000 | .042 |
| 3,001 to 10,000 | .020 | 1,001 to 10,000 | .060 | | |
| 10,001 to 30,000 | .040 | | | | |
| 30,001 to 60,000 | .080 | | | | |

MOS Digital and Linear Gate/Logic Array Die Complexity Failure Rate - $C_1$*

| Digital | | Linear | | PLA/PAL | |
|---|---|---|---|---|---|
| No. Gates | $C_1$ | No. Transistors | $C_1$ | No. Gates | $C_1$ |
| 1 to 100 | .010 | 1 to 100 | .010 | Up to 500 | .00085 |
| 101 to 1,000 | .020 | 101 to 300 | .020 | 501 to 1,000 | .0017 |
| 1,001 to 3,000 | .040 | 301 to 1,000 | .040 | 2,001 to 5,000 | .0034 |
| 3,001 to 10,000 | .080 | 1,001 to 10,000 | .060 | 5,001 to 20,000 | .0068 |
| 10,001 to 30,000 | .16 | | | | |
| 30,001 to 60,000 | .29 | | | | |

*NOTE: For CMOS gate counts above 60,000 use the VHSIC/VHSIC-Like model in Section 5.3

Microprocessor
Die Complexity Failure Rate - $C_1$

| No. Bits | Bipolar $C_1$ | MOS $C_1$ |
|---|---|---|
| Up to 8 | .060 | .14 |
| Up to 16 | .12 | .28 |
| Up to 32 | .24 | .56 |

All Other Model Parameters

| Parameter | Refer to |
|---|---|
| $\pi_T$ | Section 5.8 |
| $C_2$ | Section 5.9 |
| $\pi_E$, $\pi_Q$, $\pi_L$ | Section 5.10 |



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## 5.2 MICROCIRCUITS, MEMORIES

**DESCRIPTION**

1. Read Only Memories (ROM)
2. Programmable Read Only Memories (PROM)
3. Ultraviolet Eraseable PROMs (UVEPROM)
4. "Flash," MNOS and Floating Gate Electrically Eraseable PROMs (EEPROM). Includes both floating gate tunnel oxide (FLOTOX) and textured polysilicon type EEPROMs
5. Static Random Access Memories (SRAM)
6. Dynamic Random Access Memories (DRAM)

$$\lambda_p = (C_1 \pi_T + C_2 \pi_E + \lambda_{cyc}) \pi_Q \pi_L \text{ Failures/}10^6 \text{ Hours}$$

Die Complexity Failure Rate - $C_1$

| Memory Size, B (Bits) | MOS | | | | Bipolar | |
|---|---|---|---|---|---|---|
| | ROM | PROM, UVEPROM, EEPROM, EAPROM | DRAM | SRAM (MOS & BiMOS) | ROM, PROM | SRAM |
| Up to 16K | .00065 | .00085 | .0013 | .0078 | .0094 | .0052 |
| 16K < B ≤ 64K | .0013 | .0017 | .0025 | .016 | .019 | .011 |
| 64K < B ≤ 256K | .0026 | .0034 | .0050 | .031 | .038 | .021 |
| 256K < B ≤ 1M | .0052 | .0068 | .010 | .062 | .075 | .042 |

$A_1$ Factor for $\lambda_{cyc}$ Calculation

| Total No. of Programming Cycles Over EEPROM Life, C | Flotox[1] | Textured-Poly[2] |
|---|---|---|
| Up to 100 | .00070 | .0097 |
| 100 < C ≤ 200 | .0014 | .014 |
| 200 < C ≤ 500 | .0034 | .023 |
| 500 < C ≤ 1K | .0068 | .033 |
| 1K < C ≤ 3K | .020 | .061 |
| 3K < C ≤ 7K | .049 | .14 |
| 7K < C ≤ 15K | .10 | .30 |
| 15K < C ≤ 20K | .14 | .30 |
| 20K < C ≤ 30K | .20 | .30 |
| 30K < C ≤ 100K | .68 | .30 |
| 100K < C ≤ 200K | 1.3 | .30 |
| 200K < C ≤ 400K | 2.7 | .30 |
| 400K < C ≤ 500K | 3.4 | .30 |

1. $A_1 = 6.817 \times 10^{-6}$ (C)
2. No underlying equation for Textured-Poly.

$A_2$ Factor for $\lambda_{cyc}$ Calculation

| Total No. of Programming Cycles Over EEPROM Life, C | Textured-Poly $A_2$ |
|---|---|
| Up to 300K | 0 |
| 300K < C ≤ 400K | 1.1 |
| 400K < C ≤ 500K | 2.3 |

All Other Model Parameters

| Parameter | Refer to |
|---|---|
| $\pi_T$ | Section 5.8 |
| $C_2$ | Section 5.9 |
| $\pi_E$, $\pi_Q$, $\pi_L$ | Section 5.10 |
| $\lambda_{cyc}$ (EEPROMS only) | Page 5-5 |

$\lambda_{cyc} = 0$ For all other devices



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EEPROM Read/Write Cycling Induced Failure Rate - $\lambda_{cyc}$

| | |
|---|---|
| All Memory Devices Except Flotox and Textured-Poly EEPROMS | $\lambda_{cyc} = 0$ |
| Flotox and Textured Poly EEPROMS | $\lambda_{cyc} = \left[ A_1 B_1 + \frac{A_2 B_2}{\pi_Q} \right] \pi_{ECC}$ |

| Model Factor | Flotox | Textured-Poly |
|---|---|---|
| $A_1$ | Page 5-4 | Page 5-4 |
| $B_1$ | Page 5-6 | Page 5-6 |
| $A_2$ | $A_2 = 0$ | Page 5-5 |
| $B_2$ | $B_2 = 0$ | Page 5-6 |
| $\pi_Q$ | Section 5.10 | Section 5.10 |
| Error Correction Code (ECC) Options: | | |
| 1. No On-Chip ECC | $\pi_{ECC} = 1.0$ | $\pi_{ECC} = 1.0$ |
| 2. On-Chip Hamming Code | $\pi_{ECC} = .72$ | $\pi_{ECC} = .72$ |
| 3. Two-Needs-One Redundant Cell Approach | $\pi_{ECC} = .68$ | $\pi_{ECC} = .68$ |

NOTES:

1. See Reference 24 for modeling off-chip error detection and correction schemes at the memory system level.
2. If EEPROM type is unknown, assume Flotox.
3. Error Correction Code Options: Some EEPROM manufacturers have incorporated on-chip error correction circuitry into their EEPROM devices. This is represented by the on-chip hamming code entry. Other manufacturers have taken a redundant cell approach which incorporates an extra storage transistor in every memory cell. This is represented by the two-needs-one redundant cell entry.
4. The $A_1$ and $A_2$ factors shown in Section 5.2 were developed based on an assumed system life of 10,000 operating hours. For EEPROMs used in systems with significantly longer or shorter expected lifetimes the $A_1$ and $A_2$ factors should be multiplied by:

$$\frac{10{,}000}{\text{System Lifetime Operating Hours}}$$



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$B_1$ and $B_2$ Factors for $\lambda_{cyc}$ Calculation

| | Flotox[1] ($B_1$) | | | | | Textured-Poly[2] ($B_1$) | | | | | Textured-Poly[3] ($B_2$) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Memory Size, B(Bits) → / $T_J$ (°C) | 4K | 16K | 64K | 256K | 1M | 4K | 16K | 64K | 256K | 1M | 4K | 16K | 64K | 256K | 1M |
| 25 | .27 | 0.55 | 1.1 | 2.2 | 4.3 | .47 | .66 | .94 | 1.3 | 1.9 | .54 | 0.76 | 1.1 | 1.5 | 2.1 |
| 30 | .30 | 0.60 | 1.2 | 2.4 | 4.8 | .50 | .71 | 1.0 | 1.4 | 2.0 | .50 | 0.71 | 1.0 | 1.4 | 2.0 |
| 35 | .33 | 0.66 | 1.3 | 2.7 | 5.2 | .54 | .77 | 1.1 | 1.5 | 2.2 | .47 | 0.67 | .95 | 1.3 | 1.9 |
| 40 | .36 | 0.72 | 1.4 | 2.9 | 5.7 | .58 | .82 | 1.2 | 1.6 | 2.3 | .45 | 0.63 | .89 | 1.3 | 1.8 |
| 45 | .40 | 0.79 | 1.6 | 3.2 | 6.3 | .62 | .88 | 1.3 | 1.8 | 2.5 | .42 | 0.59 | .84 | 1.2 | 1.7 |
| 50 | .43 | 0.86 | 1.7 | 3.4 | 6.8 | .67 | .94 | 1.3 | 1.9 | 2.7 | .40 | 0.56 | .80 | 1.1 | 1.6 |
| 55 | .47 | 0.93 | 1.9 | 3.7 | 7.4 | .71 | 1.0 | 1.4 | 2.0 | 2.8 | .38 | 0.53 | .75 | 1.1 | 1.5 |
| 60 | .51 | 1.0 | 2.0 | 4.1 | 8.0 | .76 | 1.1 | 1.5 | 2.1 | 3.0 | .36 | 0.50 | .72 | 1.0 | 1.4 |
| 65 | .55 | 1.1 | 2.2 | 4.4 | 8.6 | .81 | 1.1 | 1.6 | 2.3 | 3.2 | .34 | 0.48 | .68 | .96 | 1.3 |
| 70 | .59 | 1.2 | 2.4 | 4.7 | 9.3 | .86 | 1.2 | 1.7 | 2.4 | 3.4 | .32 | 0.46 | .65 | .91 | 1.3 |
| 75 | .63 | 1.3 | 2.5 | 5.1 | 10 | .91 | 1.3 | 1.8 | 2.6 | 3.6 | .31 | 0.43 | .62 | .87 | 1.2 |
| 80 | .68 | 1.4 | 2.7 | 5.4 | 11 | .96 | 1.4 | 1.9 | 2.7 | 3.8 | .29 | 0.41 | .59 | .83 | 1.2 |
| 85 | .73 | 1.5 | 2.9 | 5.8 | 12 | 1.0 | 1.4 | 2.0 | 2.9 | 4.0 | .28 | 0.40 | .56 | .79 | 1.1 |
| 90 | .78 | 1.6 | 3.1 | 6.2 | 12 | 1.1 | 1.5 | 2.2 | 3.0 | 4.3 | .27 | 0.38 | .54 | .75 | 1.1 |
| 95 | .83 | 1.7 | 3.3 | 6.7 | 13 | 1.1 | 1.6 | 2.3 | 3.2 | 4.5 | .26 | 0.36 | .51 | .72 | 1.0 |
| 100 | .89 | 1.8 | 3.5 | 7.1 | 14 | 1.2 | 1.7 | 2.4 | 3.4 | 4.7 | .25 | 0.35 | .49 | .69 | .98 |
| 105 | .94 | 1.9 | 3.8 | 7.5 | 15 | 1.3 | 1.8 | 2.5 | 3.5 | 5.0 | .24 | 0.33 | .47 | .66 | .94 |
| 110 | 1.0 | 2.0 | 4.0 | 8.0 | 16 | 1.3 | 1.9 | 2.6 | 3.7 | 5.2 | .23 | 0.32 | .45 | .64 | .90 |
| 115 | 1.1 | 2.1 | 4.2 | 8.5 | 17 | 1.4 | 1.9 | 2.8 | 3.9 | 5.5 | .22 | 0.31 | .44 | .61 | .87 |
| 120 | 1.1 | 2.2 | 4.5 | 9.0 | 18 | 1.4 | 2.0 | 2.9 | 4.1 | 5.7 | .21 | 0.30 | .42 | .59 | .83 |
| 125 | 1.2 | 2.4 | 4.7 | 9.5 | 19 | 1.5 | 2.1 | 3.0 | 4.2 | 6.0 | .20 | 0.29 | .41 | .57 | .80 |
| 130 | 1.3 | 2.5 | 5.0 | 10 | 20 | 1.6 | 2.2 | 3.2 | 4.4 | 6.2 | .19 | 0.28 | .39 | .55 | .77 |
| 135 | 1.3 | 2.6 | 5.3 | 11 | 21 | 1.6 | 2.3 | 3.3 | 4.6 | 6.5 | .19 | 0.27 | .38 | .53 | .75 |
| 140 | 1.4 | 2.8 | 5.6 | 11 | 22 | 1.7 | 2.4 | 3.4 | 4.8 | 6.8 | .18 | 0.26 | .36 | .51 | .72 |
| 145 | 1.5 | 2.9 | 5.8 | 12 | 23 | 1.8 | 2.5 | 3.6 | 5.0 | 7.1 | .18 | 0.25 | .35 | .50 | .70 |
| 150 | 1.5 | 3.1 | 6.1 | 12 | 24 | 1.8 | 2.6 | 3.7 | 5.2 | 7.4 | .17 | 0.24 | .34 | .48 | .68 |
| 155 | 1.6 | 3.2 | 6.4 | 13 | 25 | 1.9 | 2.7 | 3.9 | 5.4 | 7.6 | .16 | 0.23 | .33 | .46 | .65 |
| 160 | 1.7 | 3.4 | 6.7 | 14 | 27 | 2.0 | 2.8 | 4.0 | 5.6 | 7.9 | .16 | 0.23 | .32 | .45 | .63 |
| 165 | 1.8 | 3.5 | 7.1 | 14 | 28 | 2.1 | 2.9 | 4.2 | 5.8 | 8.2 | .15 | 0.22 | .31 | .44 | .62 |
| 170 | 1.8 | 3.7 | 7.4 | 15 | 29 | 2.1 | 3.0 | 4.3 | 6.1 | 8.5 | .15 | 0.21 | .30 | .42 | .60 |
| 175 | 1.9 | 3.9 | 7.7 | 15 | 31 | 2.2 | 3.1 | 4.5 | 6.3 | 8.8 | .15 | 0.21 | .29 | .41 | .58 |

1. $B_1 = \left(\frac{B}{16000}\right)^{.5}\left[\exp\left(\frac{-.15}{8.63 \times 10^{-5}}\left(\frac{1}{T_J + 273} - \frac{1}{333}\right)\right)\right]$

2. $B_1 = \left(\frac{B}{64000}\right)^{.25}\left[\exp\left(\frac{-.12}{8.63 \times 10^{-5}}\left(\frac{1}{T_J + 273} - \frac{1}{303}\right)\right)\right]$

3. $B_2 = \left(\frac{B}{64000}\right)^{.25}\left[\exp\left(\frac{.1}{8.63 \times 10^{-5}}\left(\frac{1}{T_J + 273} - \frac{1}{303}\right)\right)\right]$

$T_J$ = Worse Case Junction Temperature (°C). See Section 5.11 for $T_J$ Determination

B = Number of bits. NOTE: 1K = 1024 bits

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## 5.3 MICROCIRCUITS, VHSIC/VHSIC-LIKE AND VLSI CMOS

**DESCRIPTION**
CMOS greater than 60,000 gates

$$\lambda_p = \lambda_{BD}\pi_{MFG}\pi_T\pi_{CD} + \lambda_{BP}\pi_E\pi_Q\pi_{PT} + \lambda_{EOS} \text{ Failures/}10^6 \text{ Hours}$$

Die Base Failure Rate - $\lambda_{BD}$

| Part Type | $\lambda_{BD}$ |
|---|---|
| Logic and Custom | 0.16 |
| Gate Array | 0.24 |

All Other Model Parameters

| Parameter | Refer to |
|---|---|
| $\pi_T$ | Section 5.8 |
| $\pi_E$, $\pi_Q$ | Section 5.10 |

Manufacturing Process Correction Factor - $\pi_{MFG}$

| Manufacturing Process | $\pi_{MFG}$ |
|---|---|
| QML or QPL | .55 |
| Non QML or Non QPL | 2.0 |

Package Type Correction Factor - $\pi_{PT}$

| Package Type | $\pi_{PT}$ Hermetic | $\pi_{PT}$ Nonhermetic |
|---|---|---|
| DIP | 1.0 | 1.3 |
| Pin Grid Array | 2.2 | 2.9 |
| Chip Carrier (Surface Mount Technology) | 4.7 | 6.1 |

Die Complexity Correction Factor - $\pi_{CD}$

| Feature Size (Microns) | Die Area ($cm^2$) A ≤ .4 | .4 < A ≤ .7 | .7 < A ≤ 1.0 | 1.0 < A ≤ 2.0 | 2.0 < A ≤ 3.0 |
|---|---|---|---|---|---|
| .80 | 8.0 | 14 | 19 | 38 | 58 |
| 1.00 | 5.2 | 8.9 | 13 | 25 | 37 |
| 1.25 | 3.5 | 5.8 | 8.2 | 16 | 24 |

$$\pi_{CD} = \left(\left(\frac{A}{.21}\right)\left(\frac{2}{X_s}\right)^2(.64)\right) + .36$$

A = Total Scribed Chip Die Area in $cm^2$ $X_s$ = Feature Size (microns)

Die Area Conversion: $cm^2 = MIL^2 \div 155{,}000$

Package Base Failure Rate - $\lambda_{BP}$

| Number of Pins | $\lambda_{BP}$ |
|---|---|
| 24 | .0026 |
| 28 | .0027 |
| 40 | .0029 |
| 44 | .0030 |
| 48 | .0030 |
| 52 | .0031 |
| 64 | .0033 |
| 84 | .0036 |
| 120 | .0043 |
| 124 | .0043 |
| 144 | .0047 |
| 220 | .0060 |

$$\lambda_{BP} = .0022 + \left((1.72 \times 10^{-5})(NP)\right)$$

NP = Number of Package Pins

Electrical Overstress Failure Rate - $\lambda_{EOS}$

| $V_{TH}$ (ESD Susceptibility (Volts))* | $\lambda_{EOS}$ |
|---|---|
| 0 - 1000 | .065 |
| > 1000 - 2000 | .053 |
| > 2000 - 4000 | .044 |
| > 4000 - 16000 | .029 |
| > 16000 | .0027 |

$$\lambda_{EOS} = (-\ln(1 - .00057\exp(-.0002\,V_{TH}))/.00876$$

$V_{TH}$ = ESD Susceptibility (volts)

* Voltage ranges which will cause the part to fail. If unknown, use 0 - 1000 volts.



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## 5.4 MICROCIRCUITS, GaAs MMIC AND DIGITAL DEVICES

**DESCRIPTION**
Gallium Arsenide Microwave Monolithic Integrated Circuit (GaAs MMIC) and GaAs Digital Integrated Circuits using MESFET Transistors and Gold Based Metallization

$$\lambda_p = [\, C_1\pi_T\pi_A + C_2\pi_E \,]\,\pi_L\pi_Q \quad \text{Failures}/10^6 \text{ Hours}$$

MMIC: Die Complexity Failure Rates - $C_1$

| Complexity (No. of Elements) | $C_1$ |
|---|---|
| 1 to 100 | 4.5 |
| 101 to 1000 | 7.2 |
| 1. $C_1$ accounts for the following active elements: transistors, diodes. | |

Digital: Die Complexity Failure Rates - $C_1$

| Complexity (No. of Elements) | $C_1$ |
|---|---|
| 1 to 1000 | 25 |
| 1,001 to 10,000 | 51 |
| 1. $C_1$ accounts for the following active elements: transistors, diodes. | |

Device Application Factor - $\pi_A$

| Application | $\pi_A$ |
|---|---|
| MMIC Devices | |
| Low Noise & Low Power (≤ 100 mW) | 1.0 |
| Driver & High Power (> 100 mW) | 3.0 |
| Unknown | 3.0 |
| Digital Devices | |
| All Digital Applications | 1.0 |

All Other Model Parameters

| Parameter | Refer to |
|---|---|
| $\pi_T$ | Section 5.8 |
| $C_2$ | Section 5.9 |
| $\pi_E$, $\pi_L$, $\pi_Q$ | Section 5.10 |



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## 5.5 MICROCIRCUITS, HYBRIDS

### DESCRIPTION
Hybrid Microcircuits

$$\lambda_p = [\Sigma N_C \lambda_C] (1 + .2 \pi_E) \pi_F \pi_Q \pi_L \text{ Failures/}10^6 \text{ Hours}$$

$N_C$ = Number of Each Particular Component
$\lambda_C$ = Failure Rate of Each Particular Component

The general procedure for developing an overall hybrid failure rate is to calculate an individual failure rate for each component type used in the hybrid and then sum them. This summation is then modified to account for the overall hybrid function ($\pi_F$), screening level ($\pi_Q$), and maturity ($\pi_L$). The hybrid package failure rate is a function of the active component failure modified by the environmental factor (i.e., (1 + .2 $\pi_E$) ). Only the component types listed in the following table are considered to contribute significantly to the overall failure rate of most hybrids. All other component types (e.g., resistors, inductors, etc.) are considered to contribute insignificantly to the overall hybrid failure rate, and are assumed to have a failure rate of zero. This simplification is valid for most hybrids; however, if the hybrid consists of mostly passive components then a failure rate should be calculated for these devices. If factoring in other component types, assume $\pi_Q = 1$, $\pi_E = 1$ and $T_A$ = Hybrid Case Temperature for these calculations.

Determination of $\lambda_C$

| Determine $\lambda_C$ for These Component Types | Handbook Section | Make These Assumptions When Determining $\lambda_C$ |
|---|---|---|
| Microcircuits | 5 | $C_2 = 0$, $\pi_Q = 1$, $\pi_L = 1$, $T_J$ as Determined from Section 5.12, $\lambda_{BP} = 0$ (for VHSIC). |
| Discrete Semiconductors | 6 | $\pi_Q = 1$, $T_J$ as Determined from Section 6.14, $\pi_E = 1$. |
| Capacitors | 10 | $\pi_Q = 1$, $T_A$ = Hybrid Case Temperature, $\pi_E = 1$. |

NOTE: If maximum rated stress for a die is unknown, assume the same as for a discretely package die of the same type. If the same die has several ratings based on the discrete packaged type, assume the lowest rating. Power rating used should be based on case temperature for discrete semiconductors.

Circuit Function Factor - $\pi_F$

| Circuit Type | $\pi_F$ |
|---|---|
| Digital | 1.0 |
| Video, 10 MHz < f < 1 GHz | 1.2 |
| Microwave, f > 1 GHz | 2.6 |
| Linear, f < 10 MHz | 5.8 |
| Power | 21 |

All Other Hybrid Model Parameters

| | |
|---|---|
| $\pi_L$, $\pi_Q$, $\pi_E$ | Refer to Section 5.10 |



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## 5.6 MICROCIRCUITS, SAW DEVICES

**DESCRIPTION**
Surface Acoustic Wave Devices

$$\lambda_p = 2.1\ \pi_Q\ \pi_E \text{ Failures}/10^6 \text{ Hours}$$

Quality Factor - $\pi_Q$

| Screening Level | $\pi_Q$ |
|---|---|
| 10 Temperature Cycles (-55°C to +125°C) with end point electrical tests at temperature extremes. | .10 |
| None beyond best commerical practices. | 1.0 |

Environmental Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | .5 |
| $G_F$ | 2.0 |
| $G_M$ | 4.0 |
| $N_S$ | 4.0 |
| $N_U$ | 6.0 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 5.0 |
| $A_{UC}$ | 5.0 |
| $A_{UF}$ | 8.0 |
| $A_{RW}$ | 8.0 |
| $S_F$ | .50 |
| $M_F$ | 5.0 |
| $M_L$ | 12 |
| $C_L$ | 220 |



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## 5.7 MICROCIRCUITS, MAGNETIC BUBBLE MEMORIES

The magnetic bubble memory device in its present form is a non-hermetic assembly consisting of the following two major structural segments:

1. A basic bubble chip or die consisting of memory or a storage area (e.g., an array of minor loops), and required control and detection elements (e.g., generators, various gates and detectors).

2. A magnetic structure to provide controlled magnetic fields consisting of permanent magnets, coils, and a housing.

These two structural segments of the device are interconnected by a mechanical substrate and lead frame. The interconnect substrate in the present technology is normally a printed circuit board. It should be noted that this model does not include external support microelectronic devices required for magnetic bubble memory operation. The model is based on Reference 33. The general form of the failure rate model is:

$$\lambda_p = \lambda_1 + \lambda_2 \text{ Failures/}10^6 \text{ Hours}$$

where:

$\lambda_1$ = Failure Rate of the Control and Detection Structure

$$\lambda_1 = \pi_Q [N_C C_{11} \pi_{T1} \pi_W + (N_C C_{21} + C_2) \pi_E] \pi_D \pi_L$$

$\lambda_2$ = Failure Rate of the Memory Storage Area

$$\lambda_2 = \pi_Q N_C (C_{12} \pi_{T2} + C_{22} \pi_E) \pi_L$$

Chips Per Package - $N_C$

| | | |
|---|---|---|
| $N_C$ | = | Number of Bubble Chips per Packaged Device |

Temperature Factor – $\pi_T$

$$\pi_T = (.1) \exp\left[\frac{-Ea}{8.63 \times 10^{-5}} \left(\frac{1}{T_J + 273} - \frac{1}{298}\right)\right]$$

Use:

| | | |
|---|---|---|
| $E_a$ | = | .8 to Calculate $\pi_{T1}$ |
| $E_a$ | = | .55 to Calculate $\pi_{T2}$ |
| $T_J$ | = | Junction Temperature (°C), $25 \le T_J \le 175$ |
| $T_J$ | = | $T_{CASE}$ + 10°C |

Device Complexity Failure Rates for Control and Detection Structure - $C_{11}$ and $C_{21}$

| | | |
|---|---|---|
| $C_{11}$ | = | $.00095(N_1)^{.40}$ |
| $C_{21}$ | = | $.0001(N_1)^{.226}$ |
| $N_1$ | = | Number of Dissipative Elements on a Chip (gates, detectors, generators, etc.), $N_1 \le 1000$ |



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## 5.7 MICROCIRCUIT, MAGNETIC BUBBLE MEMORIES

### Write Duty Cycle Factor - $\pi_W$

$$\pi_W = \frac{10D}{(R/W)^{.3}}$$

$\pi_W = 1$ for $D \leq .3$ or $R/W \geq 2154$

$$D = \frac{\text{Avg. Device Data Rate}}{\text{Mfg. Max. Rated Data Rate}} \leq 1$$

R/W = No. of Reads per Write

NOTE:
For seed-bubble generators, divide $\pi_W$ by 4, or use 1, whichever is greater.

### Duty Cycle Factor - $\pi_D$

$$\pi_D = .9D + .1$$

$$D = \frac{\text{Avg. Device Data Rate}}{\text{Mfg. Max. Rated Data Rate}} \leq 1$$

### Device Complexity Failure Rates for Memory Storage Structure - $C_{12}$ and $C_{22}$

$$C_{12} = .00007(N_2)^{.3}$$

$$C_{22} = .00001(N_2)^{.3}$$

$N_2$ = Number of Bits, $N_2 \leq 9 \times 10^6$

### All Other Model Parameters

| Parameter | Section |
|---|---|
| $C_2$ | 5.9 |
| $\pi_E$, $\pi_Q$, $\pi_L$ | 5.10 |



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## Temperature Factor For All Microcircuits - $\pi_T$

| | TTL, ASTTL, CML, HTTL, FTTL, DTL, ECL, ALSTTL | F, LTTL, STTL | BiCMOS, LSTTL | III, $I^2L$, ISL | Digital MOS, VHSIC CMOS | Linear (Bipolar & MOS) | Memories (Bipolar & MOS), MNOS | GaAs MMIC Active Devices, $\pi_{TA}$ | GaAs Digital Active Devices, $\pi_{TA}$ |
|---|---|---|---|---|---|---|---|---|---|
| Ea(eV) → $T_J$ (°C) | .4 | .45 | .5 | .6 | .35 | .65 | .8 | 1.5 | 1.4 |
| 25 | .10 | .10 | .10 | .10 | .10 | .10 | .10 | 3.20E-09 | 1.00E-08 |
| 30 | .13 | .13 | .14 | .15 | .13 | .15 | .15 | 8.40E-09 | 2.50E-08 |
| 35 | .17 | .18 | .19 | .21 | .16 | .23 | .21 | 2.10E-08 | 5.90E-08 |
| 40 | .21 | .23 | .25 | .31 | .19 | .34 | .31 | 5.20E-08 | 1.40E-07 |
| 45 | .27 | .3 | .34 | .43 | .24 | .49 | .43 | 1.30E-07 | 3.10E-07 |
| 50 | .33 | .39 | .45 | .61 | .29 | .71 | .61 | 2.90E-07 | 6.80E-07 |
| 55 | .42 | .50 | .59 | .85 | .35 | 1.0 | .85 | 6.70E-07 | 1.50E-06 |
| 60 | .51 | .63 | .77 | 1.2 | .42 | 1.4 | 1.2 | 1.50E-06 | 3.10E-06 |
| 65 | .63 | .80 | 1.0 | 1.6 | .50 | 2.0 | 1.6 | 3.20E-06 | 6.40E-06 |
| 70 | .77 | 1.0 | 1.3 | 2.1 | .60 | 2.8 | 2.1 | 6.80E-06 | 1.30E-05 |
| 75 | .94 | 1.2 | 1.6 | 2.9 | .71 | 3.8 | 2.9 | 1.40E-05 | 2.50E-05 |
| 80 | 1.1 | 1.5 | 2.1 | 3.8 | .84 | 5.2 | 3.8 | 2.90E-05 | 4.90E-05 |
| 85 | 1.4 | 1.9 | 2.6 | 5.0 | .98 | 7.0 | 5.0 | 5.70E-05 | 9.40E-05 |
| 90 | 1.6 | 2.3 | 3.3 | 6.6 | 1.1 | 9.3 | 6.6 | 1.10E-04 | 1.70E-04 |
| 95 | 1.9 | 2.8 | 4.1 | 8.5 | 1.3 | 12 | 8.5 | 2.10E-04 | 3.20E-04 |
| 100 | 2.3 | 3.4 | 5.0 | 11 | 1.5 | 16 | 11 | 4.00E-04 | 5.80E-04 |
| 105 | 2.7 | 4.1 | 6.2 | 14 | 1.8 | 21 | 14 | 7.50E-04 | 1.00E-03 |
| 110 | 3.2 | 4.9 | 7.5 | 18 | 2.1 | 28 | 18 | 1.40E-03 | 1.80E-03 |
| 115 | 3.7 | 5.8 | 9.2 | 23 | 2.4 | 35 | 23 | 2.40E-03 | 3.10E-03 |
| 120 | 4.3 | 6.9 | 11 | 28 | 2.7 | 45 | 28 | 4.30E-03 | 5.30E-03 |
| 125 | 5 | 8.2 | 13 | 35 | 3.1 | 58 | 35 | 7.50E-03 | 9.00E-03 |
| 130 | 5.8 | 9.6 | 16 | 44 | 3.5 | 73 | 44 | 1.30E-02 | 1.50E-02 |
| 135 | 6.7 | 11 | 19 | 54 | 3.9 | 92 | 54 | 2.20E-02 | 2.40E-02 |
| 140 | 7.7 | 13 | 23 | 67 | 4.4 | 120 | 67 | 3.70E-02 | 3.90E-02 |
| 145 | 8.8 | 15 | 27 | 82 | 5.0 | 140 | 82 | 6.10E-02 | 6.30E-02 |
| 150 | 10 | 18 | 32 | 100 | 5.6 | 180 | 100 | 1.00E-01 | 1.00E-01 |
| 155 | 11 | 20 | 37 | 120 | 6.3 | 220 | 120 | 1.60E-01 | 1.60E-01 |
| 160 | 13 | 24 | 43 | 150 | 7.0 | 270 | 150 | 2.60E-01 | 2.40E-01 |
| 165 | 15 | 27 | 50 | 180 | 7.8 | 330 | 180 | 4.10E-01 | 3.70E-01 |
| 170 | 16 | 31 | 59 | 210 | 8.7 | 400 | 210 | 6.40E-01 | 5.70E-01 |
| 175 | 18 | 35 | 68 | 250 | 9.6 | 480 | 250 | 9.90E-01 | 8.50E-01 |

$$\pi_T = .1 \exp\left(\frac{-Ea}{8.617 \times 10^{-5}}\left(\frac{1}{T_J + 273} - \frac{1}{296}\right)\right) \text{ Silicon Devices} \qquad \pi_T = .1 \exp\left(\frac{-Ea}{8.617 \times 10^{-5}}\left(\frac{1}{T_J + 273} - \frac{1}{423}\right)\right) \text{ GaAs Devices}$$

$E_a$ = Effective Activation Energy (eV) (Shown Above)

$T_J$ = Worse Case Junction Temperature (Silicon Devices) or Average Active Device Channel Temperature (GaAs Devices). See Section 5.11 (or Section 5.12 for Hybrids) for $T_J$ Determination.

NOTES:
1. $T_J = T_C + P\theta_{JC}$
   $T_C$ = Case Temperature (°C)
   P = Device Power Dissipation (W)
   $\theta_{JC}$ = Junction to Case Thermal Resistance (°C/W)
   $\theta_{JC}$ should be obtained from the device manufacturer, MIL-M-38510, or from the default values shown in Section 5.11 for the closest equivalent device.
2. Use Digital MOS column for HC, HCT, AC, ACT, C and FCT technologies.
3. Table entries should be considered valid only up to the rated temperature of the component under consideration.



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## 5.9 MICROCIRCUITS, $C_2$ TABLE FOR ALL

### Package Failure Rate for all Microcircuits - $C_2$

| Number of Functional Pins, $N_p$ | Package Type: Hermetic: DIPs w/Solder or Weld Seal, Pin Grid Array (PGA)[1], SMT (Leaded and Nonleaded) | DIPs with Glass Seal[2] | Flatpacks with Axial Leads on 50 Mil Centers[3] | Cans[4] | Nonhermetic: DIPs, PGA, SMT (Leaded and Nonleaded)[5] |
|---|---|---|---|---|---|
| 3 | .00092 | .00047 | .00022 | .00027 | .0012 |
| 4 | .0013 | .00073 | .00037 | .00049 | .0016 |
| 6 | .0019 | .0013 | .00078 | .0011 | .0025 |
| 8 | .0026 | .0021 | .0013 | .0020 | .0034 |
| 10 | .0034 | .0029 | .0020 | .0031 | .0043 |
| 12 | .0041 | .0038 | .0028 | .0044 | .0053 |
| 14 | .0048 | .0048 | .0037 | .0060 | .0062 |
| 16 | .0056 | .0059 | .0047 | .0079 | .0072 |
| 18 | .0064 | .0071 | .0058 | | .0082 |
| 22 | .0079 | .0096 | .0083 | | .010 |
| 24 | .0087 | .011 | .0098 | | .011 |
| 28 | .010 | .014 | | | .013 |
| 36 | .013 | .020 | | | .017 |
| 40 | .015 | .024 | | | .019 |
| 64 | .025 | .048 | | | .032 |
| 80 | .032 | | | | .041 |
| 128 | .053 | | | | .068 |
| 180 | .076 | | | | .098 |
| 224 | .097 | | | | .12 |

1. $C_2 = 2.8 \times 10^{-4} (N_p)^{1.08}$
2. $C_2 = 9.0 \times 10^{-5} (N_p)^{1.51}$
3. $C_2 = 3.0 \times 10^{-5} (N_p)^{1.82}$
4. $C_2 = 3.0 \times 10^{-5} (N_p)^{2.01}$
5. $C_2 = 3.6 \times 10^{-4} (N_p)^{1.08}$

NOTES:

1. SMT: Surface Mount Technology
2. DIP: Dual In-Line Package
3. If DIP Seal type is unknown, assume glass
4. The package failure rate ($C_2$) accounts for failures associated only with the package itself. Failures associated with mounting the package to a circuit board are accounted for in Section 16, Interconnection Assemblies.



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## 5.10 MICROCIRCUITS, $\pi_E$, $\lambda_L$ AND $\pi_Q$ TABLES FOR ALL

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | .50 |
| $G_F$ | 2.0 |
| $G_M$ | 4.0 |
| $N_S$ | 4.0 |
| $N_U$ | 6.0 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 5.0 |
| $A_{UC}$ | 5.0 |
| $A_{UF}$ | 8.0 |
| $A_{RW}$ | 8.0 |
| $S_F$ | .50 |
| $M_F$ | 5.0 |
| $M_L$ | 12 |
| $C_L$ | 220 |

Learning Factor - $\pi_L$

| Years in Production, Y | $\pi_L$ |
|---|---|
| ≤ .1 | 2.0 |
| .5 | 1.8 |
| 1.0 | 1.5 |
| 1.5 | 1.2 |
| ≥ 2.0 | 1.0 |

$\pi_L = .01 \exp(5.35 - .35Y)$

Y = Years generic device type has been in production

Quality Factors - $\pi_Q$

| Description | $\pi_Q$ |
|---|---|
| Class S Categories:<br>1. Procured in full accordance with MIL-M-38510, Class S requirements.<br>2. Procured in full accordance with MIL-I-38535 and Appendix B thereto (Class U).<br>3. Hybrids: (Procured to Class S requirements (Quality Level K) of MIL-H-38534. | .25 |
| Class B Categories:<br>1. Procured in full accordance with MIL-M-38510, Class B requirements.<br>2. Procured in full accordance with MIL-I-38535, (Class Q).<br>3. Hybrids: Procured to Class B requirements (Quality Level H) of MIL-H-38534. | 1.0 |
| Class B-1 Category:<br>Fully compliant with all requirements of paragraph 1.2.1 of MIL-STD-883 and procured to a MIL drawing, DESC drawing or other government approved documentation. (Does not include hybrids). For hybrids use custom screening section below. | 2.0 |



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## 5.10 MICROCIRCUITS, $\pi_E$, $\pi_L$ AND $\pi_Q$ TABLES FOR ALL

Quality Factors (cont'd): $\pi_Q$ Calculation for Custom Screening Programs

| Group | MIL-STD-883 Screen/Test (Note 3) | Point Valuation | |
|---|---|---|---|
| 1* | TM 1010 (Temperature Cycle, Cond B Minimum) and TM 2001 (Constant Acceleration, Cond B Minimum) and TM 5004 (or 5008 for Hybrids) (Final Electricals @ Temp Extremes) and TM 1014 (Seal Test, Cond A, B, or C) and TM 2009 (External Visual) | 50 | |
| 2* | TM 1010 (Temperature Cycle, Cond B Minimum) or TM 2001 (Constant Acceleration, Cond B Minimum) TM 5004 (or 5008 for Hybrids) (Final Electricals @ Temp Extremes) and TM 1014 (Seal Test, Cond A, B, or C) and TM 2009 (External Visual) | 37 | |
| 3 | Pre-Burn in Electricals TM 1015 (Burn-in B-Level/S-Level) and TM 5004 (or 5008 for Hybrids) (Post Burn-in Electricals @ Temp Extremes) | 30<br>36 | (B Level)<br>(S Level) |
| 4* | TM 2020 Pind (Particle Impact Noise Detection) | 11 | |
| 5 | TM 5004 (or 5008 for Hybrids) (Final Electricals @ Temperature Extremes) | 11 | (Note 1) |
| 6 | TM 2010/17 (Internal Visual) | 7 | |
| 7* | TM 1014 (Seal Test, Cond A, B, or C) | 7 | (Note 2) |
| 8 | TM 2012 (Radiography) | 7 | |
| 9 | TM 2009 (External Visual) | 7 | (Note 2) |
| 10 | TM 5007/5013 (GaAs) (Wafer Acceptance) | 1 | |
| 11 | TM 2023 (Non-Destructive Bond Pull) | 1 | |

$$\pi_Q = 2 + \frac{87}{\Sigma \text{ Point Valuations}}$$

*NOT APPROPRIATE FOR PLASTIC PARTS.

NOTES:
1. Point valuation only assigned if used independent of Groups 1, 2 or 3.
2. Point valuation only assigned if used independent of Groups 1 or 2.
3. Sequencing of tests within groups 1, 2 and 3 must be followed.
4. TM refers to the MIL-STD-883 Test Method.
5. Nonhermetic parts should be used only in controlled environments (i.e., $G_B$ and other temperature/humidity controlled environments).

EXAMPLES:

1. Mfg. performs Group 1 test and Class B burn-in: $\pi_Q = 2 + \frac{87}{50+30} = 3.1$
2. Mfg. performs internal visual test, seal test and final electrical test: $\pi_Q = 2 + \frac{87}{7+7+11} = 5.5$

| | |
|---|---|
| Other Commercial or Unknown Screening Levels | $\pi_Q = 10$ |



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## 5.11 MICROCIRCUITS, $T_J$ DETERMINATION, (ALL EXCEPT HYBRIDS)

Ideally, device case temperatures should be determined from a detailed thermal analysis of the equipment. Device junction temperature is then calculated with the following relationship:

$$T_J = T_C + \theta_{JC}P$$

$T_J$ = Worst Case Junction Temperature (°C).

$T_C$ = Case Temperature (°C). If not available, use the following default table.

Default Case Temperature ($T_C$) for all Environments

| Environment | $G_B$ | $G_F$ | $G_M$ | $N_S$ | $N_U$ | $A_{IC}$ | $A_{IF}$ | $A_{UC}$ | $A_{UF}$ | $A_{RW}$ | $S_F$ | $M_F$ | $M_L$ | $C_L$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $T_C$ (°C) | 35 | 45 | 50 | 45 | 50 | 60 | 60 | 75 | 75 | 60 | 35 | 50 | 60 | 45 |

$\theta_{JC}$ = Junction-to-case thermal resistance (°C/watt) for a device soldered into a printed circuit board. If $\theta_{JC}$ is not available, use a value contained in a specification for the closest equivalent device or use the following table.

| Package Type (Ceramic Only) | Die Area > 14,400 mil$^2$ $\theta_{JC}$ (°C/W) | Die Area ≤ 14,400 mil$^2$ $\theta_{JC}$ (°C/W) |
|---|---|---|
| Dual-In-Line | 11 | 28 |
| Flat Package | 10 | 22 |
| Chip Carrier | 10 | 20 |
| Pin Grid Array | 10 | 20 |
| Can | – | 70 |

P = The maximum power dissipation realized in a system application. If the applied power is not available, use the maximum power dissipation from the specification for the closest equivalent device.



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## 5.12 MICROCIRCUITS, $T_J$ DETERMINATION, (FOR HYBRIDS)

This section describes a method for estimating junction temperature ($T_J$) for integrated circuit dice mounted in a hybrid package. A hybrid is normally made up of one or more substrate assemblies mounted within a sealed package. Each substrate assembly consists of active and passive chips with thick or thin film metallization mounted on the substrate, which in turn may have multiple layers of metallization and dielectric on the surface. Figure 5-1 is a cross-sectional view of a hybrid with a single multi-layered substrate. The layers within the hybrid are made up of various materials with different thermal characteristics. The table following Figure 5-1 provides a list of commonly used hybrid materials with typical thicknesses and corresponding thermal conductivities (K). If the hybrid internal structure cannot be determined, use the following default values for the temperature rise from case to junction: microcircuits, 10°C; transistors, 25°C; diodes, 20°C. Assume capacitors are at $T_C$.

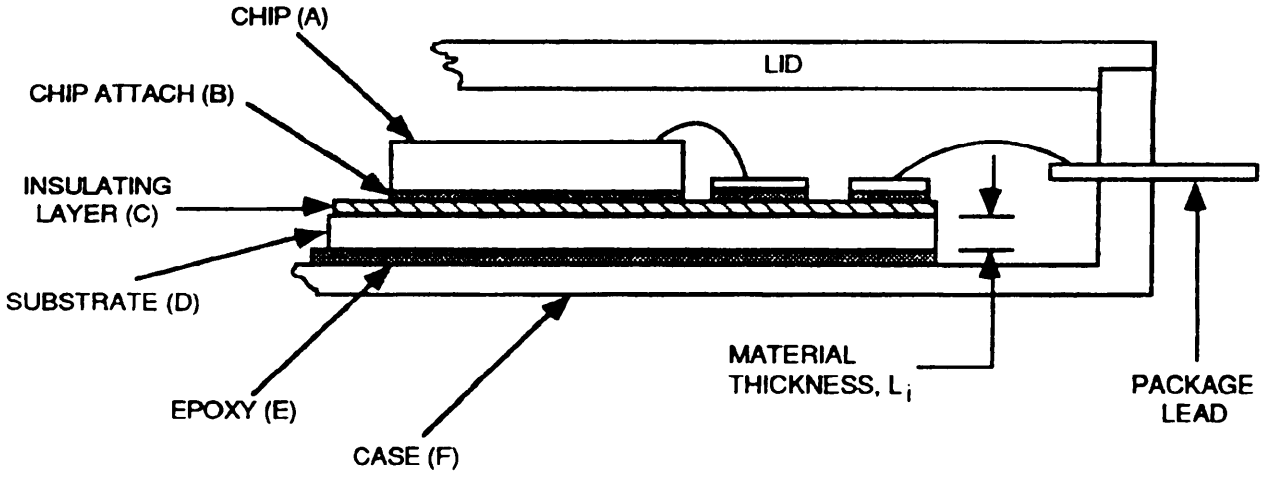



**Figure 5-1: Cross-sectional View of a Hybrid with a Single Multi-Layered Substrate**



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## 5.12 MICROCIRCUITS, $T_J$ DETERMINATION, (FOR HYBRIDS)

Typical Hybrid Characteristics

| Material | Typical Usage | Typical Thickness, $L_i$ (in.) | Feature From Figure 5-1 | Thermal Conductivity, $K_i$ $\left(\frac{W/in^2}{°C/in}\right)$ | $\left(\frac{1}{K_i}\right)\left(L_i\right)$ $\left(in^2\ °C/W\right)$ |
|---|---|---|---|---|---|
| Silicon | Chip Device | 0.010 | A | 2.20 | .0045 |
| GaAs | Chip Device | 0.0070 | A | .76 | .0092 |
| Au Eutectic | Chip Attach | 0.0001 | B | 6.9 | .000014 |
| Solder | Chip/Substrate Attach | 0.0030 | B/E | 1.3 | .0023 |
| Epoxy (Dielectric) | Chip/Substrate Attach | 0.0035 | B/E | .0060 | .58 |
| Epoxy (Conductive) | Chip Attach | 0.0035 | B | .15 | .023 |
| Thick Film Dielectric | Glass Insulating Layer | 0.0030 | C | .66 | .0045 |
| Alumina | Substrate, MHP | 0.025 | D | .64 | .039 |
| Beryllium Oxide | Substrate, PHP | 0.025 | D | 6.6 | .0038 |
| Kovar | Case, MHP | 0.020 | F | .42 | .048 |
| Aluminum | Case, MHP | 0.020 | F | 4.6 | .0043 |
| Copper | Case, PHP | 0.020 | F | 9.9 | .0020 |

NOTE: MHP: Multichip Hybrid Package, PHP: Power Hybrid Package (Pwr: ≥ 2W, Typically)

$$\theta_{JC} = \frac{\sum_{i=1}^{n}\left(\frac{1}{K_i}\right)\left(L_i\right)}{A}$$

n = Number of Material Layers

$K_i$ = Thermal Conductivity of $i^{th}$ Material $\left(\frac{W/in^2}{°C/in}\right)$ (User Provided or From Table)

$L_i$ = Thickness of $i^{th}$ Material (in) (User Provided or From Table)

A = Die Area ($in^2$). If Die Area cannot be readily determined, estimate as follows:
$A = [\ .00278\ (\text{No. of Die Active Wire Terminals}) + .0417]^2$

Estimate $T_J$ as Follows:

$$T_J = T_C + .9\ (\theta_{JC})\ (P_D)$$

$T_C$ = Hybrid Case Temperature (°C). If unknown, use the $T_C$ Default Table shown in Section 5.11.

$\theta_{JC}$ = Junction-to-Case Thermal Resistance (°C/W) (As determined above)

$P_D$ = Die Power Dissipation (W)



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## 5.13 MICROCIRCUITS, EXAMPLES

### Example 1: CMOS Digital Gate Array

Given: A CMOS digital timing chip (4046) in an airborne inhabited cargo application, case temperature 48°C, 75mW power dissipation. The device is procured with normal manufacturer's screening consisting of temperature cycling, constant acceleration, electrical testing, seal test and external visual inspection, in the sequence given. The component manufacturer also performs a B-level burn-in followed by electrical testing. All screens and tests are performed to the applicable MIL-STD-883 screening method. The package is a 24 pin ceramic DIP with a glass seal. The device has been manufactured for several years and has 1000 transistors.

$$\lambda_p = (C_1\pi_T + C_2\pi_E)\,\pi_Q\pi_L \qquad \text{Section 5.1}$$

$C_1 = .020$ — 1000 Transistors = 250 Gates, MOS $C_1$ Table, Digital Column

$\pi_T = .29$ — Determine $T_J$ from Section 5.11
$T_J = 48°C + (28°C/W)(.075W) = 50°C$
Determine $\pi_T$ from Section 5.8, Digital MOS Column.

$C_2 = .011$ — Section 5.9

$\pi_E = 4.0$ — Section 5.10

$\pi_Q = 3.1$ — Section 5.10

| | |
|---|---|
| Group 1 Tests | 50 Points |
| Group 3 Tests (B-level) | 30 Points |
| TOTAL | 80 Points |

$$\pi_Q = 2 + \frac{87}{80} = 3.1$$

$\pi_L = 1$ — Section 5.10

$$\lambda_p = [\,(.020)(.29) + (.011)\,(4)\,]\,(3.1)(1) = .15 \text{ Failure}/10^6 \text{ Hours}$$

### Example 2: EEPROM

Given: A 128K Flotox EEPROM that is expected to have a $T_J$ of 80°C and experience 10,000 read/write cycles over the life of the system. The part is procured to all requirements of Paragraph 1.2.1, MIL-STD-883, Class B screening level requirements and has been in production for three years. It is packaged in a 28 pin DIP with a glass seal and will be used in an airborne uninhabited cargo application.

$$\pi_p = (C_1\,\pi_T + C_2\,\pi_E + \lambda_{cyc})\,\pi_Q\,\pi_L \qquad \text{Section 5.2}$$

$C_1 = .0034$ — Section 5.2

$\pi_T = 3.8$ — Section 5.8

$C_2 = .014$ — Section 5.9



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$\pi_E$ = 5.0 Section 5.10
$\pi_Q$ = 2.0 Section 5.10
$\pi_L$ = 1.0 Section 5.10
$\lambda_{cyc}$ = .38 Section 5.2:

$$\lambda_{cyc} = \left[ A_1 B_1 + \frac{A_2 B_2}{\pi_Q} \right] \pi_{ECC}$$

$A_2 = B_2 = 0$ for Flotox
Assume No ECC, $\pi_{ECC} = 1$
$A_1 = .1$, 7K ≤ C ≤ 15K Entry
$B_1 = 3.8$ (Use Equation 1 at bottom of $B_1$ and $B_2$ Table)
$\lambda_{cyc} = A_1 B_1 = (.1)(3.8) = .38$

$$\lambda_p = [\,(.0034)(3.8) + (.014)(5.0) + .38]\,(2.0)(1) = .93 \text{ Failures}/10^6 \text{ Hours}$$

**Example 3: GaAs MMIC**

Given: A MA4GM212 Single Pole Double Throw Switch, DC - 12 GHz, 4 transistors, 4 inductors, 8 resistors, maximum input $P_D$ = 30 dbm, 16 pin hermetic flatpack, maximum $T_{CH}$ = 145°C in a ground benign environment. The part has been manufactured for 1 year and is screened to Paragraph 1.2.1 of MIL-STD-883, Class B equivalent screen.

$$\lambda_p = [\,C_1 \pi_T \pi_A + C_2 \pi_E\,]\,\pi_L \pi_Q \qquad \text{Section 5.4}$$

$C_1$ = 4.5 Section 5.4, MMIC Table, 4 Active Elements (See Footnote to Table)
$\pi_T$ = .061 Section 5.8, $T_J = T_{CH}$ = 145°C
$\pi_A$ = 3.0 Section 5.4, Unknown Application
$C_2$ = .0047 Section 5.9
$\pi_E$ = .50 Section 5.10
$\pi_L$ = 1.5 Section 5.10
$\pi_Q$ = 2.0 Section 5.10

$$\lambda_p = \left[(4.5)(.061)(3.0) + (.0047)(.5)\right](1.5)(2.0) = 2.5 \text{ Failures}/10^6 \text{ Hours}$$

NOTE: The passive elements are assumed to contribute negligibly to the overall device failure rate.

**Example 4: Hybrid**

Given: A linear multichip hybrid driver in a hermetically sealed Kovar package. The substrate is alumina and there are two thick film dielectric layers. The die and substrate attach materials are conductive epoxy and solder, respectively. The application environment is naval unsheltered, 65°C case temperature and the device has been in production for over two years. The device is



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## 5.13 MICROCIRCUITS, EXAMPLES

screened to MIL-STD-883, Method 5008, in accordance with Table VIII, Class B requirements. The hybrid contains the following components:

| | | | |
|---|---|---|---|
| Active Components: | 1 | - | LM106 Bipolar Comparator/Buffer Die (13 Transistors) |
| | 1 | - | LM741A Bipolar Operational Amplifier Die (24 Transistors) |
| | 2 | - | Si NPN Transistor |
| | 2 | - | Si PNP Transistor |
| | 2 | - | Si General Purpose Diodes |
| Passive Components: | 2 | - | Ceramic Chip Capacitors |
| | 17 | - | Thick Film Resistors |

$$\lambda_p = [\Sigma N_C \lambda_c] (1 + .2\pi_E) \pi_F \pi_Q \pi_L \quad \text{Section 5.5}$$

1. Estimate Active Device Junction Temperatures

If limited information is available on the specific hybrid materials and construction characteristics the default case-to-junction temperature rises shown in the introduction to Section 5.12 can be used. When detailed information becomes available the following Section 5.12 procedure should be used to determine the junction-to-case ($\theta_{JC}$) thermal resistance and $T_J$ values for each component.

$$\theta_{JC} = \frac{\sum_{i=1}^{n} \left(\frac{1}{K_i}\right)(L_i)}{A} \quad \text{(Equation 1)}$$

| Layer | Figure 5-1 Feature | | $\left(\frac{1}{K_i}\right)(L_i)$ $(in^2\ °C/W)$ |
|---|---|---|---|
| Silicon Chip | A | | .0045 |
| Conductive Epoxy | B | | .023 |
| Two Dielectric Layers | C | (2)(.0045) = | .009 |
| Alumina Substrate | D | | .039 |
| Solder Substrate Attachment | E | | .0023 |
| Kovar Case | F | | .048 |
| | | $\Sigma\left(\frac{1}{K_i}\right)(L_i)$ = | .1258 |

$$A = \text{Die Area} = [.00278\ (\text{No. Die Active Wire Terminals}) + .0417]^2 \quad \text{(Equation 2)}$$

$$T_J = T_C + \theta_{JC} P_D \quad \text{(Equation 3)}$$



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| | LM106 | LM741A | Si NPN | Si PNP | Si Diode | Source |
|---|---|---|---|---|---|---|
| No. of Pins | 8 | 14 | 3 | 3 | 2 | Vendor Spec. Sheet |
| Power Dissipation, $P_D$ (W) | .33 | .35 | .6 | .6 | .42 | Circuit Analysis |
| Area of Chip (in.$^2$) | .0041 | .0065 | .0025 | .0025 | .0022 | Equ. 2 Above |
| $\theta_{JC}$ (°C/W) | 30.8 | 19.4 | 50.3 | 50.3 | 56.3 | Equ. 1 Above |
| $T_J$ (°C) | 75 | 72 | 95 | 95 | 89 | Equ. 3 Above |

2. Calculate Failure Rates for Each Component:

A) LM106 Die, 13 Transistors (from Vendor Spec. Sheet)

$\lambda_p = [\, C_1 \pi_T + C_2 \pi_E \,] \pi_Q \pi_L$ Section 5.1

Because $C_2 = 0$;

$\lambda_p = C_1 \pi_T \pi_Q \pi_L$ $\pi_T$: Section 5.8; $\pi_Q$, $\pi_L$ Default to 1.0

$= (.01)(3.8)(1)(1) = .038$ Failures/$10^6$ Hours

B) LM741 Die, 23 Transistors. Use Same Procedure as Above.

$\lambda_p = C_1 \pi_T \pi_Q \pi_L = (.01)(3.1)(1)(1) = .031$ Failures/$10^6$ Hours

C) Silicon NPN Transistor, Rated Power = 5W (From Vendor Spec. Sheet), $V_{CE}/V_{CEO} = .6$, Linear Application

$\lambda_p = \lambda_b \pi_T \pi_A \pi_R \pi_S \pi_Q \pi_E$ Section 6.3; $\pi_Q$, $\pi_E$ Default to 1.0

$= (.00074)(3.9)(1.5)(1.8)(.29)(1)(1)$

$= .0023$ Failures/$10^6$ Hours

D) Silicon PNP Transistor, Same as C.

$\lambda_p = .0023$ Failures/$10^6$ Hours

E) Silicon General Purpose Diode (Analog), Voltage Stress = 60%, Metallurgically Bonded Construction.

$\lambda_p = \lambda_b \pi_T \pi_S \pi_C \pi_Q \pi_E$ Section 6.1; $\pi_Q$, $\pi_E$ Default to 1.0

$= (.0038)(6.3)(.29)(1)(1)(1)$

$= .0069$ Failures/$10^6$ Hours



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## 5.13 MICROCIRCUITS, EXAMPLES

F) Ceramic Chip Capacitor, Voltage Stress = 50%,
$T_A = T_{CASE}$ for the Hybrid, 1340 pF, 125°C Rated Temp.

$\lambda_p = \lambda_b \pi_{CV} \pi_Q \pi_E$ Section 10.11; $\pi_Q$, $\pi_E$ Default to 1.0
$= (.0028)(1.4)(1)(1)$
$= .0039$ Failures/$10^6$ Hours

G) Thick Film Resistors, per instructions in Section 5.5, the contribution of these devices is considered insignificant relative to the overall hybrid failure rate and they may be ignored.

Overall Hybrid Part Failure Rate Calculation:

$$\lambda_p = [\Sigma N_C \lambda_c] (1 + .2 \pi_E) \pi_F \pi_Q \pi_L$$

| | | | |
|---|---|---|---|
| $\pi_E$ | = | 6.0 | Section 5.10 |
| $\pi_F$ | = | 5.8 | Section 5.5 |
| $\pi_Q$ | = | 1 | Section 5.10 |
| $\pi_L$ | = | 1 | Section 5.10 |

$\lambda_p = [(1)(.038) + (1)(.031) + (2)(.0023) + (2)(.0023)$
$+ (2)(.0069) + (2)(.0039)](1 + .2(6.0))(5.8)(1)(1)$

$\lambda_p = 1.3$ Failures/$10^6$ Hours



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## 6.0 DISCRETE SEMICONDUCTORS, INTRODUCTION

The semiconductor transistor, diode and opto-electronic device sections present the failure rates on the basis of device type and construction. An analytical model of the failure rate is also presented for each device category. The various types of discrete semiconductor devices require different failure rate models that vary to some degree. The models apply to single devices unless otherwise noted. For multiple devices in a single package the hybrid model in Section 5.5 should be used.

The applicable MIL specification for transistors, and optoelectronic devices is MIL-S-19500. The quality levels (JAN, JANTX, JANTXV) are as defined in MIL-S-19500.

The temperature factor ($\pi_T$) is based on the device junction temperature. Junction temperature should be computed based on worse case power (or maximum power dissipation) and the device junction to case thermal resistance. Determination of junction temperatures is explained in Section 6.14.

Reference 28 should be consulted for further detailed information on the models appearing in this section.



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## 6.1 DIODES, LOW FREQUENCY

**SPECIFICATION**
MIL-S-19500

**DESCRIPTION**
Low Frequency Diodes: General Purpose Analog, Switching, Fast Recovery, Power Rectifier, Transient Suppressor, Current Regulator, Voltage Regulator, Voltage Reference

$$\lambda_p = \lambda_b \pi_T \pi_S \pi_C \pi_Q \pi_E \quad \text{Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| Diode Type/Application | $\lambda_b$ |
|---|---|
| General Purpose Analog | .0038 |
| Switching | .0010 |
| Power Rectifier, Fast Recovery | .069 |
| Power Rectifier/Schottky Power Diode | .0030 |
| Power Rectifier with High Voltage Stacks | .0050/ Junction |
| Transient Suppressor/Varistor | .0013 |
| Current Regulator | .0034 |
| Voltage Regulator and Voltage Reference (Avalanche and Zener) | .0020 |

Temperature Factor - $\pi_T$
(General Purpose Analog, Switching, Fast Recovery, Power Rectifier, Transient Suppressor)

| $T_J$ (°C) | $\pi_T$ | $T_J$ (°C) | $\pi_T$ |
|---|---|---|---|
| 25 | 1.0 | 105 | 9.0 |
| 30 | 1.2 | 110 | 10 |
| 35 | 1.4 | 115 | 11 |
| 40 | 1.6 | 120 | 12 |
| 45 | 1.9 | 125 | 14 |
| 50 | 2.2 | 130 | 15 |
| 55 | 2.6 | 135 | 16 |
| 60 | 3.0 | 140 | 18 |
| 65 | 3.4 | 145 | 20 |
| 70 | 3.9 | 150 | 21 |
| 75 | 4.4 | 155 | 23 |
| 80 | 5.0 | 160 | 25 |
| 85 | 5.7 | 165 | 28 |
| 90 | 6.4 | 170 | 30 |
| 95 | 7.2 | 175 | 32 |
| 100 | 8.0 | | |

$$\pi_T = \exp\left(-3091\left(\frac{1}{T_J + 273} - \frac{1}{298}\right)\right)$$

$T_J$ = Junction Temperature (°C)

Temperature Factor - $\pi_T$
(Voltage Regulator, Voltage Reference, and Current Regulator)

| $T_J$ (°C) | $\pi_T$ | $T_J$ (°C) | $\pi_T$ |
|---|---|---|---|
| 25 | 1.0 | 105 | 3.9 |
| 30 | 1.1 | 110 | 4.2 |
| 35 | 1.2 | 115 | 4.5 |
| 40 | 1.4 | 120 | 4.8 |
| 45 | 1.5 | 125 | 5.1 |
| 50 | 1.6 | 130 | 5.4 |
| 55 | 1.8 | 135 | 5.7 |
| 60 | 2.0 | 140 | 6.0 |
| 65 | 2.1 | 145 | 6.4 |
| 70 | 2.3 | 150 | 6.7 |
| 75 | 2.5 | 155 | 7.1 |
| 80 | 2.7 | 160 | 7.5 |
| 85 | 3.0 | 165 | 7.9 |
| 90 | 3.2 | 170 | 8.3 |
| 95 | 3.4 | 175 | 8.7 |
| 100 | 3.7 | | |

$$\pi_T = \exp\left(-1925\left(\frac{1}{T_J + 273} - \frac{1}{298}\right)\right)$$

$T_J$ = Junction Temperature (°C)



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## 6.1 DIODES, LOW FREQUENCY

Electrical Stress Factor - $\pi_S$

| Stress | $\pi_S$ |
|---|---|
| Transient Suppressor, Voltage Regulator, Voltage Reference, Current Regulator | 1.0 |
| All Others: | |
| $V_S \leq .30$ | 0.054 |
| $.3 < V_S \leq .40$ | 0.11 |
| $.4 < V_S \leq .50$ | 0.19 |
| $.5 < V_S \leq .60$ | 0.29 |
| $.6 < V_S \leq .70$ | 0.42 |
| $.7 < V_S \leq .80$ | 0.58 |
| $.8 < V_S \leq .90$ | 0.77 |
| $.9 < V_S \leq 1.00$ | 1.0 |

For All Except Transient Suppressor, Voltage Regulator, Voltage Reference, or Current Regulator

$\pi_S = .054 \quad (V_S \leq .3)$

$\pi_S = V_S^{2.43} \quad (.3 < V_S \leq 1)$

$V_S$ = Voltage Stress Ratio = $\frac{\text{Voltage Applied}}{\text{Voltage Rated}}$

Voltage is Diode Reverse Voltage

Contact Construction Factor - $\pi_C$

| Contact Construction | $\pi_C$ |
|---|---|
| Metallurgically Bonded | 1.0 |
| Non-Metallurgically Bonded and Spring Loaded Contacts | 2.0 |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| JANTXV | 0.7 |
| JANTX | 1.0 |
| JAN | 2.4 |
| Lower | 5.5 |
| Plastic | 8.0 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 6.0 |
| $G_M$ | 9.0 |
| $N_S$ | 9.0 |
| $N_U$ | 19 |
| $A_{IC}$ | 13 |
| $A_{IF}$ | 29 |
| $A_{UC}$ | 20 |
| $A_{UF}$ | 43 |
| $A_{RW}$ | 24 |
| $S_F$ | .50 |
| $M_F$ | 14 |
| $M_L$ | 32 |
| $C_L$ | 320 |



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## 6.2 DIODES, HIGH FREQUENCY (MICROWAVE, RF)

**SPECIFICATION**
MIL-S-19500

**DESCRIPTION**
Si IMPATT; Bulk Effect, Gunn; Tunnel, Back; Mixer, Detector, PIN, Schottky; Varactor, Step Recovery

$$\lambda_p = \lambda_b \pi_T \pi_A \pi_R \pi_Q \pi_E \quad \text{Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| Diode Type | $\lambda_b$ |
|---|---|
| Si IMPATT (≤ 35 GHz) | .22 |
| Gunn/Bulk Effect | .18 |
| Tunnel and Back (Including Mixers, Detectors) | .0023 |
| PIN | .0081 |
| Schottky Barrier (Including Detectors) and Point Contact (200 MHz ≤ Frequency ≤ 35 GHz) | .027 |
| Varactor and Step Recovery | .0025 |

Temperature Factor - $\pi_T$
(All Types Except IMPATT)

| $T_J$ (°C) | $\pi_T$ | $T_J$ (°C) | $\pi_T$ |
|---|---|---|---|
| 25 | 1.0 | 105 | 4.4 |
| 30 | 1.1 | 110 | 4.8 |
| 35 | 1.3 | 115 | 5.1 |
| 40 | 1.4 | 120 | 5.5 |
| 45 | 1.6 | 125 | 5.9 |
| 50 | 1.7 | 130 | 6.3 |
| 55 | 1.9 | 135 | 6.7 |
| 60 | 2.1 | 140 | 7.1 |
| 65 | 2.3 | 145 | 7.6 |
| 70 | 2.5 | 150 | 8.0 |
| 75 | 2.8 | 155 | 8.5 |
| 80 | 3.0 | 160 | 9.0 |
| 85 | 3.3 | 165 | 9.5 |
| 90 | 3.5 | 170 | 10 |
| 95 | 3.8 | 175 | 11 |
| 100 | 4.1 | | |

$$\pi_T = \exp\left(-2100\left(\frac{1}{T_J + 273} - \frac{1}{298}\right)\right)$$

$T_J$ = Junction Temperature (°C)

Temperature Factor - $\pi_T$
(IMPATT)

| $T_J$ (°C) | $\pi_T$ | $T_J$ (°C) | $\pi_T$ |
|---|---|---|---|
| 25 | 1.0 | 105 | 42 |
| 30 | 1.3 | 110 | 50 |
| 35 | 1.8 | 115 | 60 |
| 40 | 2.3 | 120 | 71 |
| 45 | 3.0 | 125 | 84 |
| 50 | 3.9 | 130 | 99 |
| 55 | 5.0 | 135 | 120 |
| 60 | 6.4 | 140 | 140 |
| 65 | 8.1 | 145 | 160 |
| 70 | 10 | 150 | 180 |
| 75 | 13 | 155 | 210 |
| 80 | 16 | 160 | 250 |
| 85 | 19 | 165 | 280 |
| 90 | 24 | 170 | 320 |
| 95 | 29 | 175 | 370 |
| 100 | 35 | | |

$$\pi_T = \exp\left(-5260\left(\frac{1}{T_J + 273} - \frac{1}{298}\right)\right)$$

$T_J$ = Junction Temperature (°C)

Application Factor - $\pi_A$

| Diodes Application | $\pi_A$ |
|---|---|
| Varactor, Voltage Control | .50 |
| Varactor, Multiplier | 2.5 |
| All Other Diodes | 1.0 |



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## 6.2 DIODES, HIGH FREQUENCY (MICROWAVE, RF)

Power Rating Factor - $\pi_R$

| Rated Power, $P_r$ (Watts) | $\pi_R$ |
|---|---|
| PIN Diodes | |
| $P_r \leq 10$ | .50 |
| $10 < P_r \leq 100$ | 1.3 |
| $100 < P_r \leq 1000$ | 2.0 |
| $1000 < P_r \leq 3000$ | 2.4 |
| All Other Diodes | 1.0 |

PIN Diodes $\pi_R = .326\ ln(P_r) - .25$

All Other Diodes $\pi_R = 1.0$

Quality Factor - $\pi_Q$
(Schottky)

| Quality* | $\pi_Q$ |
|---|---|
| JANTXV | .50 |
| JANTX | 1.0 |
| JAN | 1.8 |
| Lower | 2.5 |
| Plastic | — |

* For high frequency part classes not specified to MIL-S-19500 equipment quality classes are defined as devices meeting the same requirements as MIL-S-19500.

Quality Factor - $\pi_Q$
(All Types Except Schottky)

| Quality * | $\pi_Q$ |
|---|---|
| JANTXV | .50 |
| JANTX | 1.0 |
| JAN | 5.0 |
| Lower | 25 |
| Plastic | 50 |

* For high frequency part classes not specified to MIL-S-19500 equipment quality classes are defined as devices meeting the same requirements as MIL-S-19500.

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 5.0 |
| $N_S$ | 4.0 |
| $N_U$ | 11 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 5.0 |
| $A_{UC}$ | 7.0 |
| $A_{UF}$ | 12 |
| $A_{RW}$ | 16 |
| $S_F$ | .50 |
| $M_F$ | 9.0 |
| $M_L$ | 24 |
| $C_L$ | 250 |



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## 6.3 TRANSISTORS, LOW FREQUENCY, BIPOLAR

**SPECIFICATION**
MIL-S-19500

**DESCRIPTION**
NPN (Frequency < 200 MHz)
PNP (Frequency < 200 MHz)

$$\lambda_p = \lambda_b \pi_T \pi_A \pi_R \pi_S \pi_Q \pi_E \quad \text{Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| Type | $\lambda_b$ |
|---|---|
| NPN and PNP | .00074 |

Application Factor - $\pi_A$

| Application | $\pi_A$ |
|---|---|
| Linear Amplification | 1.5 |
| Switching | .70 |

Temperature Factor - $\pi_T$

| $T_J$ (°C) | $\pi_T$ | $T_J$ (°C) | $\pi_T$ |
|---|---|---|---|
| 25 | 1.0 | 105 | 4.5 |
| 30 | 1.1 | 110 | 4.8 |
| 35 | 1.3 | 115 | 5.2 |
| 40 | 1.4 | 120 | 5.6 |
| 45 | 1.6 | 125 | 5.9 |
| 50 | 1.7 | 130 | 6.3 |
| 55 | 1.9 | 135 | 6.8 |
| 60 | 2.1 | 140 | 7.2 |
| 65 | 2.3 | 145 | 7.7 |
| 70 | 2.5 | 150 | 8.1 |
| 75 | 2.8 | 155 | 8.6 |
| 80 | 3.0 | 160 | 9.1 |
| 85 | 3.3 | 165 | 9.7 |
| 90 | 3.6 | 170 | 10 |
| 95 | 3.9 | 175 | 11 |
| 100 | 4.2 | | |

$$\pi_T = \exp\left(-2114\left(\frac{1}{T_J + 273} - \frac{1}{298}\right)\right)$$

$T_J$ = Junction Temperature (°C)

Power Rating Factor - $\pi_R$

| Rated Power ($P_r$, Watts) | $\pi_R$ |
|---|---|
| $P_r \leq .1$ | .43 |
| $P_r = .5$ | .77 |
| $P_r = 1.0$ | 1.0 |
| $P_r = 5.0$ | 1.8 |
| $P_r = 10.0$ | 2.3 |
| $P_r = 50.0$ | 4.3 |
| $P_r = 100.0$ | 5.5 |
| $P_r = 500.0$ | 10 |

$\pi_R = .43$ Rated Power ≤ .1W

$\pi_R = (P_r)^{.37}$ Rated Power > .1W



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surfaces are not counted. See Figure B-1 for

## 6.3 TRANSISTORS, LOW FREQUENCY, BIPOLAR

### Voltage Stress Factor - $\pi_S$

| Applied $V_{CE}$/Rated $V_{CEO}$ | $\pi_S$ |
|---|---|
| $0 < V_S \leq .3$ | .11 |
| $.3 < V_S \leq .4$ | .16 |
| $.4 < V_S \leq .5$ | .21 |
| $.5 < V_S \leq .6$ | .29 |
| $.6 < V_S \leq .7$ | .39 |
| $.7 < V_S \leq .8$ | .54 |
| $.8 < V_S \leq .9$ | .73 |
| $.9 < V_S \leq 1.0$ | 1.0 |

$\pi_S$ = .045 exp (3.1(Vs)) $(0 < V_S \leq 1.0)$

$V_s$ = Applied $V_{CE}$ / Rated $V_{CEO}$

$V_{CE}$ = Voltage, Collector to Emitter

$V_{CEO}$ = Voltage, Collector to Emitter, Base Open

### Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 6.0 |
| $G_M$ | 9.0 |
| $N_S$ | 9.0 |
| $N_U$ | 19 |
| $A_{IC}$ | 13 |
| $A_{IF}$ | 29 |
| $A_{UC}$ | 20 |
| $A_{UF}$ | 43 |
| $A_{RW}$ | 24 |
| $S_F$ | .50 |
| $M_F$ | 14 |
| $M_L$ | 32 |
| $C_L$ | 320 |

### Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| JANTXV | .70 |
| JANTX | 1.0 |
| JAN | 2.4 |
| Lower | 5.5 |
| Plastic | 8.0 |



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## 6.4 TRANSISTORS, LOW FREQUENCY, SI FET

**SPECIFICATION**
MIL-S-19500

**DESCRIPTION**
N-Channel and P-Channel Si FET (Frequency ≤ 400 MHz)

$$\lambda_p = \lambda_b \pi_T \pi_A \pi_Q \pi_E \quad \text{Failures/}10^6\ \text{Hours}$$

Base Failure Rate - $\lambda_b$

| Transistor Type | $\lambda_b$ |
|---|---|
| MOSFET | .012 |
| JFET | .0045 |

Temperature Factor - $\pi_T$

| $T_J$ (°C) | $\pi_T$ | $T_J$ (°C) | $\pi_T$ |
|---|---|---|---|
| 25 | 1.0 | 105 | 3.9 |
| 30 | 1.1 | 110 | 4.2 |
| 35 | 1.2 | 115 | 4.5 |
| 40 | 1.4 | 120 | 4.8 |
| 45 | 1.5 | 125 | 5.1 |
| 50 | 1.6 | 130 | 5.4 |
| 55 | 1.8 | 135 | 5.7 |
| 60 | 2.0 | 140 | 6.0 |
| 65 | 2.1 | 145 | 6.4 |
| 70 | 2.3 | 150 | 6.7 |
| 75 | 2.5 | 155 | 7.1 |
| 80 | 2.7 | 160 | 7.5 |
| 85 | 3.0 | 165 | 7.9 |
| 90 | 3.2 | 170 | 8.3 |
| 95 | 3.4 | 175 | 8.7 |
| 100 | 3.7 | | |

$$\pi_T = \exp\left(-1925\left(\frac{1}{T_J + 273} - \frac{1}{298}\right)\right)$$

$T_J$ = Junction Temperature (°C)

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| JANTXV | .70 |
| JANTX | 1.0 |
| JAN | 2.4 |
| Lower | 5.5 |
| Plastic | 8.0 |

Application Factor - $\pi_A$

| Application ($P_r$, Rated Output Power) | $\pi_A$ |
|---|---|
| Linear Amplification ($P_r$ < 2W) | 1.5 |
| Small Signal Switching | .70 |
| Power FETs (Non-linear, $P_r \geq$ 2W) | |
| 2 ≤ $P_r$ < 5W | 2.0 |
| 5 ≤ $P_r$ < 50W | 4.0 |
| 50 ≤ $P_r$ < 250W | 8.0 |
| $P_r \geq$ 250W | 10 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 6.0 |
| $G_M$ | 9.0 |
| $N_S$ | 9.0 |
| $N_U$ | 19 |
| $A_{IC}$ | 13 |
| $A_{IF}$ | 29 |
| $A_{UC}$ | 20 |
| $A_{UF}$ | 43 |
| $A_{RW}$ | 24 |
| $S_F$ | .50 |
| $M_F$ | 14 |
| $M_L$ | 32 |
| $C_L$ | 320 |



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## 6.5 TRANSISTORS, UNIJUNCTION

**SPECIFICATION**
MIL-S-19500

**DESCRIPTION**
Unijunction Transistors

$$\lambda_p = \lambda_b \pi_T \pi_Q \pi_E \quad \text{Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| Type | $\lambda_b$ |
|---|---|
| All Unijunction | .0083 |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| JANTXV | .70 |
| JANTX | 1.0 |
| JAN | 2.4 |
| Lower | 5.5 |
| Plastic | 8.0 |

Temperature Factor - $\pi_T$

| $T_J$ (°C) | $\pi_T$ | $T_J$ (°C) | $\pi_T$ |
|---|---|---|---|
| 25 | 1.0 | 105 | 5.8 |
| 30 | 1.1 | 110 | 6.4 |
| 35 | 1.3 | 115 | 6.9 |
| 40 | 1.5 | 120 | 7.5 |
| 45 | 1.7 | 125 | 8.1 |
| 50 | 1.9 | 130 | 8.8 |
| 55 | 2.1 | 135 | 9.5 |
| 60 | 2.4 | 140 | 10 |
| 65 | 2.7 | 145 | 11 |
| 70 | 3.0 | 150 | 12 |
| 75 | 3.3 | 155 | 13 |
| 80 | 3.7 | 160 | 13 |
| 85 | 4.0 | 165 | 14 |
| 90 | 4.4 | 170 | 15 |
| 95 | 4.9 | 175 | 16 |
| 100 | 5.3 | | |

$$\pi_T = \exp\left(-2483\left(\frac{1}{T_J + 273} - \frac{1}{298}\right)\right)$$

$T_J$ = Junction Temperature (°C)

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 6.0 |
| $G_M$ | 9.0 |
| $N_S$ | 9.0 |
| $N_U$ | 19 |
| $A_{IC}$ | 13 |
| $A_{IF}$ | 29 |
| $A_{UC}$ | 20 |
| $A_{UF}$ | 43 |
| $A_{RW}$ | 24 |
| $S_F$ | .50 |
| $M_F$ | 14 |
| $M_L$ | 32 |
| $C_L$ | 320 |



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## 6.6 TRANSISTORS, LOW NOISE, HIGH FREQUENCY, BIPOLAR

**SPECIFICATION**
MIL-S-19500

**DESCRIPTION**
Bipolar, Microwave RF Transistor
(Frequency > 200 MHz, Power < 1W)

$$\lambda_p = \lambda_b \pi_T \pi_R \pi_S \pi_Q \pi_E \quad \text{Failures/}10^6 \text{ Hours}$$

**Application Note:** The model applies to a single die (for multiple die use the hybrid model). The model does ~~apply to ganged transistors on a single die~~.

Base Failure Rate - $\lambda_b$

| Type | $\lambda_b$ |
|---|---|
| All Types | .18 |

Temperature Factor - $\pi_T$

| $T_J$ (°C) | $\pi_T$ | $T_J$ (°C) | $\pi_T$ |
|---|---|---|---|
| 25 | 1.0 | 105 | 4.5 |
| 30 | 1.1 | 110 | 4.8 |
| 35 | 1.3 | 115 | 5.2 |
| 40 | 1.4 | 120 | 5.6 |
| 45 | 1.6 | 125 | 5.9 |
| 50 | 1.7 | 130 | 6.3 |
| 55 | 1.9 | 135 | 6.8 |
| 60 | 2.1 | 140 | 7.2 |
| 65 | 2.3 | 145 | 7.7 |
| 70 | 2.5 | 150 | 8.1 |
| 75 | 2.8 | 155 | 8.6 |
| 80 | 3.0 | 160 | 9.1 |
| 85 | 3.3 | 165 | 9.7 |
| 90 | 3.6 | 170 | 10 |
| 95 | 3.9 | 175 | 11 |
| 100 | 4.2 | | |

$$\pi_T = \exp\left(-2114\left(\frac{1}{T_J + 273} - \frac{1}{298}\right)\right)$$

$T_J$ = Junction Temperature (°C)

Power Rating Factor - $\pi_R$

| Rated Power ($P_r$, Watts) | $\pi_R$ |
|---|---|
| $P_r \leq .1$ | .43 |
| $.1 < P_r \leq .2$ | .55 |
| $.2 < P_r \leq .3$ | .64 |
| $.3 < P_r \leq .4$ | .71 |
| $.4 < P_r \leq .5$ | .77 |
| $.5 < P_r \leq .6$ | .83 |
| $.6 < P_r \leq .7$ | .88 |
| $.7 < P_r \leq .8$ | .92 |
| $.8 < P_r \leq .9$ | .96 |

$\pi_R = .43$ $\quad P_r \leq .1W$

$\pi_R = (P_r)^{.37}$ $\quad P_r > .1W$

Voltage Stress Factor - $\pi_S$

| Applied $V_{CE}$/Rated $V_{CEO}$ | $\pi_S$ |
|---|---|
| $0 < V_s \leq .3$ | .11 |
| $.3 < V_s \leq .4$ | .16 |
| $.4 < V_s \leq .5$ | .21 |
| $.5 < V_s \leq .6$ | .29 |
| $.6 < V_s \leq .7$ | .39 |
| $.7 < V_s \leq .8$ | .54 |
| $.8 < V_s \leq .9$ | .73 |
| $.9 < V_s \leq 1.0$ | 1.0 |

$\pi_S = .045 \exp(3.1(Vs)) \quad (0 < V_s \leq 1.0)$

$V_s$ = Applied $V_{CE}$ / Rated $V_{CEO}$

$V_{CE}$ = Voltage, Collector to Emitter

$V_{CEO}$ = Voltage, Collector to Emitter, Base Open



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## 6.6 TRANSISTORS, LOW NOISE, HIGH FREQUENCY, BIPOLAR

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| JANTXV | .50 |
| JANTX | 1.0 |
| JAN | 2.0 |
| Lower | 5.0 |

NOTE: For these devices, JANTXV quality class must include IR Scan for die attach and screen for barrier layer pinholes on gold metallized devices.

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 5.0 |
| $N_S$ | 4.0 |
| $N_U$ | 11 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 5.0 |
| $A_{UC}$ | 7.0 |
| $A_{UF}$ | 12 |
| $A_{RW}$ | 16 |
| $S_F$ | .50 |
| $M_F$ | 9.0 |
| $M_L$ | 24 |
| $C_L$ | 250 |



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## 6.7 TRANSISTORS, HIGH POWER, HIGH FREQUENCY, BIPOLAR

**SPECIFICATION**
MIL-S-19500

**DESCRIPTION**
Power, Microwave, RF Bipolar Transistors
(Average Power ≥ 1W)

$$\lambda_p = \lambda_b \pi_T \pi_A \pi_M \pi_Q \pi_E \quad \text{Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| Frequency (GHz) | Output Power (Watts) 1.0 | 5.0 | 10 | 50 | 100 | 200 | 300 | 400 | 500 | 600 |
|---|---|---|---|---|---|---|---|---|---|---|
| ≤ 0.5 | .038 | .039 | .040 | .050 | .067 | .12 | .20 | .36 | .62 | 1.1 |
| 1 | .046 | .047 | .048 | .060 | .080 | .14 | .24 | .42 | .74 | 1.3 |
| 2 | .065 | .067 | .069 | .086 | .11 | .20 | .35 | | | |
| 3 | .093 | .095 | .098 | .12 | .16 | .28 | | | | |
| 4 | .13 | .14 | .14 | .17 | .23 | | | | | |
| 5 | .19 | .19 | .20 | .25 | | | | | | |

$\lambda_b$ = .032 exp(.354(F) + .00558(P))  F = Frequency (GHz)  P = Output Power (W)

NOTE: Output power refers to the power level for the overall packaged device and not to individual transistors within the package (if more than one transistor is ganged together). The output power represents the power output from the active device and should not account for any duty cycle in pulsed applications. Duty cycle is accounted for when determining $\pi_A$.

Temperature Factor - $\pi_T$
(Gold Metallization)

| $T_J$ (°C) | $V_S$ ($V_{CE}/BV_{CES}$) ≤ .40 | .45 | .50 | .55 |
|---|---|---|---|---|
| ≤100 | .10 | .20 | .30 | .40 |
| 110 | .12 | .25 | .37 | .49 |
| 120 | .15 | .30 | .45 | .59 |
| 130 | .18 | .36 | .54 | .71 |
| 140 | .21 | .43 | .64 | .85 |
| 150 | .25 | .50 | .75 | 1.0 |
| 160 | .29 | .59 | .88 | 1.2 |
| 170 | .34 | .68 | 1.0 | 1.4 |
| 180 | .40 | .79 | 1.2 | 1.6 |
| 190 | .45 | .91 | 1.4 | 1.8 |
| 200 | .52 | 1.0 | 1.6 | 2.1 |

$$\pi_T = .1 \exp\left(-2903\left(\frac{1}{T_J + 273} - \frac{1}{373}\right)\right), \quad (V_S \le .40)$$

$$\pi_T = 2(V_S - .35) \exp\left(-2903\left(\frac{1}{T_J + 273} - \frac{1}{373}\right)\right), \quad (.4 < V_S \le .55)$$

$V_S$ = $V_{CE}$ / $BV_{CES}$
$V_{CE}$ = Operating Voltage (Volts)
$BV_{CES}$ = Collector-Emitter Breakdown Voltage with Base Shorted to Emitter (Volts)
$T_J$ = Peak Junction Temperature (°C)

Temperature Factor - $\pi_T$
(Aluminum Metallization)

| $T_J$ (°C) | $V_S$ ($V_{CE}/BV_{CES}$) ≤ .40 | .45 | .50 | .55 |
|---|---|---|---|---|
| ≤100 | .38 | .75 | 1.1 | 1.5 |
| 110 | .57 | 1.1 | 1.7 | 2.3 |
| 120 | .84 | 1.7 | 2.5 | 3.3 |
| 130 | 1.2 | 2.4 | 3.6 | 4.8 |
| 140 | 1.7 | 3.4 | 5.1 | 6.8 |
| 150 | 2.4 | 4.7 | 7.1 | 9.5 |
| 160 | 3.3 | 6.5 | 9.7 | 13 |
| 170 | 4.4 | 8.8 | 13 | 18 |
| 180 | 5.9 | 12 | 18 | 23 |
| 190 | 7.8 | 15 | 23 | 31 |
| 200 | 10 | 20 | 30 | 40 |

$$\pi_T = .38 \exp\left(-5794\left(\frac{1}{T_J + 273} - \frac{1}{373}\right)\right), \quad (V_S \le .40)$$

$$\pi_T = 7.55(V_S - .35) \exp\left(-5794\left(\frac{1}{T_J + 273} - \frac{1}{373}\right)\right), \quad (.4 < V_S \le .55)$$

$V_S$ = $V_{CE}$ / $BV_{CES}$
$V_{CE}$ = Operating Voltage (Volts)
$BV_{CES}$ = Collector-Emitter Breakdown Voltage with Base Shorted to Emitter (Volts)
$T_J$ = Peak Junction Temperature (°C)



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## 6.7 TRANSISTORS, HIGH POWER, HIGH FREQUENCY, BIPOLAR

Application Factor - $\pi_A$

| Application | Duty Factor | $\pi_A$ |
|---|---|---|
| CW | N/A | 7.6 |
| Pulsed | ≤ 1% | .46 |
| | 5% | .70 |
| | 10% | 1.0 |
| | 15% | 1.3 |
| | 20% | 1.6 |
| | 25% | 1.9 |
| | ≥ 30% | 2.2 |

$\pi_A$ = 7.6, CW

$\pi_A$ = .06 (Duty Factor %) + .40 , Pulsed

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| JANTXV | .50 |
| JANTX | 1.0 |
| JAN | 2.0 |
| Lower | 5.0 |

NOTE: For these devices, JANTXV quality class must include IR Scan for die attach and screen for barrier layer pinholes on gold metallized devices.

Matching Network Factor - $\pi_M$

| Matching | $\pi_M$ |
|---|---|
| Input and Output | 1.0 |
| Input | 2.0 |
| None | 4.0 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 5.0 |
| $N_S$ | 4.0 |
| $N_U$ | 11 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 5.0 |
| $A_{UC}$ | 7.0 |
| $A_{UF}$ | 12 |
| $A_{RW}$ | 16 |
| $S_F$ | .50 |
| $M_F$ | 9.0 |
| $M_L$ | 24 |
| $C_L$ | 250 |



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## 6.8 TRANSISTORS, HIGH FREQUENCY, GaAs FET

**SPECIFICATION**
MIL-S-19500

**DESCRIPTION**
GaAs Low Noise, Driver and Power FETs (≥ 1GHz)

$$\lambda_p = \lambda_b \pi_T \pi_A \pi_M \pi_Q \pi_E \quad \text{Failures/}10^6\text{ Hours}$$

Base Failure Rate - $\lambda_b$

| Operating Frequency (GHz) | Average Output Power (Watts) < .1 | .1 | .5 | 1 | 2 | 4 | 6 |
|---|---|---|---|---|---|---|---|
| 1 | .052 | -- | -- | -- | -- | -- | -- |
| 4 | .052 | .054 | .066 | .084 | .14 | .36 | .96 |
| 5 | .052 | .083 | .10 | .13 | .21 | .56 | 1.5 |
| 6 | .052 | .13 | .16 | .20 | .32 | .85 | 2.3 |
| 7 | .052 | .20 | .24 | .30 | .50 | 1.3 | 3.5 |
| 8 | .052 | .30 | .37 | .47 | .76 | 2.0 | |
| 9 | .052 | .46 | .56 | .72 | 1.2 | | |
| 10 | .052 | .71 | .87 | 1.1 | 1.8 | | |

$\lambda_b$ = .052 $\quad 1 \le F \le 10, \quad P < .1$

$\lambda_b$ = .0093 exp(.429(F) + .486(P)) $\quad 4 \le F \le 10, \quad .1 \le P \le 6$

F = Frequency (GHz) $\quad$ P = Average Output Power (Watts)

The average output power represents the power output from the active device and should not account for any duty cycle in pulsed applications.

Temperature Factor - $\pi_T$

| $T_C$ (°C) | $\pi_T$ | $T_C$ (°C) | $\pi_T$ |
|---|---|---|---|
| 25 | 1.0 | 105 | 24 |
| 30 | 1.3 | 110 | 28 |
| 35 | 1.6 | 115 | 33 |
| 40 | 2.1 | 120 | 38 |
| 45 | 2.6 | 125 | 44 |
| 50 | 3.2 | 130 | 50 |
| 55 | 4.0 | 135 | 58 |
| 60 | 4.9 | 140 | 66 |
| 65 | 5.9 | 145 | 75 |
| 70 | 7.2 | 150 | 85 |
| 75 | 8.7 | 155 | 97 |
| 80 | 10 | 160 | 110 |
| 85 | 12 | 165 | 120 |
| 90 | 15 | 170 | 140 |
| 95 | 18 | 175 | 150 |
| 100 | 21 | | |

$$\pi_T = \exp\left(-4485\left(\frac{1}{T_C + 273} - \frac{1}{298}\right)\right)$$

$T_C$ = Channel Temperature (°C)

Application Factor - $\pi_A$

| Application (P ≤ 6W) | $\pi_A$ |
|---|---|
| All Low Power and Pulsed | 1 |
| CW | 4 |

P = Average Output Power (Watts)



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## 6.8 TRANSISTORS, HIGH FREQUENCY, GaAs FET

Matching Network Factor - $\pi_M$

| Matching | $\pi_M$ |
|---|---|
| Input and Output | 1.0 |
| Input Only | 2.0 |
| None | 4.0 |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| JANTXV | .50 |
| JANTX | 1.0 |
| JAN | 2.0 |
| Lower | 5.0 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 5.0 |
| $N_S$ | 4.0 |
| $N_U$ | 11 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 5.0 |
| $A_{UC}$ | 7.0 |
| $A_{UF}$ | 12 |
| $A_{RW}$ | 16 |
| $S_F$ | .50 |
| $M_F$ | 7.5 |
| $M_L$ | 24 |
| $C_L$ | 250 |



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## 6.9 TRANSISTORS, HIGH FREQUENCY, SI FET

**SPECIFICATION**
MIL-S-19500

**DESCRIPTION**
Si FETs (Avg. Power < 300 mW, Freq. > 400 MHz)

$$\lambda_p = \lambda_b \pi_T \pi_Q \pi_E \quad \text{Failures/}10^6\text{ Hours}$$

Base Failure Rate - $\lambda_b$

| Transistor Type | $\lambda_b$ |
|---|---|
| MOSFET | .060 |
| JFET | .023 |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| JANTXV | .50 |
| JANTX | 1.0 |
| JAN | 2.0 |
| Lower | 5.0 |

Temperature Factor - $\pi_T$

| $T_J$ (°C) | $\pi_T$ | $T_J$ (°C) | $\pi_T$ |
|---|---|---|---|
| 25 | 1.0 | 105 | 3.9 |
| 30 | 1.1 | 110 | 4.2 |
| 35 | 1.2 | 115 | 4.5 |
| 40 | 1.4 | 120 | 4.8 |
| 45 | 1.5 | 125 | 5.1 |
| 50 | 1.6 | 130 | 5.4 |
| 55 | 1.8 | 135 | 5.7 |
| 60 | 2.0 | 140 | 6.0 |
| 65 | 2.1 | 145 | 6.4 |
| 70 | 2.3 | 150 | 6.7 |
| 75 | 2.5 | 155 | 7.1 |
| 80 | 2.7 | 160 | 7.5 |
| 85 | 3.0 | 165 | 7.9 |
| 90 | 3.2 | 170 | 8.3 |
| 95 | 3.4 | 175 | 8.7 |
| 100 | 3.7 | | |

$$\pi_T = \exp\left(-1925\left(\frac{1}{T_J + 273} - \frac{1}{298}\right)\right)$$

$T_J$ = Junction Temperature (°C)

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 5.0 |
| $N_S$ | 4.0 |
| $N_U$ | 11 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 5.0 |
| $A_{UC}$ | 7.0 |
| $A_{UF}$ | 12 |
| $A_{RW}$ | 16 |
| $S_F$ | .50 |
| $M_F$ | 9.0 |
| $M_L$ | 24 |
| $C_L$ | 250 |



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**SPECIFICATION**
MIL-S-19500

**DESCRIPTION**
Thyristors
SCRs, Triacs

$$\lambda_p = \lambda_b \pi_T \pi_R \pi_S \pi_Q \pi_E \quad \text{Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| Device Type | $\lambda_b$ |
|---|---|
| All Types | .0022 |

Temperature Factor - $\pi_T$

| $T_J$ (°C) | $\pi_T$ | $T_J$ (°C) | $\pi_T$ |
|---|---|---|---|
| 25 | 1.0 | 105 | 8.9 |
| 30 | 1.2 | 110 | 9.9 |
| 35 | 1.4 | 115 | 11 |
| 40 | 1.6 | 120 | 12 |
| 45 | 1.9 | 125 | 13 |
| 50 | 2.2 | 130 | 15 |
| 55 | 2.6 | 135 | 16 |
| 60 | 3.0 | 140 | 18 |
| 65 | 3.4 | 145 | 19 |
| 70 | 3.9 | 150 | 21 |
| 75 | 4.4 | 155 | 23 |
| 80 | 5.0 | 160 | 25 |
| 85 | 5.7 | 165 | 27 |
| 90 | 6.4 | 170 | 30 |
| 95 | 7.2 | 175 | 32 |
| 100 | 8.0 | | |

$$\pi_T = \exp\left(-3082\left(\frac{1}{T_J + 273} - \frac{1}{298}\right)\right)$$

$T_J$ = Junction Temperature (°C)

Current Rating Factor - $\pi_R$

| Rated Forward Current ($I_{f_{rms}}$ (Amps)) | $\pi_R$ |
|---|---|
| .05 | .30 |
| .10 | .40 |
| .50 | .76 |
| 1.0 | 1.0 |
| 5.0 | 1.9 |
| 10 | 2.5 |
| 20 | 3.3 |
| 30 | 3.9 |
| 40 | 4.4 |
| 50 | 4.8 |
| 60 | 5.1 |
| 70 | 5.5 |
| 80 | 5.8 |
| 90 | 6.0 |
| 100 | 6.3 |
| 110 | 6.6 |
| 120 | 6.8 |
| 130 | 7.0 |
| 140 | 7.2 |
| 150 | 7.4 |
| 160 | 7.6 |
| 170 | 7.8 |
| 175 | 7.9 |

$$\pi_R = (I_{f_{rms}})^{.40}$$

$I_{f_{rms}}$ = RMS Rated Forward Current (Amps)



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## 6.10 THYRISTORS AND SCRS

Voltage Stress Factor - $\pi_S$

| $V_S$ (Blocking Voltage Applied/ Blocking Voltage Rated) | $\pi_S$ |
|---|---|
| $V_S \leq .30$ | .10 |
| $.3 < V_S \leq .4$ | .18 |
| $.4 < V_S \leq .5$ | .27 |
| $.5 < V_S \leq .6$ | .38 |
| $.6 < V_S \leq .7$ | .51 |
| $.7 < V_S \leq .8$ | .65 |
| $.8 < V_S \leq .9$ | .82 |
| $.9 < V_S \leq 1.0$ | 1.0 |
| $\pi_S = .10$ | $(V_S \leq 0.3)$ |
| $\pi_S = (V_S)^{1.9}$ | $(V_S > 0.3)$ |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 6.0 |
| $G_M$ | 9.0 |
| $N_S$ | 9.0 |
| $N_U$ | 19 |
| $A_{IC}$ | 13 |
| $A_{IF}$ | 29 |
| $A_{UC}$ | 20 |
| $A_{UF}$ | 43 |
| $A_{RW}$ | 24 |
| $S_F$ | .50 |
| $M_F$ | 14 |
| $M_L$ | 32 |
| $C_L$ | 320 |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| JANTXV | 0.7 |
| JANTX | 1.0 |
| JAN | 2.4 |
| Lower | 5.5 |
| Plastic | 8.0 |



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## 6.11 OPTOELECTRONICS, DETECTORS, ISOLATORS, EMITTERS

**SPECIFICATION**
MIL-S-19500

**DESCRIPTION**
Photodetectors, Opto-isolators, Emitters

$$\lambda_p = \lambda_b \pi_T \pi_Q \pi_E \quad \text{Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| Optoelectronic Type | $\lambda_b$ |
|---|---|
| **Photodetectors** | |
| Photo-Transistor | .0055 |
| Photo-Diode | .0040 |
| **Opto-Isolators** | |
| Photodiode Output, Single Device | .0025 |
| Phototransistor Output, Single Device | .013 |
| Photodarlington Output, Single Device | .013 |
| Light Sensitive Resistor, Single Device | .0064 |
| Photodiode Output, Dual Device | .0033 |
| Phototransistor Output, Dual Device | .017 |
| Photodarlington Output, Dual Device | .017 |
| Light Sensitive Resistor, Dual Device | .0086 |
| **Emitters** | |
| Infrared Light Emitting Diode (IRLD) | .0013 |
| Light Emitting Diode (LED) | .00023 |

Temperature Factor - $\pi_T$

| $T_J$ (°C) | $\pi_T$ | $T_J$ (°C) | $\pi_T$ |
|---|---|---|---|
| 25 | 1.0 | 75 | 3.8 |
| 30 | 1.2 | 80 | 4.3 |
| 35 | 1.4 | 85 | 4.8 |
| 40 | 1.6 | 90 | 5.3 |
| 45 | 1.8 | 95 | 5.9 |
| 50 | 2.1 | 100 | 6.6 |
| 55 | 2.4 | 105 | 7.3 |
| 60 | 2.7 | 110 | 8.0 |
| 65 | 3.0 | 115 | 8.8 |
| 70 | 3.4 | | |

$$\pi_T = \exp\left(-2790\left(\frac{1}{T_J + 273} - \frac{1}{298}\right)\right)$$

$T_J$ = Junction Temperature (°C)

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| JANTXV | .70 |
| JANTX | 1.0 |
| JAN | 2.4 |
| Lower | 5.5 |
| Plastic | 8.0 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 8.0 |
| $N_S$ | 5.0 |
| $N_U$ | 12 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 6.0 |
| $A_{UC}$ | 6.0 |
| $A_{UF}$ | 8.0 |
| $A_{RW}$ | 17 |
| $S_F$ | .50 |
| $M_F$ | 9.0 |
| $M_L$ | 24 |
| $C_L$ | 450 |



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## 6.12 OPTOELECTRONICS, ALPHANUMERIC DISPLAYS

**SPECIFICATION**
MIL-S-19500

**DESCRIPTION**
Alphanumeric Display

$$\lambda_p = \lambda_b \pi_T \pi_Q \pi_E \quad \text{Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| Number of Characters | $\lambda_b$ Segment Display | $\lambda_b$ Diode Array Display |
|---|---|---|
| 1 | .00043 | .00026 |
| 1 w/Logic Chip | .00047 | .00030 |
| 2 | .00086 | .00043 |
| 2 w/Logic Chip | .00090 | .00047 |
| 3 | .0013 | .00060 |
| 3 w/Logic Chip | .0013 | .00064 |
| 4 | .0017 | .00077 |
| 4 w/Logic Chip | .0018 | .00081 |
| 5 | .0022 | .00094 |
| 6 | .0026 | .0011 |
| 7 | .0030 | .0013 |
| 8 | .0034 | .0015 |
| 9 | .0039 | .0016 |
| 10 | .0043 | .0018 |
| 11 | .0047 | .0020 |
| 12 | .0052 | .0021 |
| 13 | .0056 | .0023 |
| 14 | .0060 | .0025 |
| 15 | .0065 | .0026 |

$\lambda_b = .00043(C) + \lambda_{IC}$, for Segment Displays

$\lambda_b = .00009 + .00017(C) + \lambda_{IC}$, Diode Array Displays

C = Number of Characters

$\lambda_{IC}$ = .000043 for Displays with a Logic Chip

= 0.0 for Displays without Logic Chip

NOTE: The number of characters in a display is the number of characters contained in a single sealed package. For example, a 4 character display comprising 4 separately packaged single characters mounted together would be 4-one character displays, not 1-four character display.

Temperature Factor - $\pi_T$

| $T_J$ (°C) | $\pi_T$ | $T_J$ (°C) | $\pi_T$ |
|---|---|---|---|
| 25 | 1.0 | 75 | 3.8 |
| 30 | 1.2 | 80 | 4.3 |
| 35 | 1.4 | 85 | 4.8 |
| 40 | 1.6 | 90 | 5.3 |
| 45 | 1.8 | 95 | 5.9 |
| 50 | 2.1 | 100 | 6.6 |
| 55 | 2.4 | 105 | 7.3 |
| 60 | 2.7 | 110 | 8.0 |
| 65 | 3.0 | 115 | 8.8 |
| 70 | 3.4 | | |

$$\pi_T = \exp\left(-2790\left(\frac{1}{T_J + 273} - \frac{1}{298}\right)\right)$$

$T_J$ = Junction Temperature (°C)

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 8.0 |
| $N_S$ | 5.0 |
| $N_U$ | 12 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 6.0 |
| $A_{UC}$ | 6.0 |
| $A_{UF}$ | 8.0 |
| $A_{RW}$ | 17 |
| $S_F$ | .50 |
| $M_F$ | 9.0 |
| $M_L$ | 24 |
| $C_L$ | 450 |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| JANTXV | 0.7 |
| JANTX | 1.0 |
| JAN | 2.4 |
| Lower | 5.5 |
| Plastic | 8.0 |



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## 6.13 OPTOELECTRONICS, LASER DIODE

**SPECIFICATION**
MIL-S-19500

**DESCRIPTION**
Laser Diodes with Optical Flux Densities
< 3 MW/cm$^2$ and Forward Current < 25 amps

$$\lambda_p = \lambda_b \pi_T \pi_Q \pi_I \pi_A \pi_P \pi_E \quad \text{Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| Laser Diode Type | $\lambda_b$ |
|---|---|
| GaAs/Al GaAs | 3.23 |
| In GaAs/In GaAsP | 5.65 |

Temperature Factor - $\pi_T$

| $T_J$ (°C) | $\pi_T$ |
|---|---|
| 25 | 1.0 |
| 30 | 1.3 |
| 35 | 1.7 |
| 40 | 2.1 |
| 45 | 2.7 |
| 50 | 3.3 |
| 55 | 4.1 |
| 60 | 5.1 |
| 65 | 6.3 |
| 70 | 7.7 |
| 75 | 9.3 |

$$\pi_T = \exp\left(-4635\left(\frac{1}{T_J + 273} - \frac{1}{298}\right)\right)$$

$T_J$ = Junction Temperature (°C)

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| Hermetic Package | 1.0 |
| Nonhermetic with Facet Coating | 1.0 |
| Nonhermetic without Facet Coating | 3.3 |

Forward Current Factor, $\pi_I$

| Forward Peak Current (Amps) | $\pi_I$ |
|---|---|
| .050 | 0.13 |
| .075 | 0.17 |
| .1 | 0.21 |
| .5 | 0.62 |
| 1.0 | 1.0 |
| 2.0 | 1.6 |
| 3.0 | 2.1 |
| 4.0 | 2.6 |
| 5.0 | 3.0 |
| 10 | 4.8 |
| 15 | 6.3 |
| 20 | 7.7 |
| 25 | 8.9 |

$\pi_I = (I)^{.68}$

I = Forward Peak Current (Amps), $I \leq 25$

NOTE: For Variable Current Sources, use the Initial Current Value.

Application Factor $\pi_A$

| Application | Duty Cycle | $\pi_A$ |
|---|---|---|
| CW | ——— | 4.4 |
| Pulsed | .1 | .32 |
| | .2 | .45 |
| | .3 | .55 |
| | .4 | .63 |
| | .5 | .71 |
| | .6 | .77 |
| | .7 | .84 |
| | .8 | .89 |
| | .9 | .95 |
| | 1.0 | 1.00 |

$\pi_A$ = 4.4, CW

$\pi_A$ = Duty Cycle $^{0.5}$, Pulsed

NOTE: A duty cycle of one in pulsed application represents the maximum amount it can be driven in a pulsed mode. This is different from continuous wave application which will not withstand pulsed operating levels on a continuous basis.



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## 6.13 OPTOELECTRONICS, LASER DIODE

Power Degradation Factor - $\pi_P$

| Ratio $P_r/P_s$ | $\pi_P$ |
|---|---|
| 0.00 | .50 |
| .05 | .53 |
| .10 | .56 |
| .15 | .59 |
| .20 | .63 |
| .25 | .67 |
| .30 | .71 |
| .35 | .77 |
| .40 | .83 |
| .45 | .91 |
| .50 | 1.0 |
| .55 | 1.1 |
| .60 | 1.3 |
| .65 | 1.4 |
| .70 | 1.7 |
| .75 | 2.0 |
| .80 | 2.5 |
| .85 | 3.3 |
| .90 | 5.0 |
| .95 | 10 |

$$\pi_P = \frac{1}{2\left(1 - \frac{Pr}{Ps}\right)} \qquad 0 < \frac{Pr}{Ps} \leq .95$$

$P_s$ = Rated Optical Power Output (mW)

$P_r$ = Required Optical Power Output (mW)

NOTE: Each laser diode must be replaced when power output falls to Pr for failure rate prediction to be valid.

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 8.0 |
| $N_S$ | 5.0 |
| $N_U$ | 12 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 6.0 |
| $A_{UC}$ | 6.0 |
| $A_{UF}$ | 8.0 |
| $A_{RW}$ | 17 |
| $S_F$ | .50 |
| $M_F$ | 9.0 |
| $M_L$ | 24 |
| $C_L$ | 450 |



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## 6.14 DISCRETE SEMICONDUCTORS, $T_J$ DETERMINATION

Ideally, device case temperatures should be determined from a detailed thermal analysis of the equipment. Device junction temperature is then calculated with the following relationship:

$$T_J = T_C + \theta_{JC}P$$

where:

$T_J$ = Junction Temperature (°C)

$T_C$ = Case Temperature (°C). If no thermal analysis exists, the default case temperatures shown in Table 6-1 should be assumed.

$\theta_{JC}$ = Junction-to-Case Thermal Resistance (°C/W). This parameter should be determined from vendor, military specification sheets or Table 6-2, whichever is greater. It may also be estimated by taking the reciprocal of the recommended derating level. For example, a device derating recommendation of .16 W/°C would result in a $\theta_{JC}$ of 6.25 °C/W. If $\theta_{JC}$ cannot be determined assume a $\theta_{JC}$ value of 70°C/W.

P = Device Worse Case Power Dissipation (W)

The models are not applicable to devices at overstress conditions. If the calculated junction temperature is greater than the maximum rated junction temperature on the MIL slash sheets or the vendor's specifications, whichever is smaller, then the device is overstressed and these models **ARE NOT APPLICABLE.**

**Table 6-1: Default Case Temperatures ($T_C$) for All Environments**

| Environment | $T_C$ (°C) |
|---|---|
| $G_B$ | 35 |
| $G_F$ | 45 |
| $G_M$ | 50 |
| $N_S$ | 45 |
| $N_U$ | 50 |
| $A_{IC}$ | 60 |
| $A_{IF}$ | 60 |
| $A_{UC}$ | 75 |
| $A_{UF}$ | 75 |
| $A_{RW}$ | 60 |
| $S_F$ | 35 |
| $M_F$ | 50 |
| $M_L$ | 60 |
| $C_L$ | 45 |



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## 6.14 DISCRETE SEMICONDUCTORS, $T_J$ DETERMINATION

**Table 6-2: Approximate Junction-to-Case Thermal Resistance ($\theta_{JC}$) for Semiconductor Devices in Various Package Sizes***

| Package Type | $\theta_{JC}$ (°C/W) | Package Type | $\theta_{JC}$ (°C/W) |
|---|---|---|---|
| TO-1 | 70 | TO-205AD | 70 |
| TO-3 | 10 | TO-205AF | 70 |
| TO-5 | 70 | TO-220 | 5 |
| TO-8 | 70 | DO-4 | 5 |
| TO-9 | 70 | DO-5 | 5 |
| TO-12 | 70 | DO-7 | 10 |
| TO-18 | 70 | DO-8 | 5 |
| TO-28 | 5 | DO-9 | 5 |
| TO-33 | 70 | DO-13 | 10 |
| TO-39 | 70 | DO-14 | 5 |
| TO-41 | 10 | DO-29 | 10 |
| TO-44 | 70 | DO-35 | 10 |
| TO-46 | 70 | DO-41 | 10 |
| TO-52 | 70 | DO-45 | 5 |
| TO-53 | 5 | DO-204MB | 70 |
| TO-57 | 5 | DO-205AB | 5 |
| TO-59 | 5 | PA-42A,B | 70 |
| TO-60 | 5 | PD-36C | 70 |
| TO-61 | 5 | PD-50 | 70 |
| TO-63 | 5 | PD-77 | 70 |
| TO-66 | 10 | PD-180 | 70 |
| TO-71 | 70 | PD-319 | 70 |
| TO-72 | 70 | PD-262 | 70 |
| TO-83 | 5 | PD-975 | 70 |
| TO-89 | 22 | PD-280 | 70 |
| TO-92 | 70 | PD-216 | 70 |
| TO-94 | 5 | PT-2G | 70 |
| TO-99 | 70 | PT-6B | 70 |
| TO-126 | 5 | PH-13 | 70 |
| TO-127 | 5 | PH-16 | 70 |
| TO-204 | 10 | PH-56 | 70 |
| TO-204AA | 10 | PY-58 | 70 |
| | | PY-373 | 70 |

*When available, estimates must be based on military specification sheet or vendor values, whichever $\theta_{JC}$ is higher.



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## 6.15 DISCRETE SEMICONDUCTORS, EXAMPLE

**Example**

Given: Silicon dual transistor (complementary), JAN grade, rated for 0.25 W at 25°C, one side only, and 0.35 W at 25°C, both sides, with $T_{max}$ = 200°C, operating in linear service at 55°C case temperature in a sheltered naval environment. Side one, NPN, operating at 0.1 W and 50 percent of rated voltage and side two, PNP, operating at 0.05 W and 30 percent of rated voltage. The device operates at less than 200 MHz.

Since the device is a bipolar dual transistor operating at low frequency (<200 MHz), it falls into the Transistor, Low Frequency, Bipolar Group and the appropriate model is given in Section 6.3. Since the device is a dual device, it is necessary to compute the failure rate of each side separately and sum them together. Also, since $\theta_{JC}$ is unknown, $\theta_{JC}$ = 70°C/W will be assumed.

Based on the given information, the following model factors are determined from the appropriate tables shown in Section 6.3.

| | | | |
|---|---|---|---|
| $\lambda_b$ | = | .00074 | |
| $\pi_{T1}$ | = | 2.2 | Side 1, $T_J = T_C + \theta_{JC} P = 55 + 70(.1) = 62°C$ |
| $\pi_{T2}$ | = | 2.1 | Side 2, $T_J = 55 + 70(.05) = 59°C$ |
| $\pi_A$ | = | 1.5 | |
| $\pi_R$ | = | .68 | Using equation shown with $\pi_R$ table, $P_r$ = .35 W |
| $\pi_{S1}$ | = | .21 | Side 1, 50% Voltage Stress |
| $\pi_{S2}$ | = | .11 | Side 2, 30% Voltage Stress |
| $\pi_Q$ | = | 2.4 | |
| $\pi_E$ | = | 9 | |

$$\lambda_p = \underbrace{\lambda_b \pi_{T1} \pi_A \pi_R \pi_{S1} \pi_Q \pi_E}_{\text{SIDE 1}} + \underbrace{\lambda_b \pi_{T2} \pi_A \pi_R \pi_{S2} \pi_Q \pi_E}_{\text{SIDE 2}}$$

$$\lambda_p = (.00074)(2.2)(1.5)(.68)(.21)(2.4)(9) + (.00074)(2.1)(1.5)(.68)(.11)(2.4)(9)$$

$$= .011 \text{ Failures}/10^6 \text{ Hours}$$



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## 7.1 TUBES, ALL TYPES EXCEPT TWT AND MAGNETRON

**DESCRIPTION**
All Types Except Traveling Wave Tubes and Magnetrons. Includes Receivers, CRT, Thyratron, Crossed Field Amplifier, Pulsed Gridded, Transmitting, Vidicons, Twystron, Pulsed Klystron, CW Klystron

$$\lambda_p = \lambda_b \pi_L \pi_E \quad \text{Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$
**(Includes Both Random and Wearout Failures)**

| Tube Type | $\lambda_b$ |
|---|---|
| Receiver | |
| Triode, Tetrode, Pentode | 5.0 |
| Power Rectifier | 10 |
| CRT | 9.6 |
| Thyratron | 50 |
| Crossed Field Amplifier | |
| QK681 | 260 |
| SFD261 | 150 |
| Pulsed Gridded | |
| 2041 | 140 |
| 6952 | 390 |
| 7835 | 140 |
| Transmitting | |
| Triode, Peak Pwr. ≤ 200 KW, Avg. Pwr. ≤ 2KW, Freq. ≤ 200 MHz | 75 |
| Tetrode & Pentode, Peak Pwr. ≤ 200 KW, Avg. Power ≤ 2KW, Freq. ≤ 200 KW | 100 |
| If any of the above limits exceeded | 250 |
| Vidicon | |
| Antimony Trisulfide ($Sb_2S_3$) Photoconductive Material | 51 |
| Silicon Diode Array Photoconductive Material | 48 |
| Twystron | |
| VA144 | 850 |
| VA145E | 450 |
| VA145H | 490 |
| VA913A | 230 |
| Klystron, Pulsed* | |
| 4KMP10000LF | 43 |
| 8568 | 230 |
| L3035 | 66 |
| L3250 | 69 |
| L3403 | 93 |
| SAC42A | 100 |
| VA842 | 18 |
| Z5010A | 150 |
| ZM3038A | 190 |

* If the pulsed Klystron of interest is not listed above, use the Alternate Pulsed Klystron $\lambda_b$ Table on the following page.

| Tube Type | $\lambda_b$ |
|---|---|
| Klystron, Low Power, (e.g. Local Oscillator) | 30 |
| Klystron, Continuous Wave* | |
| 3K3000LQ | 9.0 |
| 3K50000LF | 54 |
| 3K210000LQ | 150 |
| 3KM300LA | 64 |
| 3KM3000LA | 19 |
| 3KM50000PA | 110 |
| 3KM50000PA1 | 120 |
| 3KM50000PA2 | 150 |
| 4K3CC | 610 |
| 4K3SK | 29 |
| 4K50000LQ | 30 |
| 4KM50LB | 28 |
| 4KM50LC | 15 |
| 4KM50SJ | 38 |
| 4KM50SK | 37 |
| 4KM3000LR | 140 |
| 4KM50000LQ | 79 |
| 4KM50000LR | 57 |
| 4KM170000LA | 15 |
| 8824 | 130 |
| 8825 | 120 |
| 8826 | 280 |
| VA800E | 70 |
| VA853 | 220 |
| VA856B | 65 |
| VA888E | 230 |

* If the CW Klystron of interest is not listed above, use the Alternate CW Klystron $\lambda_b$ Table on the following page.



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## 7.1 TUBES, ALL TYPES EXCEPT TWT AND MAGNETRON

### Alternate* Base Failure Rate for Pulsed Klystrons - $\lambda_b$

| P(MW) | F(GHz) .2 | .4 | .6 | .8 | 1.0 | 2.0 | 4.0 | 6.0 |
|---|---|---|---|---|---|---|---|---|
| .01 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 |
| .30 | 16 | 16 | 17 | 17 | 17 | 18 | 20 | 21 |
| .80 | 16 | 17 | 17 | 18 | 18 | 21 | 25 | 30 |
| 1.0 | 17 | 17 | 18 | 18 | 19 | 22 | 28 | 34 |
| 3.0 | 18 | 20 | 21 | 23 | 25 | 34 | 51 | |
| 5.0 | 19 | 22 | 25 | 26 | 31 | 45 | 75 | |
| 8.0 | 21 | 25 | 30 | 35 | 40 | 63 | 110 | |
| 10 | 22 | 28 | 34 | 40 | 45 | 75 | | |
| 25 | 31 | 45 | 60 | 75 | 90 | 160 | | |

$\lambda_b$ = 2.94 (F)(P) + 16

F = Operating Frequency in GHz, $0.2 \le F \le 6$

P = Peak Output Power in MW, $.01 \le P \le 25$ and $P \le 490\ F^{-2.95}$

*See previous page for other Klystron Base Failure Rates.

### Alternate* Base Failure Rate for CW Klystrons - $\lambda_b$

| P(KW) | F(MHz) 300 | 500 | 800 | 1000 | 2000 | 4000 | 6000 | 8000 |
|---|---|---|---|---|---|---|---|---|
| 0.1 | 30 | 31 | 33 | 34 | 38 | 47 | 57 | 66 |
| 1.0 | 31 | 32 | 33 | 34 | 39 | 48 | 57 | 66 |
| 3.0 | 32 | 33 | 34 | 35 | 40 | 49 | 58 | |
| 5.0 | 33 | 34 | 35 | 36 | 41 | 50 | | |
| 8.0 | 34 | 35 | 37 | 38 | 42 | | | |
| 10 | 35 | 36 | 38 | 39 | 43 | | | |
| 30 | 45 | 46 | 48 | 49 | | | | |
| 50 | 55 | 56 | 58 | 59 | | | | |
| 80 | 70 | 71 | 73 | | | | | |
| 100 | 80 | 81 | | | | | | |

$\lambda_b$ = 0.5P + .00046F + 29

P = Average Output Power in KW, $0.1 \le P \le 100$ and $P \le 8.0(10)^6(F)^{-1.7}$

F = Operating Frequency in MHz, $300 \le F \le 8000$

*See previous page for other Klystron Base Failure Rates.

### Learning Factor - $\pi_L$

| T (years) | $\pi_L$ |
|---|---|
| $\le 1$ | 10 |
| 2 | 2.3 |
| $\ge 3$ | 1.0 |

$\pi_L = 10(T)^{-2.1}$, $1 \le T \le 3$

$= 10$, $T \le 1$

$= 1$, $T \ge 3$

T = Number of Years since Introduction to Field Use

### Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | .50 |
| $G_F$ | 1.0 |
| $G_M$ | 14 |
| $N_S$ | 8.0 |
| $N_U$ | 24 |
| $A_{IC}$ | 5.0 |
| $A_{IF}$ | 8.0 |
| $A_{UC}$ | 6.0 |
| $A_{UF}$ | 12 |
| $A_{RW}$ | 40 |
| $S_F$ | .20 |
| $M_F$ | 22 |
| $M_L$ | 57 |
| $C_L$ | 1000 |



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## DESCRIPTION
Traveling Wave Tubes

$$\lambda_p = \lambda_b \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| Power (W) | Frequency (GHz) .1 | 1 | 2 | 4 | 6 | 8 | 10 | 14 | 18 |
|---|---|---|---|---|---|---|---|---|---|
| 100 | 11 | 12 | 13 | 16 | 20 | 24 | 29 | 42 | 61 |
| 500 | 11 | 12 | 13 | 16 | 20 | 24 | 29 | 42 | 62 |
| 1000 | 11 | 12 | 14 | 16 | 20 | 24 | 29 | 43 | 62 |
| 3000 | 12 | 13 | 14 | 17 | 21 | 25 | 30 | 44 | 65 |
| 5000 | 12 | 13 | 15 | 18 | 22 | 26 | 32 | 46 | 68 |
| 8000 | 13 | 14 | 16 | 19 | 23 | 28 | 33 | 49 | 72 |
| 10000 | 14 | 15 | 16 | 20 | 24 | 29 | 35 | 51 | 75 |
| 15000 | 15 | 16 | 18 | 22 | 26 | 32 | 39 | 56 | 83 |
| 20000 | 17 | 18 | 20 | 24 | 29 | 35 | 43 | 62 | 91 |
| 30000 | 20 | 22 | 24 | 29 | 36 | 43 | 52 | 76 | 110 |
| 40000 | 25 | 27 | 30 | 36 | 43 | 53 | 64 | 93 | 140 |

$\lambda_b = 11(1.00002)^P (1.1)^F$

P = Rated Power in Watts (Peak, if Pulsed), $.001 \leq P \leq 40{,}000$

F = Operating Frequency in GHz, $.3 \leq F \leq 18$.

If the operating frequency is a band, or two different values, use the geometric mean of the end point frequencies when using table.

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 3.0 |
| $G_M$ | 14 |
| $N_S$ | 6.0 |
| $N_U$ | 21 |
| $A_{IC}$ | 10 |
| $A_{IF}$ | 14 |
| $A_{UC}$ | 11 |
| $A_{UF}$ | 18 |
| $A_{RW}$ | 40 |
| $S_F$ | .10 |
| $M_F$ | 22 |
| $M_L$ | 66 |
| $C_L$ | 1000 |



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## 7.3 TUBES, MAGNETRON

### DESCRIPTION
Magnetrons, Pulsed and Continuous Wave (CW)

$$\lambda_p = \lambda_b \pi_U \pi_C \pi_E \quad \text{Failures}/10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| P(MW) | Frequency (GHz) .1 | .5 | 1 | 5 | 10 | 20 | 30 | 40 | 50 | 60 | 70 | 80 | 90 | 100 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| .01 | 1.4 | 4.6 | 7.6 | 24 | 41 | 67 | 91 | 110 | 130 | 150 | 170 | 190 | 200 | 220 |
| .05 | 1.9 | 6.3 | 10 | 34 | 56 | 93 | 120 | 150 | 180 | 210 | 230 | 260 | 280 | 300 |
| .1 | 2.2 | 7.2 | 12 | 39 | 64 | 110 | 140 | 180 | 210 | 240 | 270 | 290 | 320 | 350 |
| .3 | 2.8 | 9.0 | 15 | 48 | 80 | 130 | 180 | 220 | 260 | 300 | 330 | 370 | 400 | 430 |
| .5 | 3.1 | 10 | 17 | 54 | 89 | 150 | 200 | 240 | 290 | 330 | 370 | 410 | 440 | 480 |
| 1 | 3.5 | 11 | 19 | 62 | 100 | 170 | 230 | 280 | 330 | 380 | 420 | 470 | 510 | 550 |
| 3 | 4.4 | 14 | 24 | 77 | 130 | 210 | 280 | 350 | 410 | 470 | 530 | 580 | 630 | 680 |
| 5 | 4.9 | 16 | 26 | 85 | 140 | 230 | 310 | 390 | 460 | 520 | 580 | 640 | 700 | 760 |

**Pulsed Magnetrons:**

$\lambda_b = 19(F)^{.73}(P)^{.20}$

F = Operating Frequency in GHz, $.1 \le F \le 100$

P = Output Power in MW, $.01 \le P \le 5$

**CW Magnetrons (Rated Power < 5 KW):**

$\lambda_b = 18$

Utillization Factor - $\pi_U$

| Utilization (Radiate Hours/ Filament Hours) | $\pi_U$ |
|---|---|
| 0.0 | .44 |
| 0.1 | .50 |
| 0.2 | .55 |
| 0.3 | .61 |
| 0.4 | .66 |
| 0.5 | .72 |
| 0.6 | .78 |
| 0.7 | .83 |
| 0.8 | .89 |
| 0.9 | .94 |
| 1.0 | 1.0 |

$\pi_U = 0.44 + 0.56R$

R = Radiate Hours/Filament Hours

Construction Factor - $\pi_C$

| Construction | $\pi_C$ |
|---|---|
| CW (Rated Power < 5 KW) | 1.0 |
| Coaxial Pulsed | 1.0 |
| Conventional Pulsed | 5.4 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 4.0 |
| $N_S$ | 15 |
| $N_U$ | 47 |
| $A_{IC}$ | 10 |
| $A_{IF}$ | 16 |
| $A_{UC}$ | 12 |
| $A_{UF}$ | 23 |
| $A_{RW}$ | 80 |
| $S_F$ | .50 |
| $M_F$ | 43 |
| $M_L$ | 133 |
| $C_L$ | 2000 |



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## 8.0 LASERS, INTRODUCTION

The models and failure rates presented in this section apply to laser peculiar items only, i.e., those items wherein the lasing action is generated and controlled. In addition to laser peculiar items, there are other assemblies used with lasers that contain electronic parts and mechanical devices (pumps, valves, hoses, etc.). The failure rates for these parts should be determined with the same procedures as used for other electronic and mechanical devices in the equipment or system of which the laser is a part.

The laser failure rate models have been developed at the "functional," rather than "piece part" level because the available data were not sufficient for "piece part" model development. Nevertheless, the laser functional models are included in this Handbook in the interest of completeness. These laser models will be revised to include piece part models and other laser types when the data become available.

Because each laser family can be designed using a variety of approaches, the failure rate models have been structured on three basic laser functions which are common to most laser families, but may differ in the hardware implementation of a given function. These functions are the lasing media, the laser pumping mechanism (or pump), and the coupling method.

Examples of media-related hardware and reliability influencing factors are the solid state rod, gas, gas pressure, vacuum integrity, gas mix, outgassing, and tube diameter. The electrical discharge, the flashlamp, and energy level are examples of pump-related hardware and reliability influencing factors. The coupling function reliability influencing factors are the "Q" switch, mirrors, windows, crystals, substrates, coatings, and level of dust protection provided.

Some of the laser models require the number of active optical surfaces as an input parameter. An active optical surface is one with which the laser energy (or beam) interacts. Internally reflecting surfaces are not counted. Figure 8-1 below illustrates examples of active optical surfaces and count.

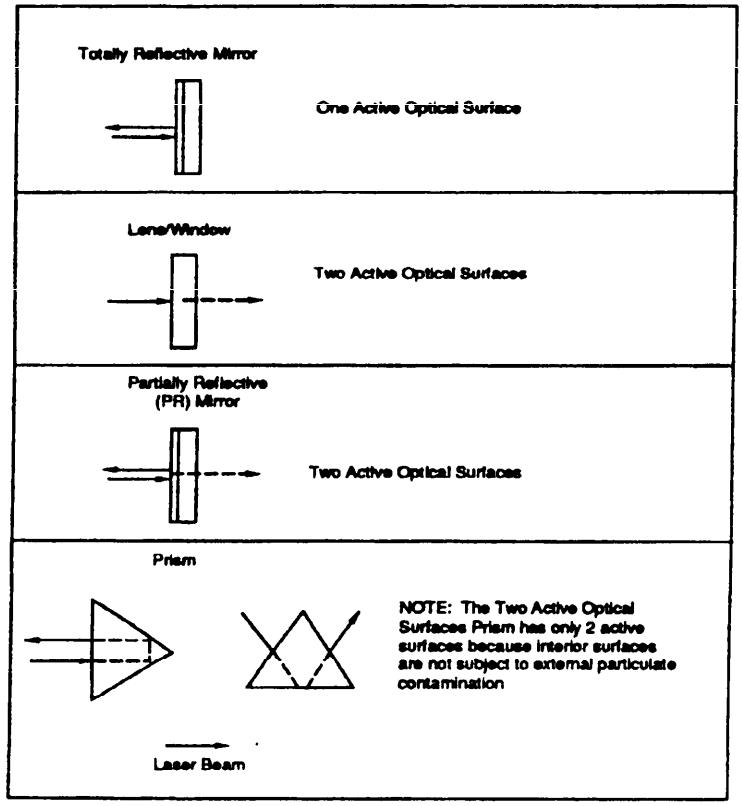



**Figure 8-1: Examples of Active Optical Surfaces**



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## 8.1 LASERS, HELIUM AND ARGON

**DESCRIPTION**
Helium Neon Lasers
Helium Cadmium Lasers
Argon Lasers

$$\lambda_p = \lambda_{MEDIA}\pi_E + \lambda_{COUPLING}\pi_E \text{ Failures/}10^6 \text{ Hours}$$

Lasing Media Failure Rate - $\lambda_{MEDIA}$

| Type | $\lambda_{MEDIA}$ |
|---|---|
| He/Ne | 84 |
| He/Cd | 228 |
| Argon | 457 |

Coupling Failure Rate - $\lambda_{COUPLING}$

| Types | $\lambda_{COUPLING}$ |
|---|---|
| Helium | 0 |
| Argon | 6 |

NOTE: The predominant argon laser failure mechanism is related to the gas media (as reflected in $\lambda_{MEDIA}$; however, when the tube is refilled periodically (preventive maintenance) the mirrors (as part of $\lambda_{COUPLING}$) can be expected to deteriorate after approximately $10^4$ hours of operation if in contact with the discharge region. $\lambda_{COUPLING}$ is negligible for helium lasers.

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | .30 |
| $G_F$ | 1.0 |
| $G_M$ | 4.0 |
| $N_S$ | 3.0 |
| $N_U$ | 4.0 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 6.0 |
| $A_{UC}$ | 7.0 |
| $A_{UF}$ | 9.0 |
| $A_{RW}$ | 5.0 |
| $S_F$ | .10 |
| $M_F$ | 3.0 |
| $M_L$ | 8.0 |
| $C_L$ | N/A |



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## 8.2 LASERS, CARBON DIOXIDE, SEALED

### DESCRIPTION
$CO_2$ Sealed Continuous Wave Lasers

$$\lambda_p = \lambda_{MEDIA}\pi_O\pi_B\pi_E + 10\,\pi_{OS}\pi_E \quad \text{Failures/}10^6\text{ Hours}$$

Lasing Media Failure Rate - $\lambda_{MEDIA}$

| Tube Current (mA) | $\lambda_{MEDIA}$ |
|---|---|
| 10 | 240 |
| 20 | 930 |
| 30 | 1620 |
| 40 | 2310 |
| 50 | 3000 |
| 100 | 6450 |
| 150 | 9900 |

$\lambda_{MEDIA} = 69(I) - 450$

I = Tube Current (mA), $10 \le I \le 150$

Gas Overfill Factor = $\pi_O$

| $CO_2$ Overfill Percent (%) | $\pi_O$ |
|---|---|
| 0 | 1.0 |
| 25 | .75 |
| 50 | .50 |

$\pi_O = 1 - .01$ (% Overfill)

Overfill percent is based on the percent increase over the optimum $CO_2$ partial pressure which is normally in the range of 1.5 to 3 $T_{orr}$ (1 $T_{orr}$ = 1 mm Hg Pressure) for most sealed $CO_2$ lasers.

Ballast Factor - $\pi_B$

| Percent of Ballast Volumetric Increase | $\pi_B$ |
|---|---|
| 0 | 1.0 |
| 50 | .58 |
| 100 | .33 |
| 150 | .19 |
| 200 | .11 |

$\pi_B = (1/3)$ (% Vol. Inc./100)

Optical Surface Factor - $\pi_{OS}$

| Active Optical Surfaces | $\pi_{OS}$ |
|---|---|
| 1 | 1 |
| 2 | 2 |

$\pi_{OS}$ = Number of Active Optical Surfaces

NOTE: Only active optical surfaces are counted. An active optical surface is one with which the laser energy or beam interacts. Internally reflecting surfaces are not counted. See Figure 8-1 for examples on determining the number of optical surfaces.

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | .30 |
| $G_F$ | 1.0 |
| $G_M$ | 4.0 |
| $N_S$ | 3.0 |
| $N_U$ | 4.0 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 6.0 |
| $A_{UC}$ | 7.0 |
| $A_{UF}$ | 9.0 |
| $A_{RW}$ | 5.0 |
| $S_F$ | .10 |
| $M_F$ | 3.0 |
| $M_L$ | 8.0 |
| $C_L$ | N/A |



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## 8.3 LASERS, CARBON DIOXIDE, FLOWING

### DESCRIPTION
$CO_2$ Flowing Lasers

$$\lambda_p = \lambda_{COUPLING}\pi_{OS}\pi_E \text{ Failures/}10^6 \text{ Hours}$$

Coupling Failure Rate - $\lambda_{COUPLING}$

| Power (KW) | $\lambda_{COUPLING}$ |
|---|---|
| .01 | 3 |
| .1 | 30 |
| 1.0 | 300 |

$\lambda_{COUPLING} = 300P$

P = Average Power Output in KW, $.01 \leq P \leq 1.0$

Beyond the 1KW range other glass failure mechanisms begin to predominate and alter the $\lambda_{COUPLING}$ values. It should also be noted that $CO_2$ flowing laser optical devices are the primary source of failure occurrence. A tailored optical cleaning preventive maintenance program on optic devices greatly extends laser life.

Optical Surface Factor - $\pi_{OS}$

| Active Optical Surfaces | $\pi_{OS}$ |
|---|---|
| 1 | 1 |
| 2 | 2 |

$\pi_{OS}$ = Number of Active Optical Surfaces

NOTE: Only active optical surfaces are counted. An active optical surface is one with which the laser energy or beam interacts. Internally reflecting surfaces are not counted. See Figure 8-1 for examples on determining the number of optical surfaces.

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | .30 |
| $G_F$ | 1.0 |
| $G_M$ | 4.0 |
| $N_S$ | 3.0 |
| $N_U$ | 4.0 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 6.0 |
| $A_{UC}$ | 7.0 |
| $A_{UF}$ | 9.0 |
| $A_{RW}$ | 5.0 |
| $S_F$ | .10 |
| $M_F$ | 3.0 |
| $M_L$ | 8.0 |
| $C_L$ | N/A |



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## 8.4 LASERS, SOLID STATE, ND:YAG AND RUBY ROD

**DESCRIPTION**
Neodymium-Yttrium-Aluminum-Garnet (ND:YAG) Rod Lasers

Ruby Rod Lasers

$$\lambda_p = (\lambda_{PUMP} + \lambda_{MEDIA} + 16.3\,\pi_C\pi_{OS})\,\pi_E \text{ Failures/}10^6 \text{ Hours}$$

### Pump Pulse Failure Rate - $\lambda_{PUMP}$ (Xenon Flashlamps)

The empirical formula used to determine $\lambda_{PUMP}$ (Failures/$10^6$ Hours) for Xenon lamps is:

$$\lambda_{PUMP} = (3600)\,(PPS)\left[2000\left(\frac{E_j}{dL\sqrt{t}}\right)^{8.58}\right][\pi_{COOL}]$$

$\lambda_{PUMP}$ is the failure rate contribution of the Xenon flashlamp or flashtube. The flashlamps evaluated herein are linear types used for military solid state laser systems. Typical default model parameters are given below.

PPS is the repetition pulse rate in pulses per second. Typical values range between 1 and 20 pulses per second.

$E_j$ is the flashlamp or flashtube input energy per pulse, in joules. Its value is determined from the actual or design input energy . For values less than 30 joules, use $E_j = 30$. Default value: $E_j = 40$.

d is the flashlamp or flashtube inside diameter, in millimeters. Default value: d = 4.

L is the flashlamp or flashtube arc length in inches. Default value: L = 2.

t is the truncated pulse width in microseconds. Use t = 100 microseconds for any truncated pulse width exceeding 100 microseconds. For shorter duration pulses, pulse width is to be measured at 10 percent of the maximum current amplitude. Default value: t = 100.

$\pi_{COOL}$ is the cooling factor due to various cooling media immediately surrounding the flashlamp or flashtube. $\pi_{COOL} = 1.0$ for any air or inert gas cooling. $\pi_{COOL} = .1$ for all liquid cooled designs. Default value: $\pi_{COOL} = .1$, liquid cooled.

### Pump Pulse Failure Rate - $\lambda_{PUMP}$ (Krypton Flashlamps)

The empirical formula used to determine $\lambda_{PUMP}$ for Krypton lamp is:

$$\lambda_{PUMP} = [625]\left[10^{(0.9\frac{P}{L})}\right]\left[\pi_{COOL}\right] \text{ Failures/}10^6 \text{ Hours}$$

$\lambda_{PUMP}$ is the failure rate contribution of the krypton flashlamp or flashtube. The flashlamps evaluted herein are the continuous wave (CW) type and are most widely used for commercial solid state applications. They are approx-imately 7mm in diameter and 5 to 6 inches long.

P is the average input power in kilowatts. Default value: P = 4.

L is the flashlamp or flashtube arc length in inches. Default value: L = 2.

$\pi_{COOL}$ is the cooling factor due to various cooling media immediately surrounding the flashlamp or flashtube. $\pi_{COOL} = 1$ for any air or inert gas cooling. $\pi_{COOL} = .1$ for all liquid designs. Default value: $\pi_{COOL} = .1$, liquid cooled.

### Media Failure Rate - $\lambda_{MEDIA}$

| Laser Type | $\lambda_{MEDIA}$ |
|---|---|
| ND:YAG | 0 |
| Ruby | (3600) (PPS) [43.5 $F^{2.52}$] |

PPS is the number of pulses per second

F is the energy density in Joules per cm.$^2$/pulse over the cross-sectional area of the laser beam, which is nominally equivalent to the cross-sectional area of the laser rod, and its value is determined from the actual design parameter of the laser rod utilized.

NOTE: $\lambda_{MEDIA}$ is negligible for ND:YAG lasers.



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## 8.4 LASERS, SOLID STATE, ND:YAG AND RUBY ROD

Coupling Cleanliness Factor - $\pi_C$

| Cleanliness Level | $\pi_C$ |
|---|---|
| Rigorous cleanliness procedures and trained maintenance personnel. Bellows provided over optical train. | 1 |
| Minimal precautions during opening, maintenance, repair, and testing. Bellows provided over optical train. | 30 |
| Minimal precautions during opening, maintenance, repair, and testing. No bellows provided over optical train. | 60 |

NOTE: Although sealed systems tend to be reliable once compatible materials have been selected and proven, extreme care must still be taken to prevent the entrance of particulates during manufacturing, field flashlamp replacement, or routine maintenance/ repair. Contamination is the major cause of solid state laser malfunction, and special provisions and vigilance must continually be provided to maintain the cleanliness level required.

Optical Surface Factor - $\pi_{OS}$

| Active Optical Surfaces | $\pi_{OS}$ |
|---|---|
| 1 | 1 |
| 2 | 2 |

$\pi_{OS}$ = Number of Active Optical Surfaces

NOTE: Only active optical surfaces are counted. An active optical surface is one with which the laser energy or beam interacts. Internally reflecting surfaces are not counted. See Figure 8-1 for examples on determining the number of optical surfaces.

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | .30 |
| $G_F$ | 1.0 |
| $G_M$ | 4.0 |
| $N_S$ | 3.0 |
| $N_U$ | 4.0 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 6.0 |
| $A_{UC}$ | 7.0 |
| $A_{UF}$ | 9.0 |
| $A_{RW}$ | 5.0 |
| $S_F$ | .10 |
| $M_F$ | 3.0 |
| $M_L$ | 8.0 |
| $C_L$ | N/A |



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## 9.0 RESISTORS, INTRODUCTION

This section includes the active resistor specifications and, in addition, some older/inactive specifications are included because of the large number of equipments still in field use which contain these parts.

The Established Reliability (ER) resistor family generally has four qualification failure rate levels when tested per the requirements of the applicable specification. These qualification failure rate levels differ by a factor of ten (from one level to the next). However, field data has shown that these failure rate levels differ by a factor of about only three, hence the $\pi_Q$ values have been set accordingly.

The use of the resistor models requires the calculation of the electrical power stress ratio, Stress = operating power/rated power, or per Section 9.16 for variable resistors. The models have been structured such that derating curves do not have to be used to find the base failure rate. The rated power for the stress ratio is equal to the full nominal rated power of the resistor. For example, a MIL-R-39008 resistor has the following derating curve:

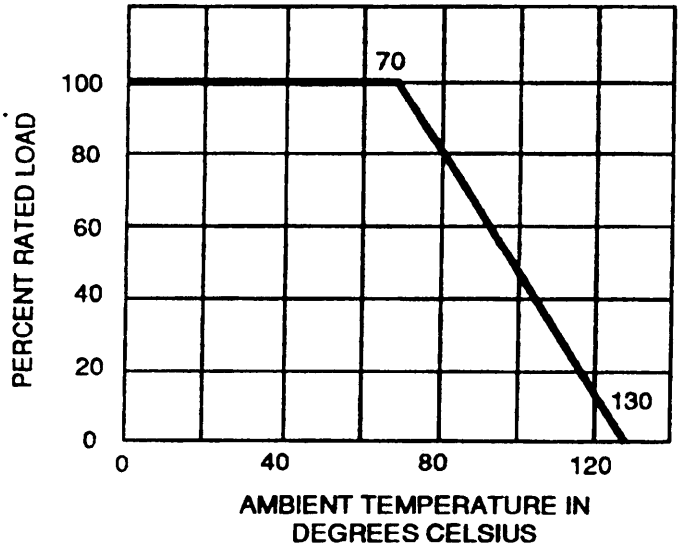



**Figure 9-1: MIL-R-39008 Derating Curve**

This particular resistor has a rating of 1 watt at 70°C ambient, or below. If it were being used in an ambient temperature of 100°C, the rated power for the stress calculation would still be 1 watt, not 45% of 1 watt (as read off the curve for 100°C). Of course, while the derating curve is not needed to determine the base failure rate, it must still be observed as the maximum operating condition. To aid in determining if a resistor is being used within rated conditions, the base failure rate tables show entries up to certain combinations of stress and temperature. If a given operating stress and temperature point falls in the blank portion of the base failure rate table, the resistor is overstressed. Such misapplication would require an analysis of the circuit and operating conditions to bring the resistor within rated conditions.



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## 9.1 RESISTORS, FIXED, COMPOSITION

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-R-39008 | RCR | Resistors, Fixed, Composition (Insulated), Established Reliability |
| MIL-R-11 | RC | Resistors, Fixed, Composition (Insulated) |

$$\lambda_p = \lambda_b \pi_R \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .00007 | .00010 | .00015 | .00020 | .00028 |
| 10 | .00011 | .00015 | .00021 | .00030 | .00043 |
| 20 | .00015 | .00022 | .00031 | .00045 | .00064 |
| 30 | .00022 | .00031 | .00046 | .00066 | .00096 |
| 40 | .00031 | .00045 | .00067 | .00098 | .0014 |
| 50 | .00044 | .00066 | .00098 | .0014 | .0021 |
| 60 | .00063 | .00095 | .0014 | .0021 | .0032 |
| 70 | .00090 | .0014 | .0021 | .0032 | .0048 |
| 80 | .0013 | .0020 | .0031 | .0047 | |
| 90 | .0018 | .0029 | .0045 | | |
| 100 | .0026 | .0041 | .0065 | | |
| 110 | .0038 | .0060 | | | |
| 120 | .0054 | | | | |

$$\lambda_b = 4.5 \times 10^{-9} \exp\left(12\left(\frac{T+273}{343}\right)\right) \exp\left(\frac{S}{.6}\left(\frac{T+273}{273}\right)\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating Power to Rated Power

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| S | .03 |
| R | 0.1 |
| P | 0.3 |
| M | 1.0 |
| MIL-R-11 | 5.0 |
| Lower | 15 |

Resistance Factor - $\pi_R$

| Resistance Range (ohms) | $\pi_R$ |
|---|---|
| < .1 M | 1.0 |
| > .1 M to 1 M | 1.1 |
| > 1.0 M to 10 M | 1.6 |
| > 10 M | 2.5 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 3.0 |
| $G_M$ | 8.0 |
| $N_S$ | 5.0 |
| $N_U$ | 13 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 5.0 |
| $A_{UC}$ | 7.0 |
| $A_{UF}$ | 11 |
| $A_{RW}$ | 19 |
| $S_F$ | .50 |
| $M_F$ | 11 |
| $M_L$ | 27 |
| $C_L$ | 490 |



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| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-R-39017 | RLR | Fixed, Film, Insulated, Established Reliability |
| MIL-R-22684 | RL | Fixed, Film, Insulated |
| MIL-R-55182 | RN (R, C, or N) | Fixed, Film, Established Reliability |
| MIL-R-10509 | RN | Fixed, Film, High Stability |

$$\lambda_p = \lambda_b \pi_R \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$
(MIL-R-22684 and MIL-R-39017)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .00059 | .00073 | .00089 | .0011 | .0013 |
| 10 | .00063 | .00078 | .00096 | .0012 | .0014 |
| 20 | .00067 | .00084 | .0010 | .0013 | .0016 |
| 30 | .00072 | .00090 | .0011 | .0014 | .0018 |
| 40 | .00078 | .00098 | .0012 | .0016 | .0019 |
| 50 | .00084 | .0011 | .0014 | .0017 | .0022 |
| 60 | .00092 | .0012 | .0015 | .0019 | .0024 |
| 70 | .0010 | .0013 | .0017 | .0021 | .0027 |
| 80 | .0011 | .0014 | .0018 | .0024 | |
| 90 | .0012 | .0016 | .0021 | .0027 | |
| 100 | .0013 | .0018 | .0023 | | |
| 110 | .0015 | .0020 | .0026 | | |
| 120 | .0017 | .0023 | | | |
| 130 | .0019 | | | | |
| 140 | .0022 | | | | |

$$\lambda_b = 3.25 \times 10^{-4} \exp\left(\frac{T+273}{343}\right)^3 \exp\left(S\left(\frac{T+273}{273}\right)\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating Power to Rated Power

Base Failure Rate - $\lambda_b$
(MIL-R-10509 and MIL-R-55182)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .00061 | .00074 | .00091 | .0011 | .0014 |
| 10 | .00067 | .00082 | .0010 | .0012 | .0015 |
| 20 | .00073 | .00091 | .0011 | .0014 | .0017 |
| 30 | .00080 | .0010 | .0013 | .0016 | .0019 |
| 40 | .00088 | .0011 | .0014 | .0017 | .0022 |
| 50 | .00096 | .0012 | .0015 | .0020 | .0025 |
| 60 | .0011 | .0013 | .0017 | .0022 | .0028 |
| 70 | .0012 | .0015 | .0019 | .0025 | .0032 |
| 80 | .0013 | .0016 | .0021 | .0028 | .0036 |
| 90 | .0014 | .0018 | .0024 | .0031 | .0040 |
| 100 | .0015 | .0020 | .0026 | .0035 | .0045 |
| 110 | .0017 | .0022 | .0029 | .0039 | .0051 |
| 120 | .0018 | .0024 | .0033 | .0043 | .0058 |
| 130 | .0020 | .0027 | .0036 | .0049 | .0065 |
| 140 | .0022 | .0030 | .0040 | .0054 | |
| 150 | .0024 | .0033 | .0045 | | |
| 160 | .0026 | .0036 | | | |
| 170 | .0029 | | | | |

$$\lambda_b = 5 \times 10^{-5} \exp\left(3.5\left(\frac{T+273}{398}\right)\right) \exp\left(S\left(\frac{T+273}{273}\right)\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating Power to Rated Power

NOTE: Do not use MIL-R-10509 (Characteristic B) below the line. Points below are overstressed.



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## 9.2 RESISTORS, FIXED, FILM

Resistance Factor - $\pi_R$

| Resistance Range (ohms) | $\pi_R$ |
|---|---|
| < .1M | 1.0 |
| ≥ 0.1 M to 1 M | 1.1 |
| > 1.0 M to 10 M | 1.6 |
| > 10 M | 2.5 |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| S | .03 |
| R | 0.1 |
| P | 0.3 |
| M | 1.0 |
| MIL-R-10509 | 5.0 |
| MIL-R-22684 | 5.0 |
| Lower | 15 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 8.0 |
| $N_S$ | 4.0 |
| $N_U$ | 14 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 8.0 |
| $A_{UC}$ | 10 |
| $A_{UF}$ | 18 |
| $A_{RW}$ | 19 |
| $S_F$ | .20 |
| $M_F$ | 10 |
| $M_L$ | 28 |
| $C_L$ | 510 |



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## 9.3 RESISTORS, FIXED, FILM, POWER

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-R-11804 | RD | Fixed, Film, Power Type |

$$\lambda_p = \lambda_b \pi_R \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .0089 | .0098 | .011 | .013 | .015 |
| 10 | .0090 | .010 | .011 | .013 | .015 |
| 20 | .0092 | .010 | .012 | .014 | .016 |
| 30 | .0094 | .010 | .012 | .014 | .017 |
| 40 | .0096 | .011 | .012 | .015 | .017 |
| 50 | .0098 | .011 | .013 | .015 | |
| 60 | .010 | .011 | .013 | .016 | |
| 70 | .010 | .012 | .014 | .016 | |
| 80 | .010 | .012 | .014 | .017 | |
| 90 | .011 | .012 | .015 | | |
| 100 | .011 | .013 | .015 | | |
| 110 | .011 | .013 | .016 | | |
| 120 | .012 | .014 | .016 | | |
| 130 | .012 | .014 | .017 | | |
| 140 | .012 | .014 | | | |
| 150 | .013 | .015 | | | |
| 160 | .013 | .016 | | | |
| 170 | .014 | .016 | | | |
| 180 | .014 | | | | |
| 190 | .015 | | | | |
| 200 | .015 | | | | |
| 210 | .016 | | | | |

$$\lambda_b = 7.33 \times 10^{-3} \exp\left(.202\left(\frac{T+273}{298}\right)^{2.6}\right) \times \exp\left(\left(\frac{S}{1.45}\right)\left(\frac{T+273}{273}\right)^{.89}\right)^{1.3}$$

T = Ambient Temperature (°C)

S = Ratio of Operating Power to Rated Power

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| MIL-SPEC | 1.0 |
| Lower | 3.0 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 10 |
| $N_S$ | 5.0 |
| $N_U$ | 17 |
| $A_{IC}$ | 6.0 |
| $A_{IF}$ | 8.0 |
| $A_{UC}$ | 14 |
| $A_{UF}$ | 18 |
| $A_{RW}$ | 25 |
| $S_F$ | .50 |
| $M_F$ | 14 |
| $M_L$ | 36 |
| $C_L$ | 660 |

Resistance Factor - $\pi_R$

| Resistance Range (ohms) | $\pi_R$ |
|---|---|
| 10 to 100 | 1.0 |
| > 100 to 100K | 1.2 |
| > 100K to 1M | 1.3 |
| > 1M | 3.5 |



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## 9.4 RESISTORS, NETWORK, FIXED, FILM

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-R-83401 | RZ | Resistor Networks, Fixed, Film |

$$\lambda_p = .00006\,\pi_T\,\pi_{NR}\pi_Q\pi_E \text{ Failures/}10^6 \text{ Hours}$$

Temperature Factor - $\pi_T$

| $T_C$ (°C) | $\pi_T$ | $T_C$ (°C) | $\pi_T$ |
|---|---|---|---|
| 25 | 1.0 | 80 | 8.3 |
| 30 | 1.3 | 85 | 9.8 |
| 35 | 1.6 | 90 | 11 |
| 40 | 1.9 | 95 | 13 |
| 45 | 2.4 | 100 | 15 |
| 50 | 2.9 | 105 | 18 |
| 55 | 3.5 | 110 | 21 |
| 60 | 4.2 | 115 | 24 |
| 65 | 5.0 | 120 | 27 |
| 70 | 6.0 | 125 | 31 |
| 75 | 7.1 | | |

$$\pi_T = \exp\left(-4056\left(\frac{1}{T_C + 273} - \frac{1}{298}\right)\right)$$

$T_C$ = Case Temperature (°C)

NOTE: If $T_C$ is unknown, it can be estimated as follows:

$T_C = T_A + 55\,(S)$

$T_A$ = Ambient Temperature (°C)

$S = \dfrac{\text{Operating Power}}{\text{Package Rated Power}}$

Any device operating at $T_C > 125$°C is overstressed.

Number of Resistors Factor - $\pi_{NR}$

$\pi_{NR}$ = Number of Film Resistors in Use

NOTE: Do not include resistors that are not used.

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| MIL-SPEC | 1 |
| Lower | 3 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 8.0 |
| $N_S$ | 4.0 |
| $N_U$ | 14 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 8.0 |
| $A_{UC}$ | 9.0 |
| $A_{UF}$ | 18 |
| $A_{RW}$ | 19 |
| $S_F$ | .50 |
| $M_F$ | 14 |
| $M_L$ | 28 |
| $C_L$ | 510 |



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## 9.5 RESISTORS, FIXED, WIREWOUND

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-R-39005 | RBR | Fixed, Wirewound, Accurate, Established Reliability |
| MIL-R-93 | RB | Fixed, Wirewound, Accurate |

$$\lambda_p = \lambda_b \pi_R \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .0033 | .0037 | .0045 | .0057 | .0075 |
| 10 | .0033 | .0038 | .0047 | .0059 | .0079 |
| 20 | .0034 | .0039 | .0048 | .0062 | .0084 |
| 30 | .0034 | .0040 | .0050 | .0066 | .0090 |
| 40 | .0035 | .0042 | .0052 | .0070 | .0097 |
| 50 | .0037 | .0043 | .0055 | .0075 | .011 |
| 60 | .0038 | .0046 | .0059 | .0081 | .012 |
| 70 | .0041 | .0049 | .0064 | .0089 | .013 |
| 80 | .0044 | .0053 | .0070 | .0099 | .015 |
| 90 | .0048 | .0059 | .0079 | .011 | .017 |
| 100 | .0055 | .0068 | .0092 | .013 | .020 |
| 110 | .0065 | .0080 | .011 | .016 | .025 |
| 120 | .0079 | .0099 | .014 | .021 | .033 |
| 130 | .010 | .013 | .018 | .028 | |
| 140 | .014 | | | | |

$$\lambda_b = .0031 \exp\left(\frac{T+273}{398}\right)^{10} \exp\left(S\left(\frac{T+273}{273}\right)\right)^{1.5}$$

T = Ambient Temperature (°C)

S = Ratio of Operating Power to Rated Power

Resistance Factor - $\pi_R$

| Resistance Range (ohms) | $\pi_R$ |
|---|---|
| Up to 10K | 1.0 |
| > 10K to 100K | 1.7 |
| > 100K to 1M | 3.0 |
| > 1M | 5.0 |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| S | .030 |
| R | .10 |
| P | .30 |
| M | 1.0 |
| MIL-R-93 | 5.0 |
| Lower | 15 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 11 |
| $N_S$ | 5.0 |
| $N_U$ | 18 |
| $A_{IC}$ | 15 |
| $A_{IF}$ | 18 |
| $A_{UC}$ | 28 |
| $A_{UF}$ | 35 |
| $A_{RW}$ | 27 |
| $S_F$ | .80 |
| $M_F$ | 14 |
| $M_L$ | 38 |
| $C_L$ | 610 |



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DESCRIPTION

## 9.6 RESISTORS, FIXED, WIREWOUND, POWER

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-R-39007 | RWR | Fixed, Wirewound, Power Type, Established Reliability |
| MIL-R-26 | RW | Fixed, Wirewound, Power Type |

$$\lambda_p = \lambda_b \pi_R \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

### Base Failure Rate - $\lambda_b$

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .0042 | .0062 | .0093 | .014 | .021 |
| 10 | .0045 | .0068 | .010 | .016 | .024 |
| 20 | .0048 | .0074 | .011 | .017 | .027 |
| 30 | .0052 | .0081 | .013 | .020 | .031 |
| 40 | .0056 | .0089 | .014 | .022 | .035 |
| 50 | .0061 | .0097 | .016 | .025 | .040 |
| 60 | .0066 | .011 | .017 | .028 | |
| 70 | .0072 | .012 | .020 | .032 | |
| 80 | .0078 | .013 | .022 | .037 | |
| 90 | .0085 | .014 | .025 | .042 | |
| 100 | .0093 | .016 | .028 | .048 | |
| 110 | .010 | .018 | .031 | .055 | |
| 120 | .011 | .020 | .036 | .063 | |
| 130 | .012 | .022 | .040 | | |
| 140 | .014 | .025 | .046 | | |
| 150 | .015 | .028 | .052 | | |
| 160 | .017 | .032 | .060 | | |
| 170 | .019 | .036 | .068 | | |
| 180 | .021 | .040 | .078 | | |
| 190 | .023 | .046 | | | |
| 200 | .026 | .052 | | | |
| 210 | .029 | .059 | | | |
| 220 | .033 | .068 | | | |
| 230 | .037 | .077 | | | |
| 240 | .042 | .088 | | | |
| 250 | .047 | .10 | | | |
| 260 | .054 | | | | |
| 270 | .061 | | | | |
| 280 | .06 | | | | |
| 290 | .079 | | | | |
| 300 | .091 | | | | |
| 310 | .10 | | | | |

$$\lambda_b = .00148 \exp\left(\frac{T+273}{298}\right)^2 \exp\left(\left(\frac{S}{.5}\right)\left(\frac{T+273}{273}\right)\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating Power to Rated Power

NOTE: Do not use MIL-R-39007 Resistors below the line. Points below are overstressed.

### Resistance Factor - $\pi_R$

(MIL-R-39007)

| MIL-R-39009 Style | Up to 500 | Resistance Range (ohms) >500 to 1K | >1K to 5K | >5K to 7.5K | >7.5 K to 10K | >10K to 15K | >15K to 20K | >20K |
|---|---|---|---|---|---|---|---|---|
| RWR 71 | 1.0 | 1.0 | 1.2 | 1.2 | 1.6 | 1.6 | 1.6 | NA |
| RWR 74 | 1.0 | 1.0 | 1.0 | 1.2 | 1.6 | 1.6 | NA | NA |
| RWR 78 | 1.0 | 1.0 | 1.0 | 1.0 | 1.2 | 1.2 | 1.2 | 1.6 |
| RWR 80 | 1.0 | 1.2 | 1.6 | 1.6 | NA | NA | NA | NA |
| RWR 81 | 1.0 | 1.6 | NA | NA | NA | NA | NA | NA |
| RWR 82 | 1.0 | 1.6 | 1.6 | NA | NA | NA | NA | NA |
| RWR 84 | 1.0 | 1.0 | 1.1 | 1.2 | 1.2 | 1.6 | NA | NA |
| RWR 89 | 1.0 | 1.0 | 1.4 | NA | NA | NA | NA | NA |

### Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| S | .03 |
| R | .10 |
| P | .30 |
| M | 1.0 |
| MIL-R-26 | 5.0 |
| Lower | 15 |



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## 9.6 RESISTORS, FIXED, WIREWOUND, POWER

### Resistance Factor - $\pi_R$

(MIL-R-26)

| MIL-R-26 Style | Resistance Range (ohms) | | | | | |
|---|---|---|---|---|---|---|
| | Up to 100 | >100 to 1K | >1K to 10K | >10K to 100K | >100K to 150K | >150K to 200K |
| RW 10 | 1.0 | 1.0 | 1.0 | 1.0 | 1.2 | 1.6 |
| RW 11 | 1.0 | 1.0 | 1.0 | 1.2 | 1.6 | NA |
| RW12 | 1.0 | 1.0 | 1.2 | 1.6 | NA | NA |
| RW 13 | 1.0 | 1.0 | 1.0 | 2.0 | NA | NA |
| RW 14 | 1.0 | 1.0 | 1.0 | 2.0 | NA | NA |
| RW 15 | 1.0 | 1.0 | 1.2 | 2.0 | NA | NA |
| RW 16 | 1.0 | 1.2 | 1.4 | NA | NA | NA |
| RW 20 | 1.0 | 1.0 | 1.6 | NA | NA | NA |
| RW 21 | 1.0 | 1.0 | 1.2 | 2.0 | NA | NA |
| RW 22 | 1.0 | 1.0 | 1.2 | 1.6 | NA | NA |
| RW 23 | 1.0 | 1.0 | 1.0 | 1.4 | NA | NA |
| RW 24 | 1.0 | 1.0 | 1.0 | 1.2 | NA | NA |
| RW 29 | 1.0 | 1.0 | 1.4 | NA | NA | NA |
| RW 30 | 1.0 | 1.2 | 1.6 | NA | NA | NA |
| RW 31 | 1.0 | 1.0 | 1.4 | NA | NA | NA |
| RW 32 | 1.0 | 1.0 | 1.2 | NA | NA | NA |
| RW 33 | 1.0 | 1.0 | 1.0 | 1.4 | NA | NA |
| RW 34 | 1.0 | 1.0 | 1.0 | 1.4 | NA | NA |
| RW 35 | 1.0 | 1.0 | 1.0 | 1.4 | NA | NA |
| RW 36 | 1.0 | 1.0 | 1.2 | 1.5 | NA | NA |
| RW 37 | 1.0 | 1.0 | 1.2 | 1.6 | NA | NA |
| RW 38 | 1.0 | 1.0 | 1.0 | 1.4 | 1.6 | NA |
| RW 39 | 1.0 | 1.0 | 1.0 | 1.4 | 1.6 | 2.0 |
| RW 47 | 1.0 | 1.0 | 1.0 | 1.4 | 1.6 | 2.0 |
| RW 55 | 1.0 | 1.0 | 1.4 | 2.4 | NA | NA |
| RW 56 | 1.0 | 1.0 | 1.2 | 2.6 | NA | NA |
| RW 67 | 1.0 | 1.0 | 1.0 | NA | NA | NA |
| RW 68 | 1.0 | 1.0 | 1.0 | NA | NA | NA |
| RW 69 | 1.0 | 1.0 | NA | NA | NA | NA |
| RW 70 | 1.0 | 1.2 | 1.4 | NA | NA | NA |
| RW 74 | 1.0 | 1.0 | 1.2 | 1.6 | NA | NA |
| RW 78 | 1.0 | 1.0 | 1.0 | 1.6 | NA | NA |
| RW 79 | 1.0 | 1.0 | 1.4 | NA | NA | NA |
| RW 80 | 1.0 | 1.2 | 1.6 | NA | NA | NA |
| RW 81 | 1.0 | 1.2 | NA | NA | NA | NA |

### Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 10 |
| $N_S$ | 5.0 |
| $N_U$ | 16 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 8.0 |
| $A_{UC}$ | 9.0 |
| $A_{UF}$ | 18 |
| $A_{RW}$ | 23 |
| $S_F$ | .30 |
| $M_F$ | 13 |
| $M_L$ | 34 |
| $C_L$ | 610 |



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## 9.7 RESISTORS, FIXED, WIREWOUND, POWER, CHASSIS MOUNTED

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-R-39009 | RER | Fixed, Wirewound, Power Type, Chassis Mounted, Established Reliability |
| MIL-R-18546 | RE | Fixed, Wirewound, Power Type, Chassis Mounted |

$$\lambda_p = \lambda_b \pi_R \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .0021 | .0032 | .0049 | .0076 | .012 |
| 10 | .0023 | .0036 | .0056 | .0087 | .014 |
| 20 | .0025 | .0040 | .0064 | .0100 | .016 |
| 30 | .0028 | .0045 | .0072 | .012 | .019 |
| 40 | .0031 | .0050 | .0082 | .013 | .022 |
| 50 | .0034 | .0056 | .0093 | .016 | .026 |
| 60 | .0037 | .0063 | .011 | .018 | |
| 70 | .0041 | .0070 | .012 | .021 | |
| 80 | .0045 | .0079 | .014 | .024 | |
| 90 | .0050 | .0088 | .016 | .028 | |
| 100 | .0055 | .0098 | .018 | .032 | |
| 110 | .0060 | .011 | .020 | | |
| 120 | .0066 | .012 | .023 | | |
| 130 | .0073 | .014 | .026 | | |
| 140 | .0081 | .015 | .030 | | |
| 150 | .0089 | .017 | .034 | | |
| 160 | .0098 | .019 | | | |
| 170 | .011 | .022 | | | |
| 180 | .012 | .024 | | | |
| 190 | .013 | .027 | | | |
| 200 | .014 | .030 | | | |
| 210 | .016 | | | | |
| 220 | .017 | | | | |
| 230 | .019 | | | | |
| 240 | .021 | | | | |
| 250 | .023 | | | | |

$$\lambda_b = .00015 \exp\left(2.64\left(\frac{T+273}{298}\right)\right)\exp\left(\frac{S}{.466}\left(\frac{T+273}{273}\right)\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating Power to Rated Power

Resistance Factor - $\pi_R$

(Characteristic G (Inductive Winding) of MIL-R-18546 and Inductively Wound Styles of MIL-R-39009)

| Style | Rated Power (W) | Resistance Range (ohms) Up to 500 | >500 to 1K | >1K to 5K | >5K to 10K | >10K to 20K | 20K |
|---|---|---|---|---|---|---|---|
| RE 60 RER60 | 5 | 1.0 | 1.2 | 1.2 | 1.6 | NA | NA |
| RE 65 RER65 | 10 | 1.0 | 1.0 | 1.2 | 1.6 | NA | NA |
| RE 70 RER70 | 20 | 1.0 | 1.0 | 1.2 | 1.2 | 1.6 | NA |
| RE 75 RER75 | 30 | 1.0 | 1.0 | 1.0 | 1.1 | 1.2 | 1.6 |
| RE 77 | 75 | 1.0 | 1.0 | 1.0 | 1.0 | 1.2 | 1.6 |
| RE 80 | 120 | 1.0 | 1.0 | 1.0 | 1.0 | 1.2 | 1.6 |

Resistance Factor - $\pi_R$

(Characteristic N (Noninductive Winding) of MIL-R-18546 and Noninductively Wound Styles of MIL-R-39009)

| Style | Rated Power (W) | Resistance Range (ohms) Up to 500 | >500 to 1K | >1K to 5K | >5K to 10K | >10K to 20K | 20K |
|---|---|---|---|---|---|---|---|
| RE 60 RER40 | 5 | 1.0 | 1.2 | 1.6 | NA | NA | NA |
| RE 65 RER45 | 10 | 1.0 | 1.2 | 1.6 | NA | NA | NA |
| RE 70 RER50 | 20 | 1.0 | 1.0 | 1.2 | 1.6 | NA | NA |
| RE 75 RER55 | 30 | 1.0 | 1.0 | 1.1 | 1.2 | 1.4 | NA |
| RE 77 | 75 | 1.0 | 1.0 | 1.0 | 1.2 | 1.6 | NA |
| RE 80 | 120 | 1.0 | 1.0 | 1.0 | 1.1 | 1.4 | NA |



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## 9.7 RESISTORS, FIXED, WIREWOUND, POWER, CHASSIS MOUNTED

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| S | .030 |
| R | .10 |
| P | .30 |
| M | 1.0 |
| MIL-R-18546 | 5.0 |
| Lower | 15 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 10 |
| $N_S$ | 5.0 |
| $N_U$ | 16 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 8.0 |
| $A_{UC}$ | 9.0 |
| $A_{UF}$ | 18 |
| $A_{RW}$ | 23 |
| $S_F$ | .50 |
| $M_F$ | 13 |
| $M_L$ | 34 |
| $C_L$ | 610 |



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## 9.8 RESISTORS, THERMISTOR

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-T-23648 | RTH | Thermally Sensitive Resistor, Insulated, Bead, Disk and Rod Types |

$$\lambda_p = \lambda_b \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| Type | $\lambda_b$ |
|---|---|
| Bead (Styles 24, 26, 28, 30, 32, 34, 36, 38, 40) | .021 |
| Disk (Styles 6, 8, 10) | .065 |
| Rod (Styles 12, 14, 16, 18, 20, 22, 42) | .105 |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| MIL-SPEC | 1 |
| Lower | 15 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 5.0 |
| $G_M$ | 21 |
| $N_S$ | 11 |
| $N_U$ | 24 |
| $A_{IC}$ | 11 |
| $A_{IF}$ | 30 |
| $A_{UC}$ | 16 |
| $A_{UF}$ | 42 |
| $A_{RW}$ | 37 |
| $S_F$ | .50 |
| $M_F$ | 20 |
| $M_L$ | 53 |
| $C_L$ | 950 |



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## 9.9 RESISTORS, VARIABLE, WIREWOUND

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-R-39015 | RTR | Variable, Wirewound, Lead Screw Actuated, Established Reliability |
| MIL-R-27208 | RT | Variable, Wirewound, Lead Screw Actuated |

$$\lambda_p = \lambda_b \pi_{TAPS} \pi_R \pi_V \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

### Base Failure Rate - $\lambda_b$

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .0089 | .011 | .013 | .016 | .020 |
| 10 | .0094 | .012 | .014 | .017 | .021 |
| 20 | .010 | .012 | .015 | .019 | .024 |
| 30 | .011 | .013 | .017 | .021 | .026 |
| 40 | .012 | .015 | .018 | .023 | .029 |
| 50 | .013 | .016 | .020 | .026 | .033 |
| 60 | .014 | .018 | .023 | .029 | .037 |
| 70 | .016 | .020 | .026 | .033 | .043 |
| 80 | .018 | .023 | .03 | .039 | .050 |
| 90 | .021 | .027 | .035 | .046 | .060 |
| 100 | .024 | .032 | .042 | .055 | |
| 110 | .029 | .038 | .051 | | |
| 120 | .035 | .047 | | | |
| 130 | .044 | .059 | | | |
| 140 | .056 | | | | |

$$\lambda_b = .0062 \exp\left(\frac{T+273}{358}\right)^5 \exp\left(S\left(\frac{T+273}{273}\right)\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating Power to Rated Power. See Section 9.16 for Calculation of S.

### Potentiometer Taps Factor - $\pi_{TAPS}$

| $N_{TAPS}$ | $\pi_{TAPS}$ | $N_{TAPS}$ | $\pi_{TAPS}$ | $N_{TAPS}$ | $\pi_{TAPS}$ |
|---|---|---|---|---|---|
| 3 | 1.0 | 13 | 2.7 | 23 | 5.2 |
| 4 | 1.1 | 14 | 2.9 | 24 | 5.5 |
| 5 | 1.2 | 15 | 3.1 | 25 | 5.8 |
| 6 | 1.4 | 16 | 3.4 | 26 | 6.1 |
| 7 | 1.5 | 17 | 3.6 | 27 | 6.4 |
| 8 | 1.7 | 18 | 3.8 | 28 | 6.7 |
| 9 | 1.9 | 19 | 4.1 | 29 | 7.0 |
| 10 | 2.1 | 20 | 4.4 | 30 | 7.4 |
| 11 | 2.3 | 21 | 4.6 | 31 | 7.7 |
| 12 | 2.5 | 22 | 4.9 | 32 | 8.0 |

$$\pi_{TAPS} = \frac{(N_{TAPS})^{\frac{3}{2}}}{25} + 0.792$$

$N_{TAPS}$ = Number of Potentiometer Taps, including the Wiper and Terminations.

### Resistance Factor - $\pi_R$

| Resistance Range (ohms) | $\pi_R$ |
|---|---|
| 10 to 2K | 1.0 |
| >2K to 5K | 1.4 |
| >5K to 20K | 2.0 |

### Voltage Factor - $\pi_V$

| Applied Voltage* / Rated Voltage | $\pi_V$ |
|---|---|
| 0 to 0.1 | 1.10 |
| >0.1 to 0.2 | 1.05 |
| >0.2 to 0.6 | 1.00 |
| >0.6 to 0.7 | 1.10 |
| >0.7 to 0.8 | 1.22 |
| >0.8 to 0.9 | 1.40 |
| >0.9 to 1.0 | 2.00 |

$$^*V_{Applied} = \sqrt{RP_{Applied}}$$

R = Nominal Total Potentiometer Resistance

$P_{Applied}$ = Power Dissipation

$V_{Rated}$ = 40 Volts for RT 26 and 27

$V_{Rated}$ = 90 Volts for RTR 12, 22 and 24; RT 12 and 22



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## 9.9 RESISTORS, VARIABLE, WIREWOUND

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| S | .020 |
| R | .060 |
| P | .20 |
| M | .60 |
| MIL-R-27208 | 3.0 |
| Lower | 10 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 12 |
| $N_S$ | 6.0 |
| $N_U$ | 20 |
| $A_{IC}$ | 5.0 |
| $A_{IF}$ | 8.0 |
| $A_{UC}$ | 9.0 |
| $A_{UF}$ | 15 |
| $A_{RW}$ | 33 |
| $S_F$ | .50 |
| $M_F$ | 18 |
| $M_L$ | 48 |
| $C_L$ | 870 |



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## 9.10 RESISTORS, VARIABLE, WIREWOUND, PRECISION

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-R-12934 | RR | Variable, Wirewound, Precision |

$$\lambda_p = \lambda_b \pi_{TAPS} \pi_C \pi_R \pi_V \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

### Base Failure Rate - $\lambda_b$

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .10 | .11 | .12 | .13 | .14 |
| 10 | .11 | .12 | .13 | .14 | .15 |
| 20 | .12 | .13 | .14 | .16 | .17 |
| 30 | .13 | .14 | .16 | .17 | .19 |
| 40 | .14 | .15 | .17 | .20 | .22 |
| 50 | .15 | .17 | .20 | .22 | .26 |
| 60 | .17 | .19 | .22 | .26 | .30 |
| 70 | .19 | .22 | .26 | .30 | .36 |
| 80 | .21 | .25 | .30 | .36 | .43 |
| 90 | .24 | .30 | .36 | .44 | .54 |
| 100 | .28 | .35 | .44 | .54 | |
| 110 | .33 | .42 | .54 | | |
| 120 | .40 | .52 | | | |
| 130 | .49 | .65 | | | |
| 140 | .60 | | | | |

$$\lambda_b = .0735 \exp\left(1.03 \left(\frac{T+273}{358}\right)^{4.45}\right) x \exp\left(\left(\frac{S}{2.74}\right)\left(\frac{T+273}{273}\right)^{3.51}\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating Power to Rated Power. See Section 9.16 for Calcuating S.

### Construction Class Factor - $\pi_C$

| Construction Class | $\pi_C$ |
|---|---|
| RR0900A2A9J103* | 2.0 |
| 3 | 1.0 |
| 4 | 3.0 |
| 5 | 1.5 |

* Sample type designation to show how construction class can be found. In this example the construction class is 2. Construction class should always appear in the eighth position.

### Resistance Factor - $\pi_R$

| Resistance Range (ohms) | $\pi_R$ |
|---|---|
| 100 to 10K | 1.0 |
| >10K to 20K | 1.1 |
| >20K to 50K | 1.4 |
| >50K to 100K | 2.0 |
| >100 K to 200K | 2.5 |
| >200K to 500K | 3.5 |

### Potentiometer Taps Factor - $\pi_{TAPS}$

| $N_{TAPS}$ | $\pi_{TAPS}$ | $N_{TAPS}$ | $\pi_{TAPS}$ | $N_{TAPS}$ | $\pi_{TAPS}$ |
|---|---|---|---|---|---|
| 3 | 1.0 | 13 | 2.7 | 23 | 5.2 |
| 4 | 1.1 | 14 | 2.9 | 24 | 5.5 |
| 5 | 1.2 | 15 | 3.1 | 25 | 5.8 |
| 6 | 1.4 | 16 | 3.4 | 26 | 6.1 |
| 7 | 1.5 | 17 | 3.6 | 27 | 6.4 |
| 8 | 1.7 | 18 | 3.8 | 28 | 6.7 |
| 9 | 1.9 | 19 | 4.1 | 29 | 7.0 |
| 10 | 2.1 | 20 | 4.4 | 30 | 7.4 |
| 11 | 2.3 | 21 | 4.6 | 31 | 7.7 |
| 12 | 2.5 | 22 | 4.9 | 32 | 8.0 |

$$\pi_{TAPS} = \frac{(N_{TAPS})^{\frac{3}{2}}}{25} + 0.792$$

$N_{TAPS}$ = Number of Potentiometer Taps, including the Wiper and Terminations.



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## 9.10 RESISTORS, VARIABLE, WIREWOUND, PRECISION

Voltage Factor - $\pi_V$

| Applied Voltage* / Rated Voltage | $\pi_V$ |
|---|---|
| 0 to 0.1 | 1.10 |
| >0.1 to 0.2 | 1.05 |
| >0.2 to 0.6 | 1.00 |
| >0.6 to 0.7 | 1.10 |
| >0.7 to 0.8 | 1.22 |
| >0.8 to 0.9 | 1.40 |
| >0.9 to 1.0 | 2.00 |

$^*V_{Applied} = \sqrt{R_P P_{Applied}}$

$R_P$ = Nominal Total Potentiometer Resistance

$P_{Applied}$ = Power Dissipation

$V_{Rated}$ = 250 Volts for RR0900, RR1100, RR1300, RR2000, RR3000, RR3100, RR3200, RR3300, RR3400, RR3500

$V_{Rated}$ = 423 Volts for RR3600, RR3700

$V_{Rated}$ = 500 Volts for RR1000, RR1400, RR2100, RR3800, RR3900

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| MIL-SPEC | 2.5 |
| Lower | 5.0 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 18 |
| $N_S$ | 8.0 |
| $N_U$ | 30 |
| $A_{IC}$ | 8.0 |
| $A_{IF}$ | 12 |
| $A_{UC}$ | 13 |
| $A_{UF}$ | 18 |
| $A_{RW}$ | 53 |
| $S_F$ | .50 |
| $M_F$ | 29 |
| $M_L$ | 76 |
| $C_L$ | 1400 |



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## 9.11 RESISTORS, VARIABLE, WIREWOUND, SEMIPRECISION

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-R-19 | RA | Variable, Wirewound, Semiprecision (Low Operating Temperature) |
| MIL-R-39002 | RK | Variable, Wirewound, Semiprecision |

$$\lambda_p = \lambda_b \pi_{TAPS} \pi_R \pi_V \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .055 | .063 | .072 | .083 | .095 |
| 10 | .058 | .069 | .081 | .095 | .11 |
| 20 | .063 | .076 | .092 | .11 | .13 |
| 30 | .069 | .086 | .11 | .13 | .17 |
| 40 | .076 | .098 | .13 | .16 | .21 |
| 50 | .085 | .11 | .15 | .20 | .27 |
| 60 | .096 | .13 | .19 | .26 | .37 |
| 70 | .11 | .16 | .24 | .35 | .52 |
| 80 | .13 | .20 | .31 | .48 | .75 |
| 90 | .16 | .26 | .42 | .69 | 1.1 |
| 100 | .19 | .34 | .59 | 1.0 | |
| 110 | .24 | .45 | .85 | | |
| 120 | .31 | | | | |
| 130 | .42 | | | | |

$$\lambda_b = .0398 \exp\left(.514 \left(\frac{T+273}{313}\right)^{5.28}\right) x \exp\left(\frac{S}{1.44}\left(\frac{T+273}{273}\right)^{4.46}\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating Power to Rated Power. See Section 9.16 for S Calculation.

NOTE: Do not use MIL-R-19 below the line. Points below are overstressed.

Resistance Factor - $\pi_R$

| Resistance Range (ohms) | $\pi_R$ |
|---|---|
| 10 to 2K | 1.0 |
| >2K to 5K | 1.4 |
| >5K to 10K | 2.0 |

Potentiometer Taps Factor - $\pi_{TAPS}$

| $N_{TAPS}$ | $\pi_{TAPS}$ | $N_{TAPS}$ | $\pi_{TAPS}$ | $N_{TAPS}$ | $\pi_{TAPS}$ |
|---|---|---|---|---|---|
| 3 | 1.0 | 13 | 2.7 | 23 | 5.2 |
| 4 | 1.1 | 14 | 2.9 | 24 | 5.5 |
| 5 | 1.2 | 15 | 3.1 | 25 | 5.8 |
| 6 | 1.4 | 16 | 3.4 | 26 | 6.1 |
| 7 | 1.5 | 17 | 3.6 | 27 | 6.4 |
| 8 | 1.7 | 18 | 3.8 | 28 | 6.7 |
| 9 | 1.9 | 19 | 4.1 | 29 | 7.0 |
| 10 | 2.1 | 20 | 4.4 | 30 | 7.4 |
| 11 | 2.3 | 21 | 4.6 | 31 | 7.7 |
| 12 | 2.5 | 22 | 4.9 | 32 | 8.0 |

$$\pi_{TAPS} = \frac{(N_{TAPS})^{\frac{3}{2}}}{25} + 0.792$$

$N_{TAPS}$ = Number of Potentiometer Taps, including the Wiper and Terminations.



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## 9.11 RESISTORS, VARIABLE, WIREWOUND, SEMIPRECISION

Voltage Factor - $\pi_V$

| Applied Voltage* / Rated Voltage | $\pi_V$ |
|---|---|
| 0 to 0.1 | 1.10 |
| >0.1 to 0.2 | 1.05 |
| >0.2 to 0.6 | 1.00 |
| >0.6 to 0.7 | 1.10 |
| >0.7 to 0.8 | 1.22 |
| >0.8 to 0.9 | 1.40 |
| >0.9 to 1.0 | 2.00 |

| | | |
|---|---|---|
| $^*V_{Applied}$ | = | $\sqrt{R_P P_{Applied}}$ |
| $R_P$ | = | Nominal Total Potentiometer Resistance |
| $P_{Applied}$ | = | Power Dissipation |
| $V_{Rated}$ | = | 50 Volts for RA10 |
| | = | 75 Volts for RA20X-XC, F |
| | = | 130 Volts for RA30X-XC, F |
| | = | 175 Volts for RA20X-XA |
| | = | 275 Volts for RK09 |
| | = | 320 Volts for RA30X-XA |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 16 |
| $N_S$ | 7.0 |
| $N_U$ | 28 |
| $A_{IC}$ | 8.0 |
| $A_{IF}$ | 12 |
| $A_{UC}$ | N/A |
| $A_{UF}$ | N/A |
| $A_{RW}$ | 38 |
| $S_F$ | .50 |
| $M_F$ | N/A |
| $M_L$ | N/A |
| $C_L$ | N/A |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| MIL-SPEC | 2.0 |
| Lower | 4.0 |



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## 9.12 RESISTORS, VARIABLE, WIREWOUND, POWER

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-R-22 | RP | Variable, Wirewound, Power Type |

$$\lambda_p = \lambda_b \pi_{TAPS} \pi_R \pi_V \pi_C \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .064 | .074 | .084 | .097 | .11 |
| 10 | .067 | .078 | .091 | .11 | .12 |
| 20 | .071 | .084 | .099 | .12 | .14 |
| 30 | .076 | .091 | .11 | .13 | .16 |
| 40 | .081 | .099 | .12 | .15 | |
| 50 | .087 | .11 | .14 | .17 | |
| 60 | .095 | .12 | .15 | | |
| 70 | .10 | .14 | .18 | | |
| 80 | .12 | .15 | | | |
| 90 | .13 | .18 | | | |
| 100 | .15 | | | | |
| 110 | .17 | | | | |
| 120 | .20 | | | | |

$$\lambda_b = .0481 \exp\left(.334 \left(\frac{T+273}{298}\right)^{4.66}\right) x \exp\left(\frac{S}{1.47}\left(\frac{T+273}{273}\right)^{2.83}\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating Power to Rated Power. See Section 9.16 for S Calculation.

Resistance Factor - $\pi_R$

| Resistance Range (ohms) | $\pi_R$ |
|---|---|
| 1 to 2K | 1.0 |
| >2K to 5K | 1.4 |
| >5K to 10K | 2.0 |

Potentiometer Taps Factor - $\pi_{TAPS}$

| $N_{TAPS}$ | $\pi_{TAPS}$ | $N_{TAPS}$ | $\pi_{TAPS}$ | $N_{TAPS}$ | $\pi_{TAPS}$ |
|---|---|---|---|---|---|
| 3 | 1.0 | 13 | 2.7 | 23 | 5.2 |
| 4 | 1.1 | 14 | 2.9 | 24 | 5.5 |
| 5 | 1.2 | 15 | 3.1 | 25 | 5.8 |
| 6 | 1.4 | 16 | 3.4 | 26 | 6.1 |
| 7 | 1.5 | 17 | 3.6 | 27 | 6.4 |
| 8 | 1.7 | 18 | 3.8 | 28 | 6.7 |
| 9 | 1.9 | 19 | 4.1 | 29 | 7.0 |
| 10 | 2.1 | 20 | 4.4 | 30 | 7.4 |
| 11 | 2.3 | 21 | 4.6 | 31 | 7.7 |
| 12 | 2.5 | 22 | 4.9 | 32 | 8.0 |

$$\pi_{TAPS} = \frac{(N_{TAPS})^{\frac{3}{2}}}{25} + 0.792$$

$N_{TAPS}$ = Number of Potentiometer Taps, including the Wiper and Terminations



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## 9.12 RESISTORS, VARIABLE, WIREWOUND, POWER

Voltage Factor - $\pi_V$

| Applied Voltage* / Rated Voltage | $\pi_V$ |
|---|---|
| 0 to 0.1 | 1.10 |
| >0.1 to 0.2 | 1.05 |
| >0.2 to 0.6 | 1.00 |
| >0.6 to 0.7 | 1.10 |
| >0.7 to 0.8 | 1.22 |
| >0.8 to 0.9 | 1.40 |
| >0.9 to 1.0 | 2.00 |

$^*V_{Applied} = \sqrt{R_P P_{Applied}}$

$R_P$ = Nominal Total Potentiometer Resistance

$P_{Applied}$ = Power Dissipation

$V_{Rated}$ = 250 Volts for RP06, RP10

= 500 Volts for Others

Construction Class Factor - $\pi_C$

| Construction Class | Style | $\pi_C$ |
|---|---|---|
| Enclosed | RP07, RP11, RP16 | 2.0 |
| Unenclosed | All Other Styles are Unenclosed | 1.0 |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| MIL-SPEC | 2.0 |
| Lower | 4.0 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 3.0 |
| $G_M$ | 16 |
| $N_S$ | 7.0 |
| $N_U$ | 28 |
| $A_{IC}$ | 8.0 |
| $A_{IF}$ | 12 |
| $A_{UC}$ | N/A |
| $A_{UF}$ | N/A |
| $A_{RW}$ | 38 |
| $S_F$ | .50 |
| $M_F$ | N/A |
| $M_L$ | N/A |
| $C_L$ | N/A |



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## 9.13 RESISTORS, VARIABLE, NONWIREWOUND

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-R-22097 | RJ | Variable, Nonwirewound (Adjustment Types) |
| MIL-R-39035 | RJR | Variable, Nonwirewound (Adjustment Types), Established Reliability |

$$\lambda_p = \lambda_b \pi_{TAPS} \pi_R \pi_V \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .021 | .023 | .024 | .026 | .028 |
| 10 | .021 | .023 | .025 | .027 | .030 |
| 20 | .022 | .024 | .026 | .029 | .031 |
| 30 | .023 | .025 | .028 | .030 | .033 |
| 40 | .024 | .026 | .029 | .032 | .036 |
| 50 | .025 | .028 | .031 | .035 | .039 |
| 60 | .026 | .030 | .033 | .038 | .043 |
| 70 | .028 | .032 | .036 | .042 | .047 |
| 80 | .030 | .035 | .040 | .046 | .053 |
| 90 | .034 | .039 | .045 | .053 | .061 |
| 100 | .038 | .044 | .052 | .061 | |
| 110 | .043 | .051 | .060 | | |
| 120 | .050 | .060 | | | |
| 130 | .060 | .073 | | | |
| 140 | .074 | | | | |

$$\lambda_b = .019 \exp\left(.445 \left(\frac{T+273}{358}\right)^{7.3}\right) x \exp\left(\frac{S}{2.69}\left(\frac{T+273}{273}\right)^{2.46}\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating Power to Rated Power. See Section 9.16 for S Calculation.

Resistance Factor - $\pi_R$

| Resistance Range (ohms) | $\pi_R$ |
|---|---|
| 10 to 50K | 1.0 |
| >50K to 100K | 1.1 |
| >100K to 200K | 1.2 |
| >200K to 500K | 1.4 |
| >500K to 1M | 1.8 |

Potentiometer Taps Factor - $\pi_{TAPS}$

| $N_{TAPS}$ | $\pi_{TAPS}$ | $N_{TAPS}$ | $\pi_{TAPS}$ | $N_{TAPS}$ | $\pi_{TAPS}$ |
|---|---|---|---|---|---|
| 3 | 1.0 | 13 | 2.7 | 23 | 5.2 |
| 4 | 1.1 | 14 | 2.9 | 24 | 5.5 |
| 5 | 1.2 | 15 | 3.1 | 25 | 5.8 |
| 6 | 1.4 | 16 | 3.4 | 26 | 6.1 |
| 7 | 1.5 | 17 | 3.6 | 27 | 6.4 |
| 8 | 1.7 | 18 | 3.8 | 28 | 6.7 |
| 9 | 1.9 | 19 | 4.1 | 29 | 7.0 |
| 10 | 2.1 | 20 | 4.4 | 30 | 7.4 |
| 11 | 2.3 | 21 | 4.6 | 31 | 7.7 |
| 12 | 2.5 | 22 | 4.9 | 32 | 8.0 |

$$\pi_{TAPS} = \frac{(N_{TAPS})^{\frac{3}{2}}}{25} + 0.792$$

$N_{TAPS}$ = Number of Potentiometer Taps, including the Wiper and Terminations.



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## 9.13 RESISTORS, VARIABLE, NONWIREWOUND

Voltage Factor - $\pi_V$

| $\frac{\text{Applied Voltage}^*}{\text{Rated Voltage}}$ | $\pi_V$ |
|---|---|
| 0 to 0.8 | 1.00 |
| >0.8 to 0.9 | 1.05 |
| >0.9 to 1.0 | 1.20 |

$^*V_{Applied} = \sqrt{R_P P_{Applied}}$

$R_P$ = Nominal Total Potentiometer Resistance

$P_{Applied}$ = Power Dissipation

$V_{Rated}$ = 200 Volts for RJ and RJR26; RJ and RJR50

= 300 Volts for All Others

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 3.0 |
| $G_M$ | 14 |
| $N_S$ | 6.0 |
| $N_U$ | 24 |
| $A_{IC}$ | 5.0 |
| $A_{IF}$ | 7.0 |
| $A_{UC}$ | 12 |
| $A_{UF}$ | 18 |
| $A_{RW}$ | 39 |
| $S_F$ | .50 |
| $M_F$ | 22 |
| $M_L$ | 57 |
| $C_L$ | 1000 |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| S | .020 |
| R | .060 |
| P | .20 |
| M | .60 |
| MIL-R-22097 | 3.0 |
| Lower | 10 |



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## 9.14 RESISTORS, VARIABLE, COMPOSITION

**SPECIFICATION**
MIL-R-94

**STYLE**
RV

**DESCRIPTION**
Variable, Composition, Low Precision

$$\lambda_p = \lambda_b \pi_{TAPS} \pi_R \pi_V \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .027 | .030 | .032 | .035 | .038 |
| 10 | .028 | .031 | .034 | .038 | .042 |
| 20 | .029 | .033 | .037 | .042 | .048 |
| 30 | .031 | .036 | .041 | .048 | .056 |
| 40 | .033 | .039 | .047 | .056 | .067 |
| 50 | .036 | .044 | .054 | .067 | .082 |
| 60 | .039 | .050 | .065 | .083 | .11 |
| 70 | .045 | .060 | .08 | .11 | .14 |
| 80 | .053 | .074 | .10 | .15 | |
| 90 | .065 | .096 | .14 | | |
| 100 | .084 | .13 | | | |
| 110 | .11 | | | | |

$$\lambda_b = .0246 \exp\left(.459 \left(\frac{T+273}{343}\right)^{9.3}\right) \times \exp\left(\frac{S}{2.32}\left(\frac{T+273}{273}\right)^{5.3}\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating Power to Rated Power. See Section 9.16 for S Calculation.

Resistance Factor - $\pi_R$

| Resistance Range (ohms) | $\pi_R$ |
|---|---|
| 50 to 50K | 1.0 |
| >50K to 100K | 1.1 |
| >100K to 200K | 1.2 |
| >200K to 500K | 1.4 |
| >500K to 1M | 1.8 |

Potentiometer Taps Factor - $\pi_{TAPS}$

| $N_{TAPS}$ | $\pi_{TAPS}$ | $N_{TAPS}$ | $\pi_{TAPS}$ | $N_{TAPS}$ | $\pi_{TAPS}$ |
|---|---|---|---|---|---|
| 3 | 1.0 | 13 | 2.7 | 23 | 5.2 |
| 4 | 1.1 | 14 | 2.9 | 24 | 5.5 |
| 5 | 1.2 | 15 | 3.1 | 25 | 5.8 |
| 6 | 1.4 | 16 | 3.4 | 26 | 6.1 |
| 7 | 1.5 | 17 | 3.6 | 27 | 6.4 |
| 8 | 1.7 | 18 | 3.8 | 28 | 6.7 |
| 9 | 1.9 | 19 | 4.1 | 29 | 7.0 |
| 10 | 2.1 | 20 | 4.4 | 30 | 7.4 |
| 11 | 2.3 | 21 | 4.6 | 31 | 7.7 |
| 12 | 2.5 | 22 | 4.9 | 32 | 8.0 |

$$\pi_{TAPS} = \frac{(N_{TAPS})^{\frac{3}{2}}}{25} + 0.792$$

$N_{TAPS}$ = Number of Potentiometer Taps, including the Wiper and Terminations.



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## 9.14 RESISTORS, VARIABLE, COMPOSITION

Voltage Factor - $\pi_V$

| Applied Voltage* / Rated Voltage | $\pi_V$ |
|---|---|
| 0 to 0.8 | 1.00 |
| >0.8 to 0.9 | 1.05 |
| >0.9 to 1.0 | 1.20 |

$^*V_{Applied} = \sqrt{R_P P_{Applied}}$

$R_P$ = Nominal Total Potentiometer Resistance

$P_{Applied}$ = Power Dissipation

$V_{Rated}$ = 500 Volts for RV4X--XA&XB
= 500 Volts for 2RV7X--XA&XB
= 350 Volts for RV2X--XA&XB
= 350 Volts for RV4X--XA&XB
= 350 Volts for RV5X--XA&XB
= 350 Volts for RV6X--XA&XB
= 250 Volts for RV1X--XA&XB
= 200 Volts for All Other Types

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 19 |
| $N_S$ | 8.0 |
| $N_U$ | 29 |
| $A_{IC}$ | 40 |
| $A_{IF}$ | 65 |
| $A_{UC}$ | 48 |
| $A_{UF}$ | 78 |
| $A_{RW}$ | 46 |
| $S_F$ | .50 |
| $M_F$ | 25 |
| $M_L$ | 66 |
| $C_L$ | 1200 |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| MIL-SPEC | 2.5 |
| Lower | 5.0 |



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## 9.15 RESISTORS, VARIABLE, NONWIREWOUND, FILM AND PRECISION

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-R-39023 | RQ | Variable, Nonwirewound, Film, Precision |
| MIL-R-23285 | RVC | Variable, Nonwirewound, Film |

$$\lambda_p = \lambda_b \pi_{TAPS} \pi_R \pi_V \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$
(RQ Style Only)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .023 | .024 | .026 | .028 | .031 |
| 10 | .024 | .026 | .029 | .031 | .034 |
| 20 | .026 | .029 | .032 | .035 | .039 |
| 30 | .028 | .032 | .036 | .040 | .045 |
| 40 | .032 | .036 | .041 | .047 | .053 |
| 50 | .037 | .042 | .049 | .057 | .065 |
| 60 | .044 | .051 | .060 | .070 | .083 |
| 70 | .053 | .064 | .076 | .091 | .11 |
| 80 | .068 | .083 | .10 | .12 | |
| 90 | .092 | .11 | .14 | | |
| 100 | .13 | .17 | | | |
| 110 | .20 | | | | |

$$\lambda_b = .018 \exp\left(\frac{T+273}{343}\right)^{7.4} \times \exp\left(\left(\frac{S}{2.55}\right)\left(\frac{T+273}{273}\right)^{3.6}\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating Power to Rated Power. See Section 9.16 for S Calculation.

Resistance Factor - $\pi_R$

| Resistance Range (Ohms) | $\pi_R$ |
|---|---|
| Up to 10K | 1.0 |
| >10K to 50K | 1.1 |
| >50K to 200K | 1.2 |
| >200K to 1M | 1.4 |
| >1M | 1.8 |

Base Failure Rate - $\lambda_b$
(RVC Style Only)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .028 | .031 | .033 | .036 | .039 |
| 10 | .029 | .032 | .035 | .038 | .042 |
| 20 | .030 | .033 | .037 | .041 | .046 |
| 30 | .031 | .035 | .040 | .045 | .051 |
| 40 | .032 | .037 | .043 | .050 | .058 |
| 50 | .034 | .040 | .047 | .056 | .066 |
| 60 | .036 | .044 | .053 | .064 | .078 |
| 70 | .039 | .049 | .060 | .075 | .093 |
| 80 | .043 | .055 | .070 | .09 | .11 |
| 90 | .048 | .063 | .083 | .11 | .15 |
| 100 | .055 | .075 | .10 | .14 | .19 |
| 110 | .064 | .091 | .13 | .18 | .26 |
| 120 | .077 | .11 | .17 | .25 | .37 |
| 130 | .096 | .15 | .23 | .36 | .55 |
| 140 | .12 | .20 | .33 | .53 | |
| 150 | .17 | .29 | .50 | | |
| 160 | .24 | .44 | | | |
| 170 | .37 | | | | |

$$\lambda_b = .0257 \exp\left(\frac{T+273}{398}\right)^{7.9} \times \exp\left(\left(\frac{S}{2.45}\right)\left(\frac{T+273}{273}\right)^{4.3}\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating Power to Rated Power. See Section 9.16 for S Calculation.



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## 9.15 RESISTORS, VARIABLE, NONWIREWOUND, FILM AND PRECISION

### Potentiometer Taps Factor - $\pi_{TAPS}$

| $N_{TAPS}$ | $\pi_{TAPS}$ | $N_{TAPS}$ | $\pi_{TAPS}$ | $N_{TAPS}$ | $\pi_{TAPS}$ |
|---|---|---|---|---|---|
| 3 | 1.0 | 13 | 2.7 | 23 | 5.2 |
| 4 | 1.1 | 14 | 2.9 | 24 | 5.5 |
| 5 | 1.2 | 15 | 3.1 | 25 | 5.8 |
| 6 | 1.4 | 16 | 3.4 | 26 | 6.1 |
| 7 | 1.5 | 17 | 3.6 | 27 | 6.4 |
| 8 | 1.7 | 18 | 3.8 | 28 | 6.7 |
| 9 | 1.9 | 19 | 4.1 | 29 | 7.0 |
| 10 | 2.1 | 20 | 4.4 | 30 | 7.4 |
| 11 | 2.3 | 21 | 4.6 | 31 | 7.7 |
| 12 | 2.5 | 22 | 4.9 | 32 | 8.0 |

$$\pi_{TAPS} = \frac{(N_{TAPS})^{\frac{3}{2}}}{25} + 0.792$$

$N_{TAPS}$ = Number of Potentiometer Taps, including the Wiper and Terminations.

### Voltage Factor - $\pi_V$

| Applied Voltage* / Rated Voltage | $\pi_V$ |
|---|---|
| 0 to 0.8 | 1.00 |
| >0.8 to 0.9 | 1.05 |
| >0.9 to 1.0 | 1.20 |

$$^*V_{Applied} = \sqrt{R_P P_{Applied}}$$

$R_P$ = Nominal Total Potentiometer Resistance

$P_{Applied}$ = Power Dissipation

$V_{Rated}$ = 250 Volts for RQ090, 110, 150, 200, 300

= 500 Volts for RQ100, 160, 210

= 350 Volts for RVC5, 6

### Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| MIL-SPEC | 2 |
| Lower | 4 |

### Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 3.0 |
| $G_M$ | 14 |
| $N_S$ | 7.0 |
| $N_U$ | 24 |
| $A_{IC}$ | 6.0 |
| $A_{IF}$ | 12 |
| $A_{UC}$ | 20 |
| $A_{UF}$ | 30 |
| $A_{RW}$ | 39 |
| $S_F$ | .50 |
| $M_F$ | 22 |
| $M_L$ | 57 |
| $C_L$ | 1000 |



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## 9.16 CALCULATION OF STRESS RATIO FOR POTENTIOMETERS

### Stress Ratio (S) Calculation for Rheostats

$$S = \frac{\left(I_{op_{max}}\right)^2}{\pi_{GANGED}\left(I_{max_{rated}}\right)^2}$$

$I_{op_{max}}$ = Maximum current which will be passed through the rheostat in the circuit.

$I_{max_{rated}}$ = Current rating of the potentiometer. If current rating is not given, use:

$I_{max_{rated}} = \sqrt{P_{rated}/R_P}$

$P_{rated}$ = Power Rating of Potentiometer

$R_P$ = Nominal Total Potentiometer Resistance

$\pi_{GANGED}$ = Factor to correct for the reduction in effective rating of the potentiometer due to the close proximity of two or more potentiometers when they are ganged together on a common shaft. See below.

### Stress Ratio (S) Calculation for Potentiometers Connected Conventionally

$$S = \frac{P_{APPLIED}}{\pi_{EFF} \times \pi_{GANGED} \times P_{RATED}}$$

$P_{Applied}$ = Equivalent power input to the potentiometer when it is not loaded (i.e., wiper lead disconnected). Calculate as follows:

$P_{Applied} = \frac{V_{in}^2}{R_P}$

$V_{in}$ = Input Voltage

$R_P$ = Nominal Total Potentiometer Resistance

$P_{RATED}$ = Power Rating of Potentiometer

$\pi_{GANGED}$ = Factor to correct for the reduction in effective rating of the potentiometer due to the close proximity of two or more potentiometers when they are ganged together on a common shaft. See below.

$\pi_{EFF}$ = Correction factor for the electrical loading effect on the wiper contact of the potentiometer. Its value is a function of the type of potentiometer, its resistance, and the load resistance. See next page.

### Ganged-Potentiometer Factor - $\pi_{GANGED}$

| Number of Sections | First Potentiometer Next to Mount | Second in Gang | Third in Gang | Fourth in Gang | Fifth in Gang | Sixth in Gang |
|---|---|---|---|---|---|---|
| Single | 1.0 | Not Applicable | | | | |
| Two | 0.75 | 0.60 | Not Applicable | | | |
| Three | 0.75 | 0.50 | 0.60 | Not Applicable | | |
| Four | 0.75 | 0.50 | 0.50 | 0.60 | Not Applicable | |
| Five | 0.75 | 0.50 | 0.40 | 0.50 | 0.60 | Not Applicable |
| Six | 0.75 | 0.50 | 0.40 | 0.40 | 0.50 | 0.60 |



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## 9.16 CALCULATION OF STRESS RATIO FOR POTENTIOMETERS

### Loaded Potentiometer Derating Factor - $\pi_{EFF}$

| $R_L/R_P$ | $K_H$ 0.2 | 0.3 | 0.5 | 1.0 |
|---|---|---|---|---|
| 0.1 | .04 | .03 | .02 | .01 |
| 0.2 | .13 | .09 | .05 | .03 |
| 0.3 | .22 | .16 | .10 | .05 |
| 0.4 | .31 | .23 | .15 | .08 |
| 0.5 | .38 | .29 | .20 | .11 |
| 0.6 | .45 | .35 | .25 | .14 |
| 0.7 | .51 | .40 | .29 | .17 |
| 0.8 | .55 | .45 | .33 | .20 |
| 0.9 | .59 | .49 | .37 | .22 |
| 1.0 | .63 | .53 | .40 | .25 |
| 1.5 | .74 | .65 | .53 | .36 |
| 2.0 | .80 | .73 | .62 | .44 |
| 3.0 | .87 | .81 | .72 | .56 |
| 4.0 | .90 | .86 | .78 | .64 |
| 5.0 | .92 | .88 | .82 | .69 |
| 10.0 | .96 | .94 | .90 | .83 |
| 100.0 | 1.00 | .99 | .99 | .98 |

$$\pi_{EFF} = \frac{R_L^2}{R_L^2 + K_H\left(R_P^2 + 2R_P R_L\right)}$$

$R_L$ = Load resistance (If $R_L$ is variable, use lowest value). $R_L$ is the total resistance between the wiper arm and one end of the potentiometer.

$R_P$ = Nominal Total Potentiometer Resistance

$K_H$ = Style Constant. See $K_H$ Table.

### Style Constant - $K_H$

| Potentiometer MIL-SPEC | Style Type | $K_H$ |
|---|---|---|
| MIL-R-19 | RA | 0.5 |
| MIL-R-22 | RP | 1.0 |
| MIL-R-94 | RV | 0.5 |
| MIL-R-12934 | RR1000, 1001, 1003, 1400, 2100, 2101, 2102, 2103 | 0.3 |
| MIL-R-12934 | All Other Types | 0.2 |
| MIL-R-22097 | RJ11, RJ12 | 0.3 |
| MIL-R-22097 | All Other Types | 0.2 |
| MIL-R-23285 | RVC | 0.5 |
| MIL-R-27208 | RT22, 24, 26, 27 | 0.2 |
| MIL-R-27208 | All Other Types | 0.3 |
| MIL-R-39002 | RK | 0.5 |
| MIL-R-39015 | RTR 22, 24 | 0.2 |
| MIL-R-39015 | RTR12 | 0.3 |
| MIL-R-39023 | RQ | 0.3 |
| MIL-R-39035 | RJR | 0.3 |



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**Example**

Given: Type RV1SAYSA505A variable 500K ohm resistor procured per MIL-R-94, rated at 0.2 watts is being used in a fixed ground environment. The resistor ambient temperature is 40°C and is dissipating 0.06 watts. The resistance connected to the wiper contact varies between 1 megohm and 3 megohms. The potentiometer is connected conventionally without ganging.

The appropriate model for RV style variable resistors is given in Section 9.14. Based on the given information the following model factors are determined from the tables shown in Section 9.14 and by following the procedure for determining electrical stress for potentiometers as described in Section 9.16.

From Section 9.16

$P_{APPLIED} = .06W$

$\pi_{EFF} = .62$ — $K_H = .5$ for MIL-R-94 (Section 9.16 Table)

$\pi_{GANGED} = 1.0$ — Not Ganged (Section 9.16 Table, Single Section, First Potentiometer)

$\pi_{RATED} = .2W$

$$S = \frac{P_{APPLIED}}{\pi_{EFF} \times \pi_{GANGED} \times \pi_{RATED}} = \frac{.06}{(.62)(1.0)(.2)} = .48$$

From Section 9.14

$\lambda_b = .047$ — $T_A = 40°C$, S Rounded to .5

$\pi_R = 1.4$ — 500K ohms

$\pi_{TAPS} = 1.0$ — 3 Taps, Basic Single Potentiometer

$\pi_V = 1.0$ — $V_{RATED} = 250$ Volts for RV1 prefix

$V_{APPLIED} = \sqrt{(500{,}000)(.06)} = 173$ volts

$V_{APPLIED}/V_{RATED} = \frac{173}{250} = .69$

$\pi_Q = 2.5$

$\pi_E = 2.0$

$\lambda_p = \lambda_b \pi_{TAPS} \pi_R \pi_V \pi_Q \pi_E$

$= (.047)(1.0)(1.4)(1.0)(2.5)(2.0) = .33$ Failures/$10^6$ Hours



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## 10.1 CAPACITORS, FIXED, PAPER, BY-PASS

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-C-25 | CP | Paper, By-pass, Filter, Blocking, DC |
| MIL-C-12889 | CA | Paper, By-pass, Radio Interference Reduction AC and DC |

$$\lambda_p = \lambda_b \pi_{CV} \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

(T = 85°C Max Rated )

(All MIL-C-12889; MIL-C-25 Styles CP25, 26, 27, 28, 29, 40, 41, 67, 69, 70, 72, 75, 76, 77, 78, 80, 81, 82; Characteristics E, F)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .00088 | .0011 | .0036 | .015 | .051 |
| 10 | .00089 | .0011 | .0036 | .016 | .052 |
| 20 | .00092 | .0011 | .0037 | .016 | .054 |
| 30 | .00097 | .0012 | .0039 | .017 | .057 |
| 40 | .0011 | .0013 | .0044 | .019 | .063 |
| 50 | .0013 | .0016 | .0052 | .022 | .075 |
| 60 | .0017 | .0021 | .0069 | .030 | .10 |
| 70 | .0027 | .0034 | .011 | .048 | .16 |
| 80 | .0060 | .0074 | .024 | .10 | .35 |

$$\lambda_b = .00086\left[\left(\frac{S}{.4}\right)^5 + 1\right]\exp\left(2.5\left(\frac{T+273}{358}\right)^{18}\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

Base Failure Rate - $\lambda_b$

(T = 125°C Max Rated)

(MIL-C-25 Styles CP 4, 5, 8, 9, 10, 11, 12 13; Characteristic K)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .00086 | .0011 | .0035 | .015 | .051 |
| 10 | .00087 | .0011 | .0035 | .015 | .051 |
| 20 | .00087 | .0011 | .0035 | .015 | .051 |
| 30 | .00088 | .0011 | .0035 | .015 | .051 |
| 40 | .00089 | .0011 | .0036 | .015 | .052 |
| 50 | .00091 | .0011 | .0037 | .016 | .053 |
| 60 | .00095 | .0012 | .0039 | .017 | .056 |
| 70 | .0010 | .0013 | .0041 | .018 | .060 |
| 80 | .0011 | .0014 | .0046 | .020 | .067 |
| 90 | .0014 | .0017 | .0056 | .024 | .081 |
| 100 | .0019 | .0023 | .0076 | .033 | .11 |
| 110 | .0030 | .0037 | .012 | .052 | .18 |
| 120 | .0063 | .0078 | .026 | .11 | .37 |

$$\lambda_b = .00086\left[\left(\frac{S}{.4}\right)^5 + 1\right]\exp\left(2.5\left(\frac{T+273}{398}\right)^{18}\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.



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## 10.1 CAPACITORS, FIXED, PAPER, BY-PASS

Capacitance Factor - $\pi_{CV}$

| Capacitance, C (μF) | $\pi_{CV}$ |
|---|---|
| MIL-C-25* | |
| .0034 | 0.7 |
| .15 | 1.0 |
| 2.3 | 1.3 |
| 16. | 1.6 |
| MIL-C-12889 | |
| All | 1.0 |

* $\pi_{CV} = 1.2C^{.095}$

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| MIL-SPEC | 3.0 |
| Lower | 7.0 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 9.0 |
| $N_S$ | 5.0 |
| $N_U$ | 15 |
| $A_{IC}$ | 6.0 |
| $A_{IF}$ | 8.0 |
| $A_{UC}$ | 17 |
| $A_{UF}$ | 32 |
| $A_{RW}$ | 22 |
| $S_F$ | .50 |
| $M_F$ | 12 |
| $M_L$ | 32 |
| $C_L$ | 570 |



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Failures/10⁸ Hours

## 10.2 CAPACITORS, FIXED, PAPER, FEED-THROUGH

**SPECIFICATION**
MIL-C-11693

**STYLE**
CZR and CZ

**DESCRIPTION**
Paper, Metallized Paper, Metallized Plastic, RFI Feed-Through Established Reliability and Non-Established Reliability

$$\lambda_p = \lambda_b \pi_{CV} \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$
(T = 85°C Max Rated)
(Characteristics E, W)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .0012 | .0014 | .0047 | .020 | .069 |
| 10 | .0012 | .0015 | .0048 | .021 | .070 |
| 20 | .0012 | .0015 | .0050 | .021 | .072 |
| 30 | .0013 | .0016 | .0053 | .023 | .076 |
| 40 | .0014 | .0018 | .0058 | .025 | .084 |
| 50 | .0017 | .0021 | .0069 | .030 | .10 |
| 60 | .0023 | .0028 | .0092 | .039 | .13 |
| 70 | .0037 | .0045 | .015 | .064 | .21 |
| 80 | .0080 | .0099 | .032 | .14 | .47 |

$$\lambda_b = .00115\left[\left(\frac{S}{.4}\right)^5 + 1\right]\exp\left(2.5\left(\frac{T+273}{358}\right)^{18}\right)$$

T = Ambient Temperature (°C)
S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

Base Failure Rate - $\lambda_b$
(T = 125°C Max Rated)
(Characteristic K)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .0012 | .0014 | .0047 | .020 | .068 |
| 10 | .0012 | .0014 | .0047 | .020 | .068 |
| 20 | .0012 | .0014 | .0047 | .020 | .068 |
| 30 | .0012 | .0014 | .0047 | .020 | .069 |
| 40 | .0012 | .0015 | .0048 | .021 | .070 |
| 50 | .0012 | .0015 | .0049 | .021 | .072 |
| 60 | .0013 | .0016 | .0052 | .022 | .075 |
| 70 | .0014 | .0017 | .0055 | .024 | .08 |
| 80 | .0015 | .0019 | .0062 | .027 | .09 |
| 90 | .0019 | .0023 | .0075 | .032 | .11 |
| 100 | .0025 | .0031 | .010 | .044 | .15 |
| 110 | .0040 | .005 | .016 | .07 | .24 |
| 120 | .0084 | .010 | .034 | .15 | .49 |

$$\lambda_b = .00115\left[\left(\frac{S}{.4}\right)^5 + 1\right]\exp\left(2.5\left(\frac{T+273}{398}\right)^{18}\right)$$

T = Ambient Temperature (°C)
S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

Base Failure Rate - $\lambda_b$
(T = 150°C Max Rated)
(Characteristic P)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .0012 | .0014 | .0047 | .020 | .068 |
| 10 | .0012 | .0014 | .0047 | .020 | .068 |
| 20 | .0012 | .0014 | .0047 | .020 | .068 |
| 30 | .0012 | .0014 | .0047 | .020 | .068 |
| 40 | .0012 | .0014 | .0047 | .020 | .068 |
| 50 | .0012 | .0015 | .0048 | .020 | .069 |
| 60 | .0012 | .0015 | .0048 | .021 | .070 |
| 70 | .0012 | .0015 | .0049 | .021 | .071 |
| 80 | .0013 | .0016 | .0051 | .022 | .074 |
| 90 | .0013 | .0017 | .0055 | .023 | .079 |
| 100 | .0015 | .0018 | .0060 | .026 | .087 |
| 110 | .0017 | .0022 | .0071 | .03 | .10 |
| 120 | .0022 | .0028 | .0091 | .039 | .13 |
| 130 | .0033 | .0040 | .013 | .057 | .19 |
| 140 | .0058 | .0072 | .024 | .10 | .34 |
| 150 | .014 | .017 | .057 | .24 | .82 |

$$\lambda_b = .00115\left[\left(\frac{S}{.4}\right)^5 + 1\right]\exp\left(2.5\left(\frac{T+273}{423}\right)^{18}\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.



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## 10.2 CAPACITORS, FIXED, PAPER, FEED-THROUGH

Capacitance Factor - $\pi_{CV}$

| Capacitance, C (μF) | $\pi_{CV}$ |
|---|---|
| 0.0031 | .70 |
| 0.061 | 1.0 |
| 1.8 | 1.5 |
| $\pi_{CV} = 1.4C^{0.12}$ | |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| M | 1.0 |
| Non-Established Reliability | 3.0 |
| Lower | 10 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 9.0 |
| $N_S$ | 7.0 |
| $N_U$ | 15 |
| $A_{IC}$ | 6.0 |
| $A_{IF}$ | 8.0 |
| $A_{UC}$ | 17 |
| $A_{UF}$ | 28 |
| $A_{RW}$ | 22 |
| $S_F$ | .50 |
| $M_F$ | 12 |
| $M_L$ | 32 |
| $C_L$ | 570 |



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## 10.3 CAPACITORS, FIXED, PAPER AND PLASTIC FILM

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-C-14157 | CPV | Paper and Plastic Film, Est. Rel. |
| MIL-C-19978 | CQR and CQ | Paper and Plastic Film, Est. Rel. and Non-Est. Rel. |

$$\lambda_p = \lambda_b \pi_{CV} \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

### Base Failure Rate - $\lambda_b$

(T = 65°C Max Rated )
(MIL-C-14157 Style CPV07;
MIL-C-19978 Characteristics P, L)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .00053 | .00065 | .0021 | .0092 | .031 |
| 10 | .00055 | .00069 | .0022 | .0096 | .032 |
| 20 | .00061 | .00075 | .0025 | .011 | .036 |
| 30 | .00071 | .00088 | .0029 | .012 | .042 |
| 40 | .00094 | .0012 | .0038 | .016 | .055 |
| 50 | .0015 | .0019 | .0061 | .026 | .088 |
| 60 | .0034 | .0042 | .014 | .059 | .20 |

$$\lambda_b = .0005\left[\left(\frac{S}{.4}\right)^5 + 1\right]\exp\left(2.5\left(\frac{T+273}{338}\right)^{18}\right)$$

T = Ambient Temperature (°C)
S = Ratio of Operating to Rated Voltage
Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

### Base Failure Rate - $\lambda_b$

(T = 85°C Max Rated)
(MIL-C-14157 Style CPV17;
MIL-C-19978 Characteristics E, F, G, M)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .00051 | .00063 | .0021 | .0089 | .030 |
| 10 | .00052 | .00064 | .0021 | .0090 | .030 |
| 20 | .00054 | .00066 | .0022 | .0093 | .031 |
| 30 | .00057 | .00070 | .0023 | .0099 | .033 |
| 40 | .00063 | .00077 | .0025 | .011 | .037 |
| 50 | .00074 | .00092 | .0030 | .013 | .043 |
| 60 | .00099 | .0012 | .0040 | .017 | .058 |
| 70 | .0016 | .0020 | .0064 | .028 | .093 |
| 80 | .0035 | .0043 | .014 | .061 | .20 |

$$\lambda_b = .0005\left[\left(\frac{S}{.4}\right)^5 + 1\right]\exp\left(2.5\left(\frac{T+273}{358}\right)^{18}\right)$$

T = Ambient Temperature (°C)
S = Ratio of Operating to Rated Voltage
Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

### Base Failure Rate - $\lambda_b$

(T = 125°C Max Rated)
(MIL-C-14157 Style CPV09 and MIL-C-19978
Characteristics K, Q, S)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .00050 | .00062 | .0020 | .0087 | .029 |
| 10 | .00050 | .00062 | .0020 | .0088 | .029 |
| 20 | .00051 | .00062 | .0020 | .0088 | .030 |
| 30 | .00051 | .00063 | .0021 | .0089 | .030 |
| 40 | .00052 | .00064 | .0021 | .009 | .030 |
| 50 | .00053 | .00066 | .0021 | .0092 | .031 |
| 60 | .00055 | .00068 | .0022 | .0096 | .032 |
| 70 | .00059 | .00073 | .0024 | .010 | .035 |
| 80 | .00067 | .00083 | .0027 | .012 | .039 |
| 90 | .00081 | .0010 | .0033 | .014 | .047 |
| 100 | .0011 | .0013 | .0044 | .019 | .064 |
| 110 | .0018 | .0022 | .0071 | .030 | .10 |
| 120 | .0037 | .0045 | .015 | .064 | .21 |

$$\lambda_b = .0005\left[\left(\frac{S}{.4}\right)^5 + 1\right]\exp\left(2.5\left(\frac{T+273}{398}\right)^{18}\right)$$

T = Ambient Temperature (°C)
S = Ratio of Operating to Rated Voltage
Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

### Base Failure Rate - $\lambda_b$

(T = 170°C Max Rated)
(MIL-C-19978 Characteristic T)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .00050 | .00062 | .0020 | .0087 | .029 |
| 10 | .00050 | .00062 | .0020 | .0087 | .029 |
| 20 | .00050 | .00062 | .0020 | .0087 | .029 |
| 30 | .00050 | .00062 | .0020 | .0087 | .029 |
| 40 | .00050 | .00062 | .0020 | .0087 | .029 |
| 50 | .00050 | .00062 | .0020 | .0088 | .030 |
| 60 | .00051 | .00063 | .0021 | .0088 | .030 |
| 70 | .00051 | .00063 | .0021 | .0089 | .030 |
| 80 | .00052 | .00065 | .0021 | .0091 | .031 |
| 90 | .00054 | .00066 | .0022 | .0093 | .031 |
| 100 | .00056 | .00069 | .0023 | .0097 | .033 |
| 110 | .00060 | .00074 | .0024 | .010 | .035 |
| 120 | .00067 | .00083 | .0027 | .012 | .039 |
| 130 | .00079 | .00098 | .0032 | .014 | .046 |
| 140 | .0010 | .0013 | .0041 | .018 | .060 |
| 150 | .0015 | .0018 | .006 | .026 | .087 |
| 160 | .0026 | .0032 | .011 | .046 | .15 |
| 170 | .0061 | .0075 | .025 | .11 | .36 |

$$\lambda_b = .0005\left[\left(\frac{S}{.4}\right)^5 + 1\right]\exp\left(2.5\left(\frac{T+273}{443}\right)^{18}\right)$$

T = Ambient Temperature (°C)
S = Ratio of Operating to Rated Voltage
Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.



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## 10.3 CAPACITORS, FIXED, PAPER AND PLASTIC FILM

Capacitance Factor - $\pi_{CV}$

| Capacitance, C (μF) | $\pi_{CV}$ |
|---|---|
| MIL-C-14157: * | |
| .0017 | .70 |
| .027 | 1.0 |
| .20 | 1.3 |
| 1.0 | 1.6 |
| MIL-C-19978: ** | |
| .00032 | .70 |
| .033 | 1.0 |
| 1.0 | 1.3 |
| 15.0 | 1.6 |

* $\pi_{CV} = 1.6C^{0.13}$

** $\pi_{CV} = 1.3C^{0.077}$

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 8.0 |
| $N_S$ | 5.0 |
| $N_U$ | 14 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 6.0 |
| $A_{UC}$ | 11.0 |
| $A_{UF}$ | 20 |
| $A_{RW}$ | 20 |
| $S_F$ | .50 |
| $M_F$ | 11 |
| $M_L$ | 29 |
| $C_L$ | 530 |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| S | .03 |
| R | .10 |
| P | .30 |
| M | 1.0 |
| L | 3.0 |
| MIL-C-19978, Non-Est. Rel. | 10 |
| Lower | 30 |



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## 10.4 CAPACITORS, FIXED, METALLIZED PAPER, PAPER-PLASTIC AND PLASTIC

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-C-18312 | CH | Metallized Paper , Paper-Plastic, Plastic |
| MIL-C-39022 | CHR | Metallized Paper, Paper-Plastic, Plastic, Established Reliability |

$$\lambda_p = \lambda_b \pi_{CV} \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

(T = 85°C Max Rated )
(MIL-C-39022 Characteristic 9 and 12 (50 Volts rated), Characteristic 49; and MIL-C-18312 Characteristic R)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .00070 | .00087 | .0029 | .012 | .041 |
| 10 | .00072 | .00089 | .0029 | .012 | .042 |
| 20 | .00074 | .00091 | .0030 | .013 | .043 |
| 30 | .00078 | .00097 | .0032 | .014 | .046 |
| 40 | .00086 | .0011 | .0035 | .015 | .051 |
| 50 | .0010 | .0013 | .0041 | .018 | .06 |
| 60 | .0014 | .0017 | .0055 | .024 | .08 |
| 70 | .0022 | .0027 | .0089 | .038 | .13 |
| 80 | .0048 | .0059 | .019 | .084 | .28 |

$$\lambda_b = .00069\left[\left(\frac{S}{.4}\right)^5 + 1\right]\exp\left(2.5\left(\frac{T+273}{358}\right)^{18}\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

Base Failure Rate - $\lambda_b$

(T = 125°C Max Rated)
(MIL-C-39022 Characteristic 9 and 12 (above 50 Volts rated), Characteristics 1, 10, 19, 29, 59; and MIL-C-18312 Characteristic N)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .00069 | .00086 | .0028 | .012 | .041 |
| 10 | .00069 | .00086 | .0028 | .012 | .041 |
| 20 | .00070 | .00086 | .0028 | .012 | .041 |
| 30 | .00070 | .00087 | .0028 | .012 | .041 |
| 40 | .00071 | .00088 | .0029 | .012 | .042 |
| 50 | .00073 | .00090 | .003. | .013 | .043 |
| 60 | .00076 | .00094 | .0031 | .013 | .045 |
| 70 | .00082 | .0010 | .0033 | .014 | .048 |
| 80 | .00092 | .0011 | .0037 | .016 | .054 |
| 90 | .0011 | .0014 | .0045 | .019 | .065 |
| 100 | .0015 | .0019 | .0061 | .026 | .088 |
| 110 | .0024 | .0030 | .0098 | .042 | .14 |
| 120 | .0051 | .0063 | .020 | .088 | .30 |

$$\lambda_b = .00069\left[\left(\frac{S}{.4}\right)^5 + 1\right]\exp\left(2.5\left(\frac{T+273}{398}\right)^{18}\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.



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## 10.4 CAPACITORS, FIXED, METALLIZED PAPER, PAPER-PLASTIC AND PLASTIC

Capacitance Factor - $\pi_{CV}$

| Capacitance, C (µF) | $\pi_{CV}$ |
|---|---|
| 0.0029 | .70 |
| 0.14 | 1.0 |
| 2.4 | 1.3 |
| $\pi_{CV} = 1.2C^{0.092}$ | |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| S | 0.03 |
| R | .10 |
| P | .30 |
| M | 1.0 |
| L | 3.0 |
| MIL-C-18312, Non-Est. Rel. | 7.0 |
| Lower | 20 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 8.0 |
| $N_S$ | 5.0 |
| $N_U$ | 14 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 6.0 |
| $A_{UC}$ | 11.0 |
| $A_{UF}$ | 20 |
| $A_{RW}$ | 20 |
| $S_F$ | .50 |
| $M_F$ | 11 |
| $M_L$ | 29 |
| $C_L$ | 530 |



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## 10.5 CAPACITORS, FIXED, PLASTIC AND METALLIZED PLASTIC

**SPECIFICATION**
MIL-C-55514

**STYLE**
CFR

**DESCRIPTION**
Plastic, Metallized Plastic, Est. Rel.

$$\lambda_p = \lambda_b \pi_{CV} \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$
(T = 85°C Max Rated )
(Characteristics M, N)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .0010 | .0012 | .0041 | .018 | .059 |
| 10 | .0010 | .0013 | .0042 | .018 | .060 |
| 20 | .0011 | .0013 | .0043 | .018 | .062 |
| 30 | .0011 | .0014 | .0045 | .020 | .066 |
| 40 | .0012 | .0015 | .0050 | .022 | .073 |
| 50 | .0015 | .0018 | .0059 | .026 | .086 |
| 60 | .0020 | .0024 | .0079 | .034 | .11 |
| 70 | .0032 | .0039 | .013 | .055 | .18 |
| 80 | .0069 | .0085 | .028 | .12 | .40 |

$$\lambda_b = .00099\left[\left(\frac{S}{.4}\right)^5 + 1\right]\exp\left(2.5\left(\frac{T+273}{358}\right)^{18}\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

Base Failure Rate - $\lambda_b$
(T = 125°C Max Rated)
(Characteristics Q, R, S)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .00099 | .0012 | .0040 | .017 | .058 |
| 10 | .0010 | .0012 | .0040 | .017 | .058 |
| 20 | .0010 | .0012 | .0041 | .017 | .059 |
| 30 | .0010 | .0012 | .0041 | .018 | .059 |
| 40 | .0010 | .0013 | .0041 | .018 | .060 |
| 50 | .0011 | .0013 | .0043 | .018 | .062 |
| 60 | .0011 | .0014 | .0044 | .019 | .064 |
| 70 | .0012 | .0015 | .0048 | .020 | .069 |
| 80 | .0013 | .0016 | .0054 | .023 | .077 |
| 90 | .0016 | .0020 | .0065 | .028 | .094 |
| 100 | .0022 | .0027 | .0087 | .038 | .13 |
| 110 | .0035 | .0043 | .014 | .06 | .20 |
| 120 | .0073 | .0090 | .029 | .13 | .43 |

$$\lambda_b = .00099\left[\left(\frac{S}{.4}\right)^5 + 1\right]\exp\left(2.5\left(\frac{T+273}{398}\right)^{18}\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.



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## 10.5 CAPACITORS, FIXED, PLASTIC AND METALLIZED PLASTIC

Capacitance Factor - $\pi_{CV}$

| Capacitance, C (μF) | $\pi_{CV}$ |
|---|---|
| 0.0049 | .70 |
| 0.33 | 1.0 |
| 7.1 | 1.3 |
| 38. | 1.5 |
| $\pi_{CV} = 1.1C^{0.085}$ | |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| S | .030 |
| R | .10 |
| P | .30 |
| M | 1.0 |
| Lower | 10 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 10 |
| $N_S$ | 5.0 |
| $N_U$ | 16 |
| $A_{IC}$ | 6 |
| $A_{IF}$ | 11 |
| $A_{UC}$ | 18 |
| $A_{UF}$ | 30 |
| $A_{RW}$ | 23 |
| $S_F$ | .50 |
| $M_F$ | 13 |
| $M_L$ | 34 |
| $C_L$ | 610 |



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## 10.6 CAPACITORS, FIXED, SUPER-METALLIZED PLASTIC

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-C-83421 | CRH | Super-Metallized Plastic, Est. Rel. |

$$\lambda_p = \lambda_b \pi_{CV} \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

(T = 125°C Max Rated)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .00055 | .00068 | .0022 | .0096 | .032 |
| 10 | .00055 | .00068 | .0022 | .0096 | .032 |
| 20 | .00056 | .00069 | .0023 | .0097 | .033 |
| 30 | .00056 | .00069 | .0023 | .0098 | .033 |
| 40 | .00057 | .00070 | .0023 | .0099 | .033 |
| 50 | .00058 | .00072 | .0024 | .010 | .034 |
| 60 | .00061 | .00075 | .0025 | .011 | .036 |
| 70 | .00065 | .00081 | .0026 | .011 | .038 |
| 80 | .00073 | .00091 | .0030 | .013 | .043 |
| 90 | .00089 | .0011 | .0036 | .015 | .052 |
| 100 | .0012 | .0015 | .0049 | .021 | .07 |
| 110 | .0019 | .0024 | .0078 | .033 | .11 |
| 120 | .0040 | .0050 | .016 | .070 | .24 |

$$\lambda_b = .00055\left[\left(\frac{S}{.4}\right)^5 + 1\right]\exp\left(2.5\left(\frac{T+273}{398}\right)^{18}\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| S | .020 |
| R | .10 |
| P | .30 |
| M | 1.0 |
| Lower | 10 |

Capacitance Factor - $\pi_{CV}$

| Capacitance, C (µF) | $\pi_{CV}$ |
|---|---|
| .001 | .64 |
| 0.14 | 1.0 |
| 2.4 | 1.3 |
| 23 | 1.6 |

$$\pi_{CV} = 1.2C^{0.092}$$

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 4.0 |
| $G_M$ | 8.0 |
| $N_S$ | 5.0 |
| $N_U$ | 14 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 6.0 |
| $A_{UC}$ | 13.0 |
| $A_{UF}$ | 20 |
| $A_{RW}$ | 20 |
| $S_F$ | .50 |
| $M_F$ | 11 |
| $M_L$ | 29 |
| $C_L$ | 530 |



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## 10.7 CAPACITORS, FIXED, MICA

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-C-5 | CM | MICA (Dipped or Molded) |
| MIL-C-39001 | CMR | MICA (Dipped), Established Reliability |

$$\lambda_p = \lambda_b \pi_{CV} \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$
(T=70°C Max Rated)
(MIL-C-5, Temp. Range M)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .00030 | .00041 | .00086 | .0019 | .0036 |
| 10 | .00047 | .00066 | .0014 | .0030 | .0058 |
| 20 | .00075 | .0011 | .0022 | .0047 | .0092 |
| 30 | .0012 | .0017 | .0035 | .0075 | .015 |
| 40 | .0019 | .0027 | .0056 | .012 | .023 |
| 50 | .0031 | .0043 | .0089 | .019 | .037 |
| 60 | .0049 | .0068 | .014 | .030 | .059 |
| 70 | .0078 | .011 | .023 | .049 | .095 |

$$\lambda_b = 8.6 \times 10^{-10} \left[ \left( \frac{S}{.4} \right)^3 + 1 \right] \exp\left( 16 \left( \frac{T+273}{343} \right) \right)$$

T = Ambient Temperature (°C)
S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

Base Failure Rate - $\lambda_b$
(T=85°C Max Rated)
(MIL-C-5, Temp. Range N)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .00017 | .00024 | .00051 | .0011 | .0021 |
| 10 | .00027 | .00038 | .00079 | .0017 | .0033 |
| 20 | .00042 | .00059 | .0012 | .0027 | .0052 |
| 30 | .00066 | .00093 | .0019 | .0042 | .0081 |
| 40 | .0010 | .0015 | .003 | .0065 | .013 |
| 50 | .0016 | .0023 | .0047 | .010 | .020 |
| 60 | .0025 | .0036 | .0074 | .016 | .031 |
| 70 | .0040 | .0056 | .012 | .025 | .048 |
| 80 | .0062 | .0087 | .018 | .039 | .076 |

$$\lambda_b = 8.6 \times 10^{-10} \left[ \left( \frac{S}{.4} \right)^3 + 1 \right] \exp\left( 16 \left( \frac{T+273}{358} \right) \right)$$

T = Ambient Temperature (°C)
S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

Base Failure Rate - $\lambda_b$
(T=125°C Max Rated)
(MIL-C-5, Temp. Range O; MIL-C-39001 Temp. Range O)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .00005 | .00007 | .00015 | .00032 | .00062 |
| 10 | .00008 | .00011 | .00022 | .00048 | .00093 |
| 20 | .00011 | .00016 | .00033 | .00071 | .0014 |
| 30 | .00017 | .00024 | .00050 | .0011 | .0021 |
| 40 | .00025 | .00036 | .00074 | .0016 | .0031 |
| 50 | .00038 | .00053 | .0011 | .0024 | .0046 |
| 60 | .00057 | .0008 | .0017 | .0036 | .0069 |
| 70 | .00085 | .0012 | .0025 | .0053 | .010 |
| 80 | .0013 | .0018 | .0037 | .008 | .016 |
| 90 | .0019 | .0027 | .0055 | .012 | .023 |
| 100 | .0028 | .0040 | .0083 | .018 | .035 |
| 110 | .0042 | .0059 | .012 | .027 | .052 |
| 120 | .0063 | .0089 | .018 | .040 | .077 |

$$\lambda_b = 8.6 \times 10^{-10} \left[ \left( \frac{S}{.4} \right)^3 + 1 \right] \exp\left( 16 \left( \frac{T+273}{398} \right) \right)$$

T = Ambient Temperature (°C)
S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

Base Failure Rate - $\lambda_b$
(T=150°C Max Rated)
(MIL-C-5, Temp. Range P; MIL-C-39001, Temp. Range P)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .00003 | .00004 | .00008 | .00017 | .00033 |
| 10 | .00004 | .00005 | .00011 | .00024 | .00047 |
| 20 | .00006 | .00008 | .00017 | .00036 | .00069 |
| 30 | .00008 | .00012 | .00024 | .00052 | .0010 |
| 40 | .00012 | .00017 | .00035 | .00076 | .0015 |
| 50 | .00018 | .00025 | .00051 | .0011 | .0022 |
| 60 | .00026 | .00036 | .00075 | .0016 | .0031 |
| 70 | .00038 | .00053 | .0011 | .0024 | .0046 |
| 80 | .00055 | .00077 | .0016 | .0034 | .0067 |
| 90 | .0008 | .0011 | .0023 | .0050 | .0098 |
| 100 | .0012 | .0016 | .0034 | .0073 | .014 |
| 110 | .0017 | .0024 | .0050 | .011 | .021 |
| 120 | .0025 | .0035 | .0073 | .016 | .030 |
| 130 | .0036 | .0051 | .011 | .023 | .044 |
| 140 | .0053 | .0074 | .015 | .033 | .065 |
| 150 | .0078 | .011 | .023 | .049 | .095 |

$$\lambda_b = 8.6 \times 10^{-10} \left[ \left( \frac{S}{.4} \right)^3 + 1 \right] \exp\left( 16 \left( \frac{T+273}{423} \right) \right)$$

T = Ambient Temperature (°C)
S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.



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## 10.7 CAPACITORS, FIXED, MICA

Capacitance Factor - $\pi_{CV}$

| Capacitance, C (pF) | $\pi_{CV}$ |
|---|---|
| 2 | .50 |
| 38 | .75 |
| 300 | 1.0 |
| 2000 | 1.3 |
| 8600 | 1.6 |
| 29000 | 1.9 |
| 84000 | 2.2 |
| $\pi_{CV} = 0.45C^{.14}$ | |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 10 |
| $N_S$ | 6.0 |
| $N_U$ | 16 |
| $A_{IC}$ | 5.0 |
| $A_{IF}$ | 7.0 |
| $A_{UC}$ | 22 |
| $A_{UF}$ | 28 |
| $A_{RW}$ | 23 |
| $S_F$ | .50 |
| $M_F$ | 13 |
| $M_L$ | 34 |
| $C_L$ | 610 |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| T | .010 |
| S | .030 |
| R | .10 |
| P | .30 |
| M | 1.0 |
| L | 1.5 |
| MIL-C-5, Non-Est. Rel. Dipped | 3.0 |
| MIL-C-5, Non-Est. Rel. Molded | 6.0 |
| Lower | 15 |



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## 10.8 CAPACITORS, FIXED, MICA, BUTTON

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-C-10950 | CB | MICA, Button Style |

$$\lambda_p = \lambda_b \pi_{CV} \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$
(T = 85°C Max Rated)
(Style CB50)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .0067 | .0094 | .019 | .042 | .082 |
| 10 | .0071 | .0099 | .021 | .044 | .086 |
| 20 | .0076 | .011 | .022 | .047 | .092 |
| 30 | .0082 | .011 | .024 | .051 | .10 |
| 40 | .009 | .013 | .026 | .056 | .11 |
| 50 | .010 | .014 | .029 | .063 | .12 |
| 60 | .012 | .016 | .033 | .072 | .14 |
| 70 | .013 | .019 | .039 | .084 | .16 |
| 80 | .016 | .023 | .047 | .10 | .20 |

$$\lambda_b = .0053\left[\left(\frac{S}{.4}\right)^3 + 1\right]\exp\left(1.2\left(\frac{T+273}{358}\right)^{6.3}\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

Base Failure Rate - $\lambda_b$
(T = 150°C Max Rated)
(All Types Except CB50)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .0058 | .0081 | .017 | .036 | .071 |
| 10 | .0059 | .0083 | .017 | .037 | .072 |
| 20 | .0061 | .0085 | .018 | .038 | .074 |
| 30 | .0062 | .0087 | .018 | .039 | .076 |
| 40 | .0064 | .009 | .019 | .040 | .079 |
| 50 | .0067 | .0094 | .019 | .042 | .082 |
| 60 | .0070 | .0098 | .020 | .044 | .086 |
| 70 | .0074 | .010 | .022 | .046 | .090 |
| 80 | .0079 | .011 | .023 | .049 | .096 |
| 90 | .0085 | .012 | .025 | .053 | .10 |
| 100 | .0093 | .013 | .027 | .058 | .11 |
| 110 | .010 | .014 | .03 | .064 | .12 |
| 120 | .011 | .016 | .033 | .072 | .14 |
| 130 | .013 | .018 | .038 | .082 | .16 |
| 140 | .015 | .021 | .044 | .095 | .18 |
| 150 | .018 | .025 | .052 | .11 | .22 |

$$\lambda_b = .0053\left[\left(\frac{S}{.4}\right)^3 + 1\right]\exp\left(1.2\left(\frac{T+273}{423}\right)^{6.3}\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.



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## 10.8 CAPACITORS, FIXED, MICA, BUTTON

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| MIL-C-10950 | 5.0 |
| Lower | 15 |

Capacitance Factor - $\pi_{CV}$

| Capacitance, C (pF) | $\pi_{CV}$ |
|---|---|
| 8 | .50 |
| 50 | .76 |
| 160 | 1.0 |
| 500 | 1.3 |
| 1200 | 1.6 |
| 2600 | 1.9 |
| 5000 | 2.2 |

$\pi_{CV} = .31C^{0.23}$

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 10 |
| $N_S$ | 5.0 |
| $N_U$ | 16 |
| $A_{IC}$ | 5.0 |
| $A_{IF}$ | 7.0 |
| $A_{UC}$ | 22 |
| $A_{UF}$ | 28 |
| $A_{RW}$ | 23 |
| $S_F$ | .50 |
| $M_F$ | 13 |
| $M_L$ | 34 |
| $C_L$ | 610 |



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## 10.9 CAPACITORS, FIXED, GLASS

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-C-11272 | CY | Glass |
| MIL-C-23269 | CYR | Glass, Established Reliability |

$$\lambda_p = \lambda_b \pi_{CV} \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$
(T=125°C Max Rated)
(All MIL-C-23296 and MIL-C-11272 Temp. Range C)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .00005 | .00005 | .00010 | .00023 | .00055 |
| 10 | .00007 | .00008 | .00014 | .00035 | .00083 |
| 20 | .00011 | .00012 | .00022 | .00052 | .0012 |
| 30 | .00016 | .00018 | .00032 | .00078 | .0018 |
| 40 | .00024 | .00027 | .00048 | .0012 | .0028 |
| 50 | .00036 | .00041 | .00072 | .0017 | .0041 |
| 60 | .00054 | .00061 | .0011 | .0026 | .0062 |
| 70 | .0008 | .00091 | .0016 | .0039 | .0092 |
| 80 | .0012 | .0014 | .0024 | .0058 | .014 |
| 90 | .0018 | .0020 | .0036 | .0087 | .021 |
| 100 | .0027 | .0030 | .0054 | .013 | .031 |
| 110 | .0040 | .0045 | .0080 | .019 | .046 |
| 120 | .0060 | .0068 | .012 | .029 | .069 |

$$\lambda_b = 8.25 \times 10^{-10} \left[ \left( \frac{S}{.5} \right)^4 + 1 \right] \exp\left( 16 \left( \frac{T+273}{398} \right) \right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

Base Failure Rate - $\lambda_b$
(T = 200°C Max Rated)
(MIL-C-11272 Temp. Range D)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .00001 | .00001 | .00002 | .00004 | .00010 |
| 10 | .00001 | .00001 | .00002 | .00006 | .00014 |
| 20 | .00002 | .00002 | .00003 | .00008 | .00019 |
| 30 | .00002 | .00003 | .00005 | .00011 | .00027 |
| 40 | .00003 | .00004 | .00007 | .00016 | .00038 |
| 50 | .00005 | .00005 | .00009 | .00022 | .00053 |
| 60 | .00006 | .00007 | .00013 | .00031 | .00074 |
| 70 | .00009 | .00010 | .00018 | .00044 | .0010 |
| 80 | .00013 | .00014 | .00025 | .00061 | .0015 |
| 90 | .00018 | .00020 | .00035 | .00086 | .0020 |
| 100 | .00025 | .00028 | .00050 | .0012 | .0029 |
| 110 | .00035 | .00039 | .00070 | .0017 | .0040 |
| 120 | .00049 | .00055 | .00098 | .0024 | .0056 |
| 130 | .00069 | .00078 | .0014 | .0033 | .0079 |
| 140 | .00096 | .0011 | .0019 | .0047 | .011 |
| 150 | .0014 | .0015 | .0027 | .0065 | .016 |
| 160 | .0019 | .0021 | .0038 | .0092 | .022 |
| 170 | .0027 | .0030 | .0053 | .013 | .031 |
| 180 | .0037 | .0042 | .0075 | .018 | .043 |
| 190 | .0052 | .0059 | .010 | .025 | .060 |
| 200 | .0073 | .0083 | .015 | .035 | .084 |

$$\lambda_b = 8.25 \times 10^{-10} \left[ \left( \frac{S}{.5} \right)^4 + 1 \right] \exp\left( 16 \left( \frac{T+273}{473} \right) \right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.



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## 10.9 CAPACITORS, FIXED, GLASS

Capacitance Factor - $\pi_{CV}$

| Capacitance, C (pF) | $\pi_{CV}$ |
|---|---|
| 1 | .62 |
| 4 | .75 |
| 30 | 1.0 |
| 200 | 1.3 |
| 900 | 1.6 |
| 3000 | 1.9 |
| 8500 | 2.2 |

$\pi_{CV} = 0.62C^{0.14}$

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| S | .030 |
| R | .10 |
| P | .30 |
| M | 1.0 |
| L | 3.0 |
| MIL-C-11272, Non-Est. Rel. | 3.0 |
| Lower | 10 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 10 |
| $N_S$ | 6.0 |
| $N_U$ | 16 |
| $A_{IC}$ | 5.0 |
| $A_{IF}$ | 7.0 |
| $A_{UC}$ | 22 |
| $A_{UF}$ | 28 |
| $A_{RW}$ | 23 |
| $S_F$ | .50 |
| $M_F$ | 13 |
| $M_L$ | 34 |
| $C_L$ | 610 |



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## 10.10 CAPACITORS, FIXED, CERAMIC, GENERAL PURPOSE

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-C-11015 | CK | Ceramic, General Purpose |
| MIL-C-39014 | CKR | Ceramic, General Purpose, Est. Rel. |

$$\lambda_p = \lambda_b \pi_{CV} \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$
(T = 85°C Max Rated)
(MIL-C-39014 Styles CKR13, 48, 64, 72;
MIL-C-11015 Type A Rated Temperature)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .00067 | .0013 | .0036 | .0088 | .018 |
| 10 | .00069 | .0013 | .0037 | .0091 | .019 |
| 20 | .00071 | .0014 | .0038 | .0093 | .019 |
| 30 | .00073 | .0014 | .0039 | .0096 | .020 |
| 40 | .00075 | .0014 | .004 | .0099 | .020 |
| 50 | .00077 | .0015 | .0042 | .010 | .021 |
| 60 | .00079 | .0015 | .0043 | .010 | .021 |
| 70 | .00081 | .0016 | .0044 | .011 | .022 |
| 80 | .00083 | .0016 | .0045 | .011 | .023 |

$$\lambda_b = .0003\left[\left(\frac{S}{.3}\right)^3 + 1\right]\exp\left(\frac{T+273}{358}\right)$$

T = Ambient Temperature (°C)
S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

Base Failure Rate - $\lambda_b$
(T = 125°C Max Rated)
(MIL-C-39014 Styles CKR05-12, 14-19, 73, 74;
MIL-C-11015 Type B Rated Temperature)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .00062 | .0012 | .0033 | .0082 | .017 |
| 10 | .00063 | .0012 | .0034 | .0084 | .017 |
| 20 | .00065 | .0013 | .0035 | .0086 | .018 |
| 30 | .00067 | .0013 | .0036 | .0088 | .018 |
| 40 | .00068 | .0013 | .0037 | .0090 | .018 |
| 50 | .00070 | .0014 | .0038 | .0093 | .019 |
| 60 | .00072 | .0014 | .0039 | .0095 | .019 |
| 70 | .00074 | .0014 | .0040 | .0097 | .020 |
| 80 | .00076 | .0015 | .0041 | .010 | .020 |
| 90 | .00077 | .0015 | .0042 | .010 | .021 |
| 100 | .00079 | .0015 | .0043 | .010 | .021 |
| 110 | .00081 | .0016 | .0044 | .011 | .022 |
| 120 | .00084 | .0016 | .0045 | .011 | .023 |

$$\lambda_b = .0003\left[\left(\frac{S}{.3}\right)^3 + 1\right]\exp\left(\frac{T+273}{398}\right)$$

T = Ambient Temperature (°C)
S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

Base Failure Rate - $\lambda_b$
(T = 150°C Max Rated)
(MIL-C-11015 Type C Rated Temperature)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .00059 | .0011 | .0032 | .0078 | .016 |
| 10 | .00061 | .0012 | .0033 | .008 | .016 |
| 20 | .00062 | .0012 | .0034 | .0082 | .017 |
| 30 | .00064 | .0012 | .0035 | .0084 | .017 |
| 40 | .00065 | .0013 | .0035 | .0086 | .018 |
| 50 | .00067 | .0013 | .0036 | .0088 | .018 |
| 60 | .00068 | .0013 | .0037 | .009 | .018 |
| 70 | .00070 | .0013 | .0038 | .0092 | .019 |
| 80 | .00072 | .0014 | .0039 | .0095 | .019 |
| 90 | .00073 | .0014 | .0040 | .0097 | .020 |
| 100 | .00075 | .0014 | .0041 | .0099 | .020 |
| 110 | .00077 | .0015 | .0042 | .010 | .021 |
| 120 | .00079 | .0015 | .0043 | .010 | .021 |
| 130 | .00081 | .0016 | .0044 | .011 | .022 |
| 140 | .00083 | .0016 | .0045 | .011 | .022 |
| 150 | .00085 | .0016 | .0046 | .011 | .023 |

$$\lambda_b = .0003\left[\left(\frac{S}{.3}\right)^3 + 1\right]\exp\left(\frac{T+273}{423}\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

**NOTE:** The rated temperature designation (type A, B, or C) is shown in the part number, e.g., CKG1**A**W22M).



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## 10.10 CAPACITORS, FIXED, CERAMIC, GENERAL PURPOSE

Capacitance Factor - $\pi_{CV}$

| Capacitance, C (pF) | $\pi_{CV}$ |
|---|---|
| 6.0 | .50 |
| 240 | .75 |
| 3300 | 1.0 |
| 36,000 | 1.3 |
| 240,000 | 1.6 |
| 1,100,000 | 1.9 |
| 4,300,000 | 2.2 |

$\pi_{CV} = .41C^{0.11}$

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 9.0 |
| $N_S$ | 5.0 |
| $N_U$ | 15 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 4.0 |
| $A_{UC}$ | 8.0 |
| $A_{UF}$ | 12 |
| $A_{RW}$ | 20 |
| $S_F$ | .40 |
| $M_F$ | 13 |
| $M_L$ | 34 |
| $C_L$ | 610 |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| S | .030 |
| R | .10 |
| P | .30 |
| M | 1.0 |
| L | 3.0 |
| MIL-C-11015, Non-Est. Rel. | 3.0 |
| Lower | 10 |



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## 10.11 CAPACITORS, FIXED, CERAMIC, TEMPERATURE COMPENSATING AND CHIP

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-C-20 | CCR and CC | Ceramic, Temperature Compensating, Est. and Non Est. Rel. |
| MIL-C-55681 | CDR | Ceramic, Chip, Est. Rel. |

$$\lambda_p = \lambda_b \pi_{CV} \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

(T = 85°C Max Rated)

(MIL-C-20 Styles CC 20, 25, 30, 32, 35, 45, 85, 95-97)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .00015 | .00028 | .00080 | .0019 | .0040 |
| 10 | .00022 | .00042 | .0012 | .0029 | .0059 |
| 20 | .00033 | .00063 | .0018 | .0043 | .0088 |
| 30 | .00049 | .00094 | .0026 | .0064 | .013 |
| 40 | .00073 | .0014 | .0039 | .0096 | .020 |
| 50 | .0011 | .0021 | .0059 | .014 | .029 |
| 60 | .0016 | .0031 | .0088 | .021 | .044 |
| 70 | .0024 | .0046 | .013 | .032 | .065 |
| 80 | .0036 | .0069 | .019 | .047 | .097 |

$$\lambda_b = 2.6 \times 10^{-9}\left[\left(\frac{S}{.3}\right)^3 + 1\right]\exp\left(14.3\left(\frac{T+273}{358}\right)\right)$$

T = Ambient Temperature (°C)
S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

Base Failure Rate - $\lambda_b$

(T = 125°C Max Rated)

(MIL-C-20 Styles CC 5-9,13-19, 21, 22, 26, 27, 31, 33, 36, 37, 47, 50-57, 75-79, 81-83, CCR 05-09,13-19, 54-57, 75-79, 81-83, 90; MIL-C-55681 All CDR Styles)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .00005 | .00009 | .00027 | .00065 | .0013 |
| 10 | .00007 | .00014 | .00038 | .00093 | .0019 |
| 20 | .00010 | .00019 | .00055 | .0013 | .0027 |
| 30 | .00014 | .00028 | .00078 | .0019 | .0039 |
| 40 | .00021 | .00040 | .0011 | .0027 | .0056 |
| 50 | .00030 | .00057 | .0016 | .0039 | .008 |
| 60 | .00042 | .00082 | .0023 | .0056 | .011 |
| 70 | .00061 | .0012 | .0033 | .008 | .016 |
| 80 | .00087 | .0017 | .0047 | .011 | .023 |
| 90 | .0012 | .0024 | .0068 | .016 | .034 |
| 100 | .0018 | .0034 | .0097 | .024 | .048 |
| 110 | .0026 | .0049 | .014 | .034 | .069 |
| 120 | .0037 | .0071 | .020 | .048 | .099 |

$$\lambda_b = 2.6 \times 10^{-9}\left[\left(\frac{S}{.3}\right)^3 + 1\right]\exp\left(14.3\left(\frac{T+273}{398}\right)\right)$$

T = Ambient Temperature (°C)
S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

Capacitance Factor - $\pi_{CV}$

| Capacitance, C (pF) | $\pi_{CV}$ |
|---|---|
| 1 | .59 |
| 7 | .75 |
| 81 | 1.0 |
| 720 | 1.3 |
| 4,100 | 1.6 |
| 17,000 | 1.9 |
| 58,000 | 2.2 |

$\pi_{CV} = .59C^{0.12}$

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| S | .030 |
| R | .10 |
| P | .30 |
| M | 1.0 |
| Non-Est. Rel. | 3.0 |
| Lower | 10 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 10 |
| $N_S$ | 5.0 |
| $N_U$ | 17 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 8.0 |
| $A_{UC}$ | 16 |
| $A_{UF}$ | 35 |
| $A_{RW}$ | 24 |
| $S_F$ | .50 |
| $M_F$ | 13 |
| $M_L$ | 34 |
| $C_L$ | 610 |



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## 10.12 CAPACITORS, FIXED, ELECTROLYTIC, TANTALUM, SOLID

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-C-39003 | CSR | Tantalum Electrolytic (Solid), Est. Rel. |

$$\lambda_p = \lambda_b \pi_{CV} \pi_{SR} \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .0042 | .0058 | .012 | .026 | .051 |
| 10 | .0043 | .0060 | .012 | .027 | .052 |
| 20 | .0045 | .0063 | .013 | .028 | .055 |
| 30 | .0048 | .0067 | .014 | .030 | .058 |
| 40 | .0051 | .0072 | .015 | .032 | .063 |
| 50 | .0057 | .0079 | .016 | .035 | .069 |
| 60 | .0064 | .009 | .019 | .040 | .078 |
| 70 | .0075 | .011 | .022 | .047 | .092 |
| 80 | .0092 | .013 | .027 | .058 | .11 |
| 90 | .012 | .017 | .034 | .074 | .14 |
| 100 | .016 | .023 | .047 | .10 | |
| 110 | .024 | .034 | .07 | .15 | |
| 120 | .039 | .054 | .11 | .24 | |

$$\lambda_b = .00375\left[\left(\frac{S}{.4}\right)^3 + 1\right]\exp\left(2.6\left(\frac{T+273}{398}\right)^9\right)$$

T = Ambient Temperature (°C)
S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

Series Resistance Factor - $\pi_{SR}$

| Circuit Resistance, CR (ohms/volt) | $\pi_{SR}$ |
|---|---|
| >0.8 | .066 |
| >0.6 to 0.8 | .10 |
| >0.4 to 0.6 | .13 |
| >0.2 to 0.4 | .20 |
| >0.1 to 0.2 | .27 |
| 0 to 0.1 | .33 |

$$CR = \frac{\text{Eff. Res. Between Cap. and Pwr. Supply}}{\text{Voltage Applied to Capacitor}}$$

Capacitance Factor - $\pi_{CV}$

| Capacitance, C (µF) | $\pi_{CV}$ |
|---|---|
| .003 | 0.5 |
| .091 | .75 |
| 1.0 | 1.0 |
| 8.9 | 1.3 |
| 50 | 1.6 |
| 210 | 1.9 |
| 710 | 2.2 |

$$\pi_{CV} = 1.0C^{0.12}$$

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| D | 0.0010 |
| C | 0.010 |
| S | 0.030 |
| B | 0.030 |
| R | 0.10 |
| P | 0.30 |
| M | 1.0 |
| L | 1.5 |
| Lower | 10 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 8.0 |
| $N_S$ | 5.0 |
| $N_U$ | 14 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 5.0 |
| $A_{UC}$ | 12 |
| $A_{UF}$ | 20 |
| $A_{RW}$ | 24 |
| $S_F$ | .40 |
| $M_F$ | 11 |
| $M_L$ | 29 |
| $C_L$ | 530 |



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## 10.13 CAPACITORS, FIXED, ELECTROLYTIC, TANTALUM, NON-SOLID

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-C-3965 | CL | Tantalum, Electrolytic (Non-Solid) |
| MIL-C-39006 | CLR | Tantalum, Electrolytic (Non-Solid), Est. Rel. |

$$\lambda_p = \lambda_b \pi_{CV} \pi_C \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$
(T = 85°C Max Rated)
(MIL-C-3965 Styles CL24-27, 34-37)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .0021 | .0029 | .0061 | .013 | .026 |
| 10 | .0023 | .0032 | .0067 | .014 | .028 |
| 20 | .0026 | .0036 | .0075 | .016 | .031 |
| 30 | .0030 | .0042 | .0087 | .019 | .036 |
| 40 | .0036 | .0051 | .011 | .023 | .044 |
| 50 | .0047 | .0066 | .014 | .029 | .057 |
| 60 | .0065 | .0091 | .019 | .041 | .079 |
| 70 | .0098 | .014 | .029 | .062 | .12 |
| 80 | .017 | .023 | .048 | .10 | .20 |

$$\lambda_b = .00165\left[\left(\frac{S}{.4}\right)^3 + 1\right]\exp\left(2.6\left(\frac{T+273}{358}\right)^{9.0}\right)$$

T = Ambient Temperature (°C)
S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

Base Failure Rate - $\lambda_b$
(T = 125°C Max Rated)
(MIL-C-3965 Styles CL20-23, 30-33, 40-43, 46-56, 64-67, 70-73; and all MIL-C-39006 Styles)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .0018 | .0026 | .0053 | .011 | .022 |
| 10 | .0019 | .0026 | .0055 | .012 | .023 |
| 20 | .0020 | .0028 | .0057 | .012 | .024 |
| 30 | .0021 | .0029 | .0061 | .013 | .026 |
| 40 | .0023 | .0032 | .0066 | .014 | .028 |
| 50 | .0025 | .0035 | .0072 | .016 | .030 |
| 60 | .0028 | .0040 | .0082 | .018 | .034 |
| 70 | .0033 | .0046 | .0096 | .021 | .040 |
| 80 | .0041 | .0057 | .012 | .025 | .049 |
| 90 | .0052 | .0073 | .015 | .033 | .064 |
| 100 | .0071 | .010 | .021 | .045 | |
| 110 | .011 | .015 | .031 | .066 | |
| 120 | .017 | .024 | .050 | .11 | |

$$\lambda_b = .00165\left[\left(\frac{S}{.4}\right)^3 + 1\right]\exp\left(2.6\left(\frac{T+273}{398}\right)^{9.0}\right)$$

T = Ambient Temperature (°C)
S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

Base Failure Rate - $\lambda_b$
(T = 175°C Max Rated)
(MIL-C-3965 Styles CL10, 13, 14, 16-18)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .0017 | .0024 | .0050 | .011 | .021 |
| 10 | .0017 | .0024 | .0051 | .011 | .021 |
| 20 | .0018 | .0025 | .0052 | .011 | .022 |
| 30 | .0018 | .0025 | .0053 | .011 | .022 |
| 40 | .0019 | .0026 | .0054 | .012 | .023 |
| 50 | .0019 | .0027 | .0056 | .012 | .023 |
| 60 | .002 | .0028 | .0058 | .013 | .024 |
| 70 | .0021 | .0030 | .0062 | .013 | .026 |
| 80 | .0023 | .0032 | .0066 | .014 | .028 |
| 90 | .0025 | .0035 | .0072 | .016 | .030 |
| 100 | .0028 | .0039 | .0080 | .017 | .034 |
| 110 | .0032 | .0044 | .0092 | .020 | .039 |
| 120 | .0037 | .0052 | .011 | .023 | |
| 130 | .0046 | .0064 | .013 | .029 | |
| 140 | .0059 | .0082 | .017 | .037 | |
| 150 | .0079 | .011 | .023 | .049 | |
| 160 | .011 | .016 | .033 | .071 | |
| 170 | .018 | .025 | .051 | | |

$$\lambda_b = .00165\left[\left(\frac{S}{.4}\right)^3 + 1\right]\exp\left(2.6\left(\frac{T+273}{448}\right)^{9.0}\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.



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## 10.13 CAPACITORS, FIXED, ELECTROLYTIC, TANTALUM, NON-SOLID

### Capacitance Factor - $\pi_{CV}$

| Capacitance, C (μF) | $\pi_{CV}$ |
|---|---|
| .091 | .70 |
| 20 | 1.0 |
| 1100 | 1.3 |

$\pi_{CV} = .82C^{0.066}$

### Construction Factor - $\pi_C$

| Construction Type | $\pi_C$ |
|---|---|
| Slug, All Tantalum | .30 |
| Foil, Hermetic * | 1.0 |
| Slug, Hermetic * | 2.0 |
| Foil, Non-Hermetic * | 2.5 |
| Slug, Non-Hermetic * | 3.0 |

*Type of Seal Identified as Follows:

1) MIL-C-3965 (CL) - Note Last Letter in Part Number:
G - Hermetic E - Non-Hermetic

Example: CL10BC700TPG is Hermetic

2) MIL-C-39006 (CLR) - Consult Individual Part Specification Sheet (slash sheet)

NOTE:
Foil Types - CL 20-25, 30-33, 40, 41, 51-54, 70-73
CLR 25, 27, 35, 37, 53, 71, 73

Slug Types - CL 10, 13, 14, 16, 17, 18, 55, 56, 64-66, 67
CLR 10, 14, 17, 65, 69, 89

All Tantalum - CL 26, 27, 34-37, 42, 43, 46-49
CLR 79

### Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| S | .030 |
| R | .10 |
| P | .30 |
| M | 1.0 |
| L | 1.5 |
| MIL-C-3965, Non-Est. Rel. | 3.0 |
| Lower | 10 |

### Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 10 |
| $N_S$ | 6.0 |
| $N_U$ | 16 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 8.0 |
| $A_{UC}$ | 14 |
| $A_{UF}$ | 30 |
| $A_{RW}$ | 23 |
| $S_F$ | .50 |
| $M_F$ | 13 |
| $M_L$ | 34 |
| $C_L$ | 610 |



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## 10.14 CAPACITORS, FIXED, ELECTROLYTIC, ALUMINUM

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-C-39018 | CUR and CU | Electrolytic, Aluminum Oxide, Est. Rel. and Non-Est. Rel. |

$$\lambda_p = \lambda_b \pi_{CV} \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$
(T = 85°C Max Rated)
(MIL-C-39018 Style 71)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .0095 | .011 | .019 | .035 | .064 |
| 10 | .012 | .015 | .024 | .046 | .084 |
| 20 | .017 | .020 | .033 | .062 | .11 |
| 30 | .023 | .028 | .046 | .087 | .16 |
| 40 | .034 | .042 | .068 | .13 | .23 |
| 50 | .054 | .065 | .11 | .20 | .36 |
| 60 | .089 | .11 | .18 | .33 | .60 |
| 70 | .16 | .19 | .31 | .58 | 1.1 |
| 80 | .29 | .35 | .58 | 1.1 | 2.0 |

$$\lambda_b = .00254\left[\left(\frac{S}{.5}\right)^3 + 1\right]\exp\left(5.09\left(\frac{T+273}{358}\right)^5\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

Base Failure Rate - $\lambda_b$
(T = 105°C Max Rated)
(MIL-C-39018 Styles 16 and 17)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .0070 | .0084 | .014 | .026 | .047 |
| 10 | .0085 | .010 | .017 | .031 | .057 |
| 20 | .011 | .013 | .021 | .040 | .072 |
| 30 | .014 | .017 | .027 | .051 | .094 |
| 40 | .019 | .022 | .037 | .069 | .13 |
| 50 | .026 | .031 | .052 | .097 | .18 |
| 60 | .038 | .046 | .076 | .14 | .26 |
| 70 | .059 | .071 | .12 | .22 | .40 |
| 80 | .095 | .11 | .19 | .35 | .64 |
| 90 | .16 | .20 | .32 | .61 | 1.1 |
| 100 | .30 | .36 | .59 | 1.1 | 2.0 |

$$\lambda_b = .00254\left[\left(\frac{S}{.5}\right)^3 + 1\right]\exp\left(5.09\left(\frac{T+273}{378}\right)^5\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

Base Failure Rate - $\lambda_b$
(T = 125°C Max Rated)
(All MIL-C-39018 Styles Except 71, 16 and 17)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .0055 | .0067 | .011 | .021 | .038 |
| 10 | .0065 | .0078 | .013 | .024 | .044 |
| 20 | .0077 | .0093 | .015 | .029 | .052 |
| 30 | .0094 | .011 | .019 | .035 | .064 |
| 40 | .012 | .014 | .023 | .044 | .080 |
| 50 | .015 | .019 | .030 | .057 | .10 |
| 60 | .021 | .025 | .041 | .077 | .14 |
| 70 | .029 | .035 | .057 | .11 | .20 |
| 80 | .042 | .050 | .083 | .16 | .28 |
| 90 | .064 | .077 | .13 | .24 | .43 |
| 100 | .10 | .12 | .20 | .38 | |
| 110 | .17 | .21 | .34 | .63 | |
| 120 | .30 | .37 | .60 | 1.1 | |

$$\lambda_b = .00254\left[\left(\frac{S}{.5}\right)^3 + 1\right]\exp\left(5.09\left(\frac{T+273}{398}\right)^5\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.



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## 10.14 CAPACITORS, FIXED, ELECTROLYTIC, ALUMINUM

Capacitance Factor - $\pi_{CV}$

| Capacitance, C (µF) | $\pi_{CV}$ |
|---|---|
| 2.5 | .40 |
| 55 | .70 |
| 400 | 1.0 |
| 1700 | 1.3 |
| 5500 | 1.6 |
| 14,000 | 1.9 |
| 32,000 | 2.2 |
| 65,000 | 2.5 |
| 120,000 | 2.8 |
| $\pi_{CV} = .34C^{0.18}$ | |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 12 |
| $N_S$ | 6.0 |
| $N_U$ | 17 |
| $A_{IC}$ | 10 |
| $A_{IF}$ | 12 |
| $A_{UC}$ | 28 |
| $A_{UF}$ | 35 |
| $A_{RW}$ | 27 |
| $S_F$ | .50 |
| $M_F$ | 14 |
| $M_L$ | 38 |
| $C_L$ | 690 |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| S | .030 |
| R | .10 |
| P | .30 |
| M | 1.0 |
| Non-Est. Rel. | 3.0 |
| Lower | 10 |



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## 10.15 CAPACITORS, FIXED, ELECTROLYTIC (DRY), ALUMINUM

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-C-62 | CE | Aluminum, Dry Electrolyte, Polarized |

$$\lambda_p = \lambda_b \pi_{CV} \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

(T = 85°C Max Rated)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .0064 | .0074 | .011 | .020 | .034 |
| 10 | .0078 | .009 | .014 | .024 | .042 |
| 20 | .0099 | .011 | .017 | .030 | .053 |
| 30 | .013 | .015 | .023 | .040 | .070 |
| 40 | .018 | .021 | .031 | .055 | .098 |
| 50 | .026 | .030 | .046 | .08 | .14 |
| 60 | .041 | .047 | .071 | .12 | .22 |
| 70 | .068 | .078 | .12 | .21 | .36 |
| 80 | .120 | .14 | .21 | .37 | .65 |

$$\lambda_b = .0028\left[\left(\frac{S}{.55}\right)^3 + 1\right]\exp\left(4.09\left(\frac{T+273}{358}\right)^{5.9}\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

Capacitance Factor - $\pi_{CV}$

| Capacitance, C (μF) | $\pi_{CV}$ |
|---|---|
| 3.2 | .40 |
| 62 | .70 |
| 400 | 1.0 |
| 1600 | 1.3 |
| 4800 | 1.6 |
| 12,000 | 1.9 |
| 26,000 | 2.2 |
| 50,000 | 2.5 |
| 91,000 | 2.8 |

$$\pi_{CV} = .32C^{0.19}$$

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| MIL-SPEC | 3.0 |
| Lower | 10 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 12 |
| $N_S$ | 6.0 |
| $N_U$ | 17 |
| $A_{IC}$ | 10 |
| $A_{IF}$ | 12 |
| $A_{UC}$ | 28 |
| $A_{UF}$ | 35 |
| $A_{RW}$ | 27 |
| $S_F$ | .50 |
| $M_F$ | 14 |
| $M_L$ | 38 |
| $C_L$ | 690 |



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## 10.16 CAPACITORS, VARIABLE, CERAMIC

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-C-81 | CV | Variable, Ceramic |

$$\lambda_p = \lambda_b \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

(T = 85°C Max Rated)

(MIL-C-81 Styles CV 11, 14, 21, 31, 32, 34, 40, 41)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .0030 | .016 | .066 | .18 | .37 |
| 10 | .0031 | .017 | .069 | .18 | .39 |
| 20 | .0033 | .018 | .073 | .20 | .41 |
| 30 | .0036 | .020 | .080 | .21 | .45 |
| 40 | .0041 | .022 | .089 | .24 | .50 |
| 50 | .0047 | .026 | .10 | .28 | .59 |
| 60 | .0058 | .031 | .13 | .34 | .72 |
| 70 | .0076 | .041 | .17 | .45 | .94 |
| 80 | .011 | .058 | .24 | .63 | 1.3 |

$$\lambda_b = .00224\left[\left(\frac{S}{.17}\right)^3 + 1\right]\exp\left(1.59\left(\frac{T+273}{358}\right)^{10.1}\right)$$

T = Ambient Temperature (°C)
S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

Base Failure Rate - $\lambda_b$

(T = 125°C Max Rated)

(MIL-C-81 Styles CV 35, 36)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .0028 | .015 | .061 | .16 | .35 |
| 10 | .0028 | .015 | .062 | .17 | .35 |
| 20 | .0029 | .016 | .064 | .17 | .36 |
| 30 | .0030 | .016 | .066 | .18 | .37 |
| 40 | .0031 | .017 | .068 | .18 | .39 |
| 50 | .0033 | .018 | .072 | .19 | .41 |
| 60 | .0035 | .019 | .077 | .21 | .44 |
| 70 | .0038 | .021 | .084 | .23 | .48 |
| 80 | .0043 | .023 | .095 | .25 | .54 |
| 90 | .0050 | .027 | .11 | .30 | .63 |
| 100 | .0062 | .033 | .14 | .36 | .76 |
| 110 | .0079 | .043 | .17 | .47 | .98 |
| 120 | .011 | .059 | .24 | .64 | 1.4 |

$$\lambda_b = .00224\left[\left(\frac{S}{.17}\right)^3 + 1\right]\exp\left(1.59\left(\frac{T+273}{398}\right)^{10.1}\right)$$

T = Ambient Temperature (°C)
S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| MIL-SPEC | 4 |
| Lower | 20 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 3.0 |
| $G_M$ | 13 |
| $N_S$ | 8.0 |
| $N_U$ | 24 |
| $A_{IC}$ | 6.0 |
| $A_{IF}$ | 10 |
| $A_{UC}$ | 37 |
| $A_{UF}$ | 70 |
| $A_{RW}$ | 36 |
| $S_F$ | .40 |
| $M_F$ | 20 |
| $M_L$ | 52 |
| $C_L$ | 950 |



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## 10.17 CAPACITORS, VARIABLE, PISTON TYPE

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-C-14409 | PC | Variable, Piston Type, Tubular Trimmer |

$$\lambda_p = \lambda_b \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

(T = 125°C Max Rated)

(MIL-C-14409 Styles G, H, J, L, T)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .0030 | .0051 | .013 | .031 | .063 |
| 10 | .0041 | .0070 | .018 | .042 | .085 |
| 20 | .0055 | .0094 | .024 | .057 | .11 |
| 30 | .0075 | .013 | .033 | .077 | .16 |
| 40 | .010 | .017 | .044 | .10 | .21 |
| 50 | .014 | .024 | .060 | .14 | .29 |
| 60 | .019 | .032 | .082 | .19 | .39 |
| 70 | .025 | .043 | .11 | .26 | .53 |
| 80 | .034 | .059 | .15 | .35 | .71 |
| 90 | .047 | .079 | .20 | .48 | .96 |
| 100 | .063 | .11 | .27 | .65 | 1.3 |
| 110 | .086 | .15 | .37 | .88 | 1.8 |
| 120 | .12 | .20 | .51 | 1.2 | 2.4 |

$$\lambda_b = 7.3 \times 10^{-7} \left[ \left( \frac{S}{.33} \right)^3 + 1 \right] \exp\left( 12.1 \left( \frac{T+273}{398} \right) \right)$$

T = Ambient Temperature (°C)
S = Ratio of Operating to Rated Voltage
Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

Base Failure Rate - $\lambda_b$

(T = 150°C Max Rated)

(MIL-C-14409 Characteristic Q)

| T°C | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .0019 | .0032 | .0081 | .019 | .038 |
| 10 | .0025 | .0042 | .011 | .025 | .051 |
| 20 | .0033 | .0056 | .014 | .034 | .068 |
| 30 | .0044 | .0074 | .019 | .045 | .09 |
| 40 | .0058 | .0099 | .025 | .060 | .12 |
| 50 | .0077 | .013 | .034 | .079 | .16 |
| 60 | .010 | .018 | .045 | .11 | .21 |
| 70 | .014 | .023 | .060 | .14 | .28 |
| 80 | .018 | .031 | .079 | .19 | .38 |
| 90 | .024 | .041 | .11 | .25 | .50 |
| 100 | .032 | .055 | .14 | .33 | .67 |
| 110 | .043 | .073 | .19 | .44 | .89 |
| 120 | .057 | .097 | .25 | .59 | 1.2 |
| 130 | .076 | .13 | .33 | .78 | 1.6 |
| 140 | .10 | .17 | .44 | 1.0 | 2.1 |
| 150 | .13 | .23 | .59 | 1.4 | 2.8 |

$$\lambda_b = 7.3 \times 10^{-7} \left[ \left( \frac{S}{.33} \right)^3 + 1 \right] \exp\left( 12.1 \left( \frac{T+273}{423} \right) \right)$$

T = Ambient Temperature (°C)
S = Ratio of Operating to Rated Voltage
Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| MIL-SPEC | 3 |
| Lower | 10 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 3.0 |
| $G_M$ | 12 |
| $N_S$ | 7.0 |
| $N_U$ | 18 |
| $A_{IC}$ | 3.0 |
| $A_{IF}$ | 4.0 |
| $A_{UC}$ | 20 |
| $A_{UF}$ | 30 |
| $A_{RW}$ | 32 |
| $S_F$ | .50 |
| $M_F$ | 18 |
| $M_L$ | 46 |
| $C_L$ | 830 |



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## 10.18 CAPACITORS, VARIABLE, AIR TRIMMER

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-C-92 | CT | Variable, Air Trimmer |

$$\lambda_p = \lambda_b \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

(T = 85°C Max Rated)

| $T_A$ (°C) | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .0074 | .013 | .032 | .076 | .15 |
| 10 | .010 | .017 | .044 | .10 | .21 |
| 20 | .014 | .023 | .059 | .14 | .28 |
| 30 | .018 | .031 | .08 | .19 | .38 |
| 40 | .025 | .042 | .11 | .26 | .52 |
| 50 | .034 | .057 | .15 | .35 | .70 |
| 60 | .046 | .078 | .20 | .47 | .94 |
| 70 | .062 | .10 | .27 | .63 | 1.3 |
| 80 | .083 | .14 | .36 | .85 | 1.7 |

$$\lambda_b = 1.92 \times 10^{-6}\left[\left(\frac{S}{.33}\right)^3 + 1\right] \exp\left(10.8\left(\frac{T+273}{358}\right)\right)$$

T = Ambient Temperature (°C)

S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 3.0 |
| $G_M$ | 13 |
| $N_S$ | 8.0 |
| $N_U$ | 24 |
| $A_{IC}$ | 6.0 |
| $A_{IF}$ | 10 |
| $A_{UC}$ | 37 |
| $A_{UF}$ | 70 |
| $A_{RW}$ | 36 |
| $S_F$ | .50 |
| $M_F$ | 20 |
| $M_L$ | 52 |
| $C_L$ | 950 |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| MIL-SPEC | 5 |
| Lower | 20 |



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## 10.19 CAPACITORS, VARIABLE AND FIXED, GAS OR VACUUM

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-C-23183 | CG | Gas or Vacuum Dielectric, Fixed and Variable, Ceramic or Glass Envelope |

$$\lambda_p = \lambda_b \pi_{CF} \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$
(T = 85°C Max Rated)
(Styles CG 20, 21, 30, 31, 32, 40-44, 51, 60-64, 67)

| T°C | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .015 | .081 | .33 | .88 | 1.9 |
| 10 | .016 | .084 | .34 | .92 | 1.9 |
| 20 | .017 | .090 | .37 | .98 | 2.1 |
| 30 | .018 | .098 | .40 | 1.1 | 2.2 |
| 40 | .020 | .11 | .45 | 1.2 | 2.5 |
| 50 | .024 | .13 | .52 | 1.4 | 2.9 |
| 60 | .029 | .16 | .64 | 1.7 | 3.6 |
| 70 | .038 | .20 | .83 | 2.2 | 4.7 |
| 80 | .054 | .29 | 1.2 | 3.2 | 6.6 |

$$\lambda_b = .0112 \left[ \left( \frac{S}{.17} \right)^3 + 1 \right] \exp\left( 1.59 \left( \frac{T+273}{358} \right)^{10.1} \right)$$

T = Ambient Temperature (°C)
S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

Base Failure Rate - $\lambda_b$
(T = 100°C Max Rated)
(Styles CG 65, 66)

| T°C | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .014 | .078 | .30 | .85 | 1.8 |
| 10 | .015 | .080 | .33 | .87 | 1.8 |
| 20 | .015 | .084 | .34 | .91 | 1.9 |
| 30 | .016 | .088 | .36 | .96 | 2.0 |
| 40 | .018 | .095 | .39 | 1.0 | 2.2 |
| 50 | .020 | .11 | .43 | 1.2 | 2.4 |
| 60 | .022 | .12 | .49 | 1.3 | 2.8 |
| 70 | .027 | .14 | .59 | 1.6 | 3.3 |
| 80 | .034 | .18 | .74 | 2.0 | 4.2 |
| 90 | .045 | .24 | .99 | 2.7 | 5.6 |
| 100 | .066 | .36 | 1.5 | 3.9 | 8.2 |

$$\lambda_b = .0112 \left[ \left( \frac{S}{.17} \right)^3 + 1 \right] \exp\left( 1.59 \left( \frac{T+273}{373} \right)^{10.1} \right)$$

T = Ambient Temperature (°C)
S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.

Base Failure Rate - $\lambda_b$
(T = 125°C Max Rated)
(Style CG 50)

| T°C | Stress .1 | .3 | .5 | .7 | .9 |
|---|---|---|---|---|---|
| 0 | .014 | .075 | .31 | .82 | 1.7 |
| 10 | .014 | .077 | .31 | .83 | 1.8 |
| 20 | .014 | .078 | .32 | .85 | 1.8 |
| 30 | .015 | .08 | .33 | .88 | 1.9 |
| 40 | .016 | .084 | .34 | .91 | 1.9 |
| 50 | .016 | .088 | .36 | .96 | 2.0 |
| 60 | .018 | .095 | .39 | 1.0 | 2.2 |
| 70 | .019 | .10 | .42 | 1.1 | 2.4 |
| 80 | .022 | .12 | .48 | 1.3 | 2.7 |
| 90 | .025 | .14 | .55 | 1.5 | 3.1 |
| 100 | .031 | .17 | .68 | 1.8 | 3.8 |
| 110 | .04 | .21 | .87 | 2.3 | 4.9 |
| 120 | .055 | .29 | 1.2 | 3.2 | 6.8 |

$$\lambda_b = .0112 \left[ \left( \frac{S}{.17} \right)^3 + 1 \right] \exp\left( 1.59 \left( \frac{T+273}{398} \right)^{10.1} \right)$$

T = Ambient Temperature (°C)
S = Ratio of Operating to Rated Voltage

Operating voltage is the sum of applied D.C. voltage and peak A.C. voltage.



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## 10.19 CAPACITORS, VARIABLE AND FIXED, GAS OR VACUUM

Configuration Factor - $\pi_{CF}$

| Configuration | $\pi_{CF}$ |
|---|---|
| Fixed | .10 |
| Variable | 1.0 |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| MIL-SPEC | 3.0 |
| Lower | 20 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 3.0 |
| $G_M$ | 14 |
| $N_S$ | 8.0 |
| $N_U$ | 27 |
| $A_{IC}$ | 10 |
| $A_{IF}$ | 18 |
| $A_{UC}$ | 70 |
| $A_{UF}$ | 108 |
| $A_{RW}$ | 40 |
| $S_F$ | .50 |
| $M_F$ | N/A |
| $M_L$ | N/A |
| $C_L$ | N/A |



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## 10.20 CAPACITORS, EXAMPLE

**Example**

Given: A 400 VDC rated capacitor type CQ09A1KE153K3 is being used in a fixed ground environment, 55°C component ambient temperature, and 200 VDC applied with 50 Vrms @ 60 Hz. The capacitor is being procured in full accordance with the applicable specification.

The letters "CQ" in the type designation indicate that the specification is MIL-C-19978 and that it is a Non-Established Reliability quality level. The 1st "K" in the designation indicates characteristic K. The "E" in the designation corresponds to a 400 volt DC rating. The "153" in the designation expresses the capacitance in picofarads. The first two digits are significant and the third is the number of zeros to follow. Therefore, this capacitor has a capacitance of 15,000 picofarads. (NOTE: Pico = $10^{-12}$, $\mu = 10^{-6}$)

The appropriate model for CQ style capacitors is given in Section 10.3. Based on the given information the following model factors are determined from the tables shown in Section 10.3. Voltage stress ratio must account for both the applied DC volts and the peak AC voltage, hence,

$S = .68$

$$S = \frac{\text{DC Volts Applied} + \sqrt{2}\,(\text{AC Volts Applied})}{\text{DC Rated Voltage}} = \frac{200 + \sqrt{2}\,(50)}{400} = .68$$

$\lambda_b = .0082$ Substitute S = .68 and $T_A$ = 55°C into equation shown with Characteristic K $\lambda_b$ Table.

$\pi_{CV} = .94$ Use Table Equation (Note 15,000 pF = .015 μF)

$\pi_Q = 10$

$\pi_E = 2.0$

$$\lambda_p = \lambda_b \pi_{CV} \pi_Q \pi_E = (.0082)(.94)(10)(2) = .15 \text{ Failures}/10^6 \text{ Hours}$$



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## 11.1 INDUCTIVE DEVICES, TRANSFORMERS

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-T-27 | TF | Audio, Power and High Power Pulse |
| MIL-T-21038 | TP | Low Power Pulse |
| MIL-T-55631 | – | IF, RF and Discriminator |

$$\lambda_p = \lambda_b \pi_Q \pi_E \text{ Failures}/10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| $T_{HS}$ (°C) | Maximum Rated Operating Temperature (°C) 85[1] | 105[2] | 130[3] | 155[4] | 170[5] | >170[6] |
|---|---|---|---|---|---|---|
| 30 | .0024 | .0023 | .0022 | .0021 | .0018 | .0016 |
| 35 | .0026 | .0023 | .0023 | .0022 | .0018 | .0016 |
| 40 | .0028 | .0024 | .0024 | .0022 | .0019 | .0016 |
| 45 | .0032 | .0025 | .0025 | .0022 | .0019 | .0016 |
| 50 | .0038 | .0027 | .0026 | .0023 | .0020 | .0017 |
| 55 | .0047 | .0029 | .0027 | .0023 | .0020 | .0017 |
| 60 | .0060 | .0032 | .0029 | .0023 | .0021 | .0017 |
| 65 | .0083 | .0035 | .0030 | .0024 | .0021 | .0017 |
| 70 | .012 | .0040 | .0033 | .0025 | .0022 | .0017 |
| 75 | .020 | .0047 | .0035 | .0026 | .0023 | .0017 |
| 80 | .036 | .0057 | .0039 | .0027 | .0024 | .0017 |
| 85 | .075 | .0071 | .0043 | .0028 | .0024 | .0017 |
| 90 | | .0093 | .0048 | .0029 | .0025 | .0018 |
| 95 | | .013 | .0054 | .0031 | .0026 | .0018 |
| 100 | | .019 | .0062 | .0033 | .0027 | .0018 |
| 105 | | .030 | .0072 | .0035 | .0028 | .0018 |
| 110 | | | .0085 | .0038 | .0030 | .0019 |
| 115 | | | .010 | .0042 | .0031 | .0019 |
| 120 | | | .013 | .0046 | .0032 | .0019 |
| 125 | | | .016 | .0052 | .0034 | .0020 |
| 130 | | | .020 | .0059 | .0036 | .0020 |
| 135 | | | | .0068 | .0038 | .0021 |
| 140 | | | | .0079 | .0040 | .0021 |
| 145 | | | | .0095 | .0042 | .0022 |
| 150 | | | | .011 | .0044 | .0023 |
| 155 | | | | .014 | .0047 | .0024 |
| 160 | | | | | .0050 | .0025 |
| 165 | | | | | .0053 | .0026 |
| 170 | | | | | .0056 | .0027 |
| 175 | | | | | | .0029 |
| 180 | | | | | | .0030 |
| 185 | | | | | | .0032 |

NOTE: The models are valid only if $T_{HS}$ is not above the temperature rating for a given insulation class.

1 $\lambda_b = .0018 \exp\left(\frac{T_{HS}+273}{329}\right)^{15.6}$ MIL-T-27 Insulation Class Q, MIL-T-21038 Insulation Class Q, and MIL-T-55631 Insulation Class Q.*

2 $\lambda_b = .002 \exp\left(\frac{T_{HS}+273}{352}\right)^{14}$ MIL-T-27 Insulation Class R, MIL-T-21038 Insulation Class R, and MIL-T-55631 Insulation Class A.*

3 $\lambda_b = .0018 \exp\left(\frac{T_{HS}+273}{364}\right)^{8.7}$ MIL-T-27 Insulation Class S, MIL-T-21038 Insulation Class S, and MIL-T-55631 Insulation Class B.*

4 $\lambda_b = .002 \exp\left(\frac{T_{HS}+273}{400}\right)^{10}$ MIL-T-27 Insulation Class V, MIL-T-21038 Insulation Class T, and MIL-T-55631 Insulation Class C.*

5 $\lambda_b = .00125 \exp\left(\frac{T_{HS}+273}{398}\right)^{3.8}$ MIL-T-27 Insulation Class T and MIL-T-21038 Insulation Class U.*

6 $\lambda_b = .00159 \exp\left(\frac{T_{HS}+273}{477}\right)^{8.4}$ MIL-T-27 Insulation Class U and MIL-T-21038 Insulation Class V.*

$T_{HS}$ = Hot Spot Temperature (°C), See Section 11.3. ***Refer to Transformer Application Note for Determination of Insulation Class**



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## 11.1 INDUCTIVE DEVICES, TRANSFORMERS

Quality Factor - $\pi_Q$

| Family Type* | MIL-SPEC | Lower |
|---|---|---|
| Pulse Transformers | 1.5 | 5.0 |
| Audio Transformers | 3.0 | 7.5 |
| Power Transformers and Filters | 8.0 | 30 |
| RF Transformers | 12 | 30 |

*** Refer to Transformer Application Note for Determination of Family Type**

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 6.0 |
| $G_M$ | 12 |
| $N_S$ | 5.0 |
| $N_U$ | 16 |
| $A_{IC}$ | 6.0 |
| $A_{IF}$ | 8.0 |
| $A_{UC}$ | 7.0 |
| $A_{UF}$ | 9.0 |
| $A_{RW}$ | 24 |
| $S_F$ | .50 |
| $M_F$ | 13 |
| $M_L$ | 34 |
| $C_L$ | 610 |

**TRANSFORMER APPLICATION NOTE: Insulation Class and Family Type Determination**

**MIL-T-27 Example Designation**

TF | 4 | R | 01 | GA | 576

TF: MIL-T-27 | 4: Grade | R: Insulation Class | 01: Family | GA: Case Symbol | 576

Family Type Codes Are:

Power Transformer and Filter: 01 thru 09, 37 thru 41

Audio Transformer: 10 thru 21, 50 thru 53

Pulse Transformer: 22 thru 36, 54

**MIL-T-21038 Example Designation**

TP | 4 | Q | X1100BC001

TP: MIL-T-21038 | 4: Grade | Q: Insulation Class | X1100BC001

**MIL-T-55631**. The Transformers are Designated with the following Types, Grades and Classes.

Type I - Intermediate Frequency Transformer
Type II - Radio Frequency Transformer
Type III - Discriminator Transformer

Grade 1 - For Use When Immersion and Moisture Resistance Tests are Required
Grade 2 - For Use When Moisture Resistance Test is Required
Grade 3 - For Use in Sealed Assemblies

Class O - 85°C Maximum Operating Temperature
Class A - 105°C Maximum Operating Temperature
Class B - 125°C Maximum Operating Temperature
Class C - > 125°C Maximum Operating Temperature

The class denotes the maximum operating temperature (temperature rise plus maximum ambient temperature).



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## 11.2 INDUCTIVE DEVICES, COILS

| SPECIFICATION | STYLE | DESCRIPTION |
|---|---|---|
| MIL-C-15305 | – | Fixed and Variable, RF |
| MIL-C-39010 | – | Molded, RF, Est. Rel. |

$$\lambda_p = \lambda_b \pi_C \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| $T_{HS}$ (°C) | Maximum Operating Temperature (°C) 85[1] | 105[2] | 125[3] | 150[4] |
|---|---|---|---|---|
| 30 | .00044 | .00043 | .00039 | .00037 |
| 35 | .00048 | .00044 | .0004 | .00037 |
| 40 | .00053 | .00046 | .00042 | .00037 |
| 45 | .0006 | .00048 | .00043 | .00038 |
| 50 | .00071 | .00051 | .00045 | .00038 |
| 55 | .00087 | .00055 | .00048 | .00039 |
| 60 | .0011 | .0006 | .00051 | .0004 |
| 65 | .0015 | .00067 | .00054 | .00041 |
| 70 | .0023 | .00076 | .00058 | .00042 |
| 75 | .0037 | .00089 | .00063 | .00043 |
| 80 | .0067 | .0011 | .00069 | .00044 |
| 85 | .014 | .0013 | .00076 | .00046 |
| 90 | | .0018 | .00085 | .00047 |
| 95 | | .0024 | .00096 | .0005 |
| 100 | | .0036 | .0011 | .00052 |
| 105 | | .0057 | .0013 | .00055 |
| 110 | | | .0015 | .00059 |
| 115 | | | .0018 | .00063 |
| 120 | | | .0022 | .00068 |
| 125 | | | .0028 | .00075 |
| 130 | | | | .00083 |
| 135 | | | | .00093 |
| 140 | | | | .0011 |
| 145 | | | | .0012 |
| 150 | | | | .0014 |

NOTE: The models are valid only if $T_{HS}$ is not above the temperature rating for a given insulation class.

1. $\lambda_b = .000335 \exp\left(\frac{T_{HS} + 273}{329}\right)^{15.6}$ — MIL-C-15305 Insulation Class O.*

2. $\lambda_b = .000379 \exp\left(\frac{T_{HS} + 273}{352}\right)^{14}$ — MIL-C-15305 Insulation Class A and MIL-C-39010 Insulation Class A.*

3. $\lambda_b = .000319 \exp\left(\frac{T_{HS} + 273}{364}\right)^{8.7}$ — MIL-C-15305 Insulation Class B and MIL-C-39010 Insulation Class B.*

4. $\lambda_b = .00035 \exp\left(\frac{T_{HS} + 273}{409}\right)^{10}$ — MIL-C-15305 Insulation Class C and MIL-C-39010 Insulation Class F.*

$T_{HS}$ = Hot Spot Temperature (°C), See Section 11.3.

*Refer to Coil Application Note for Determination of Insulation Class.

Construction Factor - $\pi_C$

| Construction | $\pi_C$ |
|---|---|
| Fixed | 1 |
| Variable | 2 |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| S | .03 |
| R | .10 |
| P | .30 |
| M | 1.0 |
| MIL-C-15305 | 4.0 |
| Lower | 20 |



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## 11.2 INDUCTIVE DEVICES, COILS

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 4.0 |
| $G_M$ | 12 |
| $N_S$ | 5.0 |
| $N_U$ | 16 |
| $A_{IC}$ | 5.0 |
| $A_{IF}$ | 7.0 |
| $A_{UC}$ | 6.0 |
| $A_{UF}$ | 8.0 |
| $A_{RW}$ | 24 |
| $S_F$ | .50 |
| $M_F$ | 13 |
| $M_L$ | 34 |
| $C_L$ | 610 |

**COIL APPLICATION NOTE: Insulation Class Determination From Part Designation**

**MIL-C-15305.** All parts in this specification are R.F. coils. An example type designation is:

| LT | 4 | K | 001 |
|---|---|---|---|
| MIL-C-15305 | Insulation Class Code | Family | |

The codes used for the Insulation Class are:

Class C: 1, 2, 3
Class B: 4, 5, 6
Class O: 7, 8, 9
Class A: 10, 11, 12

**MIL-C-39010.** An example type designation per this specification is:

| M | 39010/01 | A |
|---|---|---|
| Military Designator | Document Sheet Number | Insulation Class |



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## 11.3 INDUCTIVE DEVICES, DETERMINATION OF HOT SPOT TEMPERATURE

Hot Spot temperature can be estimated as follows:

$$T_{HS} = T_A + 1.1\ (\Delta T)$$

where:

$T_{HS}$ = Hot Spot Temperature (°C)
$T_A$ = Inductive Device Ambient Operating Temperature (°C)
$\Delta T$ = Average Temperature Rise Above Ambient (°C)

$\Delta T$ can either be determined by the appropriate "Temperature Rise" Test Method paragraph in the device base specification (e.g., paragraph 4.8.12 for MIL-T-27E), or by approximation using one of the procedures described below.

$\Delta T$ Approximation

| Information Known | $\Delta T$ Approximation |
|---|---|
| 1. MIL-C-39010 Slash Sheet Number<br>MIL-C-39010/1C-3C, 5C, 7C, 9A, 10A, 13, 14 | $\Delta T = 15°C$ |
| MIL-C-39010/4C, 6C, 8A, 11, 12 | $\Delta T = 35°C$ |
| 2. Power Loss<br>Case Radiating Surface Area | $\Delta T = 125\ W_L/A$ |
| 3. Power Loss<br>Transformer Weight | $\Delta T = 11.5\ W_L/(Wt.)^{.6766}$ |
| 4. Input Power<br>Transformer Weight<br>(Assumes 80% Efficiency) | $\Delta T = 2.1\ W_I/(Wt.)^{.6766}$ |

$W_L$ = Power Loss (W)

A = Radiating Surface Area of Case ($in^2$). See below for MIL-T-27 Case Areas

Wt. = Transformer Weight (lbs.)

$W_I$ = Input Power (W)

NOTE: Methods are listed in preferred order (i.e., most to least accurate). MIL-C-39010 are micro-miniature devices with surface areas less than 1 $in^2$. Equations 2-4 are applicable to devices with surface areas from 3 $in^2$ to 150 $in^2$. Do not include the mounting surface when determining radiating surface area.

MIL-T-27 Case Radiating Areas (Excludes Mounting Surface)

| Case | Area ($in^2$) | Case | Area ($in^2$) | Case | Area ($in^2$) |
|---|---|---|---|---|---|
| AF | 4 | GB | 33 | LB | 82 |
| AG | 7 | GA | 43 | LA | 98 |
| AH | 11 | HB | 42 | MB | 98 |
| AJ | 18 | HA | 53 | MA | 115 |
| EB | 21 | JB | 58 | NB | 117 |
| EA | 23 | JA | 71 | NA | 139 |
| FB | 25 | KB | 72 | OA | 146 |
| FA | 31 | KA | 84 | | |



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## 12.1 ROTATING DEVICES, MOTORS

The following failure-rate model applies to motors with power ratings below one horsepower. This model is applicable to polyphase, capacitor start and run and shaded pole motors. It's application may be extended to other types of fractional horsepower motors utilizing rolling element grease packed bearings. The model is dictated by two failure modes, bearing failures and winding failures. Application of the model to D.C. brush motors assumes that brushes are inspected and replaced and are not a failure mode. Typical applications include fans and blowers as well as various other motor applications. The model is based on Reference 4, which contains a more comprehensive treatment of motor life prediction methods. The reference should be reviewed when bearing loads exceed 10 percent of rated load, speeds exceed 24,000 rpm or motor loads include motor speed slip of greater than 25 percent.

The instantaneous failure rates, or hazard rates, experienced by motors are not constant but increase with time. The failure rate model in this section is an average failure rate for the motor operating over time period "t". The motor operating time period (t-hours) is selected by the analyst. Each motor must be replaced when it reaches the end of this period to make the calculated $\lambda_p$ valid. The average failure rate, $\lambda_p$, has been obtained by dividing the cumulative hazard rate by t, and can be treated as a constant failure rate and added to other part failure rates from this Handbook.

$$\lambda_p = \left[ \frac{t^2}{\alpha_B{}^3} + \frac{1}{\alpha_W} \right] \times 10^6 \text{ Failures}/10^6 \text{ Hours}$$

Bearing & Winding Characteristic Life - $\alpha_B$ and $\alpha_W$

| $T_A$ (°C) | $\alpha_B$ (Hr.) | $\alpha_W$ (Hr.) | $T_A$ (°C) | $\alpha_B$ (Hr.) | $\alpha_W$ (Hr.) |
|---|---|---|---|---|---|
| -40 | 310 | 1.9e+08 | 55 | 44000 | 2.3e+05 |
| -35 | 310 | 1.2e+08 | 60 | 35000 | 1.8e+05 |
| -30 | 330 | 7.4e+07 | 65 | 27000 | 1.4e+05 |
| -25 | 370 | 4.7e+07 | 70 | 22000 | 1.1e+05 |
| -20 | 460 | 3.1e+07 | 75 | 17000 | 8.8e+04 |
| -15 | 660 | 2.0e+07 | 80 | 14000 | 7.0e+04 |
| -10 | 1100 | 1.4e+07 | 85 | 11000 | 5.7e+04 |
| -5 | 1900 | 9.2e+06 | 90 | 9100 | 4.6e+04 |
| 0 | 3600 | 6.4e+06 | 95 | 7400 | 3.8e+04 |
| 5 | 6700 | 4.5e+06 | 100 | 6100 | 3.1e+04 |
| 10 | 13000 | 3.2e+06 | 105 | 5000 | 2.5e+04 |
| 15 | 23000 | 2.3e+06 | 110 | 4200 | 2.1e+04 |
| 20 | 39000 | 1.6e+06 | 115 | 3500 | 1.8e+04 |
| 25 | 60000 | 1.2e+06 | 120 | 2900 | 1.5e+04 |
| 30 | 78000 | 8.9e+05 | 125 | 2400 | 1.2e+04 |
| 35 | 86000 | 6.6e+05 | 130 | 2100 | 1.0e+04 |
| 40 | 80000 | 5.0e+05 | 135 | 1700 | 8.9e+03 |
| 45 | 68000 | 3.8e+05 | 140 | 1500 | 7.5e+03 |
| 50 | 55000 | 2.9e+-5 | | | |

$$\alpha_B = \left[ 10^{\left(2.534 - \frac{2357}{T_A + 273}\right)} + \frac{1}{10^{\left(20 - \frac{4500}{T_A + 273}\right)} + 300} \right]^{-1}$$

$$\alpha_W = 10^{\left[\frac{2357}{T_A + 273} - 1.83\right]}$$

$\alpha_B$ = Weibull Characteristic Life for the Motor Bearing

$\alpha_W$ = Weibull Characteristic Life for the Motor Windings

$T_A$ = Ambient Temperature (°C)

t = Motor Operating Time Period (Hours)

NOTE: See next page for method to calculate $\alpha_B$ and $\alpha_W$ when temperature is not constant.



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## 12.1 ROTATING DEVICES, MOTORS

### $\alpha_{Calculation}$ for Cycled Temperature

The following equation can be used to calculate a weighted characteristic life for both bearings and windings (e.g., for bearings substitute $\alpha_B$ for all $\alpha$'s in equation).

$$\alpha = \frac{(h_1 + h_2 + h_3 + \cdots\cdots h_m)}{\frac{h_1}{\alpha_1} + \frac{h_2}{\alpha_2} + \frac{h_3}{\alpha_3} + \cdots\cdots\cdots \frac{h_m}{\alpha_m}}$$

where:

$\alpha$ = either $\alpha_B$ or $\alpha_W$

$h_1$ = Time at Temperature $T_1$

$h_2$ = Time to Cycle From Temperature $T_1$ to $T_3$

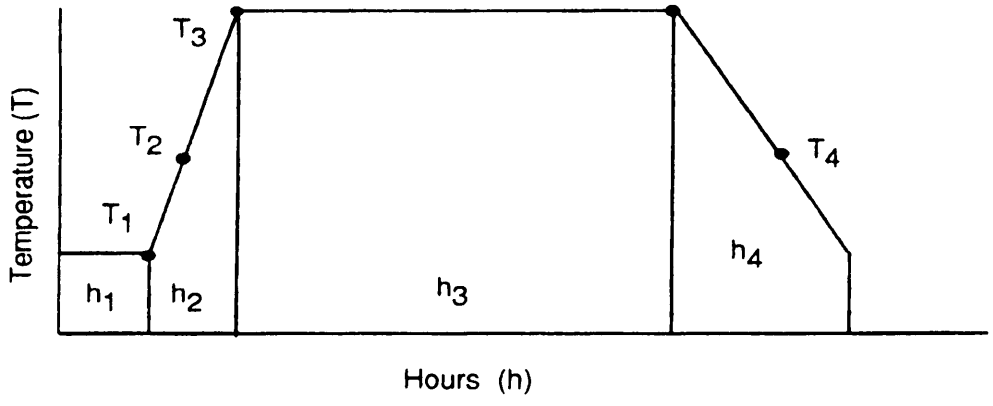
$h_3$ = Time at Temperature $T_3$

$h_m$ = Time at Temperature $T_m$

$\alpha_1$ = Bearing (or Winding) Life at $T_1$

$\alpha_2$ = Bearing (or Winding) Life at $T_2$

NOTE: $T_2 = \frac{T_1 + T_3}{2}$, $T_4 = \frac{T_3 + T_1}{2}$




Thermal Cycle



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## 12.2 ROTATING DEVICES, SYNCHROS AND RESOLVERS

### DESCRIPTION
Rotating Synchros and Resolvers

$$\lambda_p = \lambda_b \pi_S \pi_N \pi_E \text{ Failures/}10^6 \text{ Hours}$$

**NOTE:** Synchros and resolvers are predominately used in service requiring only slow and infrequent motion. Mechanical wearout problems are infrequent so that the electrical failure mode dominates, and no mechanical mode failure rate is required in the model above.

Base Failure Rate - $\lambda_b$

| $T_F$ (°C) | $\lambda_b$ | $T_F$ (°C) | $\lambda_b$ |
|---|---|---|---|
| 30 | .0083 | 85 | .032 |
| 35 | .0088 | 90 | .041 |
| 40 | .0095 | 95 | .052 |
| 45 | .010 | 100 | .069 |
| 50 | .011 | 105 | .094 |
| 55 | .013 | 110 | .13 |
| 60 | .014 | 115 | .19 |
| 65 | .016 | 120 | .29 |
| 70 | .019 | 125 | .45 |
| 75 | .022 | 130 | .74 |
| 80 | .027 | 135 | 1.3 |

$$\lambda_b = .00535 \exp\left(\frac{T+273}{334}\right)^{8.5}$$

$T_F$ = Frame Temperature (°C)

If Frame Temperature is Unknown Assume
$T_F$ = 40 °C + Ambient Temperature

Number of Brushes Factor - $\pi_N$

| Number of Brushes | $\pi_N$ |
|---|---|
| 2 | 1.4 |
| 3 | 2.5 |
| 4 | 3.2 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 12 |
| $N_S$ | 7.0 |
| $N_U$ | 18 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 6.0 |
| $A_{UC}$ | 16 |
| $A_{UF}$ | 25 |
| $A_{RW}$ | 26 |
| $S_F$ | .50 |
| $M_F$ | 14 |
| $M_L$ | 36 |
| $C_L$ | 680 |

Size Factor - $\pi_S$

| DEVICE TYPE | $\pi_S$ | | |
|---|---|---|---|
| | Size 8 or Smaller | Size 10-16 | Size 18 or Larger |
| Synchro | 2 | 1.5 | 1 |
| Resolver | 3 | 2.25 | 1.5 |



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## 12.3 ROTATING DEVICES, ELAPSED TIME METERS

### DESCRIPTION
Elapsed Time Meters

$$\lambda_p = \lambda_b \pi_T \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| Type | $\lambda_b$ |
|---|---|
| A.C. | 20 |
| Inverter Driven | 30 |
| Commutator D.C. | 80 |

Temperature Stress Factor - $\pi_T$

| Operating T (°C)/Rated T (°C) | $\pi_T$ |
|---|---|
| 0 to .5 | .5 |
| .6 | .6 |
| .8 | .8 |
| 1.0 | 1.0 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 12 |
| $N_S$ | 7.0 |
| $N_U$ | 18 |
| $A_{IC}$ | 5.0 |
| $A_{IF}$ | 8.0 |
| $A_{UC}$ | 16 |
| $A_{UF}$ | 25 |
| $A_{RW}$ | 26 |
| $S_F$ | .50 |
| $M_F$ | 14 |
| $M_L$ | 38 |
| $C_L$ | N/A |



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## 12.4 ROTATING DEVICES, EXAMPLE

**Example**

Given: Fractional Horsepower Motor operating at a thermal duty cycle of: 2 hours at 100°C, 8 hours at 20°C, 0.5 hours from 100°C to 20°C, and 0.5 hours from 20°C back to 100°C. Find the average failure rate for 4000 hours operating time.

The basic procedure is to first determine operating temperature at each time interval (or averge temperature when traversing from one temperature to another, e.g. $T_2 = (100 + 20)/2 = 60°C$. Determine $\alpha_B$ and $\alpha_W$ at each temperature and then use these values to determine a weighted average $\alpha_B$ and $\alpha_W$ to use in the $\lambda_p$ equation.

$h_1 = 2$ hr. $T_1 = 100°C$; $\alpha_B = 6100$ hours; $\alpha_W = 31000$ hours

$h_2 = h_4 = 0.5$ hr. $T_2 = 60°C$; $\alpha_B = 35000$ hours; $\alpha_W = 180000$ hours

$h_3 = 8$ hr. $T_3 = 20°C$; $\alpha_B = 39000$ hours; $\alpha_W = 1600000$ hours

$$\alpha_B = \frac{2 + 0.5 + 8 + 0.5}{\frac{2}{6100} + \frac{0.5}{35000} + \frac{8}{39000} + \frac{0.5}{35000}} = 19600 \text{ hours}$$

$$\alpha_W = \frac{2 + 0.5 + 8 + 0.5}{\frac{2}{31000} + \frac{0.5}{180000} + \frac{8}{1600000} + \frac{0.5}{180000}} = 146000 \text{ hours}$$

$$\lambda_p = \left(\frac{t^2}{\alpha_B{}^3} + \frac{1}{\alpha_w}\right) \times 10^6$$

$$\lambda_p = \left(\frac{(4000)^2}{(19600)^3} + \frac{1}{146000}\right) \times 10^6$$

$$\lambda_p = 9.0 \text{ Failures}/10^6 \text{ Hours}$$



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## 13.1 RELAYS, MECHANICAL

**SPECIFICATION**
MIL-R-5757
MIL-R-6106
MIL-R-19523
MIL-R-39016
MIL-R-19648
MIL-R-83725
MIL-R-83726 (Except Class C, Solid State Type)

**DESCRIPTION**
Mechanical Relay

$$\lambda_p = \lambda_b \pi_L \pi_C \pi_{CYC} \pi_F \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

### Base Failure Rate - $\lambda_b$

| $T_A$ (°C) | Rated Temperature 85°C[1] | 125°C[2] |
|---|---|---|
| 25 | .0060 | .0059 |
| 30 | .0061 | .0060 |
| 35 | .0063 | .0061 |
| 40 | .0065 | .0062 |
| 45 | .0068 | .0064 |
| 50 | .0072 | .0066 |
| 55 | .0077 | .0068 |
| 60 | .0084 | .0071 |
| 65 | .0094 | .0074 |
| 70 | .011 | .0079 |
| 75 | .013 | .0083 |
| 80 | .016 | .0089 |
| 85 | .020 | .0097 |
| 90 | | .011 |
| 95 | | .012 |
| 100 | | .013 |
| 105 | | .015 |
| 110 | | .018 |
| 115 | | .021 |
| 120 | | .025 |
| 125 | | .031 |

1. $\lambda_b = .00555 \exp\left(\frac{T_A + 273}{352}\right)^{15.7}$

2. $\lambda_b = .0054 \exp\left(\frac{T_A + 273}{377}\right)^{10.4}$

$T_A$ = Ambient Temperature (°C)

### Load Stress Factor - $\pi_L$

| S | Load Type Resistive[1] | Inductive[2] | Lamp[3] |
|---|---|---|---|
| .05 | 1.00 | 1.02 | 1.06 |
| .10 | 1.02 | 1.06 | 1.28 |
| .20 | 1.06 | 1.28 | 2.72 |
| .30 | 1.15 | 1.76 | 9.49 |
| .40 | 1.28 | 2.72 | 54.6 |
| .50 | 1.48 | 4.77 | |
| .60 | 1.76 | 9.49 | |
| .70 | 2.15 | 21.4 | |
| .80 | 2.72 | | |
| .90 | 3.55 | | |
| 1.00 | 4.77 | | |

1. $\pi_L = \exp\left(\frac{S}{.8}\right)^2$

2. $\pi_L = \exp\left(\frac{S}{.4}\right)^2$

3. $\pi_L = \exp\left(\frac{S}{.2}\right)^2$

$S = \frac{\text{Operating Load Current}}{\text{Rated Resistive Load Current}}$

For single devices which switch two different load types, evaluate $\pi_L$ for each possible stress load type combination and use the worse case (largest $\pi_L$).

### Cycling Factor - $\pi_{CYC}$

| Cycle Rate (Cycles per Hour) | $\pi_{CYC}$ (MIL-SPEC) |
|---|---|
| | Cycles per Hour / 10 |
| ≥ 1.0 | |
| < 1.0 | 0.1 |

| Cycle Rate (Cycles per Hour) | $\pi_{CYC}$ (Lower Quality) |
|---|---|
| > 1000 | $\left(\frac{\text{Cycles per Hour}}{100}\right)^2$ |
| 10 - 1000 | Cycles per Hour / 10 |
| < 10 | 1.0 |

NOTE: Values of $\pi_{CYC}$ for cycling rates beyond the basic design limitations of the relay are not valid. Design specifications should be consulted prior to evaluation of $\pi_{CYC}$.

### Contact Form Factor - $\pi_C$
(Applies to Active Conducting Contacts)

| Contact Form | $\pi_C$ |
|---|---|
| SPST | 1.00 |
| DPST | 1.50 |
| SPDT | 1.75 |
| 3PST | 2.00 |
| 4PST | 2.50 |
| DPDT | 3.00 |
| 3PDT | 4.25 |
| 4PDT | 5.50 |
| 6PDT | 8.00 |



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## 13.1 RELAYS, MECHANICAL

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| R | .10 |
| P | .30 |
| X | .45 |
| U | .60 |
| M | 1.0 |
| L | 1.5 |
| Non-Est. Rel. | 3.0 |

Environment Factor - $\pi_E$

| | $\pi_E$ | |
|---|---|---|
| Environment | MIL-SPEC | Lower Quality |
| $G_B$ | 1.0 | 2.0 |
| $G_F$ | 2.0 | 5.0 |
| $G_M$ | 15 | 44 |
| $N_S$ | 8.0 | 24 |
| $N_U$ | 27 | 78 |
| $A_{IC}$ | 7.0 | 15 |
| $A_{IF}$ | 9.0 | 20 |
| $A_{UC}$ | 11 | 28 |
| $A_{UF}$ | 12 | 38 |
| $A_{RW}$ | 46 | 140 |
| $S_F$ | .50 | 1.0 |
| $M_F$ | 25 | 72 |
| $M_L$ | 68 | 200 |
| $C_L$ | N/A | N/A |

Application and Construction Factor - $\pi_F$

| Contact Rating | Application Type | Construction Type | $\pi_F$ MIL-SPEC | $\pi_F$ Lower Quality |
|---|---|---|---|---|
| Signal Current (Low mv and ma) | Dry Circuit | Armature (Long) | 4 | 8 |
| | | Dry Reed | 6 | 18 |
| | | Mercury Wetted | 1 | 3 |
| | | Magnetic Latching | 4 | 8 |
| | | Balanced Armature | 7 | 14 |
| | | Solenoid | 7 | 14 |
| 0-5 Amp | General Purpose | Armature (Long) | 3 | 6 |
| | | Balanced Armature | 5 | 10 |
| | | Solenoid | 6 | 12 |
| | Sensitive (0 - 100 mw) | Armature (Long and Short) | 5 | 10 |
| | | Mercury Wetted | 2 | 6 |
| | | Magnetic Latching | 6 | 12 |
| | | Meter Movement | 100 | 100 |
| | | Balanced Armature | 10 | 20 |
| | Polarized | Armature (Short) | 10 | 20 |
| | | Meter Movement | 100 | 100 |
| | Vibrating Reed | Dry Reed | 6 | 12 |
| | | Mercury Wetted | 1 | 3 |
| | High Speed | Armature (Balanced and Short) | 25 | NA |
| | | Dry Reed | 6 | NA |
| | Thermal Time Delay | Bimetal | 10 | 20 |
| | Electronic Time Delay, Non-Thermal | | 9 | 12 |
| | Latching, Magnetic | Dry Reed | 10 | 20 |
| | | Mercury Wetted | 5 | 10 |
| | | Balanced Aramture | 5 | 10 |
| 5-20 Amp | High Voltage | Vacuum (Glass) | 20 | 40 |
| | | Vacuum (Ceramic) | 5 | 10 |
| | Medium Power | Armature (Long and Short) | 3 | 6 |
| | | Mercury Wetted | 1 | 3 |
| | | Magnetic Latching | 2 | 6 |
| | | Mechanical Latching | | |
| | | Balanced Armature | 3 | 6 |
| | | Solenoid | 2 | 6 |
| | | | 2 | 6 |
| 25-600 Amp | Contactors (High Current) | Armature (Short) | 7 | 14 |
| | | Mechanical Latching | 12 | 24 |
| | | Balanced Armature | 10 | 20 |
| | | Solenoid | 5 | 10 |



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## 13.2 RELAYS, SOLID STATE AND TIME DELAY

**SPECIFICATION**
MIL-R-28750
MIL-R-83726

**DESCRIPTION**
Relay, Solid State
Relay, Time Delay, Hybrid and Solid State

The most accurate method for predicting the failure rate of solid state (and solid state time delay) relays is to sum the failure rates for the individual components which make up the relay. The individual component failure rates can either be calculated from the models provided in the main body of this Handbook (Parts Stress Method) or from the Parts Count Method shown in Appendix A, depending upon the depth of knowledge the analyst has about the components being used. If insufficient information is available, the following default model can be used:

$$\lambda_p = \lambda_b \pi_Q \pi_E \text{ Failures}/10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| Relay Type | $\lambda_b$ |
|---|---|
| Solid State | .40 |
| Solid State Time Delay | .50 |
| Hybrid | .50 |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| MIL-SPEC | 1.0 |
| Lower | 4.0 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 3.0 |
| $G_M$ | 12 |
| $N_S$ | 6.0 |
| $N_U$ | 17 |
| $A_{IC}$ | 12 |
| $A_{IF}$ | 19 |
| $A_{UC}$ | 21 |
| $A_{UF}$ | 32 |
| $A_{RW}$ | 23 |
| $S_F$ | .40 |
| $M_F$ | 12 |
| $M_L$ | 33 |
| $C_L$ | 590 |



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## 14.1 SWITCHES, TOGGLE OR PUSHBUTTON

**SPECIFICATION**
MIL-S-3950 MIL-S-22885
MIL-S-8805 MIL-S-83731
MIL-S-8834

**DESCRIPTION**
Snap-action, Toggle or Pushbutton, Single Body

$$\lambda_p = \lambda_b \pi_{CYC} \pi_L \pi_C \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| Description | MIL-SPEC | Lower Quality |
|---|---|---|
| Snap-action | .00045 | .034 |
| Non-snap Action | .0027 | .040 |

Cycling Factor - $\pi_{CYC}$

| Switching Cycles per Hour | $\pi_{CYC}$ |
|---|---|
| ≤ 1 Cycle/Hour | 1.0 |
| > 1 Cycle/Hour | Number of Cycles/Hour |

Load Stress Factor - $\pi_L$

| Stress S | Load Type | | |
|---|---|---|---|
| | Resistive | Inductive | Lamp |
| 0.05 | 1.00 | 1.02 | 1.06 |
| 0.1 | 1.02 | 1.06 | 1.28 |
| 0.2 | 1.06 | 1.28 | 2.72 |
| 0.3 | 1.15 | 1.76 | 9.49 |
| 0.4 | 1.28 | 2.72 | 54.6 |
| 0.5 | 1.48 | 4.77 | |
| 0.6 | 1.76 | 9.49 | |
| 0.7 | 2.15 | 21.4 | |
| 0.8 | 2.72 | | |
| 0.9 | 3.55 | | |
| 1.0 | 4.77 | | |

$$S = \frac{\text{Operating Load Current}}{\text{Rated Resistive Load Current}}$$

$\pi_L = \exp(S/.8)^2$ for Resistive Load
$\pi_L = \exp(S/.4)^2$ for Inductive Load
$\pi_L = \exp(S/.2)^2$ for Lamp Load

NOTE: When the switch is rated by inductive load, then use resistive $\pi_L$.

Contact Form and Quantity Factor - $\pi_C$

| Contact Form | $\pi_C$ |
|---|---|
| SPST | 1.0 |
| DPST | 1.5 |
| SPDT | 1.7 |
| 3PST | 2.0 |
| 4PST | 2.5 |
| DPDT | 3.0 |
| 3PDT | 4.2 |
| 4PDT | 5.5 |
| 6PDT | 8.0 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 3.0 |
| $G_M$ | 18 |
| $N_S$ | 8.0 |
| $N_U$ | 29 |
| $A_{IC}$ | 10 |
| $A_{IF}$ | 18 |
| $A_{UC}$ | 13 |
| $A_{UF}$ | 22 |
| $A_{RW}$ | 46 |
| $S_F$ | .50 |
| $M_F$ | 25 |
| $M_L$ | 67 |
| $C_L$ | 1200 |



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## 14.2 SWITCHES, BASIC SENSITIVE

**SPECIFICATION**
MIL-S-8805

**DESCRIPTION**
Basic Sensitive

$$\lambda_p = \lambda_b \pi_{CYC} \pi_L \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

$\lambda_b = \lambda_{bE} + n\,\lambda_{bC}$ (if Actuation Differential is > 0.002 inches)

$\lambda_b = \lambda_{bE} + n\,\lambda_{b0}$ (if Actuation Differential is ≤ 0.002 inches)

n = Number of Active Contacts

| Description | MIL-SPEC | Lower Quality |
|---|---|---|
| $\lambda_{bE}$ | .10 | .10 |
| $\lambda_{bC}$ | .00045 | .23 |
| $\lambda_{b0}$ | .0009 | .63 |

Load Stress Factor - $\pi_L$

| Stress S | Load Type | | |
|---|---|---|---|
| | Resistive | Inductive | Lamp |
| 0.05 | 1.00 | 1.02 | 1.06 |
| 0.1 | 1.02 | 1.06 | 1.28 |
| 0.2 | 1.06 | 1.28 | 2.72 |
| 0.3 | 1.15 | 1.76 | 9.49 |
| 0.4 | 1.28 | 2.72 | 54.6 |
| 0.5 | 1.48 | 4.77 | |
| 0.6 | 1.76 | 9.49 | |
| 0.7 | 2.15 | 21.4 | |
| 0.8 | 2.72 | | |
| 0.9 | 3.55 | | |
| 1.0 | 4.77 | | |

$$S = \frac{\text{Operating Load Current}}{\text{Rated Resistive Load Current}}$$

$\pi_L = \exp(S/.8)^2$ for Resistive Load

$\pi_L = \exp(S/.4)^2$ for Inductive Load

$\pi_L = \exp(S/.2)^2$ for Lamp Load

NOTE: When the Switch is Rated by Inductive Load, then use Resistive $\pi_L$.

Cycling Factor - $\pi_{CYC}$

| Switching Cycles per Hour | $\pi_{CYC}$ |
|---|---|
| ≤ 1 Cycle/Hour | 1.0 |
| > 1 Cycle/Hour | Number of Cycles/Hour |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 3.0 |
| $G_M$ | 18 |
| $N_S$ | 8.0 |
| $N_U$ | 29 |
| $A_{IC}$ | 10 |
| $A_{IF}$ | 18 |
| $A_{UC}$ | 13 |
| $A_{UF}$ | 22 |
| $A_{RW}$ | 46 |
| $S_F$ | .50 |
| $M_F$ | 25 |
| $M_L$ | 67 |
| $C_L$ | 1200 |



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## 14.3 SWITCHES, ROTARY

**SPECIFICATION**
MIL-S-3786

**DESCRIPTION**
Rotary, Ceramic or Glass Wafer, Silver Alloy Contacts

$$\lambda_p = \lambda_b \pi_{CYC} \pi_L \pi_E \text{ Failures/}10^6 \text{ Hours}$$

### Base Failure Rate - $\lambda_b$

Base failure rate model ($\lambda_b$):

$\lambda_b = \lambda_{bE} + n\lambda_{bF}$ (for Ceramic RF Wafers)

$\lambda_b = \lambda_{bE} + n\,\lambda_{bG}$ (for Rotary Switch Medium Power Wafers)

n = Number of Active Contacts

| Description | MIL-SPEC | Lower Quality |
|---|---|---|
| $\lambda_{bE}$ | .0067 | .10 |
| $\lambda_{bF}$ | .00003 | .02 |
| $\lambda_{bG}$ | .00003 | .06 |

### Load Stress Factor - $\pi_L$

| Stress S | Load Type | | |
|---|---|---|---|
| | Resistive | Inductive | Lamp |
| 0.05 | 1.00 | 1.02 | 1.06 |
| 0.1 | 1.02 | 1.06 | 1.28 |
| 0.2 | 1.06 | 1.28 | 2.72 |
| 0.3 | 1.15 | 1.76 | 9.49 |
| 0.4 | 1.28 | 2.72 | 54.6 |
| 0.5 | 1.48 | 4.77 | |
| 0.6 | 1.76 | 9.49 | |
| 0.7 | 2.15 | 21.4 | |
| 0.8 | 2.72 | | |
| 0.9 | 3.55 | | |
| 1.0 | 4.77 | | |

$$S = \frac{\text{Operating Load Current}}{\text{Rated Resistive Load Current}}$$

$\pi_L = \exp(S/.8)^2$ for Resistive Load

$\pi_L = \exp(S/.4)^2$ for Inductive Load

$\pi_L = \exp(S/.2)^2$ for Lamp Load

NOTE: When the Switch is Rated by Inductive Load, then use Resistive $\pi_L$.

### Cycling Factor - $\pi_{CYC}$

| Switching Cycles per Hour | $\pi_{CYC}$ |
|---|---|
| ≤ 1 Cycle/Hour | 1.0 |
| > 1 Cycle/Hour | Number of Cycles/Hour |

### Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 3.0 |
| $G_M$ | 18 |
| $N_S$ | 8.0 |
| $N_U$ | 29 |
| $A_{IC}$ | 10 |
| $A_{IF}$ | 18 |
| $A_{UC}$ | 13 |
| $A_{UF}$ | 22 |
| $A_{RW}$ | 46 |
| $S_F$ | .50 |
| $M_F$ | 25 |
| $M_L$ | 67 |
| $C_L$ | 1200 |



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## 14.4 SWITCHES, THUMBWHEEL

**SPECIFICATION**
MIL-S-22710
Line

**DESCRIPTION**
Switches, Rotary (Printed Circuit) (Thumbwheel, In- and Pushbutton)

$$\lambda_p = (\lambda_{b1} + \pi_N \lambda_{b2})\, \pi_{CYC} \pi_L \pi_E \text{ Failures/}10^6 \text{ Hours}$$

CAUTION: This model applies to the switching function only. The model does not consider the contribution of any discrete components (e.g., resistors, diodes, lamp) which may be mounted on the switch. If significant (relative to the switch failure rate), the failure rate of these devices must be calculated using the appropriate section of this Handbook and added to the failure rate of the switch.

This model applies to a single switch section. This type of switch is frequently ganged to provide the required function. The model must be applied to each section individually.

Base Failure Rate - $\lambda_{b1}$ and $\lambda_{b2}$

| Description | MIL-SPEC | Lower Quality |
|---|---|---|
| $\lambda_{b1}$ | .0067 | .086 |
| $\lambda_{b2}$ | .062 | .089 |

Cycling Factor - $\pi_{CYC}$

| Switching Cycles per Hour | $\pi_{CYC}$ |
|---|---|
| ≤ 1 Cycle/Hour | 1.0 |
| > 1 Cycle/Hour | Number of Cycles/Hour |

Number of Active Contacts Factor - $\pi_N$

$\pi_N$ = Number of Active Contacts

Load Stress Factor - $\pi_L$

| Stress S | Load Type | | |
|---|---|---|---|
| | Resistive | Inductive | Lamp |
| 0.05 | 1.00 | 1.02 | 1.06 |
| 0.1 | 1.02 | 1.06 | 1.28 |
| 0.2 | 1.06 | 1.28 | 2.72 |
| 0.3 | 1.15 | 1.76 | 9.49 |
| 0.4 | 1.28 | 2.72 | 54.6 |
| 0.5 | 1.48 | 4.77 | |
| 0.6 | 1.76 | 9.49 | |
| 0.7 | 2.15 | 21.4 | |
| 0.8 | 2.72 | | |
| 0.9 | 3.55 | | |
| 1.0 | 4.77 | | |

$$S = \frac{\text{Operating Load Current}}{\text{Rated Resistive Load Current}}$$

$\pi_L = \exp(S/.8)^2$ for Resistive Load

$\pi_L = \exp(S/.4)^2$ for Inductive Load

$\pi_L = \exp(S/.2)^2$ for Lamp Load

NOTE: When the Switch is Rated by Inductive Load, then use Resistive $\pi_L$.

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 3.0 |
| $G_M$ | 18 |
| $N_S$ | 8.0 |
| $N_U$ | 29 |
| $A_{IC}$ | 10 |
| $A_{IF}$ | 18 |
| $A_{UC}$ | 13 |
| $A_{UF}$ | 22 |
| $A_{RW}$ | 46 |
| $S_F$ | .50 |
| $M_F$ | 25 |
| $M_L$ | 67 |
| $C_L$ | 1200 |



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## 14.5 SWITCHES, CIRCUIT BREAKERS

**SPECIFICATION**
MIL-C-55629
MIL-C-83383
MIL-C-39019
W-C-375

**DESCRIPTION**
Circuit Breakers, Magnetic, Unsealed, Trip-Free
Circuit Breakers, Remote Control, Thermal, Trip-Free
Circuit Breakers, Magnetic, Low Power, Sealed, Trip-Free Service
Circuit Breakers, Molded Case, Branch Circuit and Service

$$\lambda_p = \lambda_b \pi_C \pi_U \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| Description | $\lambda_b$ |
|---|---|
| Magnetic | .020 |
| Thermal | .038 |
| Thermal-Magnetic | .038 |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| MIL-SPEC | 1.0 |
| Lower | 8.4 |

Configuration Factor - $\pi_C$

| Configuration | $\pi_C$ |
|---|---|
| SPST | 1.0 |
| DPST | 2.0 |
| 3PST | 3.0 |
| 4PST | 4.0 |

Use Factor - $\pi_U$

| Use | $\pi_U$ |
|---|---|
| Not Used as a Power On/Off Switch | 1.0 |
| Also Used as a Power On/Off Switch | 10 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 15 |
| $N_S$ | 8.0 |
| $N_U$ | 27 |
| $A_{IC}$ | 7.0 |
| $A_{IF}$ | 9.0 |
| $A_{UC}$ | 11 |
| $A_{UF}$ | 12 |
| $A_{RW}$ | 46 |
| $S_F$ | .50 |
| $M_F$ | 25 |
| $M_L$ | 66 |
| $C_L$ | N/A |



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## 15.1 CONNECTORS, GENERAL (EXCEPT PRINTED CIRCUIT BOARD)

| SPECIFICATION* | DESCRIPTION | SPECIFICATION* | DESCRIPTION |
|---|---|---|---|
| MIL-C-24308 | Rack and Panel | MIL-C-3607 | Coaxial, RF |
| MIL-C-28748 | | MIL-C-3643 | |
| MIL-C-28804 | | MIL-C-3650 | |
| MIL-C-83513 | | MIL-C-3655 | |
| MIL-C-83733 | | MIL-C-25516 | |
| MIL-C-5015 | Circular | MIL-C-39012 | |
| MIL-C-26482 | | MIL-C-55235 | |
| MIL-C-28840 | | MIL-C-55339 | |
| MIL-C-38999 | | MIL-C-3767 | Power |
| MIL-C-81511 | | MIL-C-22992 | |
| MIL-C-83723 | | | |
| * NOTE: See following page for connector configurations. | | MIL-C-49142 | Triaxial, RF |

$$\lambda_p = \lambda_b \pi_K \pi_P \pi_E \text{ Failures/}10^6 \text{ Hours}$$

**APPLICATION NOTE:** The failure rate model is for a mated pair of connectors. It is sometimes desirable to assign half of the overall mated pair connector (i.e., single connector) failure rate to the line replaceable unit and half to the chassis (or backplane). An example of when this would be beneficial is for input to maintainability prediction to allow a failure rate weighted repair time to be estimated for both the LRU and chassis. This accounting procedure could be significant if repair times for the two halves of the connector are substantially different. For a single connector divide $\lambda_p$ by two.

Base Failure Rate - $\lambda_b$

| $T_o$ (°C) | Insert Material* A[1] | B[2] | C[3] | D[4] |
|---|---|---|---|---|
| 0 | .00006 | .00025 | .0021 | .0038 |
| 10 | .00008 | .00033 | .0026 | .0048 |
| 20 | .00009 | .00044 | .0032 | .0062 |
| 30 | .00011 | .00057 | .0040 | .0078 |
| 40 | .00014 | .00073 | .0048 | .0099 |
| 50 | .00016 | .00093 | .0059 | .013 |
| 60 | .00020 | .0012 | .0071 | .016 |
| 70 | .00023 | .0015 | .0087 | .020 |
| 80 | .00027 | .0019 | .011 | .026 |
| 90 | .00032 | .0023 | .013 | .033 |
| 100 | .00037 | .0029 | .016 | .043 |
| 110 | .00043 | .0036 | .020 | .056 |
| 120 | .00050 | .0045 | .024 | .074 |
| 130 | .00059 | .0056 | | |
| 140 | .00069 | .0070 | | |
| 150 | .00080 | .0087 | | |
| 160 | .00094 | .011 | | |
| 170 | .0011 | .014 | | |
| 180 | .0013 | .018 | | |
| 190 | .0016 | .022 | | |
| 200 | .0019 | .029 | | |
| 210 | .0023 | | | |
| 220 | .0028 | | | |
| 230 | .0034 | | | |
| 240 | .0042 | | | |
| 250 | .0053 | | | |

* If a mating pair of connectors uses two types of insert materials, use the average of the base failure rates for the two insert material types. See following page for insert material determination.

Base Failure Rate - $\lambda_b$ (cont'd)

1. $\lambda_b = .020 \exp\left(\left(\frac{-1592.0}{T_o + 273}\right) + \left(\frac{T_o + 273}{473}\right)^{5.36}\right)$

2. $\lambda_b = .431 \exp\left(\left(\frac{-2073.6}{T_o + 273}\right) + \left(\frac{T_o + 273}{423}\right)^{4.66}\right)$

3. $\lambda_b = .190 \exp\left(\left(\frac{-1298.0}{T_o + 273}\right) + \left(\frac{T_o + 273}{373}\right)^{4.25}\right)$

4. $\lambda_b = .770 \exp\left(\left(\frac{-1528.8}{T_o + 273}\right) + \left(\frac{T_o + 273}{358}\right)^{4.72}\right)$

$T_o$ = Internal Contact Operating Temperature (°C)

$T_o$ = Connector Ambient Temperature + Insert Temperature Rise

See following page for Insert Temperature Rise Determination.



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## 15.1 CONNECTORS, GENERAL (EXCEPT PRINTED CIRCUIT BOARD)

Insert Material Determination

| Configuration | Specification | Possible Insert Materials A | B | C | D |
|---|---|---|---|---|---|
| Rack and Panel | MIL-C-28748 | | X | | |
| | MIL-C-83733 | | X | | |
| | MIL-C-24308 | X | X | | |
| | MIL-C-28804 | X | X | | |
| | MIL-C-83513 | X | X | | |
| Circular | MIL-C-5015 | | X | | X |
| | MIL-C-26482 | X | X | | X |
| | MIL-C-28840 | X | X | | |
| | MIL-C-38999 | X | X | | |
| | MIL-C-81511 | | X | | |
| | MIL-C-83723 | | X | | |
| Power | MIL-C-3767 | | X | | X |
| | MIL-C-22992 | | X | | X |
| Coaxial | MIL-C-3607 | | | X | |
| | MIL-C-3643 | | | X | |
| | MIL-C-3650 | | | X | |
| | MIL-C-3655 | | | X | |
| | MIL-C-25516 | | | X | |
| | MIL-C-39012 | | | X | |
| | MIL-C-55235 | | | X | |
| | MIL-C-55339 | | X | X | |
| Triaxial | MIL-C-49142 | | X | X | |

| Insert Material Type | Common Insert Materials | Temperature Range (°C)* |
|---|---|---|
| A | Vitreous Glass, Alumina Ceramic, Polyimide | -55 to 250 |
| B | Diallylphtalate, Melamine, Fluorosilicione, Silicone Rubber, Polysulfone, Epoxy Resin | -55 to 200 |
| C | Polytetrafluorethylene (Teflon), Chlorotrifluorethylene (Kel-f) | -55 to 125 |
| D | Polyamide (Nylon), Polychloroprene (Neoprene), Polyethylene | -55 to 125 |

*These temperature ranges indicate maximum capability of the insert material only. Connectors using these materials generally have a reduced temperature range caused by other considerations of connector design. Applicable connector specifications contain connector operating temperature range.

Insert Temperature Rise (ΔT °C) Determination

| Amperes Per Contact | Contact Gauge 22 | 20 | 16 | 12 |
|---|---|---|---|---|
| 2 | 4 | 2 | 1 | 0 |
| 3 | 8 | 5 | 2 | 1 |
| 4 | 13 | 8 | 4 | 1 |
| 5 | 19 | 13 | 5 | 2 |
| 6 | 27 | 18 | 8 | 3 |
| 7 | 36 | 23 | 10 | 4 |
| 8 | 46 | 30 | 13 | 5 |
| 9 | 57 | 37 | 16 | 6 |
| 10 | 70 | 45 | 19 | 7 |
| 15 | | 96 | 41 | 15 |
| 20 | | | 70 | 26 |
| 25 | | | 106 | 39 |
| 30 | | | | 54 |
| 35 | | | | 72 |
| 40 | | | | 92 |

$\Delta T = 0.989\ (i)^{1.85}$ 22 Gauge Contacts
$\Delta T = 0.640\ (i)^{1.85}$ 20 Gauge Contacts
$\Delta T = 0.274\ (i)^{1.85}$ 16 Gauge Contacts
$\Delta T = 0.100\ (i)^{1.85}$ 12 Gauge Contacts

$\Delta T$ = Insert Temperature Rise
$i$ = Amperes per Contact

RF Coaxial Connectors $\Delta T = 5°C$

RF Coaxial Connectors (High Power Applications) $\Delta T = 50°C$

Mating/Unmating Factor - $\pi_K$

| Mating/Unmating Cycles* (per 1000 hours) | $\pi_K$ |
|---|---|
| 0 to .05 | 1.0 |
| > .05 to .5 | 1.5 |
| > .5 to 5 | 2.0 |
| > 5 to 50 | 3.0 |
| > 50 | 4.0 |

*One cycle includes both connect and disconnect.



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## 15.1 CONNECTORS, GENERAL (EXCEPT PRINTED CIRCUIT BOARD)

Active Pins Factor - $\pi_P$

| Number of Active Contacts | $\pi_P$ | Number of Active Contacts | $\pi_P$ |
|---|---|---|---|
| 1 | 1.0 | 65 | 13 |
| 2 | 1.4 | 70 | 15 |
| 3 | 1.6 | 75 | 16 |
| 4 | 1.7 | 80 | 18 |
| 5 | 1.9 | 85 | 19 |
| 6 | 2.0 | 90 | 21 |
| 7 | 2.2 | 95 | 23 |
| 8 | 2.3 | 100 | 25 |
| 9 | 2.4 | 105 | 27 |
| 10 | 2.6 | 110 | 30 |
| 11 | 2.7 | 115 | 32 |
| 12 | 2.9 | 120 | 35 |
| 13 | 3.0 | 125 | 37 |
| 14 | 3.1 | 130 | 40 |
| 15 | 3.3 | 135 | 43 |
| 16 | 3.4 | 140 | 46 |
| 17 | 3.6 | 145 | 50 |
| 18 | 3.7 | 150 | 53 |
| 19 | 3.9 | 155 | 57 |
| 20 | 4.0 | 160 | 61 |
| 25 | 4.8 | 165 | 65 |
| 30 | 5.6 | 170 | 69 |
| 35 | 6.5 | 175 | 74 |
| 40 | 7.4 | 180 | 78 |
| 45 | 8.4 | 185 | 83 |
| 50 | 9.5 | 190 | 89 |
| 55 | 11 | 195 | 94 |
| 60 | 12 | 200 | 100 |

$$\pi_P = \exp\left(\frac{N-1}{10}\right)^q$$

$q = 0.51064$

$N$ = Number of Active Contacts

An active contact is the conductive element in a connector which mates with another element for the purpose of transferring electrical energy. For coaxial and triaxial connectors, the shield contact is counted as an active contact.

Environment Factor - $\pi_E$

| | $\pi_E$ | |
|---|---|---|
| Environment | MIL-SPEC | Lower Quality |
| $G_B$ | 1.0 | 2.0 |
| $G_F$ | 1.0 | 5.0 |
| $G_M$ | 8.0 | 21 |
| $N_S$ | 5.0 | 10 |
| $N_U$ | 13 | 27 |
| $A_{IC}$ | 3.0 | 12 |
| $A_{IF}$ | 5.0 | 18 |
| $A_{UC}$ | 8.0 | 17 |
| $A_{UF}$ | 12 | 25 |
| $A_{RW}$ | 19 | 37 |
| $S_F$ | .50 | .80 |
| $M_F$ | 10 | 20 |
| $M_L$ | 27 | 54 |
| $C_L$ | 490 | 970 |



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## 15.2 CONNECTORS, PRINTED CIRCUIT BOARD

**SPECIFICATION**
MIL-C-21097
MIL-C-55302

**DESCRIPTION**
One-Piece Connector
Two-Piece Connector

$$\lambda_p = \lambda_b \pi_K \pi_P \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| $T_o$ (°C) | $\lambda_b$ | $T_o$ (°C) | $\lambda_b$ |
|---|---|---|---|
| 0 | .00012 | 110 | .0018 |
| 10 | .00017 | 120 | .0022 |
| 20 | .00022 | 130 | .0028 |
| 30 | .00028 | 140 | .0035 |
| 40 | .00037 | 150 | .0044 |
| 50 | .00047 | 160 | .0055 |
| 60 | .00059 | 170 | .0069 |
| 70 | .00075 | 180 | .0088 |
| 80 | .00093 | 190 | .011 |
| 90 | .0012 | 200 | .015 |
| 100 | .0015 | | |

$$\lambda_b = .216 \exp\left(\left(\frac{-2073.6}{T_o + 273}\right) + \left(\frac{T_o + 273}{423}\right)^{4.66}\right)$$

$T_o$ = Internal Contact Operating Temperature (°C)

Connector Temperature Rise ($\Delta T$ °C) Determination

| Amperes Per Contact | Contact Guage 26 | 22 | 20 |
|---|---|---|---|
| 1 | 2 | 1 | 1 |
| 2 | 8 | 4 | 2 |
| 3 | 16 | 8 | 5 |
| 4 | 27 | 13 | 8 |
| 5 | 41 | 19 | 13 |

$\Delta T = 2.100\ (i)^{1.85}$ 26 Guage Contacts
$\Delta T = 0.989\ (i)^{1.85}$ 22 Guage Contacts
$\Delta T = 0.640\ (i)^{1.85}$ 20 Guage Contacts

$\Delta T$ = Contact Temperature Rise

$i$ = Amperes per Contact

Mating/Unmating Factor - $\pi_K$

| Mating/Unmating Cycles* (Per1000 Hours) | $\pi_K$ |
|---|---|
| 0 to .05 | 1.0 |
| > .05 to .5 | 1.5 |
| > .5 to 5 | 2.0 |
| > 5 to 50 | 3.0 |
| > 50 | 4.0 |

* A cycle is defined as the mating and unmating of a connector.



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## 15.2 CONNECTORS, PRINTED CIRCUIT BOARD

Active Pins Factor - $\pi_P$

| Number of Active Contacts | $\pi_P$ | Number of Active Contacts | $\pi_P$ |
|---|---|---|---|
| 1 | 1.0 | 65 | 13 |
| 2 | 1.4 | 70 | 15 |
| 3 | 1.6 | 75 | 16 |
| 4 | 1.7 | 80 | 18 |
| 5 | 1.9 | 85 | 19 |
| 6 | 2.0 | 90 | 21 |
| 7 | 2.2 | 95 | 23 |
| 8 | 2.3 | 100 | 25 |
| 9 | 2.4 | 105 | 27 |
| 10 | 2.6 | 110 | 30 |
| 11 | 2.7 | 115 | 32 |
| 12 | 2.9 | 120 | 35 |
| 13 | 3.0 | 125 | 37 |
| 14 | 3.1 | 130 | 40 |
| 15 | 3.3 | 135 | 43 |
| 16 | 3.4 | 140 | 46 |
| 17 | 3.6 | 145 | 50 |
| 18 | 3.7 | 150 | 53 |
| 19 | 3.9 | 155 | 57 |
| 20 | 4.0 | 160 | 61 |
| 25 | 4.8 | 165 | 65 |
| 30 | 5.6 | 170 | 69 |
| 35 | 6.5 | 175 | 74 |
| 40 | 7.4 | 180 | 78 |
| 45 | 8.4 | 185 | 83 |
| 50 | 9.5 | 190 | 89 |
| 55 | 11 | 195 | 94 |
| 60 | 12 | 200 | 100 |

$$\pi_P = \exp\left(\frac{N-1}{10}\right)^q$$

$q = 0.51064$

$N$ = Number of Active Pins

An active contact is the conductive element which mates with another element for the purpose of transferring electrical energy.

Environment Factor - $\pi_E$

| | $\pi_E$ | |
|---|---|---|
| Environment | MIL-SPEC | Lower Quality |
| $G_B$ | 1.0 | 2.0 |
| $G_F$ | 3.0 | 7.0 |
| $G_M$ | 8.0 | 17 |
| $N_S$ | 5.0 | 10 |
| $N_U$ | 13 | 26 |
| $A_{IC}$ | 6.0 | 14 |
| $A_{IF}$ | 11 | 22 |
| $A_{UC}$ | 6.0 | 14 |
| $A_{UF}$ | 11 | 22 |
| $A_{RW}$ | 19 | 37 |
| $S_F$ | .50 | .80 |
| $M_F$ | 10 | 20 |
| $M_L$ | 27 | 54 |
| $C_L$ | 490 | 970 |



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## 15.3 CONNECTORS, INTEGRATED CIRCUIT SOCKETS

**SPECIFICATION**
MIL-S-83734

**DESCRIPTION**
IC Sockets, Plug-in

$$\lambda_p = \lambda_b \pi_P \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| Type | $\lambda_b$ |
|---|---|
| All MIL-S-83734 | .00042 |

Active Pins Factor - $\pi_P$

| Number of Active Contacts | $\pi_P$ |
|---|---|
| 6 | 2.0 |
| 8 | 2.3 |
| 10 | 2.6 |
| 14 | 3.1 |
| 16 | 3.4 |
| 18 | 3.7 |
| 20 | 4.0 |
| 22 | 4.3 |
| 24 | 4.6 |
| 28 | 5.3 |
| 36 | 6.7 |
| 40 | 7.4 |
| 48 | 9.1 |
| 50 | 9.5 |
| 64 | 13 |

$$\pi_P = \exp\left(\frac{N-1}{10}\right)^q$$

$q = 0.51064$

N = Number of Active Contacts

An active contact is the conductive element which mates with another element for the purpose of transferring electrical energy.

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 3.0 |
| $G_M$ | 14 |
| $N_S$ | 6.0 |
| $N_U$ | 18 |
| $A_{IC}$ | 8.0 |
| $A_{IF}$ | 12 |
| $A_{UC}$ | 11 |
| $A_{UF}$ | 13 |
| $A_{RW}$ | 25 |
| $S_F$ | .50 |
| $M_F$ | 14 |
| $M_L$ | 36 |
| $C_L$ | 650 |



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## 16.1 INTERCONNECTION ASSEMBLIES WITH PLATED THROUGH HOLES

**DESCRIPTION**
Circuit Boards, Printed (PCBs) and Discrete Wiring

$$\lambda_p = \lambda_b\left[N_1 \pi_C + N_2 (\pi_C + 13)\right] \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

**APPLICATION NOTE:** For assemblies not using Plated Through Holes (PTH), use Section 17, Connections. A discrete wiring assembly with electroless deposit plated through holes is basically a pattern of insulated wires laid down on an adhesive coated substrate. The primary cause of failure for both printed wiring and discrete wiring assemblies is associated with plated through hole problems (e.g., barrel cracking).

Base Failure Rate - $\lambda_b$

| Technology | $\lambda_b$ |
|---|---|
| Printed Wiring Assembly/Printed Circuit Boards with PTHs | .000041 |
| Discrete Wiring with Electroless Deposited PTH (≤ 2 Levels of Circuitry) | .00026 |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| MIL-SPEC or Comparable Institute for Interconnecting, and Packaging Electronic Circuits (IPC) Standards | 1 |
| Lower | 2 |

Number of PTHs Factor - $N_1$ and $N_2$

| Factor | Quantity |
|---|---|
| $N_1$ | Quantity of Wave Soldered Functional PTHs |
| $N_2$ | Quantity of Hand Soldered PTHs |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 7.0 |
| $N_S$ | 5.0 |
| $N_U$ | 13 |
| $A_{IC}$ | 5.0 |
| $A_{IF}$ | 8.0 |
| $A_{UC}$ | 16 |
| $A_{UF}$ | 28 |
| $A_{RW}$ | 19 |
| $S_F$ | .50 |
| $M_F$ | 10 |
| $M_L$ | 27 |
| $C_L$ | 500 |

Complexity Factor - $\pi_C$

| Number of Circuit Planes, P | $\pi_C$ |
|---|---|
| ≤ 2 | 1.0 |
| 3 | 1.3 |
| 4 | 1.6 |
| 5 | 1.8 |
| 6 | 2.0 |
| 7 | 2.2 |
| 8 | 2.4 |
| 9 | 2.6 |
| 10 | 2.8 |
| 11 | 2.9 |
| 12 | 3.1 |
| 13 | 3.3 |
| 14 | 3.4 |
| 15 | 3.6 |
| 16 | 3.7 |
| Discrete Wiring w/PTH | 1 |
| $\pi_C = .65\ P^{.63}$ | $2 \le P \le 16$ |



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## 17.1 CONNECTIONS

**DESCRIPTION**
Connections Used on All Assemblies Except Those Using Plated Through Holes (PTH)

**APPLICATION NOTE:** The failure rate model in this section applies to connections used on all assemblies except those using plated through holes. Use the Interconnection Assembly Model in Section 16 to account for connections to a circuit board using plated through hole technology. The failure rate of the structure which supports the connections and parts, e.g., non-plated-through hole boards and terminal straps, is considered to be zero. Solderless wrap connections are characterized by solid wire wrapped under tension around a post, whereas hand soldering with wrapping does not depend on a tension induced connection. The following model is for a single connection.

$$\lambda_p = \lambda_b \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| Connection Type | $\lambda_b$ (F/$10^6$ hrs) |
|---|---|
| Hand Solder, w/o Wrapping | .0026 |
| Hand Solder, w/Wrapping | .00014 |
| Crimp | .00026 |
| Weld | .00005 |
| Solderless Wrap | .0000035 |
| Clip Termination | .00012 |
| Reflow Solder | .000069 |

Quality Factor - $\pi_Q$

| Quality Grade | $\pi_Q$ | Comments |
|---|---|---|
| Crimp Types | | |
| Automated | 1.0 | Daily pull tests recommended. |
| Manual | | |
| Upper | 1.0 | Only MIL-SPEC or equivalent tools and terminals, pull test at beginning and end of each shift, color coded tools and terminations. |
| Standard | 2.0 | MIL-SPEC tools, pull test at beginning of each shift. |
| Lower | 20.0 | Anything less than standard criteria. |
| All Types Except Crimp | 1.0 | |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 7.0 |
| $N_S$ | 4.0 |
| $N_U$ | 11 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 6.0 |
| $A_{UC}$ | 6.0 |
| $A_{UF}$ | 8.0 |
| $A_{RW}$ | 16 |
| $S_F$ | .50 |
| $M_F$ | 9.0 |
| $M_L$ | 24 |
| $C_L$ | 420 |



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## 18.1 METERS, PANEL

**SPECIFICATION**
MIL-M-10304

**DESCRIPTION**
Meter, Electrical Indicating, Panel Type, Ruggedized

$$\lambda_p = \lambda_b \pi_A \pi_F \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| Type | $\lambda_b$ |
|---|---|
| All | .090 |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| MIL-M-10304 | 1.0 |
| Lower | 3.4 |

Application Factor - $\pi_A$

| Application | $\pi_A$ |
|---|---|
| Direct Current | 1.0 |
| Alternating Current | 1.7 |

Function Factor - $\pi_F$

| Function | $\pi_F$ |
|---|---|
| Ammeter | 1.0 |
| Voltmeter | 1.0 |
| Other* | 2.8 |

* Meters whose basic meter movement construction is an ammeter with associated conversion elements.

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 4.0 |
| $G_M$ | 25 |
| $N_S$ | 12 |
| $N_U$ | 35 |
| $A_{IC}$ | 28 |
| $A_{IF}$ | 42 |
| $A_{UC}$ | 58 |
| $A_{UF}$ | 73 |
| $A_{RW}$ | 60 |
| $S_F$ | 1.1 |
| $M_F$ | 60 |
| $M_L$ | N/A |
| $C_L$ | N/A |



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## 19.1 QUARTZ CRYSTALS

**SPECIFICATION**
MIL-C-3098

**DESCRIPTION**
Crystal Units, Quartz

$$\lambda_p = \lambda_b \pi_Q \pi_E \text{ Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| Frequency, f(MHz) | $\lambda_b$ |
|---|---|
| 0.5 | .011 |
| 1.0 | .013 |
| 5.0 | .019 |
| 10 | .022 |
| 15 | .024 |
| 20 | .026 |
| 25 | .027 |
| 30 | .028 |
| 35 | .029 |
| 40 | .030 |
| 45 | .031 |
| 50 | .032 |
| 55 | .033 |
| 60 | .033 |
| 65 | .034 |
| 70 | .035 |
| 75 | .035 |
| 80 | .036 |
| 85 | .036 |
| 90 | .037 |
| 95 | .037 |
| 100 | .037 |
| 105 | .038 |
| $\lambda_b = .013(f)^{.23}$ | |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 3.0 |
| $G_M$ | 10 |
| $N_S$ | 6.0 |
| $N_U$ | 16 |
| $A_{IC}$ | 12 |
| $A_{IF}$ | 17 |
| $A_{UC}$ | 22 |
| $A_{UF}$ | 28 |
| $A_{RW}$ | 23 |
| $S_F$ | .50 |
| $M_F$ | 13 |
| $M_L$ | 32 |
| $C_L$ | 500 |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| MIL-SPEC | 1.0 |
| Lower | 2.1 |



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**SPECIFICATION**
MIL-L-6363
W-L-111

**DESCRIPTION**
Lamps, Incandescent, Aviation Service
Lamps, Incandescent, Miniature, Tungsten-Filament

$$\lambda_p = \lambda_b \pi_U \pi_A \pi_E \text{ Failures/}10^6 \text{ Hours}$$

**APPLICATION NOTE:** The data used to develop this model included randomly occurring catastrophic failures and failures due to tungsten filament wearout.

Base Failure Rate - $\lambda_b$

| Rated Voltage, $V_r$ (Volts) | $\lambda_b$ |
|---|---|
| 5 | .59 |
| 6 | .75 |
| 12 | 1.8 |
| 14 | 2.2 |
| 24 | 4.5 |
| 28 | 5.4 |
| 37.5 | 7.9 |
| $\lambda_b = .074(V_r)^{1.29}$ | |

Utilization Factor - $\pi_U$

| Utilization (Illuminate Hours/ Equipment Operate Hours) | $\pi_U$ |
|---|---|
| < 0.10 | 0.10 |
| 0.10 to 0.90 | 0.72 |
| > 0.90 | 1.0 |

Application Factor - $\pi_A$

| Application | $\pi_A$ |
|---|---|
| Alternating Current | 1.0 |
| Direct Current | 3.3 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 3.0 |
| $N_S$ | 3.0 |
| $N_U$ | 4.0 |
| $A_{IC}$ | 4.0 |
| $A_{IF}$ | 4.0 |
| $A_{UC}$ | 5.0 |
| $A_{UF}$ | 6.0 |
| $A_{RW}$ | 5.0 |
| $S_F$ | .70 |
| $M_F$ | 4.0 |
| $M_L$ | 6.0 |
| $C_L$ | 27 |



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## 21.1 ELECTRONIC FILTERS, NON-TUNABLE

**SPECIFICATION**
MIL-F-15733
MIL-F-18327

**DESCRIPTION**
Filters, Radio Frequency Interference
Filters, High Pass, Low Pass, Band Pass, Band Suppression, and Dual Functioning (Non-tunable)

The most accurate way to estimate the failure rate for electronic filters is to sum the failure rates for the individual components which make up the filter (e.g., IC's, diodes, resistors, etc.) using the appropriate models provided in this Handbook. The Parts Stress models or the Parts Count method given in Appendix A can be used to determine individual component failure rates. If insufficient information is available then the following default model can be used.

$$\lambda_p = \lambda_b \pi_Q \pi_E \quad \text{Failures/}10^6 \text{ Hours}$$

Base Failure Rate - $\lambda_b$

| Type | $\lambda_b$ |
|---|---|
| MIL-F-15733, Ceramic-Ferrite Construction (Styles FL 10-16, 22, 24, 30-32, 34, 35, 38, 41-43, 45, 47-50, 61-65, 70, 81-93, 95, 96) | .022 |
| MIL-F-15733, Discrete LC Components, (Styles FL 37, 53, 74) | .12 |
| MIL-F-18327, Discrete LC Components (Composition 1) | .12 |
| MIL-F-18327, Discrete LC and Crystal Components (Composition 2) | .27 |

Quality Factor - $\pi_Q$

| Quality | $\pi_Q$ |
|---|---|
| MIL-SPEC | 1.0 |
| Lower | 2.9 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 6.0 |
| $N_S$ | 4.0 |
| $N_U$ | 9.0 |
| $A_{IC}$ | 7.0 |
| $A_{IF}$ | 9.0 |
| $A_{UC}$ | 11 |
| $A_{UF}$ | 13 |
| $A_{RW}$ | 11 |
| $S_F$ | .80 |
| $M_F$ | 7.0 |
| $M_L$ | 15 |
| $C_L$ | 120 |



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## 22.1 FUSES

**SPECIFICATION**
W-F-1726
W-F-1814
MIL-F-5372
ML-F-23419
MIL-F-15160

**DESCRIPTION**
Fuse, Cartridge Class H
Fuse, Cartridge, High Interrupting Capacity
Fuse, Current Limiter Type, Aircraft
Fuse, Instrument Type
Fuse, Instrument, Power and Telephone (Nonindicating), Style F01

$$\lambda_p = \lambda_b \pi_E \text{ Failures/}10^6 \text{ Hours}$$

**APPLICATION NOTE:** The reliability modeling of fuses presents a unique problem. Unlike most other components, there is very little correlation between the number of fuse replacements and actual fuse failures. Generally when a fuse opens, or "blows," something else in the circuit has created an overload condition and the fuse is simply functioning as designed. This model is based on life test data and represents fuse open and shorting failure modes due primarily to mechanical fatigue and corrosion. A short failure mode is most commonly caused by electrically conductive material shorting the fuse terminals together causing a failure to open condition when rated current is exceeded.

Base Failure Rate - $\lambda_b$

| Type | $\lambda_b$ |
|---|---|
| W-F-1726, W-F-1814, MIL-F-5372, MIL-F-23419, ML-F-15160 | .010 |

Environment Factor - $\pi_E$

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 8.0 |
| $N_S$ | 5.0 |
| $N_U$ | 11 |
| $A_{IC}$ | 9.0 |
| $A_{IF}$ | 12 |
| $A_{UC}$ | 15 |
| $A_{UF}$ | 18 |
| $A_{RW}$ | 16 |
| $S_F$ | .90 |
| $M_F$ | 10 |
| $M_L$ | 21 |
| $C_L$ | 230 |



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## 23.1 MISCELLANEOUS PARTS

$\lambda_p$ - Failure Rates for Miscellaneous Parts (Failures/$10^6$ Hours)

| Part Type | Failure Rate |
|---|---|
| Vibrators (MIL-V-95) | |
| 60-cycle | 15 |
| 120-cycle | 20 |
| 400-cycle | 40 |
| Lamps | |
| Neon Lamps | 0.20 |
| Fiber Optic Cables (Single Fiber Types Only) | 0.1 (Per Fiber Km) |
| Single Fiber Optic Connectors* | 0.10 |
| Microwave Elements (Coaxial & Waveguide) | |
| Attenuators (Fixed & Variable) | See Resistors, Type RD |
| Fixed Elements (Directional Couplers, Fixed Stubs & Cavities) | Negligible |
| Variable Elements (Tuned Stubs & Cavities) | 0.10 |
| Microwave Ferrite Devices | |
| Isolators & Circulators (≤100W) | 0.10 x $\pi_E$ |
| Isolators & Circulators (>100W) | 0.20 x $\pi_E$ |
| Phase Shifter (Latching) | 0.10 x $\pi_E$ |
| Dummy Loads | |
| < 100W | 0.010 x $\pi_E$ |
| 100W to ≤ 1000W | 0.030 x $\pi_E$ |
| > 1000W | 0.10 x $\pi_E$ |
| Terminations (Thin or Thick Film Loads Used in Stripline and Thin Film Circuits) | 0.030 x $\pi_E$ |

*Caution: Excessive Mating-Demating Cycles May Seriously Degrade Reliability



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## 23.1 MISCELLANEOUS PARTS

Environment Factor - $\pi_E$
(Microwave Ferrite Devices)

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 8.0 |
| $N_S$ | 5.0 |
| $N_U$ | 12 |
| $A_{IC}$ | 5.0 |
| $A_{IF}$ | 8.0 |
| $A_{UC}$ | 7.0 |
| $A_{UF}$ | 11 |
| $A_{RW}$ | 17 |
| $S_F$ | .50 |
| $M_F$ | 9.0 |
| $M_L$ | 24 |
| $C_L$ | 450 |

Environment Factor - $\pi_E$
(Dummy Loads)

| Environment | $\pi_E$ |
|---|---|
| $G_B$ | 1.0 |
| $G_F$ | 2.0 |
| $G_M$ | 10 |
| $N_S$ | 5.0 |
| $N_U$ | 17 |
| $A_{IC}$ | 6.0 |
| $A_{IF}$ | 8.0 |
| $A_{UC}$ | 14 |
| $A_{UF}$ | 22 |
| $A_{RW}$ | 25 |
| $S_F$ | .50 |
| $M_F$ | 14 |
| $M_L$ | 36 |
| $C_L$ | 660 |



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## APPENDIX A: PARTS COUNT RELIABILITY PREDICTION

**Parts Count Reliability Prediction** - This prediction method is applicable during bid proposal and early design phases when insufficient information is available to use the part stress analysis models shown in the main body of this Handbook. The information needed to apply the method is (1) generic part types (including complexity for microcircuits) and quantities, (2) part quality levels, and (3) equipment environment. The equipment failure rate is obtained by looking up a generic failure rate in one of the following tables, multiplying it by a quality factor, and then summing it with failure rates obtained for other components in the equipment. The general mathematical expression for equipment failure rate with this method is:

$$\lambda_{EQUIP} = \sum_{i=1}^{i=n} N_i (\lambda_g \pi_Q)_i \qquad \text{Equation 1}$$

for a given equipment environment where:

$\lambda_{EQUIP}$ = Total equipment failure rate (Failures/$10^6$ Hours)

$\lambda_g$ = Generic failure rate for the $i^{th}$ generic part (Failures/$10^6$ Hours)

$\pi_Q$ = Quality factor for the $i^{th}$ generic part

$N_i$ = Quantity of $i^{th}$ generic part

n = Number of different generic part categories in the equipment

Equation 1 applies if the entire equipment is being used in one environment. If the equipment comprises several units operating in different environments (such as avionics systems with units in airborne inhabited ($A_I$) and uninhabited ($A_U$) environments), then Equation 1 should be applied to the portions of the equipment in each environment. These "environment-equipment" failure rates should be added to determine total equipment failure rate. Environmental symbols are defined in Section 3.

The quality factors to be used with each part type are shown with the applicable $\lambda_g$ tables and are not necessarily the same values that are used in the Part Stress Analysis. Microcircuits have an additional multiplying factor, $\pi_L$, which accounts for the maturity of the manufacturing process. For devices in production two years or more, no modification is needed. For those in production less than two years, $\lambda_g$ should be multiplied by the appropriate $\pi_L$ factor (See page A-4).

It should be noted that no generic failure rates are shown for hybrid microcircuits. Each hybrid is a fairly unique device. Since none of these devices have been standardized, their complexity cannot be determined from their name or function. Identically or similarly named hybrids can have a wide range of complexity that thwarts categorization for purposes of this prediction method. If hybrids are anticipated for a design, their use and construction should be thoroughly investigated on an individual basis with application of the prediction model in Section 5.

The failure rates shown in this Appendix were calculated by assigning model default values to the failure rate models of Section 5 through 23. The specific default values used for the model parameters are shown with the $\lambda_g$ Tables for microcircuits. Default parameters for all other part classes are summarized in the tables starting on Page A-12. For parts with characteristics which differ significantly from the assumed defaults, or parts used in large quantities, the underlying models in the main body of this Handbook can be used.



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**Generic Failure Rate, $\lambda_g$ (Failures/$10^6$ Hours) for Microcircuits. See Page A-4 for $\pi_Q$ Values**

**(Defaults: $\pi_T$ Based on Ea Shown, Solder or Weld Seal DIPs/PGAs (No. Pins as Shown Below), $\pi_L$ = 1 (Device in Production ≥ 2 Yr.))**

| Section # | Part Type | Environ. → $T_J$ (°C) → | $G_B$ 50 | $G_F$ 60 | $G_M$ 65 | $N_S$ 60 | $N_U$ 65 | $A_{IC}$ 75 | $A_F$ 75 | $A_{UC}$ 90 | $A_{UF}$ 90 | $A_{RW}$ 75 | $S_F$ 50 | $M_F$ 65 | $M_L$ 75 | $C_L$ 60 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 5.1 | Bipolar Technology | | | | | | | | | | | | | | | |
| | Gate/Logic Arrays, Digital (Ea = .4) | | | | | | | | | | | | | | | |
| | 1 - 100 Gates | (16 Pin DIP) | .0036 | .012 | .024 | .024 | .035 | .025 | .030 | .032 | .049 | .047 | .0036 | .030 | .069 | 1.2 |
| | 101 - 1000 Gates | (24 Pin DIP) | .0060 | .020 | .038 | .037 | .055 | .039 | .048 | .051 | .077 | .074 | .0060 | .046 | .11 | 1.9 |
| | 1001 to 3000 Gates | (40 Pin DIP) | .011 | .035 | .066 | .065 | .097 | .070 | .085 | .091 | .14 | .13 | .011 | .082 | .19 | 3.3 |
| | 3001 to 10,000 Gates | (128 Pin PGA) | .033 | .12 | .22 | .22 | .33 | .23 | .28 | .30 | .46 | .44 | .033 | .28 | .65 | 12 |
| | 10,000 to 30,000 Gates | (180 Pin PGA) | .052 | .17 | .33 | .33 | .48 | .34 | .42 | .45 | .68 | .65 | .052 | .41 | .95 | 17 |
| | 30,000 to 60,000 Gates | (224 Pin PGA) | .075 | .23 | .44 | .43 | .63 | .46 | .56 | .61 | .90 | .85 | .075 | .53 | 1.2 | 21 |
| 5.1 | Gate/Logic Arrays, Linear (Ea = .65) | | | | | | | | | | | | | | | |
| | 1 - 100 Transistors | (14 Pin DIP) | .0095 | .024 | .039 | .034 | .049 | .057 | .062 | .12 | .13 | .076 | .0095 | .044 | .096 | 1.1 |
| | 101 - 300 Transistors | (16 Pin DIP) | .017 | .041 | .065 | .054 | .078 | .10 | .11 | .22 | .24 | .13 | .017 | .072 | .15 | 1.4 |
| | 301 - 1000 Transistors | (24 Pin DIP) | .033 | .074 | .11 | .092 | .13 | .19 | .19 | .41 | .44 | .22 | .033 | .12 | .26 | 2.0 |
| | 1001 - 10,000 Transistors | (40 Pin DIP) | .050 | .12 | .18 | .15 | .21 | .29 | .30 | .63 | .67 | .35 | .050 | .19 | .41 | 3.4 |
| 5.1 | Programmable Logic Arrays (Ea = .4) | | | | | | | | | | | | | | | |
| | Up to 200 Gates | (16 Pin DIP) | .0061 | .016 | .029 | .027 | .040 | .032 | .037 | .044 | .061 | .054 | .0061 | .034 | .076 | 1.2 |
| | 201 to 1000 Gates | (24 Pin DIP) | .011 | .028 | .048 | .045 | .065 | .054 | .063 | .077 | .10 | .089 | .011 | .057 | .12 | 1.9 |
| | 1001 to 5000 Gates | (40 Pin DIP) | .022 | .052 | .087 | .082 | .12 | .099 | .11 | .14 | .19 | .16 | .022 | .10 | .22 | 3.3 |
| | MOS Technology | | | | | | | | | | | | | | | |
| 5.1 | Gate/Logic Arrays, Digital (Ea = .35) | | | | | | | | | | | | | | | |
| | 1 to 100 Gates | (16 Pin DIP) | .0057 | .015 | .027 | .027 | .039 | .029 | .035 | .039 | .056 | .052 | .0057 | .033 | .074 | 1.2 |
| | 101 to 1000 Gates | (24 Pin DIP) | .010 | .026 | .045 | .043 | .062 | .049 | .057 | .066 | .092 | .083 | .010 | .053 | .12 | 1.9 |
| | 1001 to 3000 Gates | (40 Pin DIP) | .019 | .047 | .080 | .077 | .11 | .088 | .10 | .12 | .17 | .15 | .019 | .095 | .21 | 3.3 |
| | 3001 to 10,000 Gates | (128 Pin PGA) | .049 | .14 | .25 | .24 | .36 | .27 | .32 | .36 | .51 | .48 | .049 | .30 | .69 | 12 |
| | 10,001 to 30,000 Gates | (180 Pin PGA) | .084 | .22 | .39 | .37 | .54 | .42 | .49 | .56 | .79 | .72 | .084 | .46 | 1.0 | 17 |
| | 30,000 to 60,000 Gates | (224 Pin PGA) | .13 | .31 | .53 | .51 | .73 | .59 | .69 | .82 | 1.1 | .98 | .13 | .63 | 1.4 | 21 |
| 5.1 | Gate/Logic Arrays, Linear (Ea = .65) | | | | | | | | | | | | | | | |
| | 1 to 100 Transistors | (14 Pin DIP) | .0095 | .024 | .039 | .034 | .049 | .057 | .062 | .12 | .13 | .076 | .0095 | .044 | .096 | 1.1 |
| | 101 to 300 Transistors | (16 Pin DIP) | .017 | .041 | .065 | .054 | .078 | .10 | .11 | .22 | .24 | .13 | .017 | .072 | .15 | 1.4 |
| | 301 to 1,000 Transistors | (24 Pin DIP) | .033 | .074 | .11 | .092 | .13 | .19 | .19 | .41 | .44 | .22 | .033 | .12 | .26 | 2.0 |
| | 1001 to 10,000 Transistors | (40 Pin DIP) | .05 | .12 | .18 | .15 | .21 | .29 | .30 | .63 | .67 | .35 | .05 | .19 | .41 | 3.4 |
| 5.1 | Floating Gate Programmable Logic Array, MOS (Ea = .35) | | | | | | | | | | | | | | | |
| | Up to 16K Cells | (24 Pin DIP) | .0046 | .018 | .035 | .035 | .052 | .035 | .044 | .044 | .070 | .070 | .0046 | .044 | .10 | 1.9 |
| | 16K to 64K Cells | (28 Pin DIP) | .0058 | .021 | .042 | .042 | .062 | .042 | .052 | .053 | .084 | .083 | .0058 | .052 | .12 | 2.3 |
| | 64K to 256K Cells | (28 Pin DIP) | .0061 | .022 | .043 | .042 | .063 | .043 | .054 | .055 | .086 | .084 | .0061 | .053 | .13 | 2.3 |
| | 256K to 1M Cells | (40 Pin DIP) | .0095 | .033 | .064 | .063 | .094 | .065 | .080 | .083 | .13 | .13 | .0095 | .079 | .19 | 3.3 |
| 5.1 | Microprocessors, Bipolar (Ea = .4) | | | | | | | | | | | | | | | |
| | Up to 8 Bits | (40 Pin DIP) | .028 | .061 | .098 | .091 | .13 | .12 | .13 | .17 | .22 | .18 | .028 | .11 | .24 | 3.3 |
| | Up to 16 Bits | (64 Pin PGA) | .052 | .11 | .18 | .16 | .23 | .21 | .24 | .32 | .39 | .31 | .052 | .20 | .41 | 5.6 |
| | Up to 32 Bits | (128 Pin PGA) | .11 | .23 | .36 | .33 | .47 | .44 | .49 | .65 | .81 | .65 | .11 | .42 | .86 | 12 |
| 5.1 | Microprocessors, MOS (Ea = .35) | | | | | | | | | | | | | | | |
| | Up to 8 Bits | (40 Pin DIP) | .048 | .089 | .13 | .12 | .16 | .16 | .17 | .24 | .28 | .22 | .048 | .15 | .28 | 3.4 |
| | Up to 16 Bits | (64 Pin PGA) | .093 | .17 | .24 | .22 | .29 | .30 | .32 | .45 | .52 | .40 | .093 | .27 | .50 | 5.6 |
| | Up to 32 Bits | (128 Pin PGA) | .19 | .34 | .49 | .45 | .60 | .61 | .66 | .90 | 1.1 | .82 | .19 | .54 | 1.0 | 12 |



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Generic Failure Rate, $\lambda_g$ (Failures/$10^6$ Hours) for Microcircuits. See Page A-4 for $\pi_Q$ Values

(Defaults: $\pi_T$ Based on Ea Shown, Solder or Weld Seal DIPs/PGAs (No. Pins as Shown Below), $\pi_L$ = 1 (Device in Production ≥ 2 Yr.))

| Section # | Part Type | Environ.→ $T_J$(°C)→ | $C_B$ 50 | $C_F$ 60 | $C_M$ 65 | $N_S$ 60 | $N_U$ 65 | $A_{MC}$ 75 | $A_F$ 75 | $A_{UC}$ 90 | $A_{UF}$ 90 | $A_{RW}$ 75 | $S_F$ 50 | $M_F$ 65 | $M_L$ 75 | $C_L$ 60 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 5.2 | MOS Technology Memories, ROM (Ea = .6) | | | | | | | | | | | | | | | |
| | Up to 16K | (24 Pin DIP) | .0047 | .018 | .036 | .035 | .053 | .037 | .045 | .048 | .074 | .071 | .0047 | .044 | .11 | 1.9 |
| | 16K to 64K | (28 Pin DIP) | .0059 | .022 | .043 | .042 | .063 | .045 | .055 | .060 | .090 | .086 | .0059 | .053 | .13 | 2.3 |
| | 64K to 256K | (28 Pin DIP) | .0067 | .023 | .045 | .044 | .066 | .048 | .059 | .068 | .089 | .089 | .0067 | .055 | .13 | 2.3 |
| | 256K to 1 MB | (40 Pin DIP) | .011 | .036 | .068 | .066 | .098 | .075 | .090 | .11 | .15 | .14 | .011 | .083 | .20 | 3.3 |
| 5.2 | Memories, PROM, UVEPROM, EEPROM, EAPROM (Ea = .6) NOTE: $\lambda_{cyc}$ = 0 Assumed for EEPROM | | | | | | | | | | | | | | | |
| | Up to 16K | (24 Pin DIP) | .0049 | .018 | .038 | .036 | .053 | .037 | .046 | .049 | .075 | .072 | .0048 | .045 | .11 | 1.9 |
| | 16K to 64K | (28 Pin DIP) | .0061 | .022 | .044 | .043 | .064 | .046 | .056 | .062 | .093 | .087 | .0062 | .054 | .13 | 2.3 |
| | 64K to 256K | (28 Pin DIP) | .0072 | .024 | .046 | .045 | .067 | .051 | .061 | .073 | .10 | .092 | .0072 | .057 | .13 | 2.3 |
| | 256K to 1 MB | (40 Pin DIP) | .012 | .038 | .071 | .068 | .10 | .080 | .095 | .12 | .16 | .14 | .012 | .088 | .20 | 3.3 |
| 5.2 | Memories, DRAM (Ea = .6) | | | | | | | | | | | | | | | |
| | Up to 16K | (18 Pin DIP) | .0040 | .014 | .027 | .027 | .040 | .029 | .035 | .040 | .059 | .055 | .0040 | .034 | .080 | 1.4 |
| | 16K to 64K | (22 Pin DIP) | .0055 | .019 | .036 | .034 | .051 | .039 | .047 | .058 | .079 | .070 | .0055 | .043 | .10 | 1.7 |
| | 64K to 256K | (24 Pin DIP) | .0074 | .023 | .043 | .040 | .060 | .049 | .058 | .076 | .10 | .084 | .0074 | .051 | .12 | 1.9 |
| | 256K to 1 MB | (28 Pin DIP) | .011 | .032 | .057 | .053 | .077 | .070 | .080 | .12 | .15 | .11 | .011 | .067 | .15 | 2.3 |
| 5.2 | Memories, SRAM, (MOS & BiMOS) (Ea = .6) | | | | | | | | | | | | | | | |
| | Up to 16K | (18 Pin DIP) | .0079 | .022 | .038 | .034 | .050 | .048 | .054 | .063 | .10 | .073 | .0079 | .044 | .098 | 1.4 |
| | 16K to 64K | (22 Pin DIP) | .014 | .034 | .057 | .050 | .073 | .077 | .085 | .14 | .17 | .11 | .014 | .065 | .14 | 1.8 |
| | 64K to 256K | (24 Pin DIP) | .023 | .053 | .084 | .071 | .10 | .12 | .13 | .25 | .27 | .16 | .023 | .092 | .19 | 1.9 |
| | 256K to 1 MB | (28 Pin DIP) | .043 | .092 | .14 | .11 | .16 | .22 | .23 | .46 | .49 | .28 | .043 | .15 | .30 | 2.3 |
| 5.2 | Bipolar Technology Memories, ROM, PROM (Ea = .6) | | | | | | | | | | | | | | | |
| | Up to 16K | (24 Pin DIP) | .010 | .028 | .050 | .048 | .067 | .062 | .070 | .10 | .13 | .098 | .010 | .058 | .13 | 1.9 |
| | 16K to 64K | (28 Pin DIP) | .017 | .043 | .071 | .063 | .091 | .095 | .11 | .18 | .21 | .14 | .017 | .081 | .18 | 2.3 |
| | 64K to 256K | (28 Pin DIP) | .028 | .065 | .10 | .085 | .12 | .15 | .16 | .30 | .33 | .19 | .028 | .11 | .23 | 2.3 |
| | 256K to 1 MB | (40 Pin DIP) | .053 | .12 | .18 | .15 | .21 | .27 | .29 | .56 | .61 | .33 | .053 | .19 | .39 | 3.4 |
| 5.2 | Memories, SRAM (Ea = .6) | | | | | | | | | | | | | | | |
| | Up to 16K | (24 Pin DIP) | .0075 | .023 | .043 | .041 | .060 | .050 | .058 | .077 | .10 | .084 | .0075 | .052 | .12 | 1.9 |
| | 16K to 64K | (28 Pin DIP) | .012 | .033 | .058 | .054 | .079 | .072 | .083 | .12 | .15 | .11 | .012 | .069 | .15 | 2.3 |
| | 64K to 256K | (28 Pin DIP) | .018 | .045 | .074 | .065 | .085 | .10 | .11 | .19 | .22 | .14 | .018 | .084 | .18 | 2.3 |
| | 256K to 1 MB | (40 Pin DIP) | .033 | .079 | .13 | .11 | .16 | .18 | .20 | .35 | .39 | .24 | .033 | .14 | .30 | 3.4 |
| 5.3 | VHSIC Microcircuits, CMOS | | | | | | Refer to Section 5.3, VHSIC CMOS | | | | | | | | | |
| 5.4 | GaAs MMIC (Ea = 1.5) | | | | | | | | | | | | | | | |
| | 1 to 10 Active and/or Passive Elements | (8 Pin DIP) | .019 | .034 | .046 | .039 | .052 | .065 | .088 | .11 | .12 | .076 | .019 | .049 | .086 | .61 |
| | 11 to 100 Active and/or Passive Elements (Default: Driver and High Power (> 100 mW)) | (16 Pin DIP) | .025 | .047 | .067 | .058 | .079 | .091 | .097 | .15 | .17 | .11 | .025 | .073 | .14 | 1.3 |
| 5.4 | GaAs Digital (Ea = 1.4) | | | | | | | | | | | | | | | |
| | 1 to 1000 Active and/or Passive Elements | (36 Pin DIP) | .0085 | .030 | .057 | .057 | .084 | .060 | .073 | .080 | .12 | .11 | .0085 | .071 | .17 | 3.0 |
| | 1001 to 10,000 Active and/or Passive Elements | (64 Pin PGA) | .014 | .053 | .10 | .10 | .15 | .11 | .13 | .14 | .22 | .21 | .014 | .13 | .31 | 5.5 |



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Quality Factors - $\pi_Q$

| Description | $\pi_Q$ |
|---|---|
| **Class S Categories:**<br>1. Procured in full accordance with MIL-M-38510, Class S requirements.<br>2. Procured in full accordance with MIL-I-38535 and Appendix B thereto (Class U).<br>3. Hybrids: (Procured to Class S requirements (Quality Level K) of MIL-H-38534. | .25 |
| **Class B Categories:**<br>1. Procured in full accordance with MIL-M-38510, Class B requirements.<br>2. Procured in full accordance with MIL-I-38535, (Class Q).<br>3. Hybrids: Procured to Class B requirements (Quality Level H) of MIL-H-38534. | 1.0 |
| **Class B-1 Category:**<br>Fully compliant with all requirements of paragraph 1.2.1 of MIL-STD-883 and procured to a MIL drawing, DESC drawing or other government approved documentation. (Does not include hybrids). For hybrids use custom screening section below. | 2.0 |

Learning Factor - $\pi_L$

| Years in Production, Y | $\pi_L$ |
|---|---|
| ≤ .1 | 2.0 |
| .5 | 1.8 |
| 1.0 | 1.5 |
| 1.5 | 1.2 |
| ≥ 2.0 | 1.0 |

$\pi_L = .01 \exp(5.35 - .35Y)$

Y = Years generic device type has been in production

Quality Factors (cont'd): $\pi_Q$ Calculation for Custom Screening Programs

| Group | MIL-STD-883 Screen/Test (Note 3) | Point Valuation |
|---|---|---|
| 1* | TM 1010 (Temperature Cycle, Cond B Minimum) and TM 2001 (Constant Acceleration, Cond B Minimum) and TM 5004 (or 5008 for Hybrids) (Final Electricals @ Temp Extremes) and TM 1014 (Seal Test, Cond A, B, or C) and TM 2009 (External Visual) | 60 |
| 2* | TM 1010 (Temperature Cycle, Cond B Minimum) or TM 2001 (Constant Acceleration, Cond B Minimum)<br>TM 5004 (or 5008 for Hybrids) (Final Electricals @ Temp Extremes) and TM 1014 (Seal Test, Cond A, B, or C) and TM 2009 (External Visual) | 37 |
| 3 | Pre-Burn In Electricals<br>TM 1015 (Burn-In B-Level/S-Level) and TM 5004 (or 5008 for Hybrids)<br>(Post Burn-In Electricals @ Temp Extremes) | 30 (B Level)<br>36 (S Level) |
| 4* | TM 2020 Pind (Particle Impact Noise Detection) | 11 |
| 5 | TM 5004 (or 5008 for Hybrids) (Final Electricals @ Temperature Extremes) | 11 (Note 1) |
| 6 | TM 2010/17 (Internal Visual) | 7 |
| 7* | TM 1014 (Seal Test, Cond A, B, or C) | 7 (Note 2) |
| 8 | TM 2012 (Radiography) | 7 |
| 9 | TM 2009 (External Visual) | 7 (Note 2) |
| 10 | TM 5007/5013 (GaAs) (Wafer Acceptance) | 1 |
| 11 | TM 2023 (Non-Destructive Bond Pull) | 1 |

$$\pi_Q = 2 + \frac{87}{\Sigma \text{ Point Valuations}}$$

*NOT APPROPRIATE FOR PLASTIC PARTS.

NOTES:
1. Point valuation only assigned if used independent of Groups 1, 2 or 3.
2. Point valuation only assigned if used independent of Groups 1 or 2.
3. Sequencing of tests within groups 1, 2 and 3 must be followed.
4. TM refers to the MIL-STD-883 Test Method.
5. Nonhermetic parts should be used only in controlled environments (i.e., $G_B$ and other temperature/humidity controlled environments).

EXAMPLES:

1. Mfg. performs Group 1 test and Class B burn-in: $\pi_Q = 2 + \frac{87}{60+30} = 3.1$
2. Mfg. performs internal visual test, seal test and final electrical test: $\pi_Q = 2 + \frac{87}{7+7+11} = 5.5$

| | |
|---|---|
| Other Commercial or Unknown Screening Levels | $\pi_Q = 10$ |

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Generic Failure Rate - $\lambda_g$ (Failures/$10^6$ Hours) for Discrete Semiconductors

| Section # | Part Type | Env.→ $G_B$ | $G_F$ | $G_M$ | $N_S$ | $N_U$ | $A_{IC}$ | $A_{IF}$ | $A_{UC}$ | $A_{UF}$ | $A_{RW}$ | $S_F$ | $M_F$ | $M_L$ | $C_L$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | $T_J$ (°C) → 50 | 60 | 65 | 60 | 65 | 75 | 75 | 90 | 90 | 75 | 50 | 65 | 75 | 60 |
| | DIODES | | | | | | | | | | | | | | |
| 6.1 | General Purpose Analog | .0036 | .028 | .049 | .043 | .10 | .092 | .21 | .20 | .44 | .17 | .0018 | .076 | .23 | 1.5 |
| 6.1 | Switching | .00094 | .0075 | .013 | .011 | .027 | .024 | .054 | .054 | .12 | .045 | .00047 | .020 | .060 | .40 |
| 6.1 | Fast Recovery Pwr. Rectifier | .065 | .52 | .89 | .78 | 1.9 | 1.7 | 3.7 | 3.7 | 8.0 | 3.1 | .032 | 1.4 | 4.1 | 28 |
| 6.1 | Power Rectifier/ Schottky Pwr. | .0028 | .022 | .039 | .034 | .082 | .073 | .16 | .16 | .35 | .13 | .0014 | .060 | .18 | 1.2 |
| 6.1 | Transient Suppressor/Varistor | .0029 | .023 | .040 | .035 | .084 | .075 | .17 | .17 | .36 | .14 | .0015 | .062 | .18 | 1.2 |
| 6.1 | Voltage Ref/Reg. (Avalanche and Zener) | .0033 | .024 | .039 | .035 | .082 | .066 | .15 | .13 | .27 | .12 | .0016 | .060 | .16 | 1.3 |
| 6.1 | Current Regulator | .0056 | .040 | .066 | .060 | .14 | .11 | .25 | .22 | .46 | .21 | .0028 | .10 | .28 | 2.1 |
| 6.2 | Si Impatt (f ≤ 35 GHz) | .86 | 2.8 | 8.9 | 5.6 | 20 | 11 | 14 | 36 | 62 | 44 | .43 | 16 | 67 | 350 |
| 6.2 | Gunn/Bulk Effect | .31 | .76 | 2.1 | 1.5 | 4.6 | 2.0 | 2.5 | 4.5 | 7.6 | 7.9 | .16 | 3.7 | 12 | 94 |
| 6.2 | Tunnel and Back | .004 | .0096 | .0026 | .0019 | .058 | .025 | .032 | .057 | .097 | .10 | .002 | .048 | .15 | 1.2 |
| 6.2 | PIN | .028 | .068 | .19 | .14 | .41 | .18 | .22 | .40 | .69 | .71 | .014 | .34 | 1.1 | 8.5 |
| 6.2 | Schottky Barrier and Point Contact (200 MHz ≤ f ≤ 35 GHz) | .047 | .11 | .31 | .23 | .68 | .30 | .37 | .67 | 1.1 | 1.2 | .023 | .56 | 1.8 | 14 |
| 6.2 | Varactor | .0043 | .010 | .029 | .021 | .063 | .028 | .034 | .062 | .11 | .11 | .0022 | .052 | .17 | 1.3 |
| 6.10 | Thyristor/SCR | .0025 | .020 | .034 | .030 | .072 | .064 | .14 | .14 | .31 | .12 | .0012 | .053 | .16 | 1.1 |
| | TRANSISTORS | | | | | | | | | | | | | | |
| 6.3 | NPN/PNP (f < 200 MHz) | .00015 | .0011 | .0017 | .0017 | .0037 | .0030 | .0067 | .0060 | .013 | .0056 | .000073 | .0027 | .0074 | .056 |
| 6.3 | Power NPN/PNP (f < 200 MHz) | .0057 | .042 | .069 | .063 | .15 | .12 | .26 | .23 | .50 | .22 | .0029 | .11 | .29 | 2.2 |
| 6.4 | Si FET (f ≤ 400 MHz) | .014 | .099 | .16 | .15 | .34 | .28 | .62 | .53 | 1.1 | .51 | .0069 | .25 | .68 | 5.3 |
| 6.9 | Si FET (f > 400 MHz) | .099 | .24 | .64 | .47 | 1.4 | .61 | .76 | 1.3 | 2.3 | 2.4 | .049 | 1.2 | 3.6 | 30 |
| 6.8 | GaAs FET (P < 100 mW) | .17 | .51 | 1.5 | 1.0 | 3.4 | 1.8 | 2.3 | 5.4 | 9.2 | 7.2 | .083 | 2.8 | 11 | 63 |
| 6.8 | GaAs FET (P ≥ 100 mW) | .42 | 1.3 | 3.9 | 2.5 | 8.5 | 4.5 | 5.6 | 13 | 23 | 18 | .21 | 6.9 | 27 | 160 |
| 6.5 | Unijunction | .016 | .12 | .20 | .18 | .42 | .35 | .80 | .74 | 1.6 | .66 | .0079 | .31 | .88 | 6.4 |
| 6.6 | RF, Low Noise (f > 200 MHz, P < 1W) | .094 | .23 | .63 | .46 | 1.4 | .60 | .75 | 1.3 | 2.3 | 2.4 | .047 | 1.1 | 3.6 | 28 |
| 6.7 | RF, Power (P ≥ 1W) | .074 | .15 | .37 | .29 | .81 | .29 | .37 | .52 | .88 | .037 | .33 | .66 | 1.8 | 18 |



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**Generic Failure Rate - $\lambda_g$ (Failures/$10^6$ Hours) for Discrete Semiconductors (cont'd)**

| Section # | Part Type | Env.→ $G_B$ | $G_F$ | $G_M$ | $N_S$ | $N_U$ | $A_{IC}$ | $A_{IF}$ | $A_{UC}$ | $A_{UF}$ | $A_{RW}$ | $S_F$ | $M_F$ | $M_L$ | $C_L$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | $T_J$ (°C)→ 50 | 60 | 65 | 60 | 65 | 75 | 75 | 90 | 90 | 75 | 50 | 65 | 75 | 60 |
| | OPTO-ELECTRONICS | | | | | | | | | | | | | | |
| 6.11 | Photodetector | .011 | .029 | .083 | .059 | .18 | .084 | .11 | .21 | .35 | .34 | .0057 | .15 | .51 | 3.7 |
| 6.11 | Opto-Isolator | .027 | .070 | .20 | .14 | .43 | .20 | .25 | .49 | .83 | .80 | .013 | .35 | 1.2 | 8.7 |
| 6.11 | Emitter | .00047 | .0012 | .0035 | .0025 | .0077 | .0035 | .0044 | .0086 | .015 | .014 | .00024 | .0063 | .021 | .15 |
| 6.12 | Alphanumeric Display | .0062 | .016 | .045 | .032 | .10 | .046 | .058 | .11 | .19 | .18 | .0031 | .082 | .28 | 2.0 |
| 6.13 | Laser Diode, GaAs/Al GaAs | 5.1 | 16 | 49 | 32 | 110 | 58 | 72 | 100 | 170 | 230 | 2.6 | 87 | 350 | 2000 |
| 6.13 | Laser Diode, In GaAs/In GaAsP | 8.9 | 28 | 85 | 55 | 190 | 100 | 130 | 180 | 300 | 400 | 4.5 | 150 | 600 | 3500 |
| 7 | TUBES | See Section 7 (Includes Receivers, CRTs, Cross Field Amplifiers, Klystrons, TWTs, Magnetrons) | | | | | | | | | | | | | |
| 8 | LASERS | See Section 8 | | | | | | | | | | | | | |

**Discrete Semiconductor Quality Factors - $\pi_Q$**

| Section Number | Part Types | JANTXV | JANTX | JAN | Lower | Plastic |
|---|---|---|---|---|---|---|
| 6.1, 6.3, 6.4, 6.5, 6.10, 6.11, 6.12 | Non-RF Devices/ Opto-Electronics* | .70 | 1.0 | 2.4 | 5.5 | 8.0 |
| 6.2 | High Freq Diodes | .50 | 1.0 | 5.0 | 25 | 50 |
| 6.2 | Schottky Diodes | .50 | 1.0 | 1.8 | 2.5 | ---- |
| 6.6, 6.7, 6.8, 6.9 | RF Transistors | .50 | 1.0 | 2.0 | 5.0 | ---- |
| 6.13 | *Laser Diodes | $\pi_Q$ = 1.0 Hermetic Package<br>= 1.0 Nonhermetic with Facet Coating<br>= 3.3 Nonhermetic without Facet Coating | | | | |



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**Generic Failure Rate, $\lambda_g$ (Failure/$10^6$ Hours) For Resistors**

| Section # | Part Type | Style | MIL-R- | Env. → $G_B$ / $T_A$ (°C) → 30 | $G_F$ 40 | $G_M$ 45 | $N_S$ 40 | $N_U$ 45 | $A_{IC}$ 55 | $A_{IF}$ 55 | $A_{UC}$ 70 | $A_{UF}$ 70 | $A_{RW}$ 55 | $S_F$ 30 | $M_F$ 45 | $M_L$ 55 | $C_L$ 40 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 9.1 | Composition | RCR | 39008 | .00050 | .0022 | .0071 | .0037 | .012 | .0052 | .0065 | .016 | .025 | .025 | .00025 | .0098 | .035 | .36 |
| 9.1 | Composition | RC | 11 | .00050 | .0022 | .0071 | .0037 | .012 | .0052 | .0065 | .016 | .025 | .025 | .00025 | .0098 | .035 | .36 |
| 9.2 | Film, Insulated | RLR | 39017 | .0012 | .0027 | .011 | .0054 | .020 | .0063 | .013 | .018 | .033 | .030 | .00025 | .014 | .044 | .69 |
| 9.2 | Film, Insulated | RL | 22684 | .0012 | .0027 | .011 | .0054 | .020 | .0063 | .013 | .018 | .033 | .030 | .00025 | .014 | .044 | .69 |
| 9.2 | Film, RN (R, C or N) | RNR | 55182 | .0014 | .0031 | .013 | .0061 | .023 | .0072 | .014 | .021 | .038 | .034 | .00028 | .016 | .050 | .78 |
| 9.2 | Film | RN | 10509 | .0014 | .0031 | .013 | .0061 | .023 | .0072 | .014 | .021 | .038 | .034 | .00028 | .016 | .050 | .78 |
| 9.3 | Film, Power | RD | 11804 | .012 | .025 | .13 | .062 | .21 | .078 | .10 | .19 | .24 | .32 | .0060 | .18 | .47 | 8.2 |
| 9.4 | Film, Network | RZ | 83401 | .0023 | .0066 | .031 | .013 | .055 | .022 | .043 | .077 | .15 | .10 | .0011 | .055 | .15 | 1.7 |
| 9.5 | Wirewound, Accurate | RBR | 39005 | .0085 | .018 | .10 | .045 | .16 | .15 | .17 | .30 | .38 | .26 | .0068 | .13 | .37 | 5.4 |
| 9.5 | Wirewound, Accurate | RB | 93 | .0085 | .018 | .10 | .045 | .16 | .15 | .17 | .30 | .38 | .26 | .0068 | .13 | .37 | 5.4 |
| 9.6 | Wirewound, Power | RWR | 39007 | .014 | .031 | .16 | .077 | .26 | .073 | .15 | .19 | .39 | .42 | .0042 | .21 | .62 | 9.4 |
| 9.6 | Wirewound, Power | RW | 26 | .013 | .028 | .15 | .070 | .24 | .066 | .13 | .18 | .35 | .38 | .0038 | .19 | .56 | 8.6 |
| 9.7 | Wirewound, Power, Chassis Mounted | RER | 39009 | .0080 | .018 | .096 | .045 | .15 | .044 | .088 | .12 | .24 | .25 | .0040 | .13 | .37 | 5.5 |
| 9.7 | Wirewound, Power, Chassis Mounted | RE | 18546 | .0080 | .018 | .096 | .045 | .15 | .044 | .088 | .12 | .24 | .25 | .0040 | .13 | .37 | 5.5 |
| 9.8 | Thermistor | RTH | 23648 | .065 | .32 | 1.4 | .71 | 1.6 | .71 | 1.9 | 1.0 | 2.7 | 2.4 | .032 | 1.3 | 3.4 | 62 |
| 9.9 | Wirewound, Variable | RTR | 39015 | .025 | .055 | .35 | .16 | .58 | .16 | .26 | .35 | .59 | 1.1 | .013 | .52 | 1.6 | 24 |
| 9.9 | Wirewound, Variable | RT | 27208 | .025 | .055 | .35 | .16 | .58 | .16 | .26 | .35 | .59 | 1.1 | .013 | .52 | 1.6 | 24 |
| 9.10 | Wirewound, Variable, Precision | RR | 12934 | .33 | .73 | 7.0 | 2.9 | 12 | 3.5 | 5.3 | 7.1 | 9.8 | 23 | .16 | 11 | 33 | 510 |
| 9.11 | Wirewound, Variable, Semiprecision | RA | 19 | .15 | .35 | 3.1 | 1.2 | 5.4 | 1.9 | 2.8 | * | * | 9.0 | .075 | * | * | * |
| 9.11 | Wirewound, Variable, Semiprecision | RK | 39002 | .15 | .35 | 3.1 | 1.2 | 5.4 | 1.9 | 2.8 | * | * | 9.0 | .075 | * | * | * |
| 9.12 | Wirewound, Variable, Power | RP | 22 | .15 | .34 | 2.9 | 1.2 | 5.0 | 1.6 | 2.4 | * | * | 7.6 | .076 | * | * | * |
| 9.13 | Nonwirewound, Variable | RJR | 39035 | .033 | .10 | .50 | .21 | .87 | .19 | .27 | .52 | .79 | 1.5 | .017 | .79 | 2.2 | 35 |
| 9.13 | Nonwirewound, Variable | RJ | 22097 | .033 | .10 | .50 | .21 | .87 | .19 | .27 | .52 | .79 | 1.5 | .017 | .79 | 2.2 | 35 |
| 9.14 | Composition, Variable | RV | 94 | .050 | .11 | 1.1 | .45 | 1.7 | 2.8 | 4.6 | 4.6 | 7.5 | 3.3 | .025 | 1.5 | 4.7 | 67 |
| 9.15 | Nonwirewound, Variable Precision | RQ | 39023 | .043 | .15 | .75 | .35 | 1.3 | .39 | .78 | 1.8 | 2.8 | 2.5 | .021 | 1.2 | 3.7 | 49 |
| 9.15 | Film, Variable | RVC | 23285 | .048 | .16 | .76 | .36 | 1.3 | .36 | .72 | 1.4 | 2.2 | 2.3 | .024 | 1.2 | 3.4 | 52 |

NOTE: 1) * Not Normally used in this Environment
2) $T_A$ = Default Component Ambient Temperature (°C)

| Quality | Established Reliability Styles S | R | P | M | MIL-SPEC | Lower |
|---|---|---|---|---|---|---|
| $\pi_Q$ | .030 | .10 | .30 | 1.0 | 3.0 | 10 |



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Generic Failure Rate, $\lambda_g$ (Failures/$10^6$ Hours) for Capacitors

| Section # | Part Type or Dielectric | Style | MIL-C- | Env.→ $G_B$<br>$T_A$ (°C)→ 30 | $G_F$<br>40 | $G_M$<br>45 | $N_S$<br>40 | $N_U$<br>45 | $A_{IC}$<br>55 | $A_{IF}$<br>55 | $A_{UC}$<br>70 | $A_{UF}$<br>70 | $A_{RW}$<br>55 | $S_F$<br>30 | $M_F$<br>45 | $M_L$<br>55 | $C_L$<br>40 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 10.1 | Paper, By-Pass | CP | 25 | .0036 | .0072 | .033 | .018 | .055 | .023 | .03 | .070 | .13 | .083 | .0018 | .044 | .12 | 2.1 |
| 10.1 | Paper, By-Pass | CA | 12889 | .0039 | .0087 | .042 | .022 | .070 | .035 | .047 | .19 | .35 | .13 | .002 | .056 | .19 | 2.5 |
| 10.2 | Paper/Plastic, Feed-through | CZR | 11693 | .0047 | .0096 | .044 | .034 | .073 | .030 | .040 | .094 | .15 | .11 | .0024 | .058 | .16 | 2.7 |
| 10.3 | Paper/Plastic Film | CPV | 14157 | .0021 | .0042 | .017 | .010 | .030 | .0088 | .013 | .026 | .048 | .044 | .0010 | .023 | .063 | 1.1 |
| 10.3 | Paper/Plastic Film | CQR | 19978 | .0021 | .0042 | .017 | .010 | .030 | .0088 | .013 | .026 | .048 | .044 | .0010 | .023 | .063 | 1.1 |
| 10.4 | Metallized Paper/Plastic | CHR | 39022 | .0029 | .0058 | .023 | .014 | .041 | .012 | .018 | .037 | .066 | .060 | .0014 | .032 | .088 | 1.5 |
| 10.4 | Metallized Plastic/ Plastic | CH | 18312 | .0029 | .0058 | .023 | .014 | .041 | .012 | .018 | .037 | .066 | .060 | .0014 | .032 | .088 | 1.5 |
| 10.5 | Metallized Paper/Plastic | CFR | 55514 | .0041 | .0083 | .042 | .021 | .067 | .026 | .048 | .086 | .14 | .10 | .0020 | .054 | .15 | 2.5 |
| 10.6 | Metallized Plastic | CRH | 83421 | .0023 | .0092 | .019 | .012 | .033 | .0096 | .014 | .034 | .053 | .048 | .0011 | .026 | .07 | 1.2 |
| 10.7 | MICA (Dipped or Molded) | CMR | 39001 | .0005 | .0015 | .0091 | .0044 | .014 | .0068 | .0095 | .054 | .069 | .031 | .00025 | .012 | .046 | .45 |
| 10.7 | MICA (Dipped) | CM | 5 | .0005 | .0015 | .0091 | .0044 | .014 | .0068 | .0095 | .054 | .069 | .031 | .00025 | .012 | .046 | .45 |
| 10.8 | MICA (Button) | CB | 10950 | .018 | .037 | .19 | .094 | .31 | .10 | .14 | .47 | .60 | .46 | .0091 | .25 | .68 | 11 |
| 10.9 | Glass | CYR | 23269 | .00032 | .00096 | .0059 | .0029 | .0094 | .0044 | .0062 | .035 | .045 | .020 | .00016 | .0076 | .030 | .29 |
| 10.9 | Glass | CY | 11272 | .00032 | .00096 | .0059 | .0029 | .0094 | .0044 | .0062 | .035 | .045 | .020 | .00016 | .0076 | .030 | .29 |
| 10.10 | Ceramic (Gen. Purpose) | CK | 11015 | .0036 | .0074 | .034 | .019 | .056 | .015 | .015 | .032 | .048 | .077 | .0014 | .049 | .13 | 2.3 |
| 10.10 | Ceramic (Gen. Purpose) | CKR | 39014 | .0036 | .0074 | .034 | .019 | .056 | .015 | .015 | .032 | .048 | .077 | .0014 | .049 | .13 | 2.3 |
| 10.11 | Ceramic (Temp. Comp.) | CCR | 20 | .00078 | .0022 | .013 | .0056 | .023 | .0077 | .015 | .053 | .12 | .046 | .00039 | .017 | .065 | .68 |
| 10.11 | Ceramic Chip | CDR | 55681 | .00078 | .0022 | .013 | .0056 | .023 | .0077 | .015 | .053 | .12 | .046 | .00039 | .017 | .065 | .68 |
| 10.12 | Tantalum, Solid | CSR | 39003 | .0018 | .0039 | .016 | .0097 | .028 | .0091 | .011 | .034 | .057 | .055 | .00072 | .022 | .066 | 1.0 |
| 10.13 | Tantalum, Non-Solid | CLR | 39006 | .0061 | .013 | .069 | .039 | .11 | .031 | .061 | .13 | .29 | .18 | .0030 | .089 | .26 | 4.0 |
| 10.13 | Tantalum, Non-Solid | CL | 3965 | .0061 | .013 | .069 | .039 | .11 | .031 | .061 | .13 | .29 | .18 | .0030 | .089 | .26 | 4.0 |
| 10.14 | Aluminum Oxide | CUR | 39018 | .024 | .061 | .42 | .18 | .59 | .46 | .55 | 2.1 | 2.6 | 1.2 | .012 | .49 | 1.7 | 21 |
| 10.15 | Aluminum Dry | CE | 62 | .029 | .081 | .58 | .24 | .83 | .73 | .88 | 4.3 | 5.4 | 2.0 | .015 | .68 | 2.8 | 28 |
| 10.16 | Variable, Ceramic | CV | 81 | .08 | .27 | 1.2 | .71 | 2.3 | .69 | 1.1 | 6.2 | 12 | 4.1 | .032 | 1.9 | 5.9 | 65 |
| 10.17 | Variable, Piston | PC | 14409 | .033 | .13 | .62 | .31 | .93 | .21 | .28 | 2.2 | 3.3 | 2.2 | .016 | .93 | 3.2 | 37 |
| 10.18 | Variable, Air Trimmer | CT | 92 | .080 | .33 | 1.6 | .87 | 3.0 | 1.0 | 1.7 | 9.9 | 19 | 6.1 | .032 | 2.5 | 8.9 | 100 |
| 10.19 | Variable, Vacuum | CG | 23183 | 0.4 | 1.3 | 6.7 | 3.6 | 13 | 5.7 | 10 | 58 | 90 | 28 | .20 | • | • | •• |

NOTE: 1) • Not Normally used in this Environment
2) $T_A$ = Default Component Ambient Temperature (°C)

| Quality | Established Reliability Styles S | R | P | M | L | MIL-SPEC | Lower |
|---|---|---|---|---|---|---|---|
| $\pi_Q$ | .030 | .10 | .30 | 1.0 | 3.0 | 3.0 | 10 |

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**Generic Failure Rate, $\lambda_g$ (Failures/$10^6$ Hours) for Inductive and Electromechanical Parts**

| Section # | Part Type | MIL- | Env.→ $G_B$ / $T_A$ (°C)→30 | $G_F$ 40 | $G_M$ 45 | $N_S$ 40 | $N_U$ 45 | $A_{IC}$ 55 | $A_{IF}$ 55 | $A_{UC}$ 70 | $A_{UF}$ 70 | $A_{RW}$ 55 | $S_F$ 30 | $M_F$ 45 | $M_L$ 55 | $C_L$ 40 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | INDUCTIVE DEVICES | | | | | | | | | | | | | | | |
| 11.1 | Low Power Pulse XFMR | T-21038 | .0035 | .023 | .049 | .019 | .065 | .027 | .037 | .041 | .052 | .11 | .0018 | .053 | .16 | 2.3 |
| 11.1 | Audio XFMR | T-27 | .0071 | .046 | .097 | .038 | .13 | .055 | .073 | .081 | .10 | .22 | .0035 | .11 | .31 | 4.7 |
| 11.1 | High Pwr. Pulse and Pwr. XFMR, Filter | T-27 | .023 | .16 | .34 | .13 | .45 | .21 | .27 | .35 | .45 | .82 | .011 | .37 | 1.2 | 16 |
| 11.1 | RF XFMR | T-55631 | .028 | .18 | .39 | .15 | .52 | .22 | .29 | .33 | .42 | .88 | .014 | .42 | 1.2 | 19 |
| 11.2 | RF Coils, Fixed or Molded | C-15305 C-39010 | .0017 | .0073 | .023 | .0091 | .031 | .011 | .015 | .016 | .022 | .052 | .00083 | .25 | .073 | 1.1 |
| 11.2 | RF Coils, Variable | C-15305 | .0033 | .015 | .046 | .018 | .061 | .022 | .03 | .033 | .044 | .10 | .0017 | .05 | .15 | 2.2 |
| | ROTATING DEVICES | | | | | | | | | | | | | | | |
| 12.1 | Motors | | 1.6 | 2.4 | 3.3 | 2.4 | 3.3 | 7.1 | 7.1 | 31 | 31 | 7.1 | 1.6 | * | * | * |
| 12.2 | Synchros | | .07 | .20 | 1.5 | .70 | 2.2 | .78 | 1.2 | 7.9 | 12 | 5.1 | .035 | 1.7 | 7.1 | 68 |
| 12.2 | Resolvers | | .11 | .30 | 2.2 | 1.0 | 3.3 | 1.2 | 1.8 | 12 | 18 | 7.6 | .053 | 2.6 | 11 | 100 |
| | ELAPSED TIME METERS | | | | | | | | | | | | | | | |
| 12.3 | ETM-AC | | 10 | 20 | 120 | 70 | 180 | 50 | 80 | 160 | 250 | 260 | 5.0 | 140 | 380 | * |
| 12.3 | ETM-Inverter Driver | | 15 | 30 | 180 | 105 | 270 | 75 | 120 | 240 | 375 | 390 | 7.5 | 210 | 570 | * |
| 13.3 | ETM-Commutator DC | | 40 | 80 | 480 | 280 | 720 | 200 | 320 | 640 | 1000 | 1040 | 20 | 560 | 1520 | * |
| | RELAYS | | | | | | | | | | | | | | | |
| 13.1 | General Purpose | | .13 | .28 | 2.1 | 1.1 | 3.8 | 1.1 | 1.4 | 1.9 | 2.1 | 7.0 | .066 | 3.5 | 10 | * |
| 13.1 | Contactor, High Current | | .43 | .89 | 6.9 | 3.6 | 12 | 3.4 | 4.4 | 6.2 | 6.7 | 22 | .21 | 11 | 32 | * |
| 13.1 | Latching | | .13 | .28 | 2.1 | 1.1 | 3.8 | 1.1 | 1.4 | 1.9 | 2.1 | 7.0 | .066 | 3.5 | 10 | * |
| 13.1 | Reed | | .11 | .23 | 1.8 | .92 | 3.3 | .96 | 1.2 | 2.1 | 2.3 | 6.3 | .054 | 3.0 | 9.0 | * |
| 13.1 | Thermal, Bi-metal | | .29 | .60 | 4.6 | 2.4 | 8.2 | 2.3 | 2.9 | 4.1 | 4.5 | 15 | .14 | 7.6 | 22 | * |
| 13.1 | Meter Movement | | .88 | 1.8 | 14 | 7.4 | 26 | 7.1 | 9.1 | 13 | 14 | 46 | .44 | 24 | 67 | * |
| 13.2 | Solid State | | .40 | 1.2 | 4.8 | 2.4 | 6.8 | 4.8 | 7.6 | 8.4 | 13 | 9.2 | .16 | 4.8 | 13 | 240 |
| 13.2 | Hybrid and Solid State Time Delay | | .50 | 1.5 | 6.0 | 3.0 | 8.5 | 6.0 | 9.5 | 11 | 16 | 12 | .20 | 6.0 | 17 | 300 |
| | SWITCHES | | | | | | | | | | | | | | | |
| 14.1 | Toggle or Pushbutton | | .0010 | .0030 | .018 | .0080 | .025 | .010 | .018 | .013 | .022 | .046 | .0005 | .025 | .067 | 1.2 |
| 14.2 | Sensitive | S-8805 | .15 | .44 | 2.7 | 1.2 | 4.3 | 1.5 | 2.7 | 1.9 | 3.3 | 6.8 | .074 | 3.7 | 9.9 | 180 |
| 14.3 | Rotary Wafer | S-3786 | .33 | .99 | 5.9 | 2.6 | 9.5 | 3.3 | 5.9 | 4.3 | 7.2 | 15 | .16 | 8.2 | 22 | 390 |
| 14.4 | Thumbwheel | S-22710 | .56 | 1.7 | 10 | 4.5 | 16 | 5.6 | 10 | 7.3 | 12 | 26 | .26 | 14 | 38 | 670 |
| 14.5 | Circuit Breaker, Thermal | C-83383 | .11 | .23 | 1.7 | .91 | 3.1 | .80 | 1.0 | 1.3 | 1.4 | 5.2 | .057 | 2.8 | 7.5 | N/A |
| 14.5 | Circuit Breaker, Magnetic | C-55629 | .060 | .12 | .90 | .48 | 1.6 | .42 | .54 | .66 | .72 | 2.8 | .030 | 1.5 | 4.0 | N/A |
| | CONNECTORS | | | | | | | | | | | | | | | |
| 15.1 | Circular/Rack/Panel | | 0.011 | 0.14 | .11 | .069 | .20 | .059 | .098 | .23 | .34 | .37 | .0054 | .16 | .42 | 6.8 |
| 15.1 | Coaxial | | .012 | .015 | .13 | .075 | .21 | .060 | .10 | .22 | .32 | .38 | .0061 | .16 | .54 | 7.3 |
| 15.2 | Printed Circuit Board Connector | | .0054 | .021 | .055 | .035 | .10 | .059 | .11 | .085 | .16 | .19 | .0027 | .078 | .21 | 3.4 |
| 15.3 | IC Sockets | | .0019 | .0058 | .027 | .012 | .035 | .015 | .023 | .021 | .025 | .048 | .00097 | .027 | .070 | 1.3 |
| 16.1 | Interconnection Assemblies (PCBs) | | .053 | .11 | .37 | .69 | .27 | .27 | .43 | .85 | 1.5 | 1.0 | .027 | .53 | 1.4 | 27 |

NOTE: 1) * Not normally used in this environment
2) $T_A$ = Default Component Ambient Temperature (°C)



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Generic Failure Rate, $\lambda_g$ (Failures/$10^6$ Hours) for Miscellaneous Parts

| Section # | Part Type Dielectric | MIL- | Env.→ $G_B$ / $T_A$ (°C)→30 | $G_F$ 40 | $G_M$ 45 | $N_S$ 40 | $N_U$ 45 | $A_{IC}$ 55 | $A_{IF}$ 55 | $A_{UC}$ 70 | $A_{UF}$ 70 | $A_{RW}$ 55 | $S_F$ 30 | $M_F$ 45 | $M_L$ 55 | $C_L$ 40 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | SINGLE CONNECTIONS | | | | | | | | | | | | | | | |
| 17.1 | Hand Solder, w/o Wrapping | | .0026 | .0052 | .018 | .010 | .029 | .010 | .016 | .016 | .021 | .042 | .0013 | .023 | .062 | 1.1 |
| 17.1 | Hand Solder, w/Wrapping | | .00014 | .00028 | .00098 | .00056 | .0015 | .00056 | .00084 | .00084 | .0011 | .0022 | .00007 | .0013 | .0034 | .059 |
| 17.1 | Crimp | | .00026 | .00052 | .0018 | .0010 | .0029 | .0010 | .0016 | .0016 | .0021 | .0042 | .00013 | .0023 | .0062 | .11 |
| 17.1 | Weld | | .000050 | .000100 | .000350 | .000200 | .000550 | .000200 | .000300 | .000300 | .000400 | .000800 | .000025 | .000450 | .001200 | .021000 |
| 17.1 | Solderless Wrap | | .0000035 | .000007 | .000025 | .000014 | .000039 | .000014 | .000021 | .000021 | .000028 | .000056 | .0000018 | .000031 | .000084 | .0015 |
| 17.1 | Clip Termination | | .00012 | .00024 | .00084 | .00048 | .0013 | .00048 | .00072 | .00072 | .00096 | .0019 | .00006 | .0011 | .0029 | .050 |
| 17.1 | Reflow Solder | | .000069 | .000138 | .000483 | .000276 | .000759 | .000276 | .000414 | .000414 | .000552 | .001104 | .000035 | .000621 | .001656 | .02898 |
| | METERS, PANEL | | | | | | | | | | | | | | | |
| 18.1 | DC Ammeter or Voltmeter | M-10304 | 0.09 | 0.35 | 2.3 | 1.1 | 3.2 | 2.5 | 3.8 | 5.2 | 6.6 | 5.4 | 0.099 | 5.4 | N/A | N/A |
| 18.1 | AC Ammeter or Voltmeter | M-10304 | 0.15 | 0.61 | 3.8 | 1.8 | 5.4 | 4.3 | 6.4 | 8.9 | 11 | 9.2 | 0.17 | 9.2 | N/A | N/A |
| 19.1 | Quartz Crystals | C-3098 | .032 | .096 | .32 | .19 | .51 | .38 | .54 | .70 | .90 | .74 | .016 | .42 | 1.0 | 16 |
| 20.1 | Lamps, Incandescent, AC | | 3.9 | 7.8 | 12 | 12 | 16 | 16 | 16 | 19 | 23 | 19 | 2.7 | 16 | 23 | 100 |
| 20.1 | Lamps, Incandescent, DC | | 13 | 26 | 38 | 38 | 51 | 51 | 51 | 64 | 77 | 64 | 9.0 | 51 | 77 | 350 |
| | ELECTRONIC FILTERS | | | | | | | | | | | | | | | |
| 21.1 | Ceramic-Ferrite | F-15733 | .022 | .044 | .13 | .088 | .20 | .15 | .20 | .24 | .29 | .24 | .018 | .15 | .33 | 2.6 |
| 21.1 | Discrete LC Comp. | F-15733 | .12 | .24 | .72 | .48 | 1.1 | .84 | 1.1 | 1.3 | 1.6 | 1.3 | .096 | .84 | 1.8 | 14 |
| 21.1 | Discrete LC & Crystal Comp. | F-18327 | .27 | .54 | 1.6 | 1.1 | 2.4 | 1.9 | 2.4 | 3.0 | 3.5 | 3.0 | .22 | 1.9 | 4.1 | 32 |
| 22.1 | FUSES | | .010 | .020 | .060 | .050 | .11 | .090 | .12 | .15 | .18 | .16 | .009 | .10 | .21 | 2.3 |



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## $\pi_Q$ Factor for Use with Section 11-22 Devices

| Section # | Part Type | Established Reliability | MIL-SPEC | Non-MIL |
|---|---|---|---|---|
| 11.1, 11.2 | Inductive Devices | .25* | 1.0 | 10 |
| 12.1, 12.2, 12.3 | Rotating Devices | N/A | N/A | N/A |
| 13.1 | Relays, Mechanical | .60 | 3.0 | 9.0 |
| 13.2 | Relays, Solid State and Time Delay (Hybrid & Solid State) | N/A | 1.0 | 4 |
| 14.1, 14.2 | Switches, Toggle, Pushbutton, Sensitive | N/A | 1.0 | 20 |
| 14.3 | Switches, Rotary Wafer | N/A | 1.0 | 50 |
| 14.4 | Switches, Thumbwheel | N/A | 1.0 | 10 |
| 14.5 | Circuit Breakers, Thermal | N/A | 1.0 | 8.4 |
| 15.1, 15.2, 15.3 | Connectors | N/A | 1.0 | 2.0 |
| 16.1 | Interconnection Assemblies | N/A | 1.0 | 2.0 |
| 17.1 | Connections | N/A | N/A | N/A |
| 18.1 | Meters, Panel | N/A | 1.0 | 3.4 |
| 19.1 | Quartz Crystals | N/A | 1.0 | 2.1 |
| 20.1 | Lamps, Incandescent | N/A | N/A | N/A |
| 21.1 | Electronic Filters | N/A | 1.0 | 2.9 |
| 22.1 | Fuses | N/A | N/A | N/A |

* Category applies only to MIL-C-39010 Coils.



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## APPENDIX A: PARTS COUNT

**Default Parameters for Discrete Semiconductors**

| Section # | Part Type | $\lambda_b$ | $\pi_T$ | $\pi_M$ | $\pi_S$ | $\pi_C$ | $\pi_A$ | $\pi_R$ | Comments |
|---|---|---|---|---|---|---|---|---|---|
| 5.0 | MICROCIRCUITS | All Defaults provided with $\lambda_g$ Table | | | | | | | |
| | DIODES | | | | | | | | |
| 6.1 | General Purpose Analog | .0038 | | | .42 | 1.0 | | | Voltage Stress = .7, Metallurgically Bonded Contacts |
| 6.1 | Switching | .001 | | | .42 | 1.0 | | | Voltage Stress = .7, Metallurgically Bonded Contacts |
| 6.1 | Fast Recovery Power Rectifier | .069 | | | .42 | 1.0 | | | Voltage Stress = .7, Metallurgically Bonded Contacts |
| 6.1 | Transient Suppressor/Varistor | .0031 | | | 1.0 | 1.0 | | | Metallurgically Bonded Contacts |
| 6.1 | Power Rectifier | .003 | | | .42 | 1.0 | | | Voltage Stress = .7, Metallurgically Bonded Contacts |
| 6.1 | Voltage Ref/Reg. (Avalanche & Zener) | .002 | | | 1.0 | 1.0 | | | Metallurgically Bonded Contacts |
| 6.1 | Current Regulator | .0034 | | | 1.0 | 1.0 | | | Metallurgically Bonded Contacts |
| 6.2 | Si Impatt (≤ 35 GHz) | .22 | | | | | 1.0 | 1.0 | |
| 6.2 | Gunn/Bulk Effect | .18 | | | | | 1.0 | 1.0 | |
| 6.2 | Tunnel and Back | .0023 | | | | | 1.0 | 1.0 | |
| 6.2 | PIN | .0081 | | | | | 1.0 | 2.0 | Rated Power = 1000W |
| 6.2 | Schottky Barrier and Point Contact (200 MHz ≤ frequency ≤ 35 GHz) | .027 | | | 1.0 | 1.0 | | | |
| 6.2 | Varactor | .0025 | | | 2.5 | 1.0 | | | Multiplier Application |
| 6.10 | Thyristor/SCR | .0022 | | | .51 | | | 1.0 | Voltage Stress = .7, Rated Forward Current = 1 Amp |
| | TRANSISTORS | | | | | | | | |
| 6.3 | NPN/PNP (f < 200 MHz) | .00074 | | | .21 | | .70 | .77 | Voltage Stress = .5, Switching Application, Rated Power = .5W |
| 6.3 | Power NPN/PNP (f < 200 MHz) | .00074 | | | .54 | | 1.5 | 5.5 | Voltage Stress = .8, Linear Application, Rated Power = 100W |
| 6.4 | Si FET (f ≤ 400 MHz) | .012 | | | | | .70 | | MOSFET, Small Signal Switching |
| 6.9 | Si FET (f > 400 MHz) | .060 | | | | | | | MOSFET |
| 6.8 | GaAs FET (P < 100 mW) | .052 | | 1.0 | | | 1.0 | | Low Noise Application, 1 ≤ f ≤ 10 GHz, Input and Output Matching |
| 6.8 | GaAs FET (P ≥ 100 mW) | .13 | | 1.0 | | | 1.0 | | CW Application, 5 GHz, 1W Average Output Power, Input and Output Matching |
| 6.5 | Unijunction | .0083 | | | | | | | |
| 6.6 | RF, Low Noise, Bipolar (f > 200 MHz, P < 1W) | .18 | | | .39 | | | .77 | Voltage Stress = .7, Rated Power = .5W |
| 6.7 | RF, Power (P ≥ 1W) | .08 | .36 | 1.0 | | | 1.6 | | 1 GHz, 100W, $T_J$ = 130°C for all Environments, Voltage Stress = .45, Gold Metallization Pulsed Application, 20% Duty Factor, Pulse Width = 5ms, Input and Output Matching |



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is procured in complete accordance with the applicable MIL-SPEC and Lower. If the part

**Default Parameters for Discrete Semiconductors**

| Section # | Part Type | $\lambda_b$ | $\pi_T$ | $\pi_M$ | $\pi_S$ | $\pi_C$ | $\pi_A$ | $\pi_R$ | Comments |
|---|---|---|---|---|---|---|---|---|---|
| | OPTO-ELECTRONICS | | | | | | | | |
| 6.11 | Photodetector | .0055 | | | | | | | Phototransistor |
| 6.11 | Opto-Isolator | .013 | | | | | | | Phototransistor, Single Device |
| 6.11 | Emitter | .00023 | | | | | | | LED |
| 6.12 | Alphanumeric Display | .0030 | | | | | | | 7 Character Segment Display |
| 6.13 | Laser Diode, GaAs/Al GaAs | 3.23 | | | 1.0 ($\pi_P$) | | .77 | | GaAs/Al GaAs, Hermetic, for Environments with $T_J$ > 75°C, assume $T_J$ = 75°C, Forward Peak Current = .5 Amps ($\pi_I$ = .62) Duty Cycle = .6, Pr/Ps = .5 ($\pi_P$ = 1) |
| 6.13 | Laser Diode, In/GaAs/In GaAsP | 5.65 | | | 1.0 ($\pi_P$) | | .77 | | GaAs/Al GaAs, Hermetic, for Environments with $T_J$ > 75°C, assume $T_J$ = 75°C, Forward Peak Current = .5 Amps ($\pi_I$ = .62) Duty Cycle = .6, Pr/Ps = .5 ($\pi_P$ = 1) |



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**Default Parameters for Resistors**

| Section # | Part Type | Style | MIL-R-SPEC | $\pi_R$ | $\pi_V$ | $\pi_{TAPS}$ | Comments |
|---|---|---|---|---|---|---|---|
| 9.1 | Composition | RCR | 39008 | 1.1 | | | Pwr. Stress = .5, 1M ohm |
| 9.1 | Composition | RC | 11 | 1.1 | | | Pwr. Stress = .5, 1M ohm |
| 9.2 | Film, Insulated | RLR | 39017 | 1.1 | | | Pwr. Stress = .5, 1M ohm |
| 9.2 | Film, Insulated | RL | 22684 | 1.1 | | | Pwr. Stress = .5, 1M ohm |
| 9.2 | Film, RN (R, C or N) | RNR | 55182 | 1.1 | | | Pwr. Stress = .5, 1M ohm |
| 9.2 | Film | RN | 10509 | 1.1 | | | Pwr. Stress = .5, 1M ohm |
| 9.3 | Film, Power | RD | 11804 | 1.0 | | | Pwr. Stress = .5, 100 ohm |
| 9.4 | Fixed, Network | RZ | 83401 | | | | Pwr. Stress = .5, $T_C = T_A + 28°C$, 10 Film Resistors |
| 9.5 | Wirewound, Accurate | RBR | 39005 | 1.7 | | | Pwr. Stress = .5,100K ohms |
| 9.5 | Wirewound, Accurate | RB | 93 | 1.7 | | | Pwr. Stress = .5, 100K ohms |
| 9.6 | Wirewound, Power | RWR | 39007 | 1.1 | | | Pwr. Stress = .5, 5K ohms, RWR 84 |
| 9.6 | Wirewound, Power | RW | 26 | 1.0 | | | Pwr. Stress = .5, 5K ohms, RW10 |
| 9.7 | Wirewound, Power, Chassis Mounted | RER | 39009 | 1.1 | | | Pwr. Stress = .5, Noninductively Wound, 5K ohm, RER 55 |
| 9.7 | Wirewound, Power, Chassis Mounted | RE | 18546 | 1.1 | | | Pwr. Stress = .5, MIL-R-18546, Char. N, 5K ohm, RE75 |
| 9.8 | Thermistor | RTH | 23648 | | | | Disk Type |
| 9.9 | Wirewound, Variable | RTR | 39015 | 1.4 | 1.1 | 1.0 | Pwr. Stress = .5, 5K ohms, 3 Taps, Voltage Stress = .5 |
| 9.9 | Wirewound, Variable | RT | 27208 | 1.4 | 1.1 | 1.0 | Pwr. Stress = .5, 3 Taps, Voltage Stress = .5 |
| 9.10 | Wirewound, Variable, Precision | RR | 12934 | 1.4 | 1.1 | 1.0 | Pwr. Stress = .5, Construction Class S ($\pi_C$ = 1.5), 50K ohm, 3 Taps, Voltage Stress = .5 |
| 9.11 | Wirewound, Variable, Semiprecision | RA | 19 | 1.4 | 1.0 | 1.0 | Pwr. Stress = .5, 5K ohms, 3 Taps, Voltage Stress = .5 |
| 9.11 | Wirewound, Semiprecision | RK | 39002 | 1.4 | 1.0 | 1.0 | Pwr. Stress = .5, 3 Taps, Voltage Stress = .5 |
| 9.12 | Wirewound, Variable, Power | RP | 22 | 1.4 | 1.0 | 1.0 | Pwr. Stress = .5, 3 Taps, Voltage Stress = .5, Unenclosed ($\pi_C$ = 1) |
| 9.13 | Nonwirewound, Variable | RJR | 39035 | 1.2 | 1.0 | 1.0 | Pwr. Stress = .5, 200K ohm, 3 Taps, Voltage Stress = .5 |
| 9.13 | Nonwirewound, Variable | RJ | 22097 | 1.2 | 1.0 | 1.0 | Pwr. Stress = .5, 200K ohm, 3 Taps, Voltage Stress = .5 |
| 9.14 | Composition, Variable | RV | 94 | 1.2 | 1.0 | 1.0 | Pwr. Stress = .5, 200K ohm, 3 Taps, Voltage Stress = .5 |
| 9.15 | Nonwirewound, Variable Precision | RQ | 39023 | 1.2 | 1.0 | 1.0 | Pwr. Stress = .5, 200K ohm, 3 Taps, Voltage Stress = .5 |
| 9.15 | Film, Variable | RVC | 23285 | 1.2 | 1.0 | 1.0 | Pwr. Stress = .5, 200K ohm, 3 Taps, Voltage Stress = .5 |



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**Default Parameters for Capacitors**

| Section # | Part Type or Dielectric | Style | MIL-C-SPEC | $\pi_{CV}$ | Temp. Rating | Comments |
|---|---|---|---|---|---|---|
| 10.1 | Paper, By-Pass | CP | 25 | 1.0 | 125 | Voltage Stress = .5, .15 µF |
| 10.1 | Paper, By-Pass | CA | 12889 | 1.0 | 85 | Voltage Stress = .5, .15 µF |
| 10.2 | Paper/Plastic, Feed-through | CZR | 11693 | 1.0 | 125 | Voltage Stress = .5, .061 µF |
| 10.3 | Paper/Plastic Film | CPV | 14157 | 1.0 | 125 | Voltage Stress = .5, .027 µF |
| 10.3 | Paper/Plastic Film | CQR | 19978 | 1.0 | 125 | Voltage Stress = .5, .033 µF |
| 10.4 | Metallized Paper/Plastic | CHR | 39022 | 1.0 | 125 | Voltage Stress = .5, .14 µF |
| 10.4 | Metallized Plastic/Plastic | CH | 18312 | 1.0 | 125 | Voltage Stress = .5, .14 µF |
| 10.5 | Metallized Paper/Plastic | CFR | 55514 | 1.0 | 125 | Voltage Stress = .5, .33 µF |
| 10.6 | Metallized Plastic | CRH | 83421 | 1.0 | 125 | Voltage Stress = .5, .14 µF |
| 10.7 | MICA (Dipped or Molded) | CMR | 39001 | 1.0 | 125 | Voltage Stress = .5, 300 pF |
| 10.7 | MICA (Dipped) | CM | 5 | 1.0 | 125 | Voltage Stress = .5, 300 pF |
| 10.8 | MICA (Button) | CB | 10950 | 1.0 | 150 | Voltage Stress = .5, 160 pF |
| 10.9 | Glass | CYR | 23269 | 1.0 | 125 | Voltage Stress = .5, 30 pF |
| 10.9 | Glass | CY | 11272 | 1.0 | 125 | Voltage Stress = .5, 30 pF |
| 10.10 | Ceramic (Gen. Purpose) | CK | 11015 | 1.0 | 125 | Voltage Stress = .5, 3300 pF |
| 10.10 | Ceramic (Gen. Purpose) | CKR | 39014 | 1.0 | 125 | Voltage Stress = .5, 3300 pF |
| 10.11 | Ceramic (Temp. Comp.) | CCR | 20 | 1.0 | 125 | Voltage Stress = .5, 81 pF |
| 10.11 | Ceramic Chip | CDR | 55681 | 1.0 | 125 | Voltage Stress = .5, 81 pF |
| 10.12 | Tantalum, Solid | CSR | 39003 | 1.0 | 125 | Voltage Stress = .5, 1.0 µF, .6 ohms/volt, series resistance, $\pi_{SR}$ = .13 |
| 10.13 | Tantalum, Non-Solid | CLR | 39006 | 1.0 | 125 | Voltage Stress = .5, Foil, Hermetic, 20 µF, $\pi_C$ = 1 |
| 10.13 | Tantalum, Non-Solid | CL | 3965 | 1.0 | 125 | Voltage Stress = .5, Foil, Hermetic, 20 µF, $\pi_C$ = 1 |
| 10.14 | Aluminum Oxide | CUR | 39018 | 1.3 | 125 | Voltage Stress = .5, 1700 µF |
| 10.15 | Aluminum Dry | CE | 62 | 1.3 | 85 | Voltage Stress = .5, 1600 µF |
| 10.16 | Variable, Ceramic | CV | 81 | | 85 | Voltage Stress = .5 |
| 10.17 | Variable, Piston | PC | 14409 | | 125 | Voltage Stress = .5 |
| 10.18 | Variable, Air Trimmer | CT | 92 | | 85 | Voltage Stress = .5 |
| 10.19 | Variable, Vacuum | CG | 23183 | | 85 | Voltage Stress = .5, Variable Configuration |



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## Default Parameters for Inductive and Electromechanical Parts

| Section # | Part Type | MIL-SPEC | $\pi_C$ | $\pi_{CYC}$ | $\pi_F$ | Comments |
|---|---|---|---|---|---|---|
| | INDUCTIVE | | | | | |
| 11.1 | Low Pwr. Pulsed, XFMR | MIL-T-21038 | | | | Max. Rated Temp. = 130°C, ΔT = 10 |
| 11.1 | Audio XFMR | MIL-T-27 | | | | Max. Rated Temp. = 130°C, ΔT = 10 |
| 11.1 | High Pwr. Pulse and Pwr. XFMR, Filter | MIL-T-27 | | | | Max. Rated Temp. = 130°C, ΔT = 30 |
| 11.1 | RF Transformers | MIL-T-55631 | | | | Max. Rated Temp. = 130°C, ΔT = 10 |
| 11.2 | RF Coils, Fixed or Molded | MIL-C-15305 | 1 | | | Max. Rated Temp. = 125°C, ΔT = 10 |
| 11.2 | RF Coils, Variable | MIL-C-15305 | 2 | | | Max. Rated Temp., = 125°C, ΔT = 10 |
| | ROTATING DEVICES | | | | | |
| 12.1 | Motors | | | | | t = 15,000 hours (Assumed Replacement Time) |
| 12.2 | Synchros | | | | | $T_F = T_A + 40$, Size 10 - 16, 3 Brushes |
| 12.2 | Resolvers | | | | | $T_F = T_A + 40$, Size 10 - 16, 3 Brushes |
| | Elapsed Time Meters (ETM) | | | | | |
| 12.3 | ETM-AC | | | | | Op. Temp/Rated Temp. = .5 ($\pi_T$ = .5) |
| 12.3 | ETM-Inverter Driver | | | | | Op. Temp/Rated Temp. = .5 ($\pi_T$ = .5) |
| 12.3 | ETM-Commutater DC | | | | | Op. Temp/Rated Temp. = .5 ($\pi_T$ = .5) |
| | RELAYS | | | | | |
| 13.1 | General Purpose | | 3 | 1 | 5 | Max. Rated Temp. = 125°C , DPDT, MIL-SPEC, 10 Cycles/Hour, 4 Amp., General Purpose, Balanced Armature, Resistive Load, s = .5 |
| 13.1 | Contactor, High Current | | 3 | 1 | 5 | Max. Rated Temp. = 125°C, DPDT, MIL-SPEC, 10 Cycles/Hour, 600 Amp., Solenoid, Inductive Load, s = .5 |
| 13.1 | Latching | | 3 | 1 | 5 | Max. Rated Temp. = 125°C, MIL-SPEC, 4 Amp., Mercury Wetted, 10 Cyles/Hour, DPDT, Resistive Load, s = .5 |
| 13.1 | Reed | | 1 | 2 | 6 | Max. Rated Temp. = 85°C, MIL-SPEC, Signal Current, Dry Reed, 20 Cycles/Hour, SPST, Resistive Load, s = .5 |
| 13.1 | Thermal Bi-Metal | | 1 | 1 | 10 | Max. Rated Temp. = 125°C, MIL-SPEC, Bi-Metal, 10 Cycles/Hour, SPST, Inductive Load, 5 Amp., s = .5 |
| 13.1 | Meter Movement | | 1 | 1 | 100 | Max. Rated Temp. = 125°C, MIL-SPEC, Polarized Meter Movement, 10 Cycles/Hour, SPST, Resistive Load, s = .5 |
| 13.2 | Solid State | MIL-R-28750 | | | | No Defaults |
| 13.2 | Time Delay Hybrid and Solid State | MIL-R-83726 | | | | No Defaults |

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**Default Parameters for Inductive and Electromechanical Parts**

| Section # | Part Type | MIL-SPEC | $\lambda_b$ | $\pi_U$ | $\pi_C$ | $\pi_{cyc}$ | $\pi_L$ | $\pi_p$ | Comments |
|---|---|---|---|---|---|---|---|---|---|
| | SWITCHES | | | | | | | | |
| 14.1 | Toggle & Pushbutton | | .00045 | | 1.5 | 1.0 | 1.48 | | Snap-action, MIL-SPEC, ≤ 1 Cycle/Hour, Resistive Load, Current Stress = .5, DPST |
| 14.2 | Sensitive | MIL-S-8805 | .10 | | | 1.0 | 1.48 | | Actuation Differential > .002 inches, 1 Active Contact, MIL-SPEC, ≤ 1 Cycle/Hour, Resistive Load, Current Stress = .5 |
| 14.3 | Rotary Wafer | MIL-S-3786 | .0074 | | | 30 | 1.48 | | MIL-SPEC, Resistive Load, Current Stress = .5, 30 Cycles/Hour, 24 Active Contact |
| 14.4 | Thumbwheel | MIL-S-22710 | .38 | | | 1.0 | 1.48 | | MIL-SPEC, Resistive Load, .Current Stress = .5, ≤ 1 Cycle/Hour, 6 Active Contacts |
| 14.5 | Circuit Breaker, Thermal | MIL-C-83383 | .038 | 1.0 | 3.0 | | | | 3PST, Not Used as a Power On/Off Switch |
| 14.5 | Circuit Breaker, Magnetic | MIL-C-55629 | .020 | 1.0 | 3.0 | | | | 3PST, Not Used as a Power On/Off Switch |
| | CONNECTORS | | | | | | | | |
| 15.1 | Circular/Rack/Panel | | | | | | | 7.4 | $T_o = T_A + 10°C$, Insert Material B, 3 Mating/Unmating Cycles per 1000 Hours, 40 Active Contacts, MIL-SPEC $\pi_E$ |
| 15.1 | Coaxial | | | | | | | 1.4 | $T_o = T_A + 5°C$, Insert Material C, 3 Mating/Unmating Cycles per 1000 Hours, 2 Active Contacts, MIL-SPEC $\pi_E$ |
| 15.2 | Printed Circuit Board | | | | | | | 7.4 | $T_o = T_A + 10°C$, 3 Mating/Unmating Cycles per 1000 Hours, 40 Active Pins, MIL-SPEC $\pi_E$ |
| 15.3 | IC Sockets | | .00042 | | | | | 4.6 | 24 Active Contacts |
| 16.1 | Interconnection Assemblies (PCBs) | | .000041 | | | | | | Printed Wiring Assembly, 1000 Wave Soldered Functional PTHs, 3 Circuit Planes, No Hand Soldering, $\pi_E$ |



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**Default Parameters for Miscellaneous Parts**

| Section # | Part Type | MIL-SPEC | $\lambda_b$ | $\pi_U$ | $\pi_A$ | Comments |
|---|---|---|---|---|---|---|
| 17.1 | Connections | | | | | No Defaults |
| 18.1 | Meters, Panel | | | | | No Defaults |
| 19.1 | Quartz Crystals | MIL-C-3098 | .032 | | | 50 MHz |
| | LAMPS, INCANDESCENT | | | | | |
| 20.1 | AC Applications | | 5.4 | .72 | 1 | Rated Voltage 28 Volts, Utilization Rate .5, Alternating Current |
| 20.1 | DC Applications | | 5.4 | .72 | 3.3 | Rated Voltage 28 Volts, Utilization Rate .5, Direct Current |
| | ELECTRONIC FILTERS | | | | | |
| 21.1 | Ceramic-Ferrite | MIL-F-15733 | .022 | | | MIL-SPEC |
| 21.1 | Discrete LC Comp | MIL-F-15733 | .12 | | | MIL-SPEC |
| 21.1 | Discrete LC & Crystal Comp. | MIL-F-18327 | .27 | | | MIL-SPEC |
| 22.1 | FUSES | | .010 | | | |



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## APPENDIX B: VHSIC/VHSIC-LIKE AND VLSI CMOS (DETAILED MODEL)

This appendix contains the detailed version of the VHSIC/VLSI CMOS model contained in Section 5.3. It is provided to allow more detailed device level design trade-offs to be accomplished for predominate failure modes and mechanisms exhibited in CMOS devices. Reference 30 should be consulted for a detailed derivation of this model.

### VHSIC/VHSIC-LIKE FAILURE RATE MODEL

$$\lambda_P(t) = \lambda_{OX}(t) + \lambda_{MET}(t) + \lambda_{HC}(t) + \lambda_{CON}(t) + \lambda_{PAC} + \lambda_{ESD} + \lambda_{MIS}(t)$$

$\lambda_P(t)$ = Predicted Failure Rate as a Function of Time

$\lambda_{OX}(t)$ = Oxide Failure Rate

$\lambda_{MET}(t)$ = Metallization Failure Rate

$\lambda_{HC}(t)$ = Hot Carrier Failure Rate

$\lambda_{CON}(t)$ = Contamination Failure Rate

$\lambda_{PAC}$ = Package Failure Rate

$\lambda_{ESD}$ = EOS/ESD Failure Rate

$\lambda_{MIS}(t)$ = Miscellaneous Failure Rate

The equations for each of the above failure mechanism failure rates are as follows:

### OXIDE FAILURE RATE EQUATION

$$\lambda_{OX}\ (\text{in } F/10^6) = \frac{A\, A_{TYPE_{OX}}}{A_R}\left(\frac{D_{0_{OX}}}{D_R}\right)\left[(.0788\, e^{-7.7\, t_0})\,(A_{T_{OX}})\,(e^{-7.7\, A_{T_{OX}} t}) + \frac{.399}{(t+t_0)\sigma_{OX}} \exp\left(\frac{-.5}{\sigma_{OX}^2}\left(\ln\,(t + t_0) - \ln t_{50_{OX}}\right)^2\right)\right]$$

$A$ = Total Chip Area (in $cm^2$)

$A_{TYPE_{OX}}$ = .77 for Custom and Logic Devices, 1.23 for Memories and Gate Arrays



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## APPENDIX B: VHSIC-VHSIC-LIKE AND VLSI CMOS (DETAILED MODEL)

### OXIDE FAILURE RATE EQUATION (CONTINUED)

$A_R$ = .21 $cm^2$

$D0_{ox}$ = Oxide Defect Density (If unknown, use $\left(\frac{X_0}{X_S}\right)^2$ where $X_0$ = 2 µm and $X_S$ is the feature size of the device)

$D_R$ = 1 Defect/$cm^2$

$t_0$ = Effective Screening Time

= (Actual Time of Test (in $10^6$ hrs.)) * ($A_{T_{ox}}$ (at junction screening temp.) (in °K))*

$A_{T_{ox}}$ = Temperature Acceleration Factor, = $\exp\left[\frac{-.3}{8.617\text{x}10^{-5}}\left(\frac{1}{T_J} - \frac{1}{298}\right)\right]$

(where $T_J = T_C + \theta_{JC}P$ (in °K))

$A_{V_{ox}}$ = $e^{-192\left(\frac{1}{E_{ox}} - \frac{1}{2.5}\right)}$

$E_{ox}$ = Maximum Power Supply Voltage $V_{DD}$, divided by the gate oxide thickness (in MV/cm)

$t_{50_{ox}}$ = $\frac{1.3\text{x}10^{22}\ (QML)}{A_{T_{ox}}\ A_{V_{ox}}}$ (in $10^6$ hrs.)

(QML) = 2 if on QML, .5 if not.

$\sigma_{ox}$ = Sigma obtained from test data of oxide failures from the same or similar process. If not available, use a $\sigma_{ox}$ value of 1.

t = time (in $10^6$ Hours)



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## APPENDIX B: VHSIC/VHSIC-LIKE AND VLSI CMOS (DETAILED MODEL)

### METAL FAILURE RATE EQUATION

$$\lambda_{MET} = \left[\frac{A\, A_{TYPE_{MET}}}{A_R}\frac{D_{0_{MET}}}{D_R}\ (.00102\, e^{-1.18\, t_0})(A_{T_{MET}})(e^{-1.18\, A_{T_{MET}} t})\right]$$

$$+\left[\frac{.399}{(t+t_0)\sigma_{MET}}\exp\left(\frac{-.5}{\sigma_{MET}^2}\left(\ln(t+t_0) - \ln t_{50_{MET}}\right)^2\right)\right]$$

$A$ = Total Chip Area (in $cm^2$)

$A_{TYPE_{MET}}$ = .88 for Custom and Logic Devices, 1.12 for Memory and Gate Arrays

$A_R$ = .21 $cm^2$

$D_{0_{MET}}$ = Metal Defect Density (If unknown use $(\frac{X_0}{X_S})^2$ where $X_0 = 2\ \mu m$ and $X_S$ is the feature size of the device)

$D_R$ = 1 Defect/$cm^2$

$A_{T_{MET}}$ = Temperature Acceleration Factor

$$= \exp\left[\frac{-.55}{8.617\times10^{-5}}\left(\frac{1}{T_J} - \frac{1}{298}\right)\right] \left(T_J = T_{CASE} + \theta_{JC} P \quad (\text{in } ^\circ K)\right)$$

$t_0$ = Effective Screening Time (in $10^6$ hrs.)

= $A_{T_{MET}}$ (at Screening Temp. (in °K)) * (Actual Screening Time (in $10^6$ hrs))

$t_{50_{MET}}$ = $(QML)\frac{.388 * (\text{Metal Type})}{J^2 A_{T_{MET}}}$ (in $10^6$ hrs.)

(QML) = 2 if on QML, .5 if not.

Metal Type = 1 for Al, 37.5 for Al-Cu or for Al-Si-Cu

$J$ = The mean absolute value of Metal Current Density (in $10^6$ Amps/$cm^2$)

$\sigma_{MET}$ = sigma obtained from test data on electromigration failures from the same or a similar process. If this data is not available use $\sigma_{MET} = 1$.

$t$ = time (in $10^6$ hrs.)



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## APPENDIX B: VHSIC-VHSIC-LIKE AND VLSI CMOS (DETAILED MODEL)

### HOT CARRIER FAILURE RATE EQUATION

$$\lambda_{HC} = \frac{.399}{(t+t_0)\sigma_{HC}} \exp\left[\frac{-.5}{\sigma_{HC}^2}\left(\ln (t + t_0) - \ln t_{50_{HC}}\right)^2\right]$$

$$t_{50_{HC}} = \frac{(QML)3.74\times10^{-5}}{A_{T_{HC}} I_d}\left(\frac{I_{sub}}{I_d}\right)^{-2.5}$$

(QML) = 2 if on QML, .5 if not

$$A_{T_{HC}} = \exp\left[\frac{.039}{8.617\times10^{-5}}\left(\frac{1}{T_J} - \frac{1}{298}\right)\right] \text{ (where } T_J = T_C + \theta_{JC}P \text{ (in °K))}$$

$I_d$ = Drain Current at Operating Temperature. If unknown use $I_d = 3.5\, e^{-.00157\, T_J \text{ (in °K)}}$ (mA)

$I_{sub}$ = Substrate Current at Operating Temperature. If unknown use $I_{sub} = .0058\, e^{-.00689\, T_J \text{ (in °K)}}$ (mA)

$\sigma_{HC}$ = sigma derived from test data, if not available use 1.

$t_0$ = $A_{T_{HC}}$ (at Screening Temp.(in °K)) • (Test Duration in $10^6$ hours)

$t$ = time (in $10^6$ hrs.)

### CONTAMINATION FAILURE RATE EQUATION

$$\lambda_{CON} = .000022\, e^{-.0028\, t_0 A_{T_{CON}}}\, e^{-.0028\, A_{T_{CON}}\, t}$$

$$A_{T_{CON}} = \exp\left[\frac{-1.0}{8.617\times10^{-5}}\left(\frac{1}{T_J} - \frac{1}{298}\right)\right] \text{ (where } T_J = T_C + \theta_{JC}P \text{ (in °K))}$$

$t_0$ = Effective Screening Time

= $A_{T_{con}}$ (at screening junction temperature (in °K)) • (actual screening time in $10^6$ hrs.)

$t$ = time (in $10^6$ hrs.)



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## APPENDIX B: VHSIC/VHSIC-LIKE AND VLSI CMOS (DETAILED MODEL)

### PACKAGE FAILURE RATE EQUATION

$$\lambda_{PAC} = (.0024 + 1.85 \times 10^{-5} (\#Pins))\, \pi_E\, \pi_Q\, \pi_{PT} + \lambda_{PH}$$

$\pi_E$ = See Section 5.10

$\pi_Q$ = See Section 5.10

Package Type Factor ($\Pi_{PT}$)

| Package Type | $\Pi_{PT}$ |
|---|---|
| DIP | 1.0 |
| Pin Grid Array | 2.2 |
| Chip Carrier (Surface Mount Technology) | 4.7 |

$\lambda_{PH}$ = Package Hermeticity Factor

$\lambda_{PH}$ = 0 for Hermetic Packages

$$\lambda_{PH} = \frac{.399}{t\sigma_{PH}} \exp\left[\frac{-.5}{\sigma_{PH}^2}\left(\ln(t) - \ln(t_{50_{PH}})\right)^2\right] \text{ for plastic packages}$$

$$t_{50_{PH}} = 86 \times 10^{-6} \exp\left[\frac{.2}{8.617 \times 10^{-5}}\left(\frac{1}{T_A} - \frac{1}{298}\right)\right] \exp\left[\frac{2.96}{RH_{EFF}}\right]$$

$T_A$ = Ambient Temp. (in °K)

$$RH_{eff} = (DC)(RH)\left[e^{5230\left(\frac{1}{T_J} - \frac{1}{T_A}\right)}\right] + (1\text{-}DC)(RH) \quad \text{where } T_J = T_C + \theta_{JC}P \text{ (in °K)}$$

(for example, for 50% Relative Humidity, use RH = .50)

$\sigma_{PH}$ = .74

t = time (in $10^6$ hrs.)



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## APPENDIX B: VHSIC-VHSIC-LIKE AND VLSI CMOS (DETAILED MODEL)

### EOS/ESD FAILURE RATE EQUATION

$$\lambda_{EOS} = \frac{-\ln\left(1 - .00057\, e^{-.0002\, V_{TH}}\right)}{.00876}$$

$V_{TH}$ = ESD Threshold of the device using a 100 pF, 1500 ohm discharge model

### MISCELLANEOUS FAILURE RATE EQUATION

$$\lambda_{MIS} = (.01\, e^{-2.2\, t_o})\,(A_{T_{MIS}})\,(e^{-2.2\, A_{T_{MIS}}\, t})$$

$A_{T_{MIS}}$ = Temperature Acceleration Factor

$$= \exp\left[\frac{-.423}{8.6317 \times 10^{-5}}\left(\frac{1}{T_J} - \frac{1}{298}\right)\right]$$

where $T_J = T_C + \theta_{JC} P$ (in °K)

$t_o$ = Effective Screening Time

= $A_{T_{MIS}}$ (at Screening Temp. (in °K)) * Actual Screening Time (in $10^6$ hours)

$t$ = time (in $10^6$ hrs.)



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The year of publication of the Rome Laboratory (RL) (formerly Rome Air Development Center (RADC)) documents is part of the RADC (or RL) number, e.g., RADC-TR-88-97 was published in 1988.

1. "Laser Reliability Prediction," RADC-TR-75-210, AD A016437.

2. "Reliability Model for Miniature Blower Motors Per MIL-B-23071B," RADC-TR-75-178, AD A013735.

3. "High Power Microwave Tube Reliability Study," FAA-RD-76-172, AD A0033612.

4. "Electric Motor Reliability Model," RADC-TR-77-408, AD A050179.

5. "Development of Nonelectronic Part Cyclic Failure Rates," RADC-TR-77-417, AD A050678.

   This study developed new failure rate models for relays, switches, and connectors.

6. "Passive Device Failure Rate Models for MIL-HDBK-217B," RADC-TR-77-432, AD A050180.

   This study developed new failure rate models for resistors, capacitors and inductive devices.

7. "Quantification of Printed Circuit Board Connector Reliability," RADC-TR-77-433, AD A049980.

8. "Crimp Connection Reliability," RADC-TR-78-15, AD A050505.

9. "LSI/Microprocessor Reliability Prediction Model Development," RADC-TR-79-97, AD A068911.

10. "A Redundancy Notebook," RADC-TR-77-287, AD A050837.

11. "Revision of Environmental Factors for MIL-HDBK-217B," RADC-TR-80-299, AD A091837.



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12. "Traveling Wave Tube Failure Rates," RADC-TR-80-288, AD A096055.

13. "Reliability Prediction Modeling of New Devices," RADC-TR-80-237, AD A090029.

This study developed failure rate models for magnetic bubble memories and charge-coupled memories.

14. "Failure Rates for Fiber Optic Assemblies," RADC-TR-80-322, AD A092315.

15. "Printed Wiring Assembly and Interconnection Reliability," RADC-TR-81-318, AD A111214.

This study developed failure rate models for printed wiring assemblies, solderless wrap assemblies, wrapped and soldered assemblies and discrete wiring assemblies with electroless deposited plated through holes.

16. "Avionic Environmental Factors for MIL-HDBK-217," RADC-TR-81-374, AD B064430L.

17. "RADC Thermal Guide for Reliability Engineers," RADC-TR-82-172, AD A118839.

18. "Reliability Modeling of Critical Electronic Devices," RADC-TR-83-108, AD A135705.

This report developed failure rate prediction procedures for magnetrons, vidicions, cathode ray tubes, semiconductor lasers, helium-cadmium lasers, helium-neon lasers, Nd: YAG lasers, electronic filters, solid state relays, time delay relays (electronic hybrid), circuit breakers, I.C. Sockets, thumbwheel switches, electromagnetic meters, fuses, crystals, incandescent lamps, neon glow lamps and surface acoustic wave devices.

19. "Impact of Nonoperating Periods on Equipment Reliability," RADC-TR-85-91, AD A158843.

This study developed failure rate models for nonoperating periods.

20. "RADC Nonelectronic Reliability Notebook," RADC-TR-85-194, AD A163900.

This report contains failure rate data on mechanical and electromechanical parts.

21. "Reliability Prediction for Spacecraft," RADC-TR-85-229, AD A149551.

This study investigated the reliability performance histories of 300 Satellite vehicles and is the basis for the halving of all model $\pi_E$ factors for MIL-HDBK-217E to MIL-HDKB-217E, Notice 1.

22. "Surface Mount Technology: A Reliability Review," 1986, Available from Reliability Analysis Center, PO Box 4700, Rome, NY 13440-8200, 800-526-4802.

23. "Thermal Resistances of Joint Army Navy (JAN) Certified Microcircuit Packages," RADC-TR-86-97, AD B108417.

24. "Large Scale Memory Error Detection and Correction," RADC-TR-87-92, AD B117765L.

This study developed models to calculate memory system reliability for memories incorporating error detecting and correcting codes. For a summary of the study see 1989 IEEE Reliability and Maintainability Symposium Proceedings, page 197, "Accounting for Soft Errors in Memory Reliability Prediction."

25. "Reliability Analysis of a Surface Mounted Package Using Finite Element Simulation," RADC-TR-87-177, AD A189488.



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26. "VHSIC Impact on System Reliability," RADC-TR-88-13, AD B122629.

27. "Reliability Assessment of Surface Mount Technology," RADC-TR-88-72, AD A193759.

28. "Reliability Prediction Models for Discrete Semiconductor Devices," RADC-TR-88-97, AD A200529.

This study developed new failure rate prediction models for GaAs Power FETS, Transient Suppressor Diodes, Infrared LEDs, Diode Array Displays and Current Regulator Diodes.

29. "Impact of Fiber Optics on System Reliability and Maintainability," RADC-TR-88-124, AD A201946.

30. "VHSIC/VHSIC Like Reliability Prediction Modeling," RADC-TR-89-171, AD A214601.

This study provides the basis for the VHSIC model appearing in MIL-HDBK-217F, Section 5.

31. "Reliability Assessment Using Finite Element Techniques," RADC-TR-89-281, AD A216907.

This study addresses surface mounted solder interconnections and microwire board's plated-thru-hole (PTH) connections. The report gives a detailed account of the factors to be considered when performing an FEA and the procedure used to transfer the results to a reliability figure-of-merit.

32. "Reliability Analysis/Assessment of Advanced Technologies," RADC-TR-90-72, ADA 223647.

This study provides the basis for the revised microcircuit models (except VHSIC and Bubble Memories) appearing in MIL-HDBK-217F, Section 5.

33. "Improved Reliability Prediction Model for Field-Access Magnetic Bubble Devices," AFWAL-TR-81-1052.

34. "Reliability/Design Thermal Applications," MIL-HDBK-251.

35. "NASA Parts Application Handbook," MIL-HDBK-978-B (NASA).
This handbook is a five volume series which discusses a full range of electrical, electronic and electromechanical component parts. It provides extensive detailed technical information for each component part such as: definitions, construction details, operating characteristics, derating, failure mechanisms, screening techniques, standard parts, environmental considerations, and circuit application.

36. "Nonelectronic Parts Reliability Data 1991," NPRD-91.
This report contains field failure rate data on a variety of electrical, mechanical, electromechanical and microwave parts and assemblies (1400 different part types). It is available from the Reliability Analysis Center, PO Box 4700, Rome, NY 13440-8200, Phone: (315) 337-0900.

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Preparing Activity:
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User Activities:
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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instruction - Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-HDBK-217F | Reliability Prediction of Electronic Equipment |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| | ☐ USER |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

**5. PROBLEM AREAS**

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

**6. REMARKS**

| 7a. NAME OF SUBMITTER *(Last, First, MI)* - Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* - Optional |
|---|---|
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